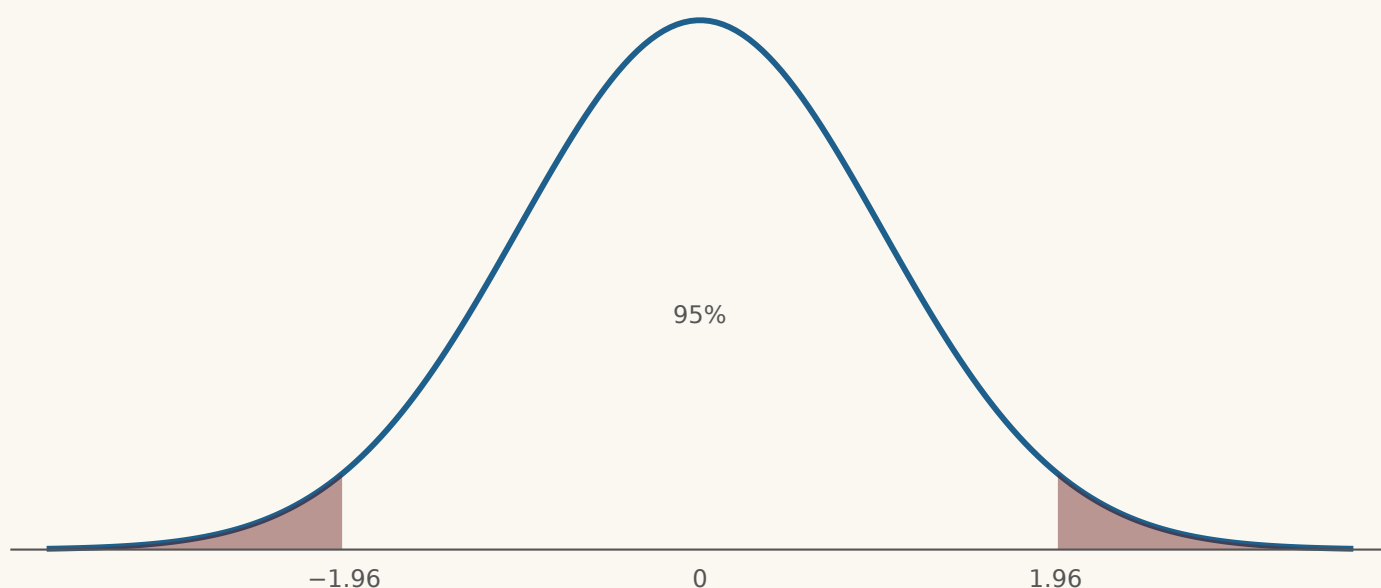


Probability & Statistics

Mock Test Series · 10 Full Tests · 300 Questions

MCQ · MSQ · NAT in the GATE DA pattern, with fully worked solutions



Inside this book

- Part A — Formula handbook & concept review with diagrams
- Part B — 10 timed mock tests (30 questions, 48 marks each), answer keys & detailed solutions
- Part C — Standard normal, t and χ^2 tables · performance tracker
- Mapped to the official GATE DA syllabus, Section 1: Probability and Statistics

Contents

How to Use This Book	3
Syllabus Coverage Map	5
Part A · Formula Handbook & Concept Review	7
Part B · Mock Tests with Answer Keys and Detailed Solutions	19
Mock Test 01 — Foundation Full-Syllabus Test I	21
Mock Test 02 — Foundation Full-Syllabus Test II	47
Mock Test 03 — Standard GATE-Level Test I (Data-Science Contexts)	71
Mock Test 04 — Standard GATE-Level Test II (Data-Science Contexts)	97
Mock Test 05 — Topic Test: Probability, Counting, Bayes & Conditional Expectation	122
Mock Test 06 — Topic Test: Random Variables, Distributions, PDFs & CDFs	146
Mock Test 07 — Topic Test: Sampling, CLT, Confidence Intervals & Hypothesis Testing	171
Mock Test 08 — Topic Test: Descriptive Statistics, Covariance, Correlation & Mixed Review	196
Mock Test 09 — Advanced Full-Syllabus Test I	221
Mock Test 10 — Advanced Full-Syllabus Test II (Grand Mock)	247
Part C · Appendix: Statistical Tables	270
C1. Standard normal CDF	270
C2. Student t critical values	271
C3. Chi-squared critical values	272
Overall Performance Tracker	273

How to Use This Book

This book contains **10 mock tests with 300 original questions (480 marks)** dedicated to Section 1 of the GATE DA syllabus — Probability and Statistics — the section that also underpins much of the Machine Learning and AI sections. Every question follows the GATE question formats (MCQ, MSQ and NAT, with 1-mark and 2-mark weightage) and comes with a fully worked solution.

The GATE DA examination at a glance

Feature	Details
Duration	3 hours, computer-based test
Questions / marks	65 questions, 100 marks
General Aptitude (GA)	10 questions, 15 marks (5 × 1-mark, 5 × 2-mark)
Subject (DA)	55 questions, 85 marks (≈ 25 × 1-mark, 30 × 2-mark)
MCQ	4 options, exactly one correct; negative marks: $-1/3$ (1-mark), $-2/3$ (2-mark)
MSQ	4 options, ONE OR MORE correct; full marks only for the exact set; no negative, no partial marking
NAT	Numerical answer typed on a virtual keypad; accepted within a range; no negative
Calculator	On-screen virtual scientific calculator only

Probability & Statistics is one of the heaviest-weighted subject sections of the DA paper, and its ideas reappear in the Machine Learning (naive Bayes, LDA, bias-variance) and AI (reasoning under uncertainty) sections. A strong score here lifts the whole paper.

Structure of each mock test

- **30 questions, 48 marks:** Q1-Q12 carry 1 mark, Q13-Q30 carry 2 marks — roughly the proportion of 1- and 2-mark questions in the real paper.
- Mixture of about 14 MCQ, 7 MSQ and 9 NAT questions per test.
- Suggested time: 90 minutes — the same time per mark as the real exam (≈ 1.8 min per mark).
- Every test is followed by an **answer key**, a **score tracker** and **detailed solutions** that state the concept, show every step, analyse each MSQ option and flag traps and shortcuts.

Scoring rules used in this book

$$\text{Score} = \sum_{\text{correct}} m_i - \sum_{\text{wrong MCQ}} \frac{m_i}{3}$$

MSQ answers are marked correct only when the selected set matches the key exactly. NAT answers are correct when they fall in the accepted range shown in the key.

Suggested 5-week plan

Week	Work	Goal
1	Part A §A1–A5; Mock 01; review every solution	Foundations & formats

Week	Work	Goal
2	Part A §A6–A9; Mock 02, Mock 05	Counting, Bayes, conditional expectation
3	Part A §A10–A11; Mock 03, Mock 06	Distributions, joint pdfs, CDFs
4	Part A §A12–A14; Mock 04, Mock 07, Mock 08	Inference & descriptive stats
5	Mock 09, Mock 10 under strict timing; redo wrong questions	Exam readiness

Exam-day strategy for this section

- First pass: attempt all 1-mark questions and every 2-mark question you can finish in under 3 minutes.
- Guess on MCQs only if you can eliminate at least two options (expected value of a blind guess is zero: $\frac{1}{4} \cdot 1 - \frac{3}{4} \cdot \frac{1}{3} = 0$).
- MSQs carry no negative marking — never leave one blank, but tick only options you can justify.
- NAT: keep at least 4 significant figures in intermediate steps; round only at the end as instructed.
- Sketch: a Venn diagram, a tree, the support of a joint pdf or the rejection region costs 20 seconds and prevents most errors.

Syllabus Coverage Map

Number of questions in each mock test by syllabus area (classified by each question's primary topic; most questions also use ideas from neighbouring areas).

Syllabus area	M01	M02	M03	M04	M05	M06	M07	M08	M09	M10	Total
Counting	2	3	3	3	9	1	0	4	3	4	32
Cond. expectation & variance	2	1	1	2	5	2	0	1	3	3	20
Bayes / conditional / joint prob.	5	4	5	3	10	2	0	2	3	3	37
Axioms, events, independence	3	4	2	3	4	1	2	1	1	2	23
Covariance & correlation	2	0	1	2	0	1	0	8	1	2	17
Hypothesis tests (z , t , χ^2)	3	3	3	2	0	0	13	0	5	3	32
Confidence intervals	1	1	3	1	0	0	6	0	2	1	15
CLT & sampling	2	2	1	2	0	0	6	0	0	2	15
Discrete distributions	3	2	5	3	1	6	0	2	3	5	30
Continuous distributions, PDF/CDF	5	6	5	7	0	17	3	5	8	5	61
Descriptive statistics	2	4	1	2	1	0	0	7	1	0	18
Total	30	30	30	30	30	30	30	30	30	30	300

Mock test overview

Test	Title	Types (MCQ/MSQ/NAT)	Marks
01	Foundation Full-Syllabus Test I	14 / 7 / 9	48
02	Foundation Full-Syllabus Test II	14 / 7 / 9	48
03	Standard GATE-Level Test I (Data-Science Contexts)	14 / 7 / 9	48
04	Standard GATE-Level Test II (Data-Science Contexts)	14 / 7 / 9	48
05	Topic Test: Probability, Counting, Bayes & Conditional Expectation	14 / 7 / 9	48
06	Topic Test: Random Variables, Distributions, PDFs & CDFs	14 / 7 / 9	48
07	Topic Test: Sampling, CLT, Confidence Intervals & Hypothesis Testing	14 / 7 / 9	48
08	Topic Test: Descriptive Statistics, Covariance, Correlation & Mixed Review	14 / 7 / 9	48
09	Advanced Full-Syllabus Test I	14 / 7 / 9	48
10	Advanced Full-Syllabus Test II (Grand Mock)	14 / 7 / 9	48

Official syllabus — Section 1: Probability and Statistics

GATE DA 2027 syllabus: Counting (permutation and combinations), probability axioms, sample space, events, independent events, mutually exclusive events, marginal, conditional and joint probability, Bayes Theorem, conditional expectation and variance, mean, median, mode and standard deviation, correlation, and covariance, random variables, discrete random variables and probability mass functions, uniform, Bernoulli, binomial distribution, continuous random variables and probability distribution function, uniform, exponential, Poisson, normal, standard normal, t-distribution, chi-squared distributions, cumulative distribution function, conditional PDF, central limit theorem, confidence interval, z-test, t-test, chi-squared test.

Part A · Formula Handbook & Concept Review

This handbook condenses every syllabus item of GATE DA Section 1 into formulas, diagrams and the traps that examiners exploit. Revise it before each mock and return to it whenever a solution refers to a result you do not remember.

A1. Counting: permutations and combinations

Situation	Count	Example
Ordered, no repetition (arrange r of n)	${}^n P_r = n!/(n-r)!$	Podium of 3 from 8: 336
Unordered, no repetition (choose r of n)	${}^n C_r = n!/(r!(n-r)!)$	Committee of 3 from 8: 56
Ordered, with repetition	n^r	4-digit PINs: 10^4
Unordered, with repetition (multisets)	${}^{n+r-1} C_r$	3 scoops from 5 flavours: 35
Arrangements with identical items	$n!/(n_1! n_2! \dots n_k!)$	MISSISSIPPI: 34650
Circular arrangements of n distinct	$(n-1)!$	6 people at round table: 120
Integer solutions $x_1 + \dots + x_k = n$, $x_i \geq 0$	${}^{n+k-1} C_{k-1}$	$x+y+z=5$: 21
Derangements of n	$D_n = n! \sum (-1)^k/k!$	$D_4 = 9$, $D_n/n! \rightarrow 1/e$

Multiplication principle: a task in k stages with $n_1, \dots, n_k$ choices gives $n_1 n_2 \dots n_k$ outcomes.

Addition principle: disjoint alternatives add.

$$\binom{n}{r} = \binom{n}{n-r}, \quad \binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}, \quad \sum_{r=0}^n \binom{n}{r} = 2^n$$

Inclusion-exclusion:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Hypergeometric probability (drawing without replacement): choosing n from N items of which K are 'successes',

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

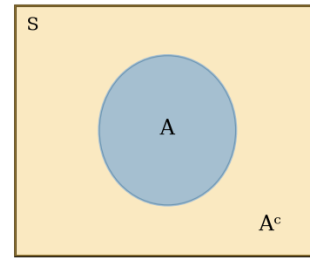
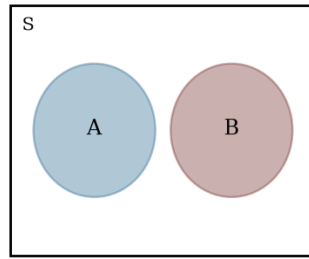
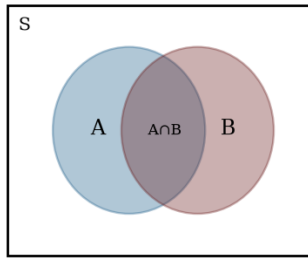
Trap: Decide first whether ORDER matters and whether REPETITION is allowed. Most counting errors come from double-counting ordered selections when the question asks for unordered groups (divide by $r!$) or from forgetting identical objects.

A2. Sample space, events and the axioms of probability

A **sample space** S is the set of all outcomes; an **event** is a subset of S. Kolmogorov's axioms:

(i) $P(A) \geq 0$, (ii) $P(S) = 1$, (iii) for pairwise disjoint $A_1, A_2, \dots$, $P(\cup A_i) = \sum P(A_i)$.

- $P(\emptyset) = 0$, $P(A^c) = 1 - P(A)$, $A \subset B \Rightarrow P(A) \leq P(B)$.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (addition rule).
- Boole: $P(A \cup B) \leq P(A) + P(B)$; Bonferroni: $P(A \cap B) \geq P(A) + P(B) - 1$.
- De Morgan: $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$.
- Equally likely outcomes: $P(A) = |A| / |S|$.

Overlapping: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ Mutually exclusive: $A \cap B = \emptyset$ Complement: $P(A^c) = 1 - P(A)$ 

Independent vs mutually exclusive

Property	Independent	Mutually exclusive
Definition	$P(A \cap B) = P(A)P(B)$	$A \cap B = \emptyset$, so $P(A \cap B) = 0$
$P(A \cup B)$	$P(A) + P(B) - P(A)P(B)$	$P(A) + P(B)$
$P(A B)$	$P(A)$	0
Both at once?	Only if $P(A) = 0$ or $P(B) = 0$	—
Complements	A^c, B and A, B^c also independent	Not preserved

Mutual independence of A, B, C requires all pairwise product rules AND $P(A \cap B \cap C) = P(A)P(B)P(C)$. Pairwise independence does not imply mutual independence (classic example: two fair coins, A = first head, B = second head, C = exactly one head).

Trap: Disjoint events with positive probability are always DEPENDENT: knowing A occurred tells you B did not.

A3. Joint, marginal and conditional probability; Bayes' theorem

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0 \quad P(A \cap B) = P(B)P(A | B)$$

Chain rule:

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2 | A_1)P(A_3 | A_1 \cap A_2) \dots$$

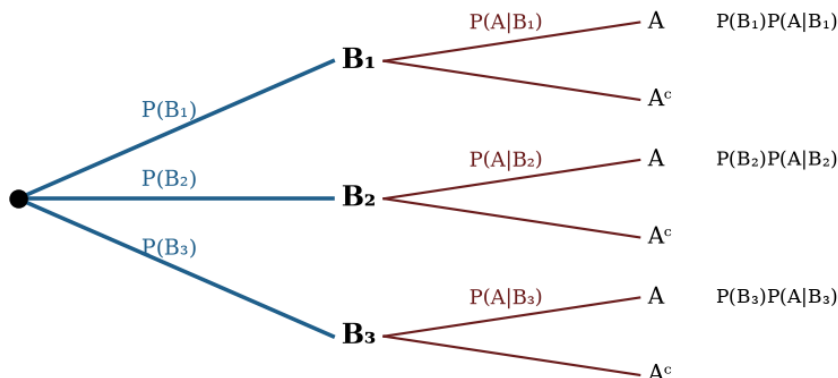
Total probability ($B_1, \dots, B_k$ partition S):

$$P(A) = \sum_{i=1}^k P(B_i)P(A | B_i)$$

Bayes' theorem:

$$P(B_j | A) = \frac{P(B_j)P(A | B_j)}{\sum_i P(B_i)P(A | B_i)}$$

Tree diagram: $P(A) = \sum P(B_i)P(A|B_i)$; $P(B_j|A) = \text{branch product} / P(A)$



Joint probability table. For two categorical variables, cell entries are joint probabilities; row/column totals are marginals; dividing a row by its total gives a conditional distribution.

	B	B ^c	Total
A	$P(A \cap B)$	$P(A \cap B^c)$	$P(A)$
A ^c	$P(A^c \cap B)$	$P(A^c \cap B^c)$	$P(A^c)$
Total	$P(B)$	$P(B^c)$	1

Odds form of Bayes: posterior odds = prior odds $\times$ likelihood ratio,

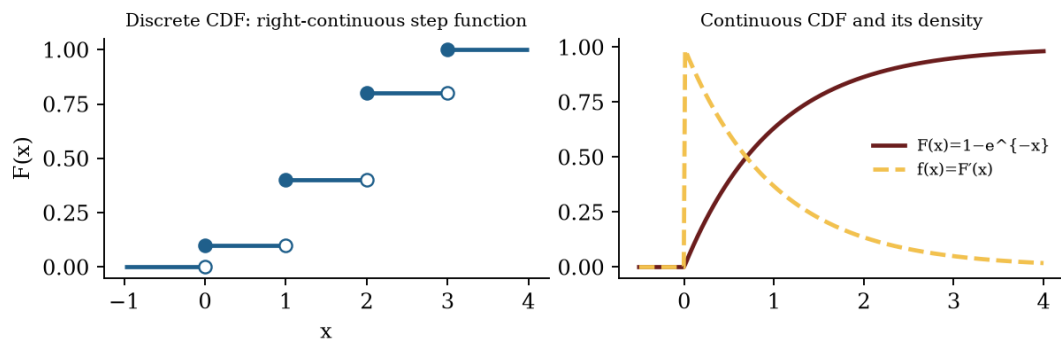
$$\frac{P(H|E)}{P(H^c|E)} = \frac{P(H)}{P(H^c)} \times \frac{P(E|H)}{P(E|H^c)}$$

Trap: Base-rate fallacy: with a rare condition, even an accurate test gives a modest $P(\text{condition} | \text{positive})$. Always weigh the likelihood by the prior.

A4. Random variables, PMF, PDF and CDF

A random variable X maps outcomes to real numbers. Its **CDF** $F(x) = P(X \leq x)$ is non-decreasing, right-continuous, with $F(-\infty) = 0$ and $F(\infty) = 1$.

	Discrete	Continuous
Description	PMF $p(x) = P(X = x)$, $\sum p(x) = 1$	PDF $f(x) \geq 0$, $\int f(x) dx = 1$
CDF	$F(x) = \sum_{t \leq x} p(t)$	$F(x) = \int_{-\infty}^x f(t) dt$
$P(a < X \leq b)$	$F(b) - F(a)$	$F(b) - F(a) = \int_a^b f$
$P(X = a)$	jump of F at a	0 for every a
From CDF to pmf/pdf	$p(a) = F(a) - F(a^-)$	$f(x) = F'(x)$



Mixed distributions have a CDF with both jumps and continuous increases; probabilities at jump points equal jump sizes.

Quantiles: the p -th quantile is the smallest x with $F(x) \geq p$; the median solves $F(m) = 1/2$. The **mode** maximises the pmf/pdf.

Transformations. If $Y = g(X)$ with g monotone and differentiable,

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

In general use the CDF method: $F_Y(y) = P(g(X) \leq y)$, then differentiate.

Trap: A valid pdf may exceed 1 (e.g. Uniform(0, 0.5) has $f = 2$); only its integral must be 1. A pmf can never exceed 1.

A5. Expectation, variance and their rules

$$E[X] = \sum_x x p(x) \quad \text{or} \quad E[X] = \int x f(x) dx, \quad E[g(X)] = \sum_x g(x) p(x)$$

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2, \quad \sigma = \sqrt{\text{Var}(X)}$$

Rule	Statement
Linearity	$E[aX + bY + c] = aE[X] + bE[Y] + c$ (always, even if dependent)
Scaling	$\text{Var}(aX + b) = a^2 \text{Var}(X)$; $\text{SD}(aX + b) = a \text{SD}(X)$
Sum	$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2 \text{Cov}(X, Y)$
Independent	$E[XY] = E[X]E[Y]$; $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
n i.i.d. copies	$\text{Var}(X_1 + \dots + X_n) = n\sigma^2$; $\text{Var}(\bar{X}) = \sigma^2/n$
Tail formula	$X \geq 0$ integer: $E[X] = \sum_{k \geq 1} P(X \geq k)$; continuous: $\int_0^\infty P(X > x) dx$
Indicators	$E[1_A] = P(A)$; count = $\sum$ indicators $\Rightarrow E[\text{count}] = \sum P(A_i)$

Conditional expectation and variance

$$E[X | Y = y] = \sum_x x p_{X|Y}(x | y) \quad \text{or} \quad \int x f_{X|Y}(x | y) dx$$

Law of total expectation (tower rule):

$$E[X] = E[E[X | Y]] = \sum_y E[X | Y = y] P(Y = y)$$

Law of total variance (Eve's law):

$$\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y])$$

Random sums: if $S = X_1 + \dots + X_N$ with N independent of i.i.d. X_i ,

$$E[S] = E[N]\mu, \quad \text{Var}(S) = E[N]\sigma^2 + \mu^2\text{Var}(N)$$

Key idea: $E[X | Y]$ is a random variable (a function of Y); $E[X | Y = y]$ is a number. The 'within-group' variance $E[\text{Var}(X|Y)]$ plus the 'between-group' variance $\text{Var}(E[X|Y])$ gives the total.

A6. Descriptive statistics: mean, median, mode, standard deviation

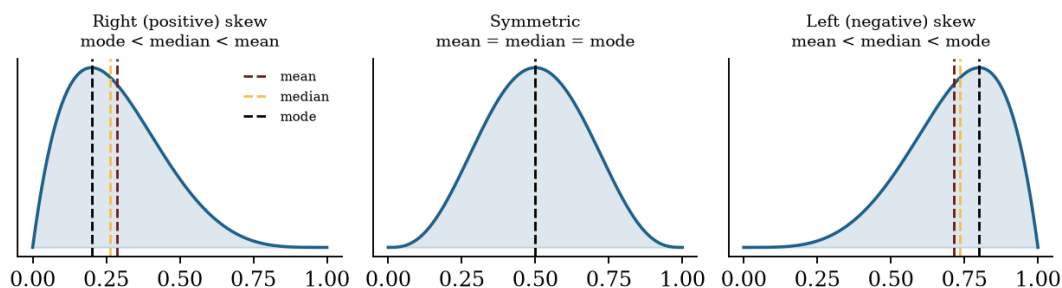
Measure	Ungrouped data	Notes
Mean	$\bar{x} = \sum x_i / n$	Sensitive to outliers; $\sum (x_i - \bar{x}) = 0$
Median	middle value of sorted data	Robust; minimises $\sum x_i - c $
Mode	most frequent value	May be non-unique
Population variance	$\sigma^2 = \sum (x_i - \mu)^2 / N$	divide by N
Sample variance	$s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$	unbiased for σ^2 (Bessel)
Shortcut	$\sum (x_i - \bar{x})^2 = \sum x_i^2 - n \bar{x}^2$	use for hand computation
Range, IQR	$\max - \min, Q_3 - Q_1$	IQR is robust
Coefficient of variation	$CV = \sigma / \mu$	unit-free spread

Grouped data with class marks m_i and frequencies f_i : $\bar{x} = \sum f_i m_i / \sum f_i$. Median = $L + ((N/2 - C)/f) \cdot h$ and Mode = $L + (f_1 - f_0)/(2f_1 - f_0 - f_2) \cdot h$, where L is the lower limit of the median/modal class, C the cumulative frequency before it and h the class width.

Linear transformation $y = ax + b$: mean, median, mode transform as $ax + b$; SD and IQR scale by $|a|$; variance by a^2 ; adding a constant does not change spread.

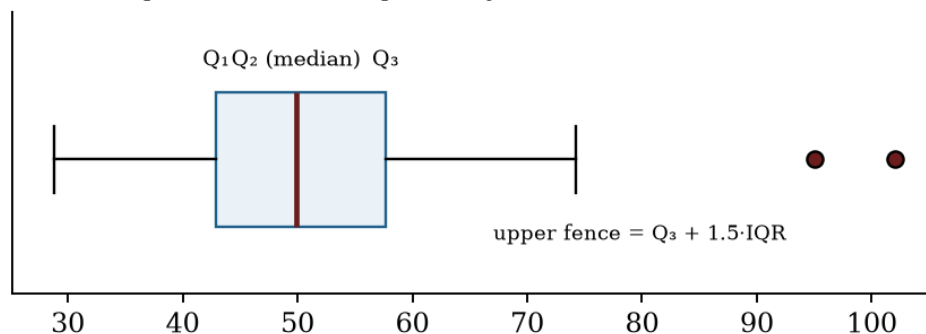
Combining two groups (sizes n_1, n_2 , means $\bar{x}_1, \bar{x}_2$, population-style variances σ_1^2, σ_2^2):

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}, \quad \sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}, \quad d_i = \bar{x}_i - \bar{x}$$



Empirical relation for moderately skewed data: $\text{Mode} \approx 3 \cdot \text{Median} - 2 \cdot \text{Mean}$.

Box plot: $\text{IQR} = Q_3 - Q_1$; points beyond $1.5 \cdot \text{IQR}$ fences are outliers

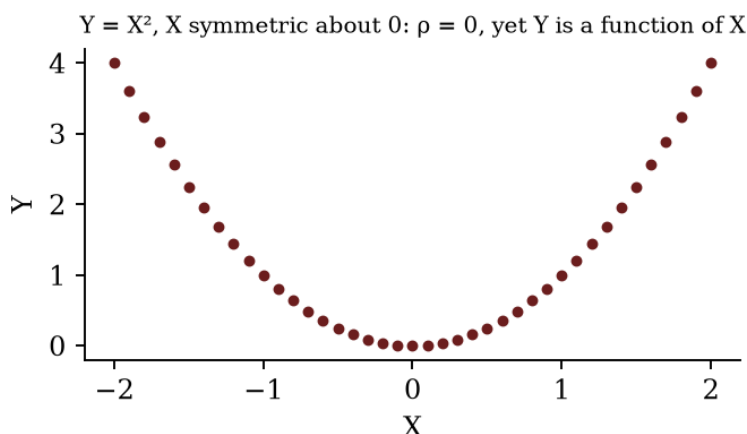
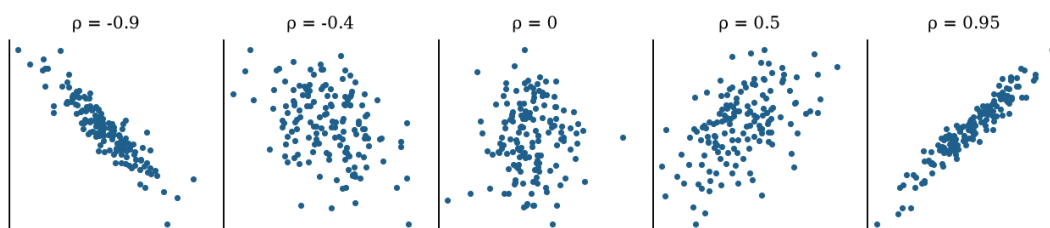


A7. Covariance and correlation

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - E[X]E[Y]$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \rho \leq 1$$

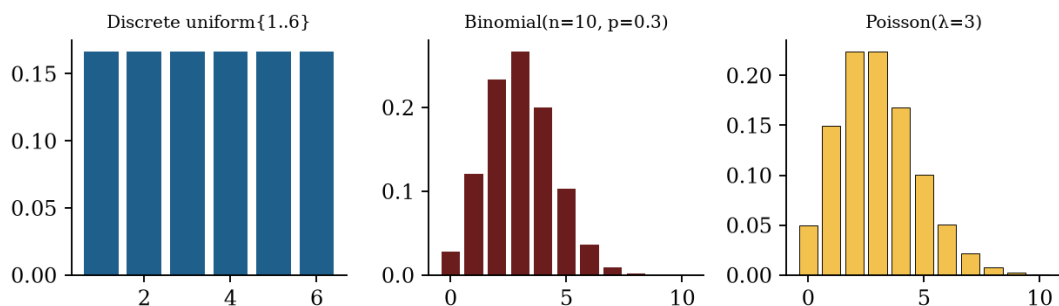
- $\text{Cov}(X, X) = \text{Var}(X)$; $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$; covariance is bilinear.
- $\rho(aX + b, cY + d) = \text{sign}(ac) \rho(X, Y)$: correlation is unit-free and shift/scale invariant.
- $|\rho| = 1$ iff $Y = aX + b$ exactly ($a \neq 0$).
- Independent $\Rightarrow \text{Cov} = 0$, but $\text{Cov} = 0$ does NOT imply independence (except for jointly normal variables).
- Sample: $r = S_{xy} / \sqrt{(S_{xx} S_{yy})}$, with $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$.
- Least-squares line of y on x : slope $b = S_{xy} / S_{xx} = r \cdot s_y / s_x$; it passes through $(\bar{x}, \bar{y})$. $R^2 = r^2$ for simple linear regression.
- Covariance matrix Σ is symmetric positive semi-definite; $\text{Var}(a^T X) = a^T \Sigma a$.



Trap: Correlation measures LINEAR association only. A perfect quadratic relation can have $\rho = 0$.

A8. Discrete distributions

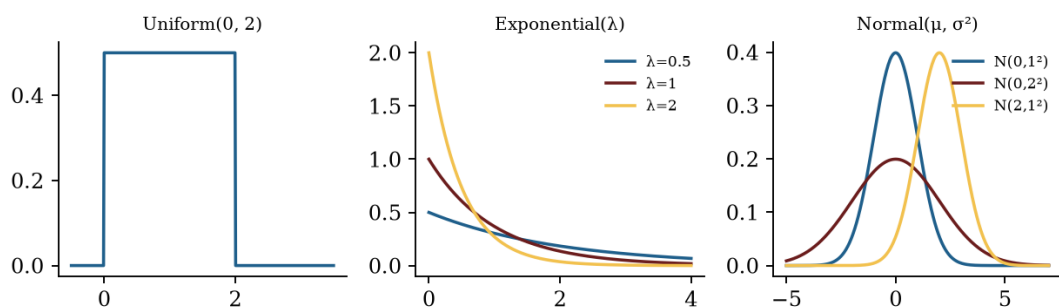
Distribution	PMF	Mean	Variance
Discrete uniform $\{a, \dots, b\}$	$1/(b - a + 1)$	$(a + b)/2$	$((b - a + 1)^2 - 1)/12$
Bernoulli(p)	$p^x(1 - p)^{1-x}, x \in \{0, 1\}$	p	$p(1 - p)$
Binomial(n, p)	${}^nC_x p^x (1 - p)^{n-x}$	np	$np(1 - p)$
Poisson(λ)	$e^{-\lambda} \lambda^x / x!$	λ	λ
Geometric(p), trials to 1st success	$(1 - p)^{x-1} p, x \geq 1$	$1/p$	$(1 - p)/p^2$
Hypergeometric(N, K, n)	${}^KC_x {}^{N-K}C_{n-x} / {}^NC_n$	nK/N	$n(K/N)(1 - K/N)(N - n)/(N - 1)$



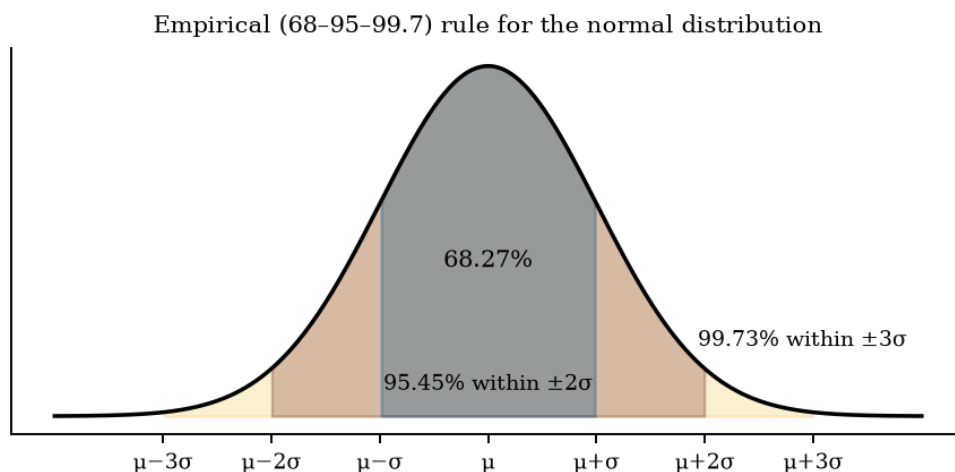
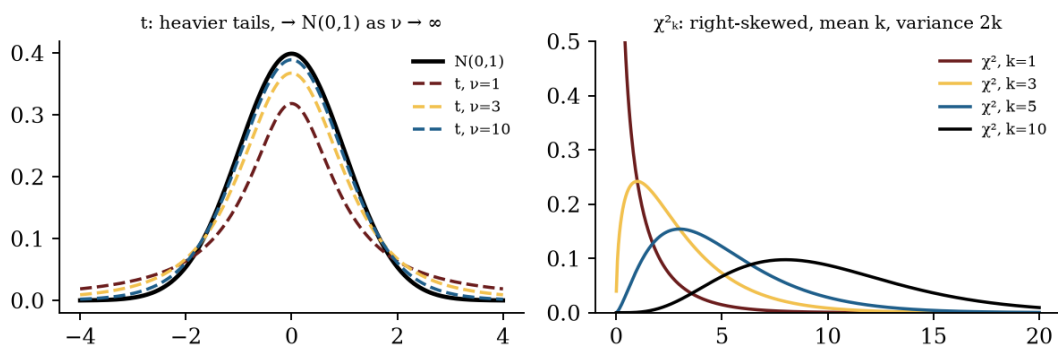
- Sum of n i.i.d. Bernoulli(p) is Binomial(n, p). Sum of independent Poissons is Poisson($\lambda_1 + \lambda_2$).
- Binomial mode: $\lfloor (n+1)p \rfloor$. Poisson mode: $\lfloor \lambda \rfloor$ (both $\lambda - 1$ and λ if λ is an integer).
- Poisson approximation to binomial: n large, p small, $\lambda = np$.
- Poisson process with rate λ : count in time t is Poisson(λt); gaps are Exponential(λ).
- Poisson thinning: if each event is kept with probability p , kept events form Poisson(λp).
- Memoryless (discrete): geometric, $P(X > m+n \mid X > m) = P(X > n)$.

A9. Continuous distributions

Distribution	PDF	Mean	Variance	CDF / notes
Uniform(a, b)	$1/(b-a), a < x < b$	$(a+b)/2$	$(b-a)^2/12$	$(x-a)/(b-a)$
Exponential(λ)	$\lambda e^{-\lambda x}, x > 0$	$1/\lambda$	$1/\lambda^2$	$1 - e^{-\lambda x}$; median $\ln 2/\lambda$
Normal(μ, σ^2)	$(1/\sigma\sqrt{2\pi}) e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2	$\Phi((x-\mu)/\sigma)$
Standard normal	$\phi(z) = e^{-z^2/2}/\sqrt{2\pi}$	0	1	$\Phi(-z) = 1 - \Phi(z)$
t with ν d.f.	symmetric, bell-shaped	0 ($\nu > 1$)	$\nu/(\nu-2)$ ($\nu > 2$)	heavier tails
Chi-squared, k d.f.	on $x > 0$, right-skewed	k	$2k$	Σ of k squared $N(0,1)$



- **Memoryless**: $X \sim \text{Exp}(\lambda) \Rightarrow P(X > s+t \mid X > s) = P(X > t)$. Only continuous memoryless law.
- **Minimum of exponentials**: $\min(X_1, \dots, X_n)$ with rates λ_i is $\text{Exp}(\sum \lambda_i)$; $P(X_i \text{ is smallest}) = \lambda_i / \sum \lambda_j$.
- **Normal closure**: independent normals add: $aX + bY \sim N(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2)$.
- **Standardising**: $Z = (X - \mu)/\sigma \sim N(0, 1)$.
- **t-distribution**: $T = Z / \sqrt{V/\nu}$ with $Z \sim N(0,1)$, $V \sim \chi^2_\nu$ independent.
- **Chi-squared**: if $Z_1, \dots, Z_k$ i.i.d. $N(0,1)$, $\sum Z_i^2 \sim \chi^2_k$; independent χ^2 add their d.f.; $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ for normal samples.
- **Uniform order statistics**: the k -th smallest of n i.i.d. $U(0,1)$ has mean $k/(n+1)$.

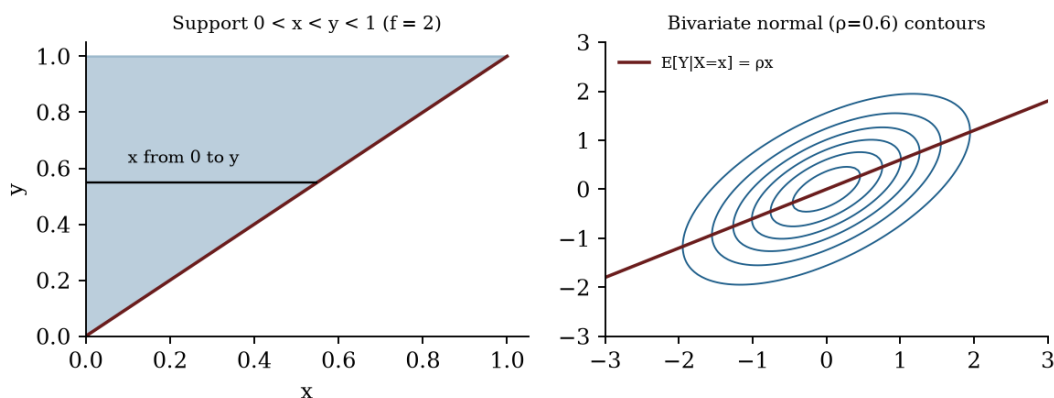


Key z values	$\Phi(z)$	Use
1.282	0.90	one-sided 10%; 80% two-sided CI
1.645	0.95	one-sided 5%; 90% two-sided CI
1.960	0.975	two-sided 5%; 95% CI
2.326	0.99	one-sided 1%; 98% CI
2.576	0.995	two-sided 1%; 99% CI

A10. Joint, marginal and conditional distributions

	Discrete	Continuous
Joint	$p(x, y) = P(X = x, Y = y)$	$f(x, y), P((X, Y) \in R) = \iint_R f$
Marginal	$p_X(x) = \sum_y p(x, y)$	$f_X(x) = \int f(x, y) dy$
Conditional	$p(x y) = p(x, y)/p_Y(y)$	$f(x y) = f(x, y)/f_Y(y)$
Independence	$p(x, y) = p_X(x)p_Y(y) \forall x, y$	$f(x, y) = f_X(x)f_Y(y) \forall x, y$

For independence of continuous X, Y the joint support must be a **rectangle** (product set) and the density must factor. A triangular support such as $0 < x < y < 1$ immediately implies dependence.



Recipe for joint-pdf questions: (1) sketch the support; (2) find the constant from $\iint f = 1$; (3) set integration limits from the sketch; (4) for a conditional pdf divide by the marginal and keep the conditional support.

Bivariate normal: $E[Y | X = x] = \mu_Y + \rho(\sigma_Y/\sigma_X)(x - \mu_X)$, $\text{Var}(Y | X) = \sigma_Y^2(1 - \rho^2)$.

A11. Sampling distributions and the Central Limit Theorem

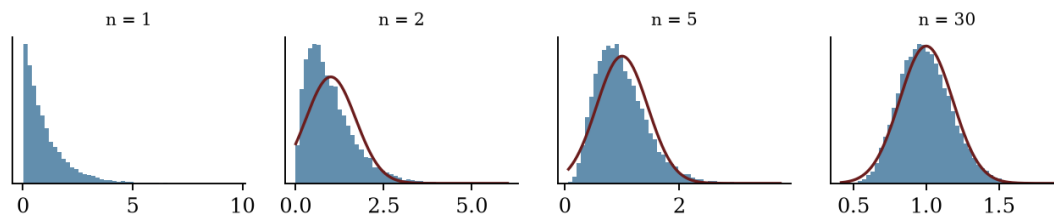
For i.i.d. $X_1, \dots, X_n$ with mean μ and variance σ^2 :

$$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

CLT: for large n (rule of thumb $n \geq 30$),

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1), \quad \sum_{i=1}^n X_i \approx N(n\mu, n\sigma^2)$$

CLT: sample means of Exponential(1) data $\rightarrow N(1, 1/n)$



Normal approximation to Binomial(n, p) ($np \geq 5$ and $n(1 - p) \geq 5$): $X \approx N(np, np(1 - p))$ with **continuity correction**: $P(X \leq k) \approx \Phi((k + 0.5 - np)/\sqrt{np(1 - p)})$, $P(X \geq k) \approx 1 - \Phi((k - 0.5 - np)/\sqrt{np(1 - p)})$.

Sample proportion: $\hat{p} \approx N(p, p(1 - p)/n)$.

If the population itself is normal, $\bar{X}$ is exactly normal for every n . With σ unknown and normal data, $(\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$ exactly.

Shortcut: Halving the standard error needs FOUR times the sample size, because $\text{SE} \propto 1/\sqrt{n}$.

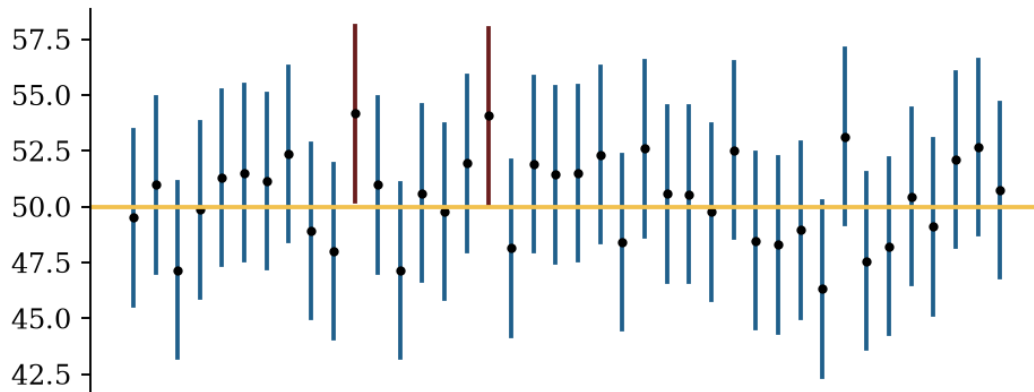
A12. Confidence intervals

Parameter	Conditions	100(1 - α)% interval
Mean μ	σ known (or n large)	$\bar{x} \pm z_{\alpha/2} \sigma/\sqrt{n}$
Mean μ	σ unknown, normal data	$\bar{x} \pm t_{\alpha/2, n-1} s/\sqrt{n}$

Parameter	Conditions	100(1 - α)% interval
Proportion p	$n\hat{p}, n(1 - \hat{p}) \geq 5$	$\hat{p} \pm z_{\alpha/2} \sqrt{\hat{p}(1 - \hat{p})/n}$
$\mu_1 - \mu_2$	σ 's known	$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
$\mu_1 - \mu_2$	equal unknown σ (pooled)	$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1+n_2-2} s_p \sqrt{(1/n_1 + 1/n_2)}$
Variance σ^2	normal data	$[(n - 1)s^2/\chi^2_{\alpha/2, n-1}, (n - 1)s^2/\chi^2_{1-\alpha/2, n-1}]$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \quad n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2 \text{ for margin } E$$

40 independent 95% CIs for $\mu = 50$: 2 miss (red). μ is fixed; intervals are random.

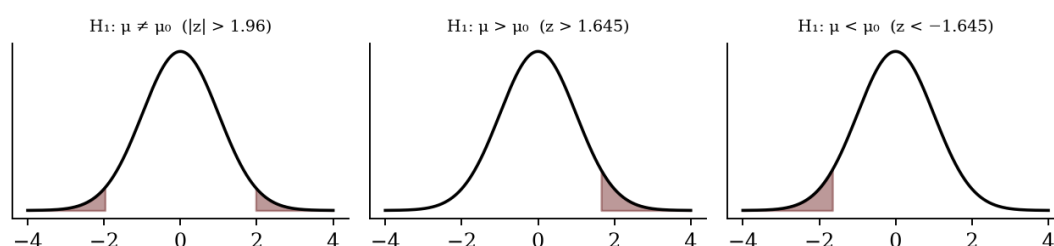


- Width = $2 \times$ margin of error. Width $\uparrow$ with confidence level and σ ; $\downarrow$ with $\sqrt{n}$.
- Interpretation: 95% of intervals built this way contain μ . A computed interval either contains μ or not — it is NOT ' μ lies in it with probability 0.95'.
- A two-sided level- α test rejects $H_0: \mu = \mu_0$ exactly when μ_0 lies outside the 100(1 - α)% CI.
- For a proportion with no prior estimate, use $\hat{p} = 0.5$ (maximises $p(1 - p)$) in sample-size formulas.

A13. Hypothesis testing: z-test, t-test, chi-squared test

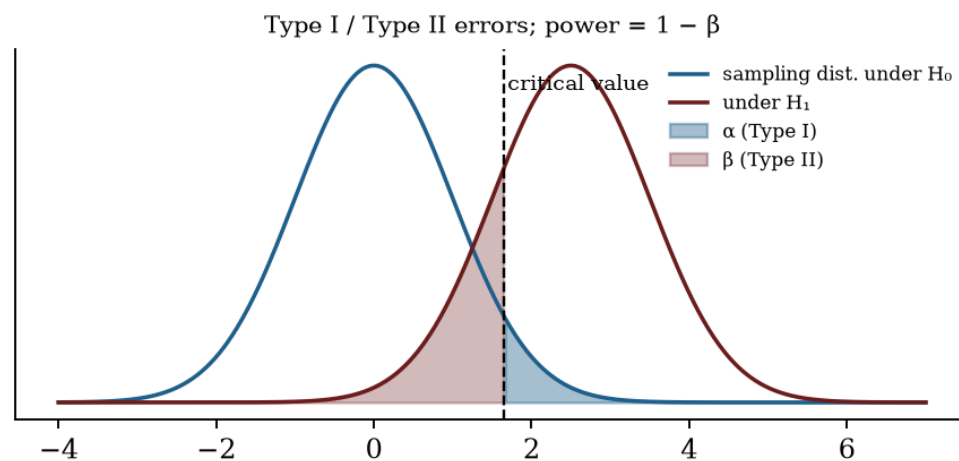
- **Steps:** state H_0 and $H_1 \rightarrow$ choose $\alpha \rightarrow$ compute the test statistic $\rightarrow$ find the critical value or p-value $\rightarrow$ decide (reject H_0 if statistic is in the rejection region, equivalently p-value $\leq \alpha$).
- **p-value:** probability, under H_0 , of a statistic at least as extreme as observed. Two-sided: $p = 2(1 - \Phi(|z|))$.
- **Type I error** $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$; **Type II** $\beta = P(\text{fail to reject} \mid H_1 \text{ true})$; **power** = $1 - \beta$.

Rejection regions at $\alpha = 0.05$ (shaded)



Test	Statistic	Distribution under H_0
One-sample z	$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$	$N(0, 1)$
Proportion z	$z = (\hat{p} - p_0)/\sqrt{p_0(1 - p_0)/n}$	$\approx N(0, 1)$

Test	Statistic	Distribution under H_0
One-sample t	$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$	t_{n-1}
Paired t	$t = d/(s_d/\sqrt{n})$ on differences	t_{n-1}
Two-sample pooled t	$t = (\bar{x}_1 - \bar{x}_2)/(s_p \sqrt{1/n_1 + 1/n_2})$	$t_{n_1+n_2-2}$
χ^2 goodness of fit	$\chi^2 = \sum (O - E)^2/E$ over k cells	χ^2_{k-1-m} (m = fitted params)
χ^2 independence ($r \times c$)	$\chi^2 = \sum (O - E)^2/E$, $E = \text{row} \cdot \text{col}/N$	$\chi^2_{(r-1)(c-1)}$
χ^2 for variance	$\chi^2 = (n - 1)s^2/\sigma_0^2$	χ^2_{n-1}



- Increasing n (for fixed α) increases power; decreasing α (stricter test) decreases power.
- χ^2 tests: use counts, not percentages; each expected count should be ≥ 5 (merge cells otherwise). The χ^2 independence/GOF test is right-tailed.
- For a 2×2 table the χ^2 statistic equals z^2 of the two-proportion z-test.
- Use t (not z) when σ is unknown and n is small; as d.f. $\rightarrow \infty$, t critical values $\rightarrow z$ critical values.

Trap: 'Fail to reject H_0 ' is not 'accept H_0 is true'. And a p-value is NOT the probability that H_0 is true.

A14. Twenty high-frequency exam traps

- Confusing 'mutually exclusive' with 'independent'.
- Using n instead of $n - 1$ for the sample standard deviation (or vice versa) — read the question.
- Forgetting continuity correction when approximating a discrete count by a normal.
- Using $\text{Var}(X - Y) = \text{Var}(X) - \text{Var}(Y)$: variances of independent terms ALWAYS add.
- Writing $\text{SD}(X + Y) = \text{SD}(X) + \text{SD}(Y)$ — add variances, then take the square root.
- Treating $P(A | B)$ as $P(B | A)$ (prosecutor's fallacy).
- Missing that 'at least one' is easiest via the complement.
- Using one-sided critical values for two-sided alternatives (1.645 vs 1.96).
- Wrong χ^2 degrees of freedom: $(r - 1)(c - 1)$ for independence, $k - 1 - m$ for goodness of fit.
- Integrating a joint pdf over the wrong region — always sketch the support first.
- Assuming zero covariance implies independence.
- Forgetting that $P(X = a) = 0$ for continuous X , so $\leq$ and $<$ give the same probability.
- For Poisson over a different time window, rescale λ (λt).
- Exponential parameter: rate λ (mean $1/\lambda$) vs scale β (mean β) — check which is given.

- The median of an $\text{Exponential}(\lambda)$ is $\ln 2/\lambda$, not $1/\lambda$.
- $\text{Var}(aX) = a^2\text{Var}(X)$, not $a\text{Var}(X)$.
- Sampling without replacement $\rightarrow$ hypergeometric, not binomial.
- Normal tables give $\Phi(z) = P(Z \leq z)$; for $P(Z > z)$ subtract from 1.
- A CI for μ is about μ , not about individual observations (that would be a prediction interval).
- The t-statistic uses $s/\sqrt{n}$; the d.f. for a paired test is (number of pairs $- 1$).

Part B · Mock Tests

Test	Title	Focus
01	Foundation Full-Syllabus Test I	Every syllabus head appears at least once: counting (circular seating, committees), axioms and Venn reasoning, independence vs mutual exclusivity, conditional probability with dice and urns, Bayes theorem (tree), joint/marginal/conditional PMFs and PDFs, law of total variance, mean/median/mode, covariance and correlation, Bernoulli/binomial/discrete uniform, Poisson, uniform/exponential/normal, CDF reading, CLT, t-interval, z-test, t-test and a χ^2 goodness-of-fit test. Five concept-check MSQs test properties of independence, CDFs, variance, correlation and the t/χ^2 families.
02	Foundation Full-Syllabus Test II	Covers the full syllabus at warm-up difficulty: counting with cards and seating, sample spaces and axioms, mutually exclusive vs independent events, contingency tables (joint, marginal, conditional), Bayes with a two-urn tree, conditional expectation/variance (random sums), conditional and joint PDFs, sample variance and grouped standard deviation, Var of linear combinations, Bernoulli/binomial, Poisson, uniform, exponential, normal, CDF-based median, CLT for sums, z-interval, z-test (p-value), paired t-test and χ^2 test of independence. Concept-check MSQs probe PDF properties, expectation rules, independence implications, sampling distributions and the logic of tests and intervals.
03	Standard GATE-Level Test I (Data-Science Contexts)	Covers the entire Probability & Statistics syllabus at exam difficulty ($\approx 30\%$ easy, 50% medium, 20% hard): counting via hashing and cross-validation folds; axioms, independence and mutual exclusivity of filter rules; Bayes theorem for classifier precision and data-source attribution; joint/marginal/conditional PMFs and PDFs; law of total variance for bursty server traffic; summary statistics and skewness; covariance matrices of features; Bernoulli dropout, binomial click-through, Poisson/exponential request logs, uniform, normal sensor noise, t and χ^2 distributions; CDFs; the CLT for mini-batch means; confidence intervals for model accuracy and latency; z-test for an A/B experiment, one-sample t-test, and χ^2 tests of independence and goodness-of-fit.
04	Standard GATE-Level Test II (Data-Science Contexts)	Covers the whole syllabus at exam difficulty ($\approx 30\%$ easy, 50% medium, 20% hard): permutations with restrictions, hash-collision (birthday) probabilities and hypergeometric mini-batch sampling; mutually exclusive vs independent events; total probability, two-stage Bayes for fraud alerts and a naive Bayes spam posterior; joint/conditional PDFs (exponential pair, $x + y$ density), random-sum variance via conditional expectation; sample and grouped-data summaries, correlation invariance and the regression slope; Bernoulli/inverted dropout, discrete uniform, Poisson, exponential, Gamma-type CDFs, normal comparisons, χ^2 of the sample variance; CLT for dropout counts; sample size for a CTR interval; two-sample z-test for an A/B experiment, paired t-test comparing models across datasets, and a $2 \times 3 \chi^2$ test of independence.
05	Topic Test: Probability, Counting, Bayes & Conditional Expectation	Counting (permutations, combinations, multinomial, stars and bars, derangements and inclusion–exclusion), probability axioms and set algebra, mutual exclusivity vs independence (pairwise vs mutual), joint/marginal/conditional probability, total probability and multi-stage Bayes, random variables and indicator methods, conditional expectation $E[X Y]$, the laws of total expectation and total variance, and joint PMF tables.

Test	Title	Focus
06	Topic Test: Random Variables, Distributions, PDFs & CDFs	Discrete distributions (uniform, Bernoulli, binomial, Poisson and the Poisson approximation), continuous distributions (uniform, exponential and memorylessness, normal and standardisation, t and χ^2 properties), CDF properties and mixed CDFs, pdf from CDF, transformations of random variables, joint/marginal/conditional pdfs, moments and covariance from densities, and quantiles, medians and modes.
07	Topic Test: Sampling, CLT, Confidence Intervals & Hypothesis Testing	Sampling distribution of the mean and S^2 ; CLT for sums and means; normal approximation to the binomial with continuity correction; z and t confidence intervals for means and proportions; sample-size planning; z -tests, one-sample/paired/pooled t -tests; χ^2 goodness-of-fit and independence; χ^2 interval and test for a variance; Type I/II errors and power; properties of the t and χ^2 distributions.
08	Topic Test: Descriptive Statistics, Covariance, Correlation & Mixed Review	Mean, median, mode, variance and SD for grouped and ungrouped data; linear transformations; combining groups; sample vs population variance; quartiles, box plots and outliers; skewness; covariance and correlation (properties, $\text{Var}(aX + bY)$, linear transforms, zero correlation vs independence, covariance matrices); regression slopes via covariance. Mixed review: binomial, exponential, Poisson, normal percentiles, Bayes theorem, conditional variance, joint PDFs, mixed CDFs and counting.
09	Advanced Full-Syllabus Test I	A final-stage readiness check covering the WHOLE syllabus at the top end of GATE DA difficulty: counting with restrictions, independence subtleties, mixed CDFs, order statistics of uniforms, minima of exponentials, Poisson thinning, random sums and conditional variance, joint pdfs on triangular supports, geometric probability, normal mixtures, covariance of indicator sums, Bayes with binomial and naive-Bayes likelihoods, CLT-based sample-size design, z -test power, paired t -test, t -based CI from raw sums, and a chi-squared goodness-of-fit test whose expected counts must first be derived (Hardy-Weinberg). Almost every 2-mark question chains two or more ideas — attempt it under strict 90-minute conditions and treat any slip as a revision flag.
10	Advanced Full-Syllabus Test II (Grand Mock)	The Grand Mock: a final-stage readiness check that sweeps the ENTIRE syllabus at peak GATE DA difficulty. Counting with upper bounds (inclusion-exclusion), probability bounds, indicator expectations, range of uniform order statistics, half-normal conditional mean, memoryless lifetimes, CLT for sums, reliability block diagrams with a Bayes twist, random sums with a geometric count, joint PMF tables, non-uniform joint pdfs over sub-regions, queue waiting times via minima of exponentials, hypergeometric variance through indicator covariances, Poisson thinning, a two-proportion z -test for A/B testing, t -test/CI duality, Poisson-mixture Bayes, pooled standard deviations, CLT sample-size design, carrier-risk genetics, and χ^2 tests of independence and goodness of fit with model-derived expected counts. Score above 60% here and you are exam-ready; anything less tells you exactly which chapters to revisit.

Mock Test 01 — Foundation

Full-Syllabus Test I

Warm-up paper covering the entire GATE DA Probability & Statistics syllabus through classic contexts: dice, coins, urns, committees, quality control, exam scores and rainfall.

Item	Details
Questions	30 (12 × 1-mark, 18 × 2-mark)
Maximum marks	48
Question types	MCQ: 14 · MSQ: 7 · NAT: 9
Suggested time	90 minutes
Negative marking	MCQ only: $-1/3$ (1-mark), $-2/3$ (2-mark). None for MSQ & NAT.
Calculator	Use only the GATE virtual scientific calculator

Topic focus of this set

Every syllabus head appears at least once: counting (circular seating, committees), axioms and Venn reasoning, independence vs mutual exclusivity, conditional probability with dice and urns, Bayes theorem (tree), joint/marginal/conditional PMFs and PDFs, law of total variance, mean/median/mode, covariance and correlation, Bernoulli/binomial/discrete uniform, Poisson, uniform/exponential/normal, CDF reading, CLT, t-interval, z-test, t-test and a χ^2 goodness-of-fit test. Five concept-check MSQs test properties of independence, CDFs, variance, correlation and the t/χ^2 families.

Instructions

- Questions 1 onwards are arranged as in GATE: all 1-mark questions first, then 2-mark questions.
- **MCQ** – exactly one correct option. **MSQ** – ONE OR MORE options are correct; credit only if you select all correct options and no wrong option (no partial marks, no negative marks).
- **NAT** – enter a numerical value using the virtual keypad. The accepted range is given in the key; round as instructed in the question.
- Statistical tables (Z, t, χ^2) are provided in the Appendix; values in solutions use those tables.
- Attempt in one sitting, then check the answer key and read every solution — including those you got right — for the alternative methods and traps.

Mock Test 01 · Question Paper

1-mark questions

Q.1 MCQ · 1 mark · -1/3

Six friends, including Asha and Bimal, sit around a circular table (only relative positions matter). The number of seating arrangements in which Asha and Bimal do **not** sit next to each other is

- (A) 48
- (B) 120
- (C) 480
- (D) 72

Q.2 MCQ · 1 mark · -1/3

For a weekend, let A = 'it rains on Saturday' and B = 'it rains on Sunday'. Suppose $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cup B) = 0.8$. The probability that it rains on **exactly one** of the two days is

- (A) 0.3
- (B) 0.2
- (C) 0.5
- (D) 0.8

Q.3 MSQ · 1 mark

Let A and B be events in a probability space. Which of the following statements is/are **always TRUE**?

- (A) If $P(A \cap B) = 0$, then A and B are independent.
- (B) If A and B are independent, then A and B^c are independent.
- (C) If $P(A) > 0$, $P(B) > 0$ and A , B are mutually exclusive, then A and B are NOT independent.
- (D) An event A with $P(A) = 1$ is independent of every event B .

Q.4 NAT · 1 mark

Two fair six-sided dice are rolled. Given that at least one die shows a 5, the probability that the sum of the two faces is 8 is _____ (round off to 2 decimal places).

Answer: _____

Q.5 MCQ · 1 mark · -1/3

The marks (out of 30) of eight students in a quiz are: 12, 15, 15, 18, 20, 22, 15, 25. Which ordering of the mean, median and mode is correct?

- (A) mean < median < mode
- (B) mode < median < mean
- (C) median < mode < mean
- (D) mean = median = mode

Q.6 MCQ · 1 mark · -1/3

A machine produces bolts, each defective independently with probability 0.1. In a random sample of 5 bolts, the probability that exactly one is defective is closest to

- (A) 0.066
- (B) 0.328
- (C) 0.410
- (D) 0.590

Q.7 MSQ · 1 mark

Let F be the cumulative distribution function of a random variable X , i.e. $F(x) = P(X \leq x)$. Which of the following is/are TRUE for **every** random variable X ?

- (A) F is non-decreasing.
- (B) F is right-continuous.
- (C) $P(X = a) = F(a) - F(a^-)$, where $F(a^-)$ is the left limit at a .
- (D) $P(a < X < b) = F(b) - F(a)$ for all $a < b$.

Q.8 NAT · 1 mark

The number of thunderstorms in a district during July follows a Poisson distribution with mean 2. The probability that the district experiences at least one thunderstorm in July is _____ (round off to 2 decimal places).

Answer: _____

Q.9 MCQ · 1 mark · -1/3

The daily rainfall X (in mm) at a hill station on a rainy day is modelled as uniform on $[2, 10]$. Given that the rainfall on a day exceeds 4 mm, the probability that it exceeds 7 mm is

- (A) 0.500
- (B) 0.375
- (C) 0.625
- (D) 0.750

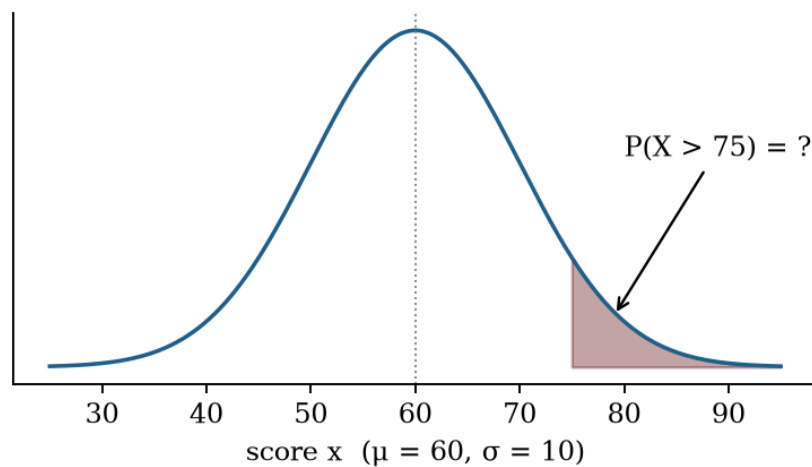
Q.10 MCQ · 1 mark · -1/3

The lifetime of a bulb is exponentially distributed with mean 400 hours. Given that a bulb has already worked for 200 hours, the probability that it works for at least another 400 hours is

- (A) $e^{-1.5}$
- (B) $1 - e^{-1}$
- (C) $e^{-0.5}$
- (D) e^{-1}

Q.11 NAT · 1 mark

Scores in a state-level exam are normally distributed with mean 60 and standard deviation 10. The proportion of candidates scoring above 75 is _____ (round off to 4 decimal places). [Use $\Phi(1.5) = 0.9332$.]



Answer: _____

Q.12 MSQ · 1 mark

Let X and Y be random variables with finite variances and let a, b be constants. Which of the following is/are TRUE?

- (A) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ for all X and Y
- (B) $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- (C) If X and Y are independent, $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$
- (D) $\text{Var}(X) = E[X^2] - (E[X])^2$

2-mark questions**Q.13 NAT · 2 marks**

A committee of 4 is chosen at random from 6 men and 5 women, all selections being equally likely. The probability that the committee contains at least 2 women is _____ (round off to 2 decimal places).

Answer: _____

Q.14 MCQ · 2 marks · -2/3

Three machines M_1, M_2, M_3 produce 50%, 30% and 20% of a factory's output, with defective rates 2%, 3% and 5% respectively. An item picked at random is found to be defective. The probability that it was produced by M_3 is closest to

- (A) 0.200
- (B) 0.310
- (C) 0.345
- (D) 0.050

Q.15 MCQ · 2 marks · -2/3

An urn contains 4 red and 6 blue balls. Two balls are drawn one after another without replacement. Given that the **second** ball drawn is red, the probability that the **first** ball drawn was red is

- (A) $2/5$
- (B) $2/15$
- (C) $1/3$
- (D) $4/9$

Q.16 NAT · 2 marks

In a quality audit, $X = 1$ if a randomly chosen item comes from the night shift (0 otherwise) and $Y =$ number of minor flaws on the item. The joint PMF is:

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	0.10	0.20	0.10
$X = 1$	0.15	0.15	0.30

$\text{Cov}(X, Y)$ is _____ (round off to 2 decimal places).

Answer: _____

Q.17 NAT · 2 marks

A fair die is rolled; if it shows N , a fair coin is then tossed N times. Let Y be the total number of heads. $\text{Var}(Y)$ is _____ (round off to 2 decimal places).

Answer: _____

Q.18 MCQ · 2 marks · -2/3

The joint PDF of (X, Y) is $f(x, y) = 2$ for $0 < y < x < 1$ and 0 otherwise. The value of $E[Y | X = 0.6]$ is

- (A) 0.30
- (B) 0.333
- (C) 0.40
- (D) 0.60

Q.19 NAT · 2 marks

The joint PDF of (X, Y) is $f(x, y) = x + y$ for $0 \leq x \leq 1, 0 \leq y \leq 1$, and 0 elsewhere. $P(X + Y \leq 1)$ is _____ (round off to 2 decimal places).

Answer: _____

Q.20 MSQ · 2 marks

Let X and Y be random variables with positive, finite variances, and let $\rho(X, Y)$ denote their correlation coefficient. Which of the following is/are TRUE?

- (A) $-1 \leq \rho(X, Y) \leq 1$
- (B) If $\text{Cov}(X, Y) = 0$, then X and Y are independent.
- (C) $\rho(2X + 3, -Y) = -\rho(X, Y)$
- (D) $\text{Cov}(X, X) = \text{Var}(X)$

Q.21 MCQ · 2 marks · -2/3

Seasonal rainfall x (cm) and crop yield y (quintals/acre) on five farms are:

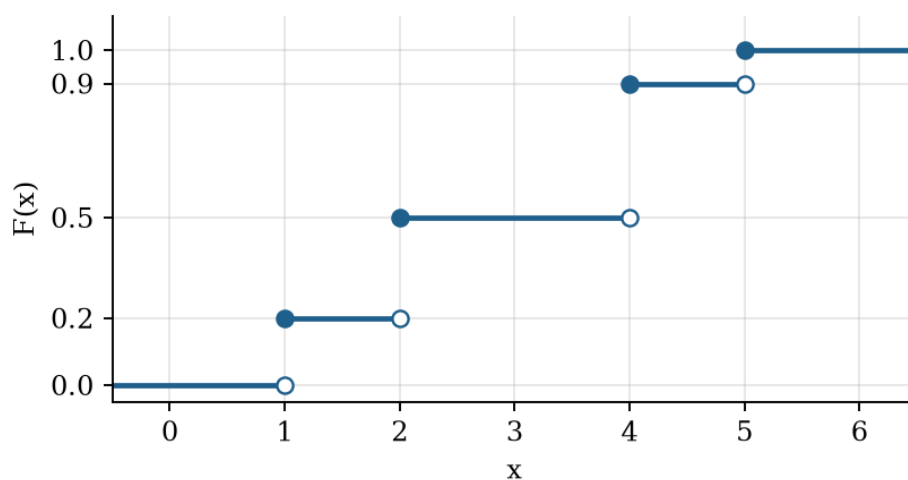
x	2	4	6	8	10
y	3	7	5	10	12

The Pearson correlation coefficient between x and y is closest to

- (A) 0.83
- (B) 0.91
- (C) 1.05
- (D) 0.47

Q.22 MSQ · 2 marks

The CDF of a discrete random variable X (the number of warning sirens sounded at a dam during a storm) is plotted below:



$F(x) = 0$ for $x < 1$, 0.2 for $1 \leq x < 2$, 0.5 for $2 \leq x < 4$, 0.9 for $4 \leq x < 5$ and 1 for $x \geq 5$. Which of the following is/are TRUE?

- (A) $P(X = 2) = 0.3$
- (B) $P(2 \leq X < 5) = 0.7$
- (C) $P(X > 4) = 0.1$
- (D) $E[X] = 3$

Q.23 MCQ · 2 marks · -2/3

Daily rainfall readings at a station during the monsoon have mean 50 mm and standard deviation 15 mm (distribution unknown). For a random sample of 36 independent days, the probability that the sample mean exceeds 54 mm is approximately [$\Phi(1.6) = 0.9452$, $\Phi(0.27) = 0.6064$]

- (A) 0.3936
- (B) 0.9452
- (C) 0.1096
- (D) 0.0548

Q.24 NAT · 2 marks

A sample of 25 bolts from a production line has mean length 50.2 mm and sample standard deviation 0.5 mm. Assuming lengths are normal, the **upper** limit of the 95% confidence interval for the mean length is _____ mm (round off to 2 decimal places). [Use $t_{0.025, 24} = 2.064$.]

Answer: _____

Q.25 MCQ · 2 marks · -2/3

A bottling plant claims its bottles contain on average 500 ml; fill volumes have known standard deviation 4 ml. An inspector suspects under-filling and measures 64 bottles, getting a mean of 498.9 ml. Testing $H_0: \mu = 500$ against $H_1: \mu < 500$ at the 5% level ($z_{0.05} = 1.645$), which is correct?

- (A) $z = -2.20$; reject H_0
- (B) $z = -2.20$; do not reject H_0
- (C) $z = -0.28$; do not reject H_0
- (D) $z = -17.6$; reject H_0

Q.26 MCQ · 2 marks · -2/3

Historically, students in a course scored a mean of 64 in the final exam. After a new tutoring programme, a random sample of 10 students scored a mean of 67.3 with sample standard deviation 5.8. Assume normality. For $H_0: \mu = 64$ vs $H_1: \mu > 64$ at the 5% level ($t_{0.05, 9} = 1.833$, $z_{0.05} = 1.645$), which is correct?

- (A) $t = 1.80$; reject H_0
- (B) $t = 1.80$; do not reject H_0
- (C) $t = 5.69$; reject H_0
- (D) $t = 1.80$; reject H_0 because $1.80 > 1.645$

Q.27 MCQ · 2 marks · -2/3

A die is rolled 120 times to test whether it is fair. The observed face counts are:

Face	1	2	3	4	5	6
Observed	15	25	18	22	16	24

Using the χ^2 goodness-of-fit test at the 5% level ($\chi^2_{0.05, 5} = 11.07$, $\chi^2_{0.05, 6} = 12.59$), which is correct?

- (A) $\chi^2 = 4.5$ with 5 df; reject fairness
- (B) $\chi^2 = 4.5$ with 6 df; do not reject fairness
- (C) $\chi^2 = 90$ with 5 df; reject fairness
- (D) $\chi^2 = 4.5$ with 5 df; do not reject fairness

Q.28 MSQ · 2 marks

Which of the following statements about the t and χ^2 distributions is/are TRUE?

- (A) The t distribution with ν degrees of freedom has heavier tails than $N(0, 1)$.
- (B) As $\nu \rightarrow \infty$, the t distribution with ν degrees of freedom approaches $N(0, 1)$.
- (C) If $Z_1, \dots, Z_k$ are i.i.d. $N(0, 1)$, then $Z_1^2 + \dots + Z_k^2$ has a χ^2 distribution with k degrees of freedom.
- (D) A χ^2 random variable with k degrees of freedom has mean k and variance k.

Q.29 NAT · 2 marks

A buyer inspects a random sample of 10 items from a large lot and accepts the lot if at most one sampled item is defective. If 5% of the items in the lot are defective (independently), the probability that the lot is accepted is _____ (round off to 3 decimal places).

Answer: _____

Q.30 MSQ · 2 marks

A fair die is rolled once; let X be the face shown. Define indicator variables $I = 1$ if X is even (0 otherwise), $J = 1$ if $X \leq 2$ (0 otherwise), and $K = 1$ if $X \leq 3$ (0 otherwise). Which of the following is/are TRUE?

- (A) $E[X] = 3.5$
- (B) $\text{Var}(X) = 35/12$
- (C) I and J are independent
- (D) I and K are independent

Mock Test 01 · Answer Key

Q	Type	Marks	Answer	Topic
1	MCQ	1	D	Counting: circular arrangements
2	MCQ	1	C	Probability axioms, union
3	MSQ	1	B, C, D	Independence vs mutual exclusivity (concept check)
4	NAT	1	0.18 (range 0.18 to 0.19)	Conditional probability (dice)
5	MCQ	1	B	Mean, median, mode
6	MCQ	1	B	Binomial distribution
7	MSQ	1	A, B, C	Properties of a CDF (concept check)
8	NAT	1	0.86 (range 0.86 to 0.87)	Poisson distribution
9	MCQ	1	A	Continuous uniform, conditional probability
10	MCQ	1	D	Exponential distribution, memorylessness
11	NAT	1	0.0668 (range 0.0665 to 0.067)	Normal distribution
12	MSQ	1	B, C, D	Variance rules (concept check)
13	NAT	2	0.65 (range 0.65 to 0.66)	Counting: committees (hypergeometric)
14	MCQ	2	C	Bayes theorem (quality control)
15	MCQ	2	C	Conditional probability (urn, without replacement)
16	NAT	2	0.06	Joint PMF, covariance
17	NAT	2	1.60 (range 1.6 to 1.61)	Conditional expectation and variance
18	MCQ	2	A	Conditional PDF, conditional expectation
19	NAT	2	0.33 (range 0.33 to 0.34)	Joint PDF, probability over a region
20	MSQ	2	A, C, D	Covariance and correlation (concept check)
21	MCQ	2	B	Sample correlation coefficient
22	MSQ	2	A, B, C	Reading a discrete CDF
23	MCQ	2	D	Central limit theorem
24	NAT	2	50.41 (range 50.4 to 50.41)	Confidence interval (t, σ unknown)
25	MCQ	2	A	One-sample z-test (left-tailed)
26	MCQ	2	B	One-sample t-test
27	MCQ	2	D	χ^2 goodness-of-fit test
28	MSQ	2	A, B, C	t and χ^2 distributions (concept check)
29	NAT	2	0.914 (range 0.913 to 0.915)	Binomial: acceptance sampling
30	MSQ	2	A, B, C	Discrete uniform, Bernoulli indicators, independence

Score tracker

	Attempted	Correct	Wrong (MCQ)	Marks
1-mark				
2-mark				
Total				/ 48

Mock Test 01 · Detailed Solutions

Q.1 [MCQ · 1 mark · -1/3]

Topic: Counting: circular arrangements · Easy

Correct answer: D

Concept: n distinct people can sit around a circle in $(n - 1)!$ ways; count the complement (A and B together) by gluing them into one block.

Total circular arrangements of 6 people:

$$\text{Total} = (6 - 1)! = 5! = 120$$

Treat Asha-Bimal as one block: 5 units around a circle, and the block can be internally ordered in 2 ways:

$$\text{Together} = (5 - 1)! \times 2! = 24 \times 2 = 48$$

$$\text{Not together} = 120 - 48 = 72$$

Why not the others? 48 is the 'together' count; 120 is the total; $480 = 6! - 2 \cdot 5!$ is the answer for a straight row, not a circle.

Shortcut: In a circle, fix one person's seat to kill rotational symmetry — that is where $(n - 1)!$ comes from. Always use the complement for 'not together'.

Q.2 [MCQ · 1 mark · -1/3]

Topic: Probability axioms, union · Easy

Correct answer: C

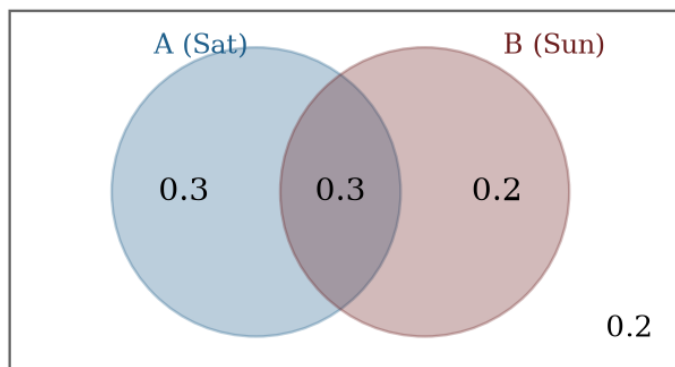
Concept: Inclusion–exclusion gives $P(A \cap B)$; 'exactly one' = union minus intersection.

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.6 + 0.5 - 0.8 = 0.3$$

$$P(\text{exactly one}) = P(A \cup B) - P(A \cap B) = 0.8 - 0.3 = 0.5$$

Equivalently $P(A \text{ only}) + P(B \text{ only}) = (0.6 - 0.3) + (0.5 - 0.3) = 0.3 + 0.2 = 0.5$.

$$\text{Exactly one} = 0.3 + 0.2 = 0.5$$



0.3 is $P(\text{both})$; 0.2 is $P(\text{neither}) = 1 - 0.8$; 0.8 is $P(\text{at least one})$.

Key idea: Fill a Venn diagram from the inside out: intersection first, then the 'only' regions, then the outside.

Topic: Independence vs mutual exclusivity (concept check) · Easy

Q.3 [MSQ · 1 mark]

Correct answer: B, C, D

Concept: Independence means $P(A \cap B) = P(A)P(B)$; mutual exclusivity means $A \cap B = \emptyset$.

(A) FALSE. Counter-example with one die: $A = \{1\}$, $B = \{2\}$. $P(A \cap B) = 0$ but $P(A)P(B) = 1/36 \neq 0$. (It would be true only if $P(A) = 0$ or $P(B) = 0$.)

(B) TRUE.

$$P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)P(B^c)$$

(C) TRUE. Mutually exclusive $\Rightarrow P(A \cap B) = 0$, but $P(A)P(B) > 0$. The product rule fails, so they are dependent (knowing A happened tells you B did not).

(D) TRUE. $P(A^c \cap B) \leq P(A^c) = 0$, so $P(A \cap B) = P(B) - 0 = P(B) = 1 \cdot P(B) = P(A)P(B)$.

Trap: Mutually exclusive events with positive probabilities are the most dependent events possible — never confuse 'disjoint' with 'independent'.

Q.4 [NAT · 1 mark]

Topic: Conditional probability (dice) · Easy

Correct answer: 0.18 (range 0.18 to 0.19)

Concept: With equally likely outcomes, $P(E | C) = |E \cap C| / |C|$ — the condition shrinks the sample space.

Outcomes with at least one 5: 6 with first die 5, 6 with second die 5, minus (5,5) counted twice:

$$|C| = 6 + 6 - 1 = 11$$

Among these, sum 8 needs the other die to show 3: (5,3) and (3,5).

$$P(\text{sum} = 8 | C) = \frac{2}{11} \approx 0.1818$$

Answer $\approx$ **0.18**.

Trap: Unconditionally $P(\text{sum} = 8) = 5/36 \approx 0.139$. Conditioning on a 5 removes (2,6), (6,2), (4,4) and raises the chance — always rebuild the reduced sample space.

Q.5 [MCQ · 1 mark · -1/3]

Topic: Mean, median, mode · Easy

Correct answer: B

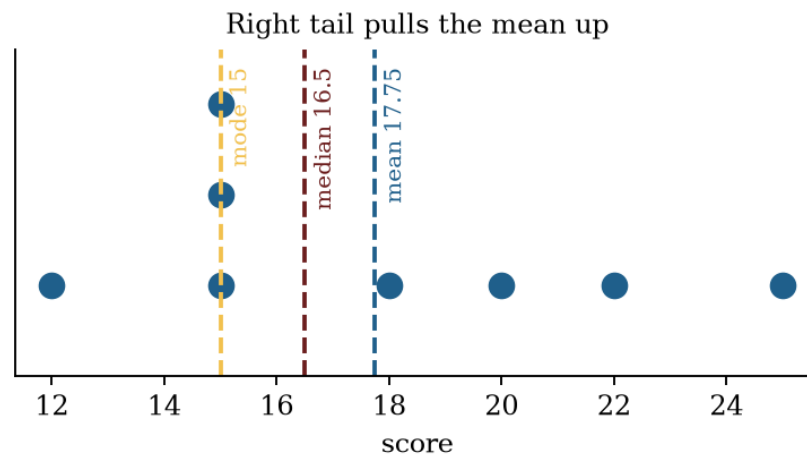
Concept: Sort the data; median = average of the two middle values for even n ; mode = most frequent value.

Sorted: 12, 15, 15, 15, 18, 20, 22, 25.

$$\bar{x} = \frac{12 + 15 + 15 + 15 + 18 + 20 + 22 + 25}{8} = \frac{142}{8} = 17.75$$

$$\text{median} = \frac{x_{(4)} + x_{(5)}}{2} = \frac{15 + 18}{2} = 16.5, \quad \text{mode} = 15$$

So $15 < 16.5 < 17.75$: mode < median < mean.



Key idea: For right-skewed data (long upper tail) the usual pattern is $\text{mode} < \text{median} < \text{mean}$; the mean is the measure most affected by extreme values.

Q.6 [MCQ · 1 mark · -1/3]

Topic: Binomial distribution · Easy

Correct answer: B

Concept: Number of defectives $X \sim \text{Bin}(n = 5, p = 0.1)$, $P(X = k) = C(n, k) p^k (1 - p)^{n-k}$.

$$P(X = 1) = {}^5_1(0.1)^1(0.9)^4 = 5 \times 0.1 \times 0.6561 = 0.32805$$

Distractors: $0.066 \approx 0.1 \times 0.9^4$ forgets the factor $C(5, 1) = 5$; $0.590 = 0.9^5 = P(X = 0)$; $0.410 = 1 - 0.9^5 = P(X \geq 1)$.

Trap: Do not forget the binomial coefficient: the single defective can be any one of the 5 positions.

Q.7 [MSQ · 1 mark]

Topic: Properties of a CDF (concept check) · Easy

Correct answer: A, B, C

Concept: A CDF is non-decreasing, right-continuous, with limits 0 at $-\infty$ and 1 at $+\infty$; jumps equal point masses.

(A) TRUE. For $x < y$, $\{X \leq x\} \subset \{X \leq y\}$, so $F(x) \leq F(y)$.

(B) TRUE. $\{X \leq x + 1/n\}$ decreases to $\{X \leq x\}$; continuity of probability gives $F(x + 1/n) \rightarrow F(x)$.

(C) TRUE. $F(a) - F(a^-) = P(X \leq a) - P(X < a) = P(X = a)$.

(D) FALSE. $F(b) - F(a) = P(a < X \leq b)$, which includes the point b. If X has a point mass at b (e.g. $X = \text{number on a die}$, $b = 3$), the two sides differ by $P(X = b)$.

Trap: For a continuous X all four interval types (a, b) , $[a, b]$, $(a, b]$, $[a, b)$ have the same probability — but for discrete X the endpoints matter.

Q.8 [NAT · 1 mark]**Topic: Poisson distribution · Easy****Correct answer: 0.86 (range 0.86 to 0.87)****Concept:** For $X \sim \text{Poisson}(\lambda)$, $P(X = k) = e^{-\lambda} \lambda^k / k!$; use the complement for 'at least one'.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-2} = 1 - 0.1353 = 0.8647$$

Answer $\approx$ **0.86**.**Shortcut:** 'At least one' $\Rightarrow 1 - P(\text{none})$. Summing $P(1) + P(2) + \dots$ directly is endless for Poisson.**Q.9** [MCQ · 1 mark · -1/3]**Topic: Continuous uniform, conditional probability · Easy****Correct answer: A****Concept:** For $X \sim U(a, b)$, probabilities are proportional to lengths; conditioning on an interval gives another uniform.

$$P(X > 7) = \frac{10 - 7}{10 - 2} = \frac{3}{8}, \quad P(X > 4) = \frac{10 - 4}{8} = \frac{6}{8}$$

$$P(X > 7 | X > 4) = \frac{P(X > 7)}{P(X > 4)} = \frac{3/8}{6/8} = 0.5$$

Shortcut: given $X > 4$, $X \sim U(4, 10)$, so $P(X > 7) = 3/6 = 0.5$.0.375 is the unconditional $P(X > 7)$; 0.750 is $P(X > 4)$; $0.625 = P(X < 7)$.**Key idea:** $\{X > 7\} \subset \{X > 4\}$, so the intersection in the numerator is just $\{X > 7\}$.**Q.10** [MCQ · 1 mark · -1/3]**Topic: Exponential distribution, memorylessness · Easy****Correct answer: D****Concept:** $X \sim \text{Exp}(\lambda)$ with mean $1/\lambda$ has $P(X > t) = e^{-\lambda t}$ and is memoryless: $P(X > s + t | X > s) = P(X > t)$.

$$\lambda = \frac{1}{400} \text{ per hour}$$

$$P(X > 600 | X > 200) = \frac{e^{-600/400}}{e^{-200/400}} = e^{-400/400} = e^{-1} \approx 0.368$$

 $e^{-1.5}$ is the unconditional $P(X > 600)$ (ignoring the condition); $1 - e^{-1}$ is a failure probability.**Key idea:** Memorylessness: a working exponential component is 'as good as new' — the 200 hours already used are irrelevant.

Q.11 [NAT · 1 mark]**Topic: Normal distribution · Easy****Correct answer: 0.0668 (range 0.0665 to 0.067)****Concept:** Standardise: $Z = (X - \mu)/\sigma \sim N(0, 1)$.

$$z = \frac{75 - 60}{10} = 1.5$$

$$P(X > 75) = P(Z > 1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668$$

Answer $\approx$ **0.0668**.**Trap:** Upper-tail area = $1 - \Phi(z)$. Reporting $\Phi(1.5)$ itself (0.9332) is the most common slip.**Q.12** [MSQ · 1 mark]**Topic: Variance rules (concept check) · Easy****Correct answer: B, C, D****Concept:** $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2\text{Cov}(X, Y)$; shifting does not change spread; scaling by a multiplies variance by a^2 .**(A) FALSE.** It needs $\text{Cov}(X, Y) = 0$. E.g. $Y = X$ gives $\text{Var}(2X) = 4\text{Var}(X) \neq 2\text{Var}(X)$.**(B) TRUE.** $\text{Var}(aX + b) = E[(aX + b - a\mu - b)^2] = a^2 E[(X - \mu)^2]$.**(C) TRUE.** Independence $\Rightarrow \text{Cov} = 0$, and $\text{Var}(-Y) = (-1)^2 \text{Var}(Y)$:

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y) = \text{Var}(X) + \text{Var}(Y)$$

(D) TRUE. Expand $E[(X - \mu)^2] = E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2$.**Trap:** Variances ADD for $X - Y$ as well as $X + Y$ (independent case). Writing $\text{Var}(X) - \text{Var}(Y)$ is a classic error — it can even be negative!**Q.13** [NAT · 2 marks]**Topic: Counting: committees (hypergeometric) · Medium****Correct answer: 0.65 (range 0.65 to 0.66)****Concept:** Equally likely subsets $\Rightarrow P = \text{favourable} / \text{total}$; split 'at least 2' into disjoint cases (or use the complement).

$$\text{Total} = \binom{11}{4} = 330$$

Count each case (women, men):

Women	Men	Count
2	2	$C(5,2) \cdot C(6,2) = 10 \times 15 = 150$
3	1	$C(5,3) \cdot C(6,1) = 10 \times 6 = 60$
4	0	$C(5,4) = 5$
Total		215

$$P(\geq 2 \text{ women}) = \frac{215}{330} = \frac{43}{66} \approx 0.6515$$

Check by complement: 0 women: $C(6,4) = 15$; 1 woman: $5 \times C(6,3) = 100$; $1 - 115/330 = 215/330$. (verified)Answer $\approx$ **0.65**.

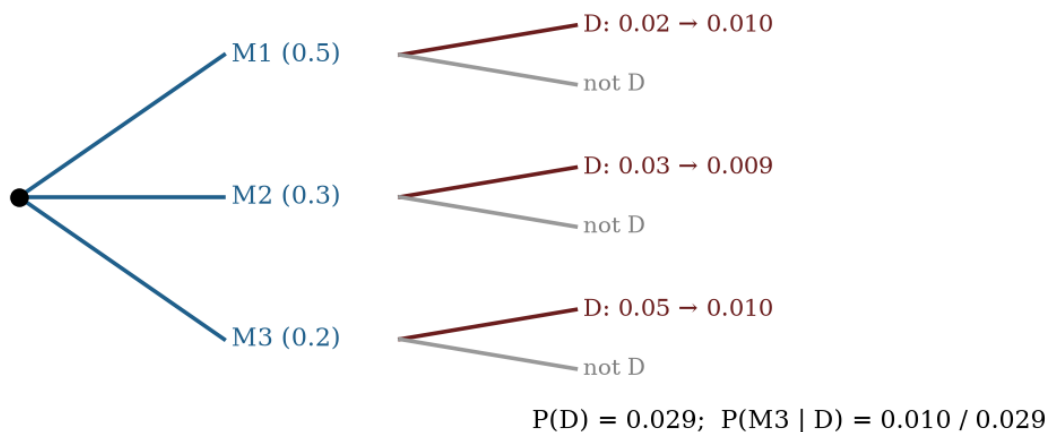
Trap: Do NOT count 'choose 2 women, then any 2 of the remaining 9' = $C(5,2)C(9,2) = 360$ — it counts committees with 3 or 4 women several times.

Q.14 [MCQ · 2 marks · -2/3]

Topic: Bayes theorem (quality control) · Medium

Correct answer: C

Concept: Bayes theorem: posterior $\propto$ prior $\times$ likelihood; the denominator is the total probability of a defect.



Total probability of a defect:

$$P(D) = 0.5(0.02) + 0.3(0.03) + 0.2(0.05) = 0.010 + 0.009 + 0.010 = 0.029$$

Bayes theorem:

$$P(M_3 | D) = \frac{P(M_3)P(D | M_3)}{P(D)} = \frac{0.010}{0.029} \approx 0.3448$$

0.200 is the prior $P(M_3)$ (ignores the evidence); $0.310 \approx 0.009/0.029$ is $P(M_2 | D)$; 0.050 is $P(D | M_3)$ — the inverse conditional.

Key idea: Although M3 makes only 20% of items, it contributes about 34% of defects because its defect rate is highest. Posterior shares always add to 1: $0.345 + 0.345 + 0.310 = 1$.

Q.15 [MCQ · 2 marks · -2/3]

Topic: Conditional probability (urn, without replacement) · Medium

Correct answer: C

Concept: $P(R_1 | R_2) = P(R_1 \cap R_2)/P(R_2)$, with $P(R_2)$ from the law of total probability.

$$P(R_1 \cap R_2) = \frac{4}{10} \cdot \frac{3}{9} = \frac{12}{90}$$

$$P(R_2) = \frac{4}{10} \cdot \frac{3}{9} + \frac{6}{10} \cdot \frac{4}{9} = \frac{12 + 24}{90} = \frac{36}{90} = \frac{2}{5}$$

$$P(R_1 | R_2) = \frac{12/90}{36/90} = \frac{1}{3}$$

Symmetry check: given ball 2 is red, ball 1 is a random draw from the other 9 balls, of which 3 are red $\Rightarrow 3/9 = 1/3$. (verified)

$2/5$ is $P(R_2)$ (equal to $P(R_1)$ by exchangeability); $4/9$ is $P(R_2 | B_1)$; $2/15 = 12/90$ is the joint probability.

Key idea: Draws without replacement are exchangeable: the 2nd ball is just as likely to be red as the 1st. Conditioning 'backwards in time' is perfectly legitimate.

Q.16 [NAT · 2 marks]

Topic: Joint PMF, covariance · Medium

Correct answer: 0.06

Concept: $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$; marginals come from row/column sums.

Marginals: $P(X = 1) = 0.15 + 0.15 + 0.30 = 0.6$; $P(Y = 0, 1, 2) = 0.25, 0.35, 0.40$.

$$E[X] = 0.6, \quad E[Y] = 0(0.25) + 1(0.35) + 2(0.40) = 1.15$$

Only cells with $X = 1$ and $Y \geq 1$ contribute to $E[XY]$:

Cells contributing to $E[XY]$: $X=1$ row only

$X=0$	0.10	0.20	0.10
$X=1$	0.15	0.15	0.30
	$Y=0$	$Y=1$	$Y=2$

$$E[XY] = 1 \cdot 1 \cdot 0.15 + 1 \cdot 2 \cdot 0.30 = 0.75$$

$$\text{Cov}(X, Y) = 0.75 - 0.6 \times 1.15 = 0.75 - 0.69 = 0.06$$

Positive covariance: night-shift items tend to carry more flaws ($E[Y | X = 1] = 0.75/0.6 = 1.25$ vs $E[Y | X = 0] = 0.40/0.4 = 1.0$).

Shortcut: For a 0/1 variable X , $\text{Cov}(X, Y) = P(X = 1) \cdot (E[Y | X = 1] - E[Y]) = 0.6 \times (1.25 - 1.15) = 0.06$ — a fast cross-check.

Q.17 [NAT · 2 marks]

Topic: Conditional expectation and variance · Medium

Correct answer: 1.60 (range 1.6 to 1.61)

Concept: Law of total variance: $\text{Var}(Y) = E[\text{Var}(Y | N)] + \text{Var}(E[Y | N])$.

Given $N = n$, $Y \sim \text{Bin}(n, 1/2)$:

$$E[Y | N] = \frac{N}{2}, \quad \text{Var}(Y | N) = N \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{N}{4}$$

For a fair die (discrete uniform on 1-6):

$$E[N] = \frac{7}{2}, \quad \text{Var}(N) = \frac{6^2 - 1}{12} = \frac{35}{12}$$

Combine:

$$E[\text{Var}(Y|N)] = \frac{E[N]}{4} = \frac{7}{8} = 0.875$$

$$\text{Var}(E[Y|N]) = \frac{\text{Var}(N)}{4} = \frac{35}{48} \approx 0.7292$$

$$\text{Var}(Y) = \frac{7}{8} + \frac{35}{48} = \frac{77}{48} \approx 1.6042$$

(Also $E[Y] = E[N]/2 = 1.75$.) Answer ≈ 1.60 .

Trap: Using only $E[\text{Var}(Y|N)] = 0.875$ misses the extra spread caused by the random number of tosses. Both pieces are needed.

Topic: Conditional PDF, conditional expectation · Medium

Q.18 [MCQ · 2 marks · -2/3]

Correct answer: A

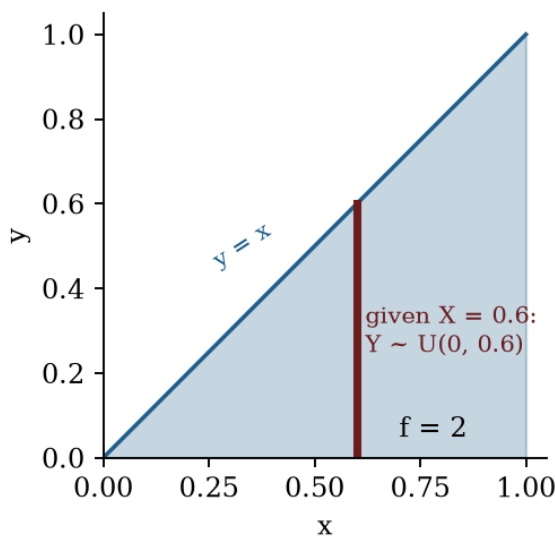
Concept: $f_{Y|X}(y|x) = f(x, y)/f_X(x)$; then integrate y against it.

Marginal of X (integrate over $0 < y < x$):

$$f_X(x) = \int_0^x 2 \, dy = 2x, \quad 0 < x < 1$$

$$f_{Y|X}(y|x) = \frac{2}{2x} = \frac{1}{x}, \quad 0 < y < x$$

So given $X = x$, $Y \sim U(0, x)$.



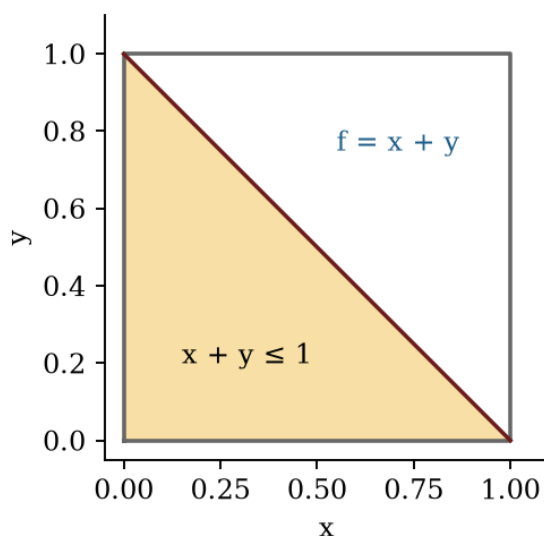
$$E[Y|X = 0.6] = \frac{0 + 0.6}{2} = 0.30$$

0.333 is the unconditional $E[Y] = \iint 2y = 1/3$; $0.40 = 2(0.6)/3$ comes from wrongly taking the conditional density $\propto y$; 0.60 is x itself.

Shortcut: A constant joint density on a region gives uniform conditionals along each slice — draw the region and read off the slice.

Q.19 [NAT · 2 marks]

Topic: Joint PDF, probability over a region · Medium

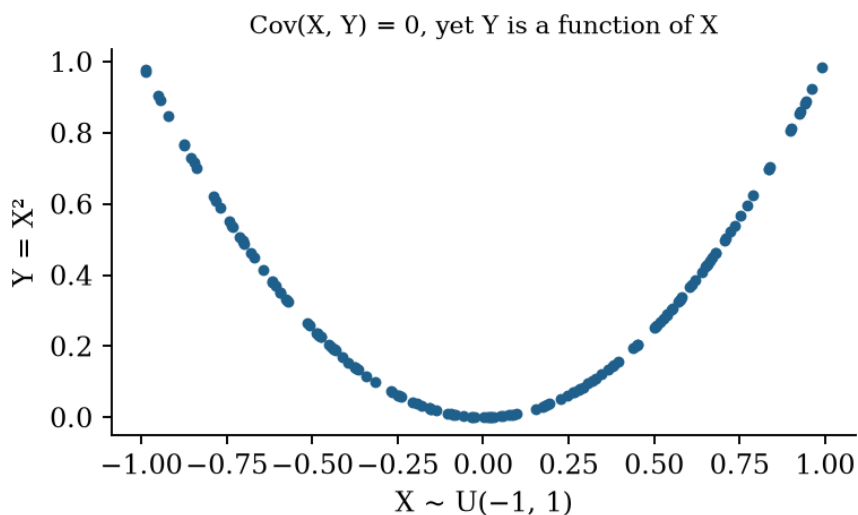
Correct answer: 0.33 (range 0.33 to 0.34)**Concept:** $P((X, Y) \in R) = \iint_R f(x, y) dy dx$, with limits read from a sketch.Region: $0 \leq x \leq 1, 0 \leq y \leq 1 - x$.

$$\begin{aligned}
 P &= \int_0^1 \int_0^{1-x} (x + y) dy dx = \int_0^1 \left[x(1 - x) + \frac{(1 - x)^2}{2} \right] dx \\
 &= \int_0^1 \frac{(1 - x)(1 + x)}{2} dx = \frac{1}{2} \int_0^1 (1 - x^2) dx = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}
 \end{aligned}$$

Answer $\approx$ **0.33**.Sanity check: the triangle is half the square, but density is smaller there ($x + y \leq 1$) than in the other half, so the answer should be below 0.5. (verified)**Trap:** Area of the region (1/2) is NOT the probability unless the density is uniform.**Q.20** [MSQ · 2 marks]

Topic: Covariance and correlation (concept check) · Medium

Correct answer: A, C, D**Concept:** $\rho = \text{Cov}(X, Y)/(\sigma_X \sigma_Y)$ is scale-free, bounded by Cauchy-Schwarz, and flips sign when one variable is negated.**(A) TRUE.** Cauchy-Schwarz: $|\text{Cov}(X, Y)| \leq \sigma_X \sigma_Y$.**(B) FALSE.** Zero covariance means no linear relation only. Example: $X \sim U(-1, 1)$, $Y = X^2$: $\text{Cov} = E[X^3] - E[X]E[X^2] = 0$, yet Y is completely determined by X .



(C) **TRUE.** $\text{Cov}(2X + 3, -Y) = -2\text{Cov}(X, Y)$; SDs are $2\sigma_X$ and σ_Y :

$$\rho(2X + 3, -Y) = \frac{-2 \text{Cov}(X, Y)}{2\sigma_X \sigma_Y} = -\rho(X, Y)$$

(D) **TRUE.** $\text{Cov}(X, X) = E[X^2] - (E[X])^2 = \text{Var}(X)$.

Key idea: Independent $\Rightarrow$ uncorrelated, but NOT conversely (the converse holds for jointly normal variables).

Q.21 [MCQ · 2 marks · -2/3]

Topic: Sample correlation coefficient · Medium

Correct answer: B

Concept: $r = S_{xy} / \sqrt{(S_{xx} S_{yy})}$ with $S_{xy} = \sum (x - \bar{x})(y - \bar{y})$.

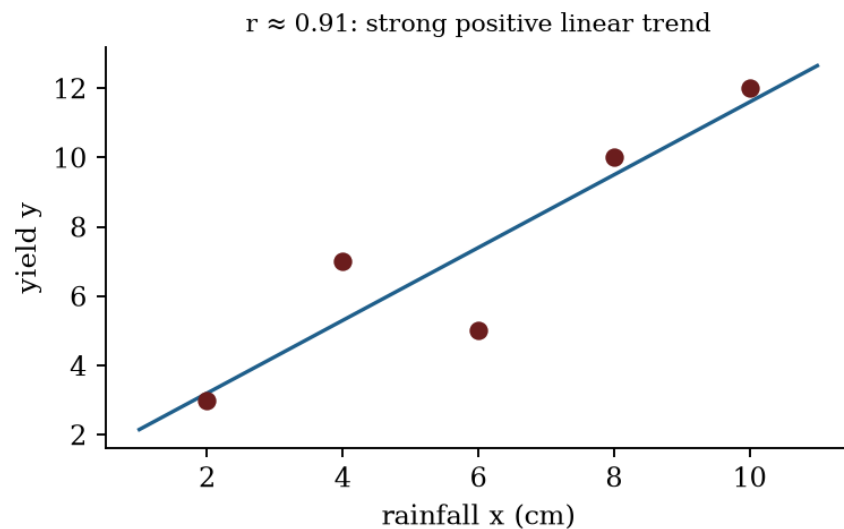
$\bar{x} = 30/5 = 6$, $\bar{y} = 37/5 = 7.4$.

$x - \bar{x}$	-4	-2	0	2	4
$y - \bar{y}$	-4.4	-0.4	-2.4	2.6	4.6
product	17.6	0.8	0	5.2	18.4

$$S_{xy} = 42, \quad S_{xx} = 16 + 4 + 0 + 4 + 16 = 40$$

$$S_{yy} = 19.36 + 0.16 + 5.76 + 6.76 + 21.16 = 53.2$$

$$r = \frac{42}{\sqrt{40 \times 53.2}} = \frac{42}{46.13} \approx 0.910$$



$0.83 \approx r^2$ (coefficient of determination); $1.05 = S_{xy}/S_{xx}$ is the regression slope, not r — and any value above 1 is impossible for r .

Shortcut: Reject options outside $[-1, 1]$ immediately; then check the sign from the scatter.

Q.22 [MSQ · 2 marks]

Topic: Reading a discrete CDF · Medium

Correct answer: A, B, C

Concept: For a discrete X , the PMF equals the jump sizes of the CDF.

x	1	2	4	5
$P(X = x)$ (jump)	0.2	0.3	0.4	0.1

(A) TRUE. Jump at 2: $0.5 - 0.2 = 0.3$.

(B) TRUE. $P(2 \leq X < 5) = F(5^-) - F(2^-) = 0.9 - 0.2 = 0.7$ ($= P(X=2) + P(X=4) = 0.3 + 0.4$).

(C) TRUE. $P(X > 4) = 1 - F(4) = 1 - 0.9 = 0.1$.

(D) FALSE.

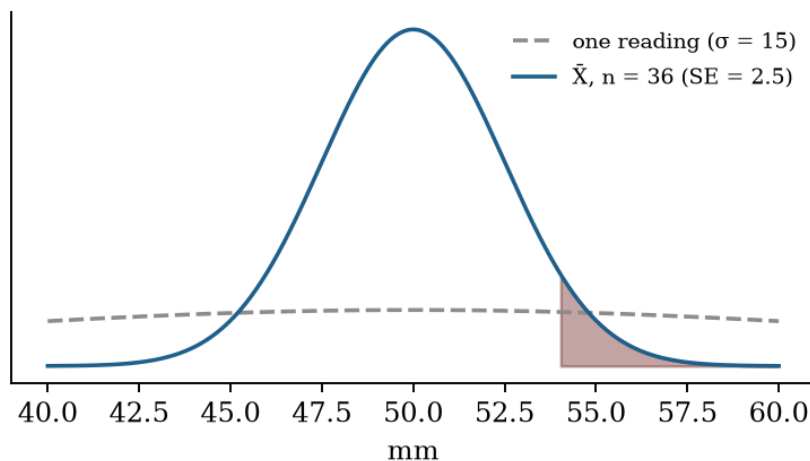
$$E[X] = 1(0.2) + 2(0.3) + 4(0.4) + 5(0.1) = 0.2 + 0.6 + 1.6 + 0.5 = 2.9$$

Trap: Filled dots mark F 's value at a jump (right-continuity). For strict/non-strict inequalities, decide whether the jump at the endpoint is included.

Q.23 [MCQ · 2 marks · -2/3]**Topic: Central limit theorem · Medium****Correct answer: D****Concept:** CLT: $\bar{X} \approx N(\mu, \sigma^2/n)$ for large n , whatever the population shape.

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{36}} = 2.5$$

$$P(\bar{X} > 54) \approx P\left(Z > \frac{54 - 50}{2.5}\right) = P(Z > 1.6) = 1 - 0.9452 = 0.0548$$



0.3936 uses $\sigma = 15$ instead of the standard error ($z \approx 0.27$); 0.9452 is $\Phi(1.6)$ itself; 0.1096 is a two-tailed area.

Trap: The spread of an average shrinks by $\sqrt{n}$. Using σ instead of $\sigma/\sqrt{n}$ is the #1 CLT mistake.

Q.24 [NAT · 2 marks]**Topic: Confidence interval (t, σ unknown) · Medium****Correct answer: 50.41 (range 50.4 to 50.41)****Concept:** σ unknown, normal data $\Rightarrow \bar{x} \pm t_{\alpha/2, n-1} \cdot s/\sqrt{n}$.

$$\frac{s}{\sqrt{n}} = \frac{0.5}{\sqrt{25}} = 0.1$$

$$\text{margin} = 2.064 \times 0.1 = 0.2064$$

$$CI = 50.2 \pm 0.2064 = (49.9936, 50.4064)$$

Upper limit $\approx$ **50.41** mm.

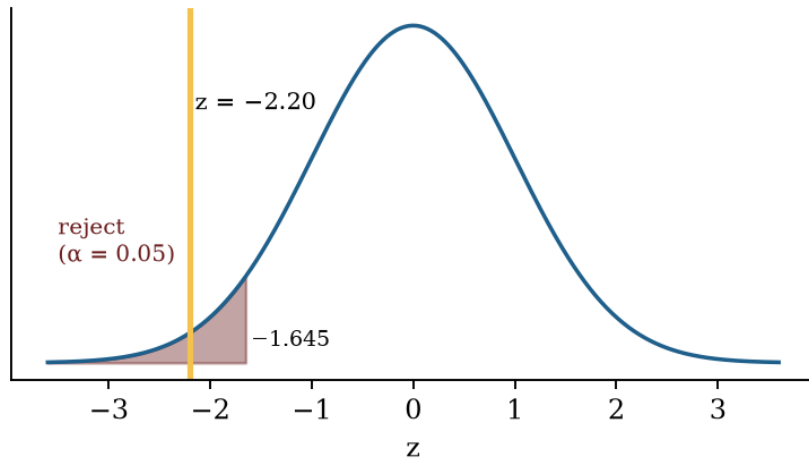
With $z = 1.96$ instead of t you would get $50.396 \approx 50.40$ — slightly narrower; GATE expects t when σ is unknown and n is small.

Key idea: Degrees of freedom = $n - 1 = 24$, and the t critical value is always larger than the matching z value.

Q.25 [MCQ · 2 marks · -2/3]**Topic: One-sample z-test (left-tailed) · Medium****Correct answer: A****Concept:** $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$; left-tailed test rejects when $z \leq -z_\alpha$.

$$\frac{\sigma}{\sqrt{n}} = \frac{4}{8} = 0.5$$

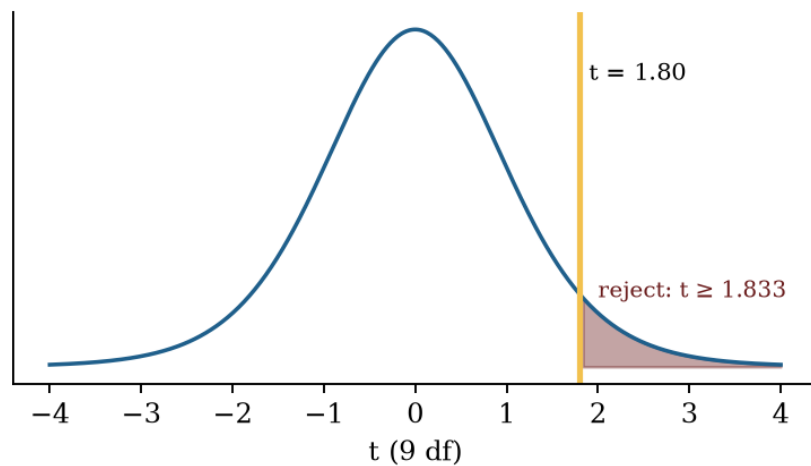
$$z = \frac{498.9 - 500}{0.5} = \frac{-1.1}{0.5} = -2.20$$

Rejection region: $z \leq -1.645$. Since $-2.20 < -1.645$, reject H_0 : evidence of under-filling.(C) divides by σ instead of $\sigma/\sqrt{n}$; (D) multiplies by $\sqrt{n}$ twice ($-1.1 \times 64/4$). p-value = $\Phi(-2.2) \approx 0.0139 < 0.05$. (verified)**Key idea:** The direction of H_1 fixes which tail holds the whole α . Here all 5% sits in the left tail.**Q.26** [MCQ · 2 marks · -2/3]**Topic: One-sample t-test · Medium****Correct answer: B****Concept:** σ unknown $\Rightarrow t = (\bar{x} - \mu_0)/(s/\sqrt{n})$ with $n - 1$ df; reject if $t \geq t_{\alpha, n-1}$.

$$\frac{s}{\sqrt{n}} = \frac{5.8}{\sqrt{10}} = \frac{5.8}{3.1623} = 1.8341$$

$$t = \frac{67.3 - 64}{1.8341} = \frac{3.3}{1.8341} \approx 1.799$$

Critical value $t_{0.05, 9} = 1.833$. Since $1.80 < 1.833$, we **do not reject** H_0 at 5%.



(C) forgets $\sqrt{n}$ ($3.3/0.58$); (A)/(D) use the z cut-off 1.645, which is wrong when σ is estimated from only 10 observations — the heavier-tailed t requires stronger evidence.

Trap: A borderline case: z-logic would reject, t-logic does not. Always use t with s and small n.

Q.27 [MCQ · 2 marks · -2/3]

Topic: χ^2 goodness-of-fit test · Medium

Correct answer: D

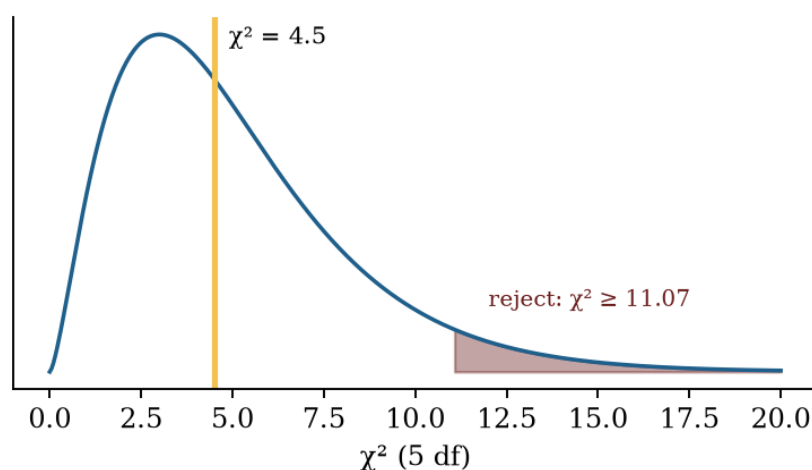
Concept: $\chi^2 = \sum (O - E)^2/E$ with $k - 1$ df (no parameters estimated).

Under H_0 (fair die), $E = 120/6 = 20$ for each face.

Face	1	2	3	4	5	6
$O - E$	-5	5	-2	2	-4	4
$(O - E)^2$	25	25	4	4	16	16

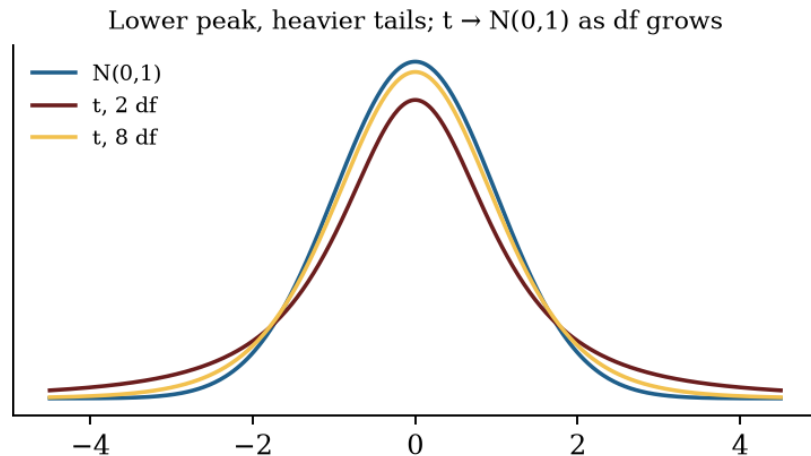
$$\chi^2 = \frac{25 + 25 + 4 + 4 + 16 + 16}{20} = \frac{90}{20} = 4.5$$

df = 6 - 1 = 5; critical value 11.07. Since $4.5 < 11.07$, do not reject H_0 .



(C) forgets to divide by E; (B) uses k instead of $k - 1$ df (the conclusion happens to agree but the df is wrong); (A) has the right statistic but the wrong decision.

Key idea: The counts must total n, which costs one degree of freedom: $df = k - 1 - (\text{number of estimated parameters})$.

Q.28 [MSQ · 2 marks]**Topic: t and χ^2 distributions (concept check) · Medium****Correct answer: A, B, C****Concept:** $t_\nu = Z/\sqrt{V/\nu}$ with $V \sim \chi^2_\nu$; χ^2_k is a sum of k squared standard normals.**(A) TRUE.** Extra randomness in the denominator (estimated σ) spreads mass into the tails.**(B) TRUE.** $V/\nu \rightarrow 1$ as $\nu \rightarrow \infty$ (law of large numbers), so $t_\nu \rightarrow Z$.**(C) TRUE.** This is the definition of χ^2_k .**(D) FALSE.** Each Z_i^2 has mean 1 and variance $E[Z^4] - 1 = 3 - 1 = 2$, so

$$E[\chi_k^2] = k, \quad \text{Var}(\chi_k^2) = 2k$$

Key idea: Remember 'mean k, variance 2k' for χ^2 ; and t-critical values shrink toward z-values as the df grow.

Q.29 [NAT · 2 marks]**Topic: Binomial: acceptance sampling · Medium****Correct answer: 0.914 (range 0.913 to 0.915)****Concept:** Number of defectives $X \sim \text{Bin}(10, 0.05)$ (large lot $\Rightarrow$ independence); $P(\text{accept}) = P(X \leq 1)$.

$$P(X=0) = (0.95)^{10} = 0.5987$$

$$P(X=1) = \binom{10}{1}(0.05)(0.95)^9 = 10 \times 0.05 \times 0.6302 = 0.3151$$

$$P(\text{accept}) = 0.5987 + 0.3151 = 0.9139$$

Answer $\approx$ **0.914**.Poisson check ($\lambda = np = 0.5$): $e^{-0.5}(1 + 0.5) = 0.9098$ — close, but the exact binomial is required here.

Trap: 'At most one' = $P(0) + P(1)$. Forgetting $P(0)$ gives only 0.315.

Topic: Discrete uniform, Bernoulli indicators,
independence · Medium

Q.30 [MSQ · 2 marks]

Correct answer: A, B, C

Concept: X is discrete uniform on $\{1, \dots, 6\}$: mean $(n + 1)/2$, variance $(n^2 - 1)/12$. Indicators are Bernoulli; two indicators are independent iff $P(\text{both} = 1) = \text{product of their means}$.

(A) TRUE. $E[X] = (6 + 1)/2 = 3.5$.

(B) TRUE.

$$\text{Var}(X) = E[X^2] - 3.5^2 = \frac{91}{6} - \frac{49}{4} = \frac{182 - 147}{12} = \frac{35}{12}$$

(C) TRUE. $I \sim \text{Bernoulli}(1/2)$, $J \sim \text{Bernoulli}(1/3)$. $\{I = 1, J = 1\} = \{X = 2\}$:

$$P(I = 1, J = 1) = \frac{1}{6} = \frac{1}{2} \cdot \frac{1}{3} = P(I = 1)P(J = 1)$$

For 0/1 variables, this single equality implies the other three cells factor as well.

(D) FALSE. $\{I = 1, K = 1\} = \{X = 2\}$:

$$P(I = 1, K = 1) = \frac{1}{6} \neq \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Key idea: Independence is a numerical property, not a 'physical' one: events about the SAME roll can still be independent.

Mock Test 02 — Foundation

Full-Syllabus Test II

A second warm-up paper over the whole syllabus — cards, coins, dice, urns, seating, factory shifts, exam scores and weather.

Item	Details
Questions	30 (12 × 1-mark, 18 × 2-mark)
Maximum marks	48
Question types	MCQ: 14 · MSQ: 7 · NAT: 9
Suggested time	90 minutes
Negative marking	MCQ only: $-1/3$ (1-mark), $-2/3$ (2-mark). None for MSQ & NAT.
Calculator	Use only the GATE virtual scientific calculator

Topic focus of this set

Covers the full syllabus at warm-up difficulty: counting with cards and seating, sample spaces and axioms, mutually exclusive vs independent events, contingency tables (joint, marginal, conditional), Bayes with a two-urn tree, conditional expectation/variance (random sums), conditional and joint PDFs, sample variance and grouped standard deviation, Var of linear combinations, Bernoulli/binomial, Poisson, uniform, exponential, normal, CDF-based median, CLT for sums, z-interval, z-test (p-value), paired t-test and χ^2 test of independence. Concept-check MSQs probe PDF properties, expectation rules, independence implications, sampling distributions and the logic of tests and intervals.

Instructions

- Questions 1 onwards are arranged as in GATE: all 1-mark questions first, then 2-mark questions.
- **MCQ** – exactly one correct option. **MSQ** – ONE OR MORE options are correct; credit only if you select all correct options and no wrong option (no partial marks, no negative marks).
- **NAT** – enter a numerical value using the virtual keypad. The accepted range is given in the key; round as instructed in the question.
- Statistical tables (Z, t, χ^2) are provided in the Appendix; values in solutions use those tables.
- Attempt in one sitting, then check the answer key and read every solution — including those you got right — for the alternative methods and traps.

Mock Test 02 · Question Paper

1-mark questions

Q.1 MCQ · 1 mark · -1/3

The number of 5-card hands that can be dealt from a standard 52-card deck containing **exactly** two aces is

- (A) 132 600
- (B) 117 600
- (C) 622 656
- (D) 103 776

Q.2 MCQ · 1 mark · -1/3

A fair coin is tossed three times. Let A = 'at least two heads' and B = 'the first toss is a tail'. $P(A \cup B)$ equals

- (A) $1/2$
- (B) $3/4$
- (C) $7/8$
- (D) $5/8$

Q.3 MSQ · 1 mark

Events A and B are mutually exclusive with $P(A) = 0.3$ and $P(B) = 0.5$. Which of the following is/are TRUE?

- (A) $P(A \cup B) = 0.8$
- (B) $P(A | B) = 0$
- (C) A and B are independent
- (D) $P(A^c \cap B^c) = 0.2$

Q.4 NAT · 1 mark

Two cards are drawn at random without replacement from a standard 52-card deck. Given that at least one of them is a king, the probability that both are kings is _____ (round off to 3 decimal places).

Answer: _____

Q.5 MCQ · 1 mark · -1/3

The rainfall (in cm) recorded at a station in five consecutive months is 4, 7, 7, 9, 13. The **sample variance** (with divisor $n - 1$) is

- (A) 8.8
- (B) 11
- (C) 3.32
- (D) 44

Q.6 MCQ · 1 mark · -1/3

In a series of independent free throws, the number of successes X follows a binomial distribution with mean 6 and variance 2.4. The parameters (n, p) are

- (A) (15, 0.4)
- (B) (12, 0.5)
- (C) (10, 0.6)
- (D) (10, 0.4)

Q.7 MSQ · 1 mark

Let f be the probability density function of a continuous random variable X . Which of the following is/are necessarily TRUE?

- (A) $f(x) \leq 1$ for all x
- (B) $f(x) \geq 0$ for all x
- (C) $P(X = c) = 0$ for every real c
- (D) $\int_{-\infty}^{\infty} f(x) dx = 1$

Q.8 NAT · 1 mark

The number of bubbles (defects) in a glass sheet follows a Poisson distribution with mean 1.5. A sheet is graded 'premium' if it has at most one bubble. The probability that a randomly chosen sheet is premium is _____ (round off to 3 decimal places).

Answer: _____

Q.9 MCQ · 1 mark · -1/3

The waiting time (in hours) until the next rain shower in a monsoon week is exponentially distributed with rate 0.25 per hour. The **median** waiting time is closest to

- (A) 4.00 h
- (B) 2.00 h
- (C) 2.77 h
- (D) 0.17 h

Q.10 MCQ · 1 mark · -1/3

If $Z \sim N(0, 1)$, then $P(-1 < Z < 2)$ is closest to $[\Phi(1) = 0.8413, \Phi(2) = 0.9772]$

- (A) 0.6826
- (B) 0.8185
- (C) 0.9544
- (D) 0.1815

Q.11 NAT · 1 mark

The rainfall X (in cm) on a stormy day has CDF $F(x) = 0$ for $x < 0$, $F(x) = x^2/9$ for $0 \leq x < 3$, and $F(x) = 1$ for $x \geq 3$. The median of X is _____ cm (round off to 2 decimal places).

Answer: _____

Q.12 MSQ · 1 mark

Let X and Y be random variables with finite means (and $X > 0$ where $1/X$ appears). Which of the following is/are TRUE **in general**?

- (A) $E[XY] = E[X] E[Y]$
- (B) $E[X + Y] = E[X] + E[Y]$
- (C) $E[E[X | Y]] = E[X]$
- (D) $E[1/X] = 1/E[X]$

2-mark questions**Q.13 NAT · 2 marks**

Four boys and four girls take seats at random in a row of 8 chairs. The probability that boys and girls sit alternately is _____ (round off to 3 decimal places).

Answer: _____

Q.14 MCQ · 2 marks · -2/3

Four fair dice are rolled. The probability that at least two of them show the same face is

- (A) $5/18$
- (B) $13/18$
- (C) $671/1296$
- (D) $1/216$

Q.15 MCQ · 2 marks · -2/3

Urn I contains 3 white and 2 black balls; Urn II contains 2 white and 4 black balls. A fair coin is tossed: heads = Urn I, tails = Urn II. One ball is drawn from the chosen urn and it is white. The probability that Urn I was chosen is

- (A) $3/5$
- (B) $1/2$
- (C) $9/14$
- (D) $5/14$

Q.16 MSQ · 2 marks

A factory's output for one week, classified by shift and quality, is:

Shift	Good	Defective	Total
Day	360	40	400
Night	180	20	200
Total	540	60	600

An item is chosen at random from the 600. Which of the following is/are TRUE?

- (A) $P(\text{Defective}) = 0.1$
- (B) $P(\text{Night} | \text{Defective}) = 1/3$
- (C) The events 'Night shift' and 'Defective' are independent
- (D) $P(\text{Day} \cap \text{Defective}) = 0.4$

Q.17 NAT · 2 marks

The number of severe storms N in a season follows a Poisson distribution with mean 4. Each storm, independently, causes a flood with probability 0.25. Let X be the number of floods in the season. $\text{Var}(X)$ is _____ (round off to 2 decimal places).

Answer: _____

Q.18 MCQ · 2 marks · -2/3

The joint PDF of (X, Y) is $f(x, y) = 6x$ for $0 < x < y < 1$ and 0 otherwise. The conditional probability $P(X < 0.25 \mid Y = 0.5)$ is

- (A) 0.25
- (B) 0.50
- (C) 0.125
- (D) 0.0625

Q.19 NAT · 2 marks

The joint PDF of (X, Y) is $f(x, y) = c(x + 2y)$ for $0 \leq x \leq 1$, $0 \leq y \leq 1$, and 0 elsewhere, where c is a constant. $P(X > Y)$ is _____ (round off to 3 decimal places).

Answer: _____

Q.20 MSQ · 2 marks

Let X and Y be **independent** random variables with finite, positive variances. Which of the following is/are necessarily TRUE?

- (A) $\text{Cov}(X, Y) = 0$
- (B) $E[XY] = E[X] E[Y]$
- (C) $\text{Var}(XY) = \text{Var}(X) \text{Var}(Y)$
- (D) X^2 and Y^2 are independent

Q.21 MCQ · 2 marks · -2/3

Daily rainfalls X and Y (in cm) at two nearby stations have $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$ and correlation coefficient 0.5. $\text{Var}(2X - Y)$ equals

- (A) 25
- (B) 37
- (C) 19
- (D) 13

Q.22 NAT · 2 marks

The scores (out of 10) of 20 students in a weekly quiz are summarised below:

Score	2	4	6	8	10
Frequency	3	5	8	3	1

The standard deviation of the scores, using divisor n (population formula), is _____ (round off to 2 decimal places).

Answer: _____

Q.23 MCQ · 2 marks · -2/3

Diameters of bolts are normally distributed with mean 10.00 mm and standard deviation 0.02 mm. A bolt is rejected if its diameter differs from 10.00 mm by more than 0.03 mm. The proportion of bolts rejected is closest to $[\Phi(1.5) = 0.9332]$

- (A) 0.0668
- (B) 0.1336
- (C) 0.8664
- (D) 0.0456

Q.24 NAT · 2 marks

Each of 48 rain-gauge readings is truncated (rounded down) to a whole millimetre. The truncation errors are independent and uniformly distributed on $[0, 1]$ mm. Using the central limit theorem (no continuity correction), the probability that the total truncation error exceeds 26 mm is _____ (round off to 3 decimal places). $[\Phi(1) = 0.8413]$

Answer: _____

Q.25 MCQ · 2 marks · -2/3

Annual rainfall in a region is normally distributed with known standard deviation 8 cm. Records for 16 randomly chosen years give a mean of 92 cm. The 99% confidence interval for the mean annual rainfall is $[z_{0.005} = 2.576, z_{0.025} = 1.96, z_{0.05} = 1.645]$

- (A) (88.08, 95.92)
- (B) (86.85, 97.15)
- (C) (88.71, 95.29)
- (D) (71.39, 112.61)

Q.26 NAT · 2 marks

To test whether a coin is fair, it is tossed 400 times and shows 224 heads. Using the normal approximation (no continuity correction) for $H_0: p = 0.5$ versus $H_1: p \neq 0.5$, the two-sided p-value is _____ (round off to 3 decimal places). $[\Phi(2.4) = 0.9918]$

Answer: _____

Q.27 MCQ · 2 marks · -2/3

Two revision methods are compared on 100 students:

Method	Pass	Fail	Total
A	40	10	50
B	30	20	50
Total	70	30	100

Using the χ^2 test of independence without continuity correction at the 5% level ($\chi^2_{0.05, 1} = 3.841$, $\chi^2_{0.05, 3} = 7.815$), which is correct?

- (A) $\chi^2 \approx 4.76$; reject independence of method and result
- (B) $\chi^2 \approx 4.76$; do not reject, since the critical value (3 df) is 7.815
- (C) $\chi^2 \approx 2.38$; do not reject independence
- (D) $\chi^2 \approx 0.95$; do not reject independence

Q.28 MCQ · 2 marks · -2/3

Six students take a test before and after a revision workshop:

Student	1	2	3	4	5	6
Before	52	60	58	45	70	63
After	55	65	57	49	72	68

Assume the differences (After – Before) are normal. For $H_0: \mu_d = 0$ vs $H_1: \mu_d > 0$ at the 5% level ($t_{0.05, 5} = 2.015$), which is correct?

- (A) $t \approx 3.53$; reject H_0
- (B) $t \approx 3.22$; do not reject H_0
- (C) $t \approx 1.32$; do not reject H_0
- (D) $t \approx 3.22$; reject H_0

Q.29 MSQ · 2 marks

Let $X_1, \dots, X_n$ be i.i.d. with mean μ and finite variance σ^2 (population shape unknown), and let $\bar{X}$ be their sample mean. Which of the following is/are TRUE?

- (A) $E[\bar{X}] = \mu$ for every n
- (B) $\text{Var}(\bar{X}) = \sigma^2/n$
- (C) For large n , the distribution of $\bar{X}$ is approximately normal
- (D) For large n , each individual X_i is approximately normal

Q.30 MSQ · 2 marks

Which of the following statements about hypothesis tests and confidence intervals is/are TRUE?

- (A) If the 95% two-sided z -interval for μ excludes μ_0 , the two-sided z -test of $H_0: \mu = \mu_0$ rejects H_0 at the 5% level.
- (B) A p -value of 0.03 means there is a 3% probability that H_0 is true.
- (C) For the same data, a 99% confidence interval is wider than a 95% confidence interval.
- (D) In a χ^2 goodness-of-fit test with k categories and no estimated parameters, the degrees of freedom are $k - 1$.

Mock Test 02 · Answer Key

Q	Type	Marks	Answer	Topic
1	MCQ	1	D	Counting: card hands
2	MCQ	1	C	Sample space, union of events
3	MSQ	1	A, B, D	Mutually exclusive events
4	NAT	1	0.030 (range 0.03 to 0.031)	Conditional probability (cards)
5	MCQ	1	B	Sample variance
6	MCQ	1	C	Binomial mean and variance
7	MSQ	1	B, C, D	Properties of a PDF (concept check)
8	NAT	1	0.558 (range 0.557 to 0.559)	Poisson distribution
9	MCQ	1	C	Exponential distribution: median
10	MCQ	1	B	Standard normal probabilities
11	NAT	1	2.12 (range 2.12 to 2.13)	CDF and median of a continuous RV
12	MSQ	1	B, C	Expectation rules (concept check)
13	NAT	2	0.029 (range 0.028 to 0.029)	Counting: seating in a row
14	MCQ	2	B	Complement rule (dice)
15	MCQ	2	C	Bayes theorem (two urns)
16	MSQ	2	A, B, C	Contingency table: joint, marginal, conditional
17	NAT	2	1.00 (range 0.99 to 1.01)	Conditional expectation and variance (random sum)
18	MCQ	2	A	Conditional PDF
19	NAT	2	0.444 (range 0.444 to 0.445)	Joint PDF: normalisation and region probability
20	MSQ	2	A, B, D	Implications of independence (concept check)
21	MCQ	2	D	Variance of a linear combination
22	NAT	2	2.11 (range 2.1 to 2.11)	Grouped data: mean and standard deviation
23	MCQ	2	B	Normal distribution: tolerance limits
24	NAT	2	0.159 (range 0.158 to 0.16)	CLT for a sum of uniforms
25	MCQ	2	B	Confidence interval (z , σ known)
26	NAT	2	0.016 (range 0.016 to 0.017)	z -test for a proportion: p -value
27	MCQ	2	A	χ^2 test of independence (2×2)
28	MCQ	2	D	Paired t -test
29	MSQ	2	A, B, C	Sampling distribution of the mean, CLT (concept check)
30	MSQ	2	A, C, D	Logic of tests and intervals (concept check)

Score tracker

	Attempted	Correct	Wrong (MCQ)	Marks
1-mark				
2-mark				
Total				/ 48

Mock Test 02 · Detailed Solutions

Q.1 [MCQ · 1 mark · -1/3]

Topic: Counting: card hands · Easy

Correct answer: D

Concept: Multiplication principle for unordered selections: choose the aces, then choose the non-aces.

$$\binom{4}{2} \times \binom{48}{3} = 6 \times 17296 = 103776$$

Distractors: 132 600 = $C(4,2) \cdot C(52,3)$ lets the other three cards include aces (overcounts hands with 3 or 4 aces); 117 600 = $C(4,2) \cdot C(50,3)$ removes only the chosen aces from the pool; 622 656 = $6 \cdot 48 \cdot 47 \cdot 46$ orders the three non-aces.

Trap: 'Exactly k of a type' $\Rightarrow$ the remaining cards must come only from the OTHER types.

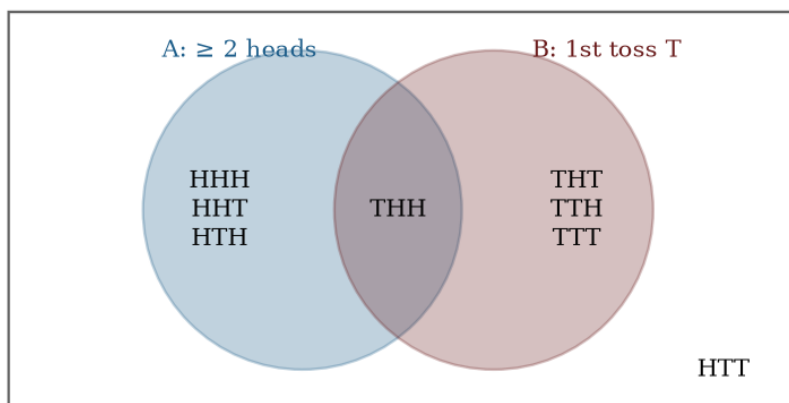
Q.2 [MCQ · 1 mark · -1/3]

Topic: Sample space, union of events · Easy

Correct answer: C

Concept: List the 8 equally likely outcomes and use $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

$A = \{HHH, HHT, HTH, THH\}$ (4 outcomes); $B = \{THH, THT, TTH, TTT\}$ (4 outcomes); $A \cap B = \{THH\}$.



$$P(A \cup B) = \frac{4}{8} + \frac{4}{8} - \frac{1}{8} = \frac{7}{8}$$

Only HTT lies outside both events: $1 - 1/8 = 7/8$. 'Adding without subtracting' gives 1 — not even an option, a useful warning sign.

Shortcut: When the sample space is small, the complement (outcomes in neither event) is often quickest.

Q.3 [MSQ · 1 mark]**Topic: Mutually exclusive events · Easy****Correct answer: A, B, D****Concept:** Mutually exclusive $\Rightarrow P(A \cap B) = 0$, so probabilities of A and B simply add.**(A) TRUE.** $P(A \cup B) = 0.3 + 0.5 - 0 = 0.8$.**(B) TRUE.** $P(A | B) = P(A \cap B)/P(B) = 0/0.5 = 0$.**(C) FALSE.** $P(A)P(B) = 0.15 \neq 0 = P(A \cap B)$.**(D) TRUE.** De Morgan: $P(A^c \cap B^c) = 1 - P(A \cup B) = 0.2$.**Key idea:** $P(A | B) = 0 \neq P(A) = 0.3$ shows directly that learning B changes the chance of A — the hallmark of dependence.**Q.4** [NAT · 1 mark]**Topic: Conditional probability (cards) · Easy****Correct answer: 0.030 (range 0.03 to 0.031)****Concept:** $P(\text{both} | \text{at least one}) = P(\text{both})/P(\text{at least one})$, since 'both' $\subset$ 'at least one'.

$$P(\text{both}) = \frac{\binom{4}{2}}{\binom{52}{2}} = \frac{6}{1326}$$

$$P(\geq 1 \text{ king}) = 1 - \frac{\binom{48}{2}}{\binom{52}{2}} = 1 - \frac{1128}{1326} = \frac{198}{1326}$$

$$P(\text{both} | \geq 1) = \frac{6}{198} = \frac{1}{33} \approx 0.0303$$

Answer $\approx$ **0.030**.**Trap:** This is NOT $P(\text{second is king} | \text{first is king}) = 3/51 \approx 0.059$. 'At least one' does not say which card is the king.**Q.5** [MCQ · 1 mark · -1/3]**Topic: Sample variance · Easy****Correct answer: B****Concept:** $s^2 = \sum(x_i - \bar{x})^2/(n - 1)$.

$$\bar{x} = \frac{4 + 7 + 7 + 9 + 13}{5} = \frac{40}{5} = 8$$

x	4	7	7	9	13
$x - \bar{x}$	-4	-1	-1	1	5
$(x - \bar{x})^2$	16	1	1	1	25

$$s^2 = \frac{16 + 1 + 1 + 1 + 25}{5 - 1} = \frac{44}{4} = 11$$

8.8 = 44/5 uses divisor n (population variance); 3.32 = $\sqrt{11}$ is the sample SD; 44 is the sum of squares itself.**Trap:** Read the divisor carefully: n - 1 (sample, unbiased) vs n (population).

Q.6 [MCQ · 1 mark · -1/3]**Topic: Binomial mean and variance · Easy****Correct answer: C****Concept:** $X \sim \text{Bin}(n, p) \Rightarrow E[X] = np, \text{Var}(X) = np(1 - p).$

$$1 - p = \frac{np(1 - p)}{np} = \frac{2.4}{6} = 0.4 \Rightarrow p = 0.6$$

$$n = \frac{6}{0.6} = 10$$

(15, 0.4) has mean 6 but variance 3.6; (12, 0.5) has variance 3; (10, 0.4) swaps p and $1 - p$ (mean 4).

Shortcut: Dividing variance by mean gives $q = 1 - p$ directly.

Q.7 [MSQ · 1 mark]**Topic: Properties of a PDF (concept check) · Easy****Correct answer: B, C, D****Concept:** A PDF is non-negative and integrates to 1; it is a density, not a probability.**(A) FALSE.** $X \sim U(0, 0.5)$ has $f(x) = 2$ on $(0, 0.5)$. Only areas must be ≤ 1 , not heights.**(B) TRUE.** Otherwise some interval would get negative probability.**(C) TRUE.** $P(X = c) = \int_c^c f(x) dx = 0.$ **(D) TRUE.** Total probability is 1.

Trap: $f(x)$ is probability per unit length; a narrow distribution must have a tall density.

Q.8 [NAT · 1 mark]**Topic: Poisson distribution · Easy****Correct answer: 0.558 (range 0.557 to 0.559)****Concept:** $P(X = k) = e^{-\lambda} \lambda^k / k!$ with $\lambda = 1.5$.

$$P(X \leq 1) = e^{-1.5}(1 + 1.5) = 0.2231 \times 2.5 = 0.5578$$

Answer $\approx$ **0.558**.

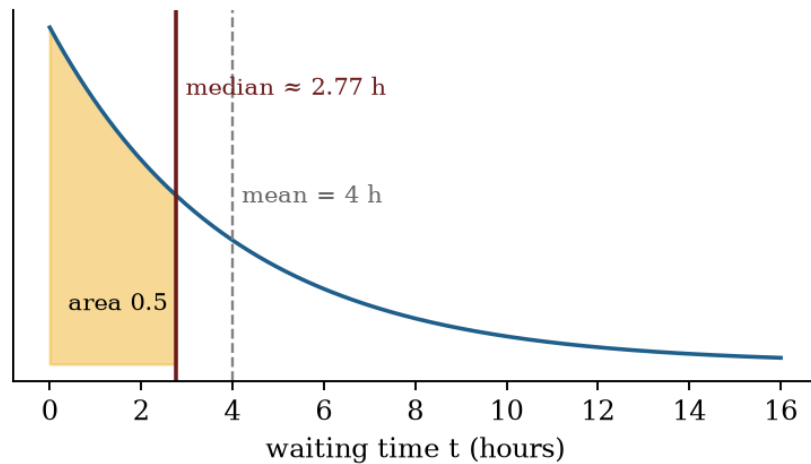
Trap: 'At most one' includes zero: $P(0) = 0.2231$ and $P(1) = 0.3347$ — forgetting $P(0)$ is the usual slip.

Q.9 [MCQ · 1 mark · -1/3]

Topic: Exponential distribution: median · Easy

Correct answer: C**Concept:** For $\text{Exp}(\lambda)$, $F(t) = 1 - e^{-\lambda t}$; the median m solves $F(m) = 1/2$.

$$1 - e^{-0.25m} = 0.5 \Rightarrow m = \frac{\ln 2}{0.25} = 4 \ln 2 \approx 2.77 \text{ h}$$



4.00 h is the mean ($1/\lambda$); 2.00 h is half the mean; $0.17 \text{ h} = 0.25 \ln 2$ multiplies instead of divides.

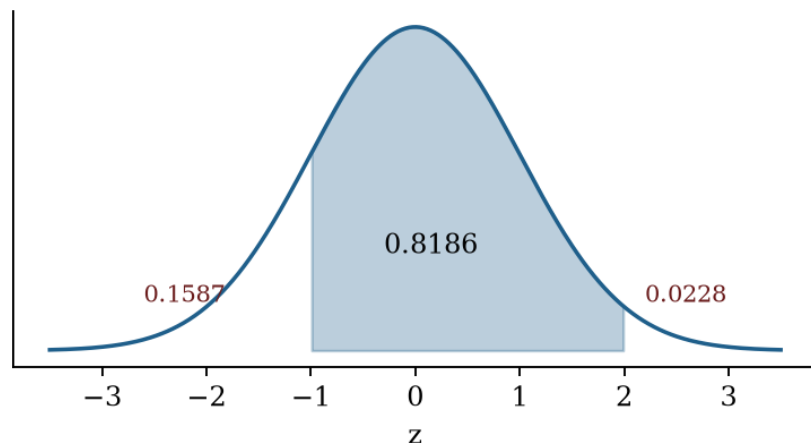
Key idea: Exponential is right-skewed, so median ($0.693/\lambda$) < mean ($1/\lambda$).

Q.10 [MCQ · 1 mark · -1/3]

Topic: Standard normal probabilities · Easy

Correct answer: B**Concept:** $P(a < Z < b) = \Phi(b) - \Phi(a)$, and $\Phi(-a) = 1 - \Phi(a)$.

$$P(-1 < Z < 2) = \Phi(2) - \Phi(-1) = 0.9772 - (1 - 0.8413) = 0.9772 - 0.1587 = 0.8185$$

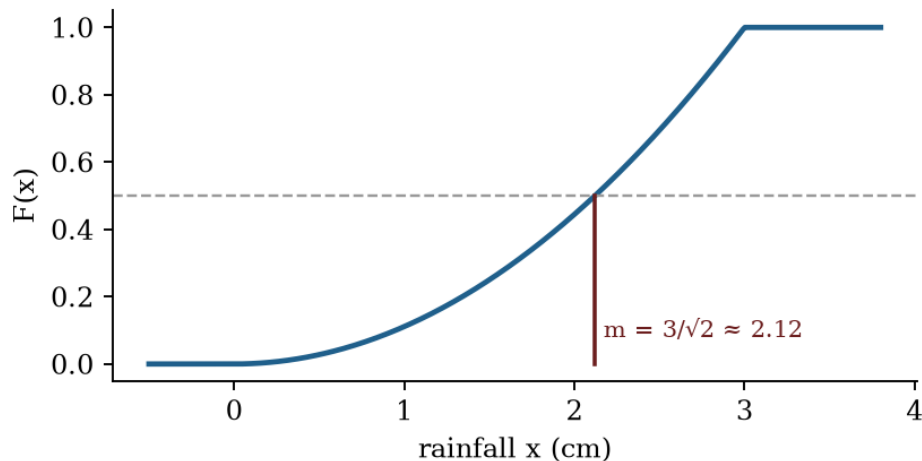


$0.6826 = P(|Z| < 1)$, $0.9544 = P(|Z| < 2)$ (symmetric intervals); 0.1815 is the complement.

Shortcut: Draw the curve and shade first; asymmetric intervals need two separate tail look-ups.

Q.11 [NAT · 1 mark]**Topic: CDF and median of a continuous RV · Easy****Correct answer: 2.12 (range 2.12 to 2.13)****Concept:** The median m of a continuous RV satisfies $F(m) = 0.5$.

$$\frac{m^2}{9} = \frac{1}{2} \Rightarrow m^2 = 4.5 \Rightarrow m = \frac{3}{\sqrt{2}} \approx 2.1213$$



Answer ≈ 2.12 cm. (For comparison, $f(x) = 2x/9$ gives mean $\int 2x^2/9 \, dx = 2$ cm $<$ median: a left-skewed density.)

Shortcut: Work directly with the CDF — no need to differentiate to the PDF for a quantile.

Q.12 [MSQ · 1 mark]**Topic: Expectation rules (concept check) · Easy****Correct answer: B, C**

Concept: Linearity of expectation needs no independence; products and non-linear functions do not pass through E .

(A) FALSE. Needs uncorrelatedness. Take $Y = X$ with $X \sim \text{Bernoulli}(1/2)$: $E[X^2] = 1/2$ but $(E[X])^2 = 1/4$.

(B) TRUE. Linearity holds for any X, Y (dependent or not).

(C) TRUE. Law of total expectation (tower property).

(D) FALSE. X uniform on $\{1, 2\}$:

$$E\left[\frac{1}{X}\right] = \frac{1}{2}\left(1 + \frac{1}{2}\right) = 0.75 \neq \frac{1}{E[X]} = \frac{1}{1.5} \approx 0.667$$

Key idea: Jensen: for convex g , $E[g(X)] \geq g(E[X])$; $1/x$ is convex on $x > 0$, so $E[1/X] \geq 1/E[X]$.

Q.13 [NAT · 2 marks]

Topic: Counting: seating in a row · Medium

Correct answer: 0.029 (range 0.028 to 0.029)**Concept:** Probability = favourable arrangements / all arrangements of 8 distinct people.

$$\text{Total} = 8! = 40320$$

Alternating patterns: BGBGBGBG or GBGBGBGB (2 patterns). In each, boys fill their 4 seats in $4!$ ways and girls in $4!$ ways:

$$\text{Favourable} = 2 \times 4! \times 4! = 2 \times 24 \times 24 = 1152$$

$$P = \frac{1152}{40320} = \frac{1}{35} \approx 0.0286$$

Answer $\approx$ **0.029**.

Check via genders only: choose which 4 of 8 seats get boys, $C(8,4) = 70$ equally likely patterns, 2 alternate $\Rightarrow 2/70 = 1/35$. (verified)

Trap: Forgetting the factor 2 (starting with a girl) halves the answer to 0.014.

Q.14 [MCQ · 2 marks · -2/3]

Topic: Complement rule (dice) · Medium

Correct answer: B**Concept:** 'At least two the same' is the complement of 'all faces different'.

$$P(\text{all different}) = \frac{6 \times 5 \times 4 \times 3}{6^4} = \frac{360}{1296} = \frac{5}{18}$$

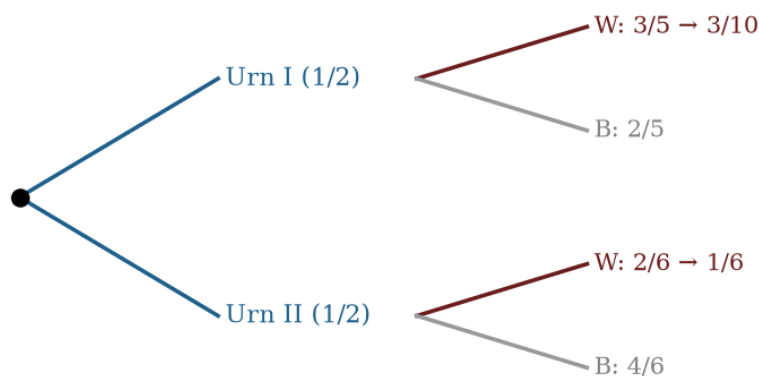
$$P(\text{at least two same}) = 1 - \frac{5}{18} = \frac{13}{18} \approx 0.722$$

$5/18$ is the complement itself; $671/1296 = 1 - (5/6)^4$ is $P(\text{at least one six})$ — a different event; $1/216 = 6/1296$ is $P(\text{all four equal})$.

Key idea: This is the birthday problem with 6 'days': with only 4 dice, a repeat is already 72% likely.

Q.15 [MCQ · 2 marks · -2/3]

Topic: Bayes theorem (two urns) · Medium

Correct answer: C**Concept:** Bayes: $P(I | W) = P(I)P(W | I) / [P(I)P(W | I) + P(II)P(W | II)]$.

$$P(W) = \frac{1}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{2}{6} = \frac{3}{10} + \frac{1}{6} = \frac{14}{30} = \frac{7}{15}$$

$$P(I | W) = \frac{3/10}{7/15} = \frac{3}{10} \cdot \frac{15}{7} = \frac{9}{14} \approx 0.643$$

3/5 is $P(W | I)$ (the reversed conditional); 1/2 is the prior; 5/14 is $P(II | W)$.

Key idea: White is more common in Urn I (60% vs 33%), so seeing white pushes the 50:50 prior towards Urn I — but not all the way.

Topic: Contingency table: joint, marginal, conditional · Medium

Q.16 [MSQ · 2 marks]

Correct answer: A, B, C

Concept: Joint = cell/total; marginal = row or column total/total; conditional = cell/row (or column) total.

(A) TRUE. $P(D) = 60/600 = 0.1$.

(B) TRUE. $P(N | D) = 20/60 = 1/3$.

(C) TRUE. $P(N) = 200/600 = 1/3$ and

$$P(N \cap D) = \frac{20}{600} = \frac{1}{30} = \frac{1}{3} \times \frac{1}{10} = P(N)P(D)$$

Equivalently $P(D | \text{Night}) = 20/200 = 0.1 = P(D | \text{Day}) = 40/400$ — defect rate does not depend on shift.

(D) FALSE. $P(\text{Day} \cap D) = 40/600 \approx 0.067$. The value 0.4 is not even a probability of this event: $40/400 = 0.1$ is $P(D | \text{Day})$, and $40/60 \approx 0.667$ is $P(\text{Day} | D)$.

Trap: Night has fewer defectives (20 vs 40) only because it makes fewer items. Compare RATES, not counts.

Topic: Conditional expectation and variance (random sum) · Medium

Q.17 [NAT · 2 marks]

Correct answer: 1.00 (range 0.99 to 1.01)

Concept: Given N , $X \sim \text{Bin}(N, 0.25)$; apply $\text{Var}(X) = E[\text{Var}(X | N)] + \text{Var}(E[X | N])$.

$$E[X | N] = 0.25N, \quad \text{Var}(X | N) = N(0.25)(0.75) = 0.1875N$$

For $N \sim \text{Poisson}(4)$: $E[N] = \text{Var}(N) = 4$.

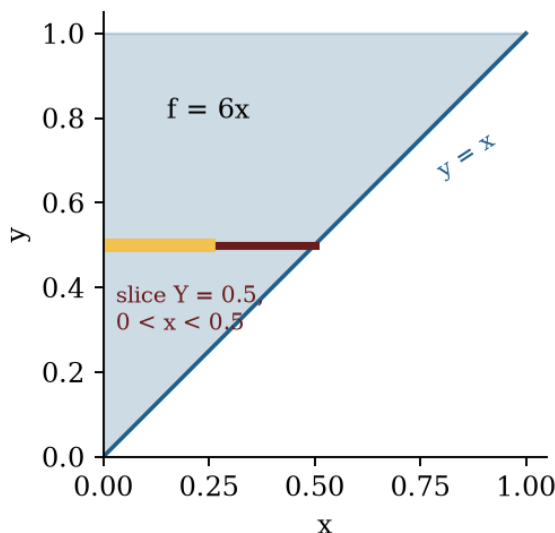
$$E[\text{Var}(X | N)] = 0.1875 \times 4 = 0.75$$

$$\text{Var}(E[X | N]) = 0.25^2 \times \text{Var}(N) = 0.0625 \times 4 = 0.25$$

$$\text{Var}(X) = 0.75 + 0.25 = 1.00$$

Also $E[X] = E[E[X | N]] = 0.25 \times 4 = 1$. Mean = variance = 1 is no coincidence: a Poisson count 'thinned' with probability p is again Poisson, here $\text{Poisson}(\lambda p) = \text{Poisson}(1)$.

Shortcut: Thinning property: $\text{Poisson}(\lambda)$ arrivals, each kept independently w.p. $p = \text{Poisson}(\lambda p)$. A quick check of any random-sum calculation.

Q.18 [MCQ · 2 marks · -2/3]**Topic: Conditional PDF · Medium****Correct answer: A****Concept:** $f_{X|Y}(x | y) = f(x, y)/f_Y(y)$, then integrate over the required x-range on the slice $Y = y$.

$$f_Y(y) = \int_0^y 6x \, dx = 3y^2, \quad 0 < y < 1$$

$$f_{X|Y}(x | y) = \frac{6x}{3y^2} = \frac{2x}{y^2}, \quad 0 < x < y$$

At $y = 0.5$: $f_{X|Y}(x | 0.5) = 2x/0.25 = 8x$ on $(0, 0.5)$. (Check: $\int_0^{0.5} 8x \, dx = 1$.)

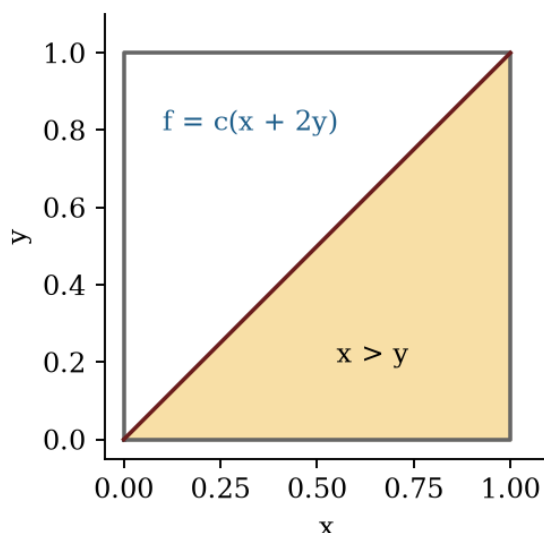
$$P(X < 0.25 | Y = 0.5) = \int_0^{0.25} 8x \, dx = 4(0.25)^2 = 0.25$$

0.50 assumes X is uniform on the slice (but the density $6x$ is not constant); $0.0625 = \int_0^{0.25} 2x \, dx$ forgets the division by $y^2 = 0.25$ in the conditional density.

Key idea: A conditional density must integrate to 1 over the slice — use this to check normalisation.

Q.19 [NAT · 2 marks]**Topic: Joint PDF: normalisation and region probability · Medium****Correct answer: 0.444 (range 0.444 to 0.445)****Concept:** First find c from total probability = 1, then integrate over the region $x > y$.

$$\int_0^1 \int_0^1 c(x + 2y) \, dy \, dx = c \left(\frac{1}{2} + 1 \right) = \frac{3c}{2} = 1 \Rightarrow c = \frac{2}{3}$$



Region below the diagonal: $0 \leq y \leq x \leq 1$.

$$\int_0^1 \int_0^x (x + 2y) dy dx = \int_0^1 (x^2 + x^2) dx = \frac{2}{3}$$

$$P(X > Y) = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9} \approx 0.4444$$

Answer ≈ 0.444 . It is below 0.5 because the density puts more weight on large y .

Trap: Forgetting c gives $2/3$ — always normalise before computing probabilities.

Topic: Implications of independence (concept check) · Medium

Q.20 [MSQ · 2 marks]

Correct answer: A, B, D

Concept: Independence factorises the joint distribution, hence expectations of products of functions: $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$.

(B) TRUE (take $g, h = \text{identity}$), hence **(A) TRUE**: $\text{Cov} = E[XY] - E[X]E[Y] = 0$.

(C) FALSE. Using independence,

$$\text{Var}(XY) = E[X^2]E[Y^2] - (E[X]E[Y])^2$$

which equals $\text{Var}(X)\text{Var}(Y)$ only when means vanish. Counter-example: X, Y i.i.d. Bernoulli($1/2$): $XY \sim \text{Bernoulli}(1/4)$, $\text{Var}(XY) = 3/16$, but $\text{Var}(X)\text{Var}(Y) = 1/16$.

(D) TRUE. Functions of independent random variables are independent.

Key idea: Independence is preserved by functions; zero covariance is not (that is why uncorrelated $\neq$ independent).

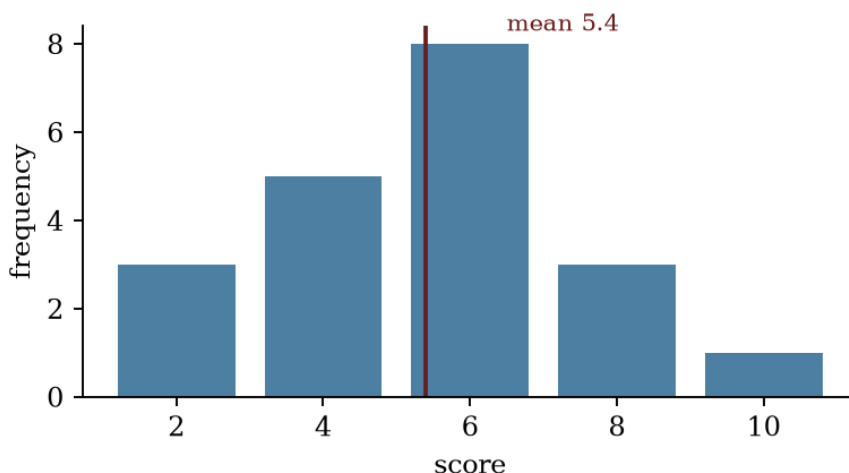
Q.21 [MCQ · 2 marks · -2/3]**Topic: Variance of a linear combination · Medium****Correct answer: D****Concept:** $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab \text{Cov}(X, Y)$, with $\text{Cov} = \rho\sigma_X\sigma_Y$.

$$\text{Cov}(X, Y) = 0.5 \times 2 \times 3 = 3$$

$$\text{Var}(2X - Y) = 4(4) + (-1)^2(9) + 2(2)(-1)(3) = 16 + 9 - 12 = 13$$

25 ignores the covariance; 37 adds it with the wrong sign (+12); 19 uses $2 \times 2 \times 3 \times \frac{1}{2} = 6$ instead of 12 for the cross term.

Key idea: Positively correlated variables partly cancel in a difference, so $\text{Var}(2X - Y)$ is smaller than the independent case (25).

Q.22 [NAT · 2 marks]**Topic: Grouped data: mean and standard deviation · Medium****Correct answer: 2.11 (range 2.1 to 2.11)****Concept:** $\sigma^2 = (\sum f x^2)/n - \bar{x}^2$ for frequency data.

$$\bar{x} = \frac{3(2) + 5(4) + 8(6) + 3(8) + 1(10)}{20} = \frac{108}{20} = 5.4$$

$$\frac{\sum f x^2}{n} = \frac{3(4) + 5(16) + 8(36) + 3(64) + 1(100)}{20} = \frac{672}{20} = 33.6$$

$$\sigma^2 = 33.6 - 5.4^2 = 33.6 - 29.16 = 4.44$$

$$\sigma = \sqrt{4.44} \approx 2.1071$$

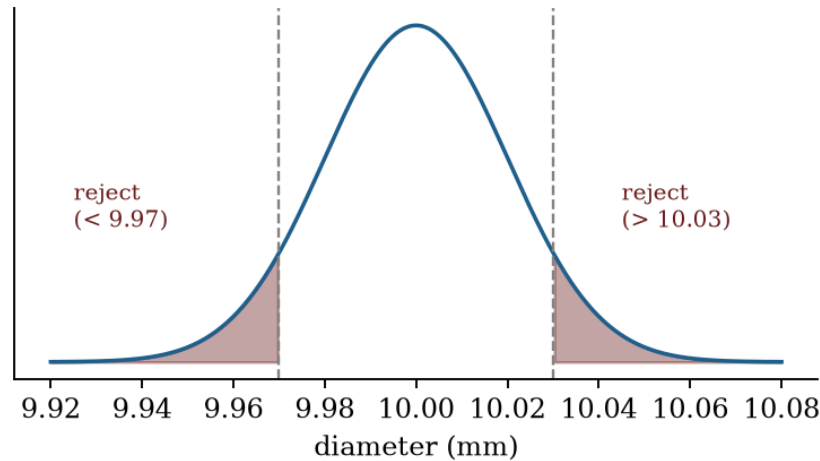
Answer ≈ 2.11 . (With divisor $n - 1$ one would get $\sqrt{(88.8/19)} \approx 2.16$ — not asked here.)

Trap: Multiply each value (and each square) by its frequency — averaging the five distinct scores alone gives a wrong mean of 6.

Q.23 [MCQ · 2 marks · -2/3]**Topic: Normal distribution: tolerance limits · Medium****Correct answer: B****Concept:** $P(|X - \mu| > d) = 2[1 - \Phi(d/\sigma)]$ by symmetry.

$$z = \frac{0.03}{0.02} = 1.5$$

$$P(\text{reject}) = P(|Z| > 1.5) = 2(1 - 0.9332) = 2(0.0668) = 0.1336$$



0.0668 counts only one tail; 0.8664 is the acceptance proportion; $0.0456 = P(|Z| > 2)$.

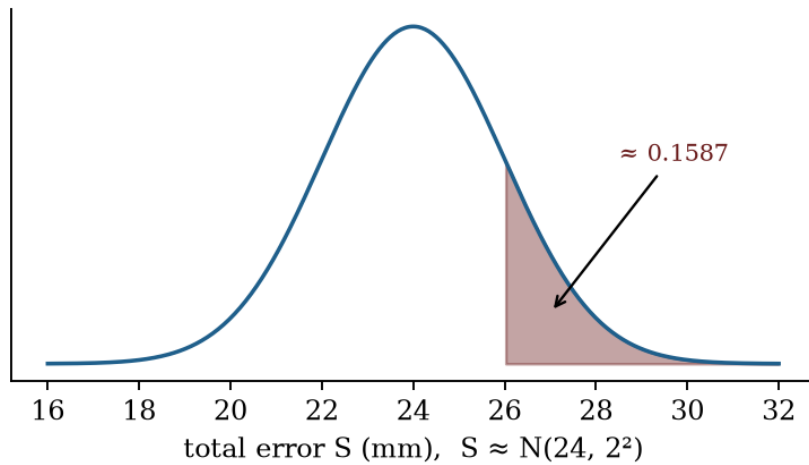
Trap: Two-sided tolerance = two tails. Draw both rejection regions.

Q.24 [NAT · 2 marks]**Topic: CLT for a sum of uniforms · Medium****Correct answer: 0.159 (range 0.158 to 0.16)****Concept:** For $S =$ sum of n i.i.d. variables with mean μ and variance σ^2 , $S \approx N(n\mu, n\sigma^2)$ for large n .

$$U \sim U(0, 1): \quad E[U] = \frac{1}{2}, \quad \text{Var}(U) = \frac{(1-0)^2}{12} = \frac{1}{12}$$

$$E[S] = 48 \times \frac{1}{2} = 24, \quad \text{Var}(S) = \frac{48}{12} = 4, \quad \text{SD}(S) = 2$$

$$P(S > 26) \approx P\left(Z > \frac{26 - 24}{2}\right) = 1 - \Phi(1) = 0.1587$$



Answer ≈ 0.159 .

Trap: Variances add for a sum ($n \sigma^2$); the SD grows like $\sqrt{n}$, not n . Using $SD = 48/\sqrt{12} \approx 13.9$ is a common blunder.

Q.25 [MCQ · 2 marks · -2/3]

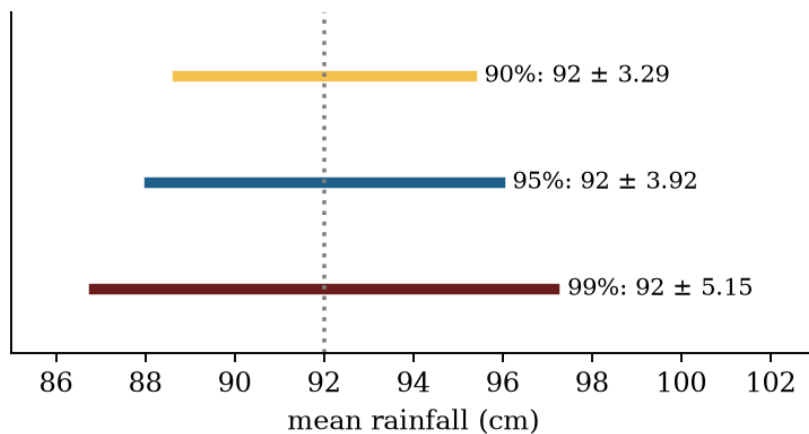
Topic: Confidence interval (z , σ known) · Medium

Correct answer: B

Concept: σ known $\Rightarrow \bar{x} \pm z_{\alpha/2} \sigma/\sqrt{n}$; for 99%, $\alpha/2 = 0.005$.

$$\frac{\sigma}{\sqrt{n}} = \frac{8}{4} = 2, \quad 2.576 \times 2 = 5.152$$

$$92 \pm 5.152 = (86.85, 97.15)$$



(A) is the 95% interval, (C) the 90% interval; (D) uses σ instead of $\sigma/\sqrt{n}$ (92 ± 20.61).

Key idea: Higher confidence $\Rightarrow$ wider interval (for the same data). Width = $2 z \sigma/\sqrt{n}$.

Q.26 [NAT · 2 marks]

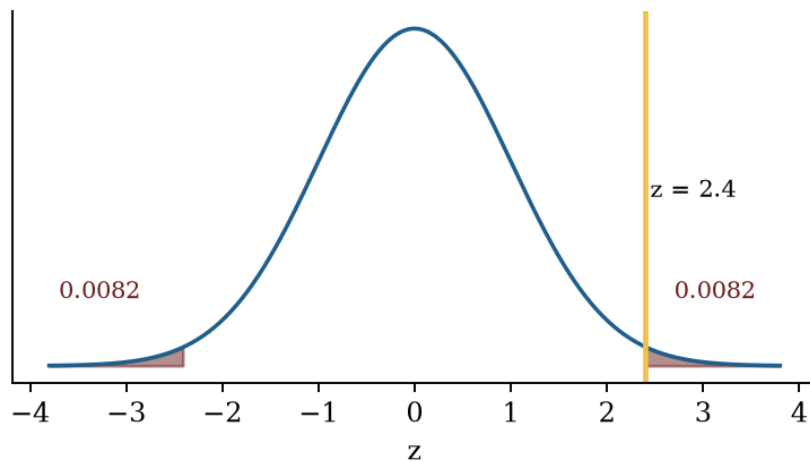
Topic: z-test for a proportion: p-value · Medium

Correct answer: 0.016 (range 0.016 to 0.017)**Concept:** Under H_0 , $X \sim \text{Bin}(400, 0.5) \approx N(np, np(1 - p))$; two-sided p-value = $2P(Z \geq |z|)$.

$$np = 200, \quad \sqrt{np(1 - p)} = \sqrt{100} = 10$$

$$z = \frac{224 - 200}{10} = 2.4$$

$$p\text{-value} = 2[1 - \Phi(2.4)] = 2(0.0082) = 0.0164$$

Answer $\approx$ **0.016**. Since $0.016 < 0.05$, fairness is rejected at 5% (but not at 1%).

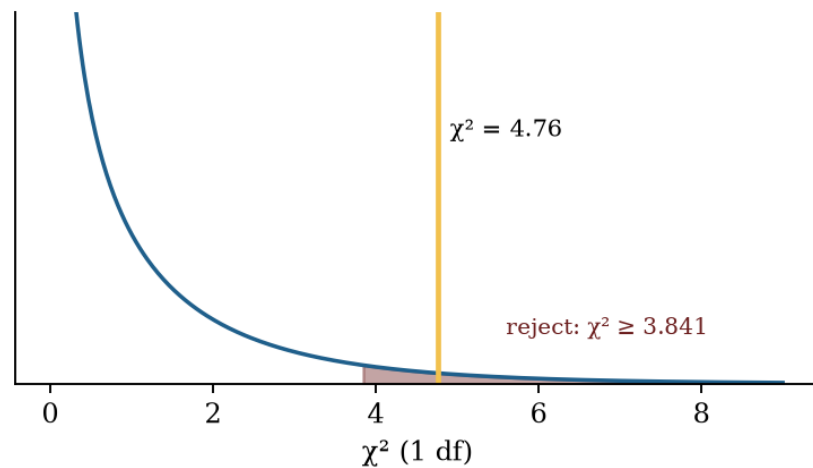
Trap: For a two-sided test, double the one-tail area. Reporting 0.0082 is the classic half-p-value error.

Q.27 [MCQ · 2 marks · -2/3]Topic: χ^2 test of independence (2×2) · Medium**Correct answer: A****Concept:** $E_{ij} = (\text{row total})(\text{column total})/n$; $\chi^2 = \sum (O - E)^2/E$ with $(r - 1)(c - 1)$ df.

Cell	O	E	$(O - E)^2/E$
A, Pass	40	$50 \cdot 70/100 = 35$	$25/35 = 0.714$
A, Fail	10	$50 \cdot 30/100 = 15$	$25/15 = 1.667$
B, Pass	30	35	0.714
B, Fail	20	15	1.667

$$\chi^2 = 2(0.7143) + 2(1.6667) = 4.762$$

df = $(2 - 1)(2 - 1) = 1$; critical value 3.841. Since $4.76 > 3.841$, reject H_0 : pass rate depends on the method (80% vs 60%).



(B) uses $rc - 1 = 3$ df; (C) sums only two cells; (D) is not a valid computation of the statistic.

Trap: Degrees of freedom for an $r \times c$ table are $(r - 1)(c - 1)$, not $rc - 1$.

Q.28 [MCQ · 2 marks · -2/3]

Topic: Paired t-test · Medium

Correct answer: D

Concept: Paired data $\Rightarrow$ one-sample t-test on the differences: $t = \bar{d}/(s_d/\sqrt{n})$, $n - 1$ df.

d	3	5	-1	4	2	5
$d - \bar{d}$	0	2	-4	1	-1	2
$(d - \bar{d})^2$	0	4	16	1	1	4

$$\bar{d} = \frac{18}{6} = 3, \quad s_d^2 = \frac{26}{5} = 5.2, \quad s_d = 2.2804$$

$$t = \frac{3}{2.2804/\sqrt{6}} = \frac{3}{0.9309} \approx 3.223$$

With 5 df, critical value 2.015. Since $3.22 > 2.015$, reject H_0 : the workshop improves scores on average.

$t \approx 1.32$ forgets $\sqrt{n}$ ($3/2.28$); $t \approx 3.53$ uses divisor n for s_d ($\sqrt{(26/6)} = 2.08$) — the conclusion agrees but the statistic is wrong; (B) has the right statistic but the wrong decision.

Key idea: Do not run a two-sample test on paired data — pairing removes student-to-student variation, which is exactly why it is powerful.

Topic: Sampling distribution of the mean, CLT (concept check) · Medium

Q.29 [MSQ · 2 marks]

Correct answer: A, B, C

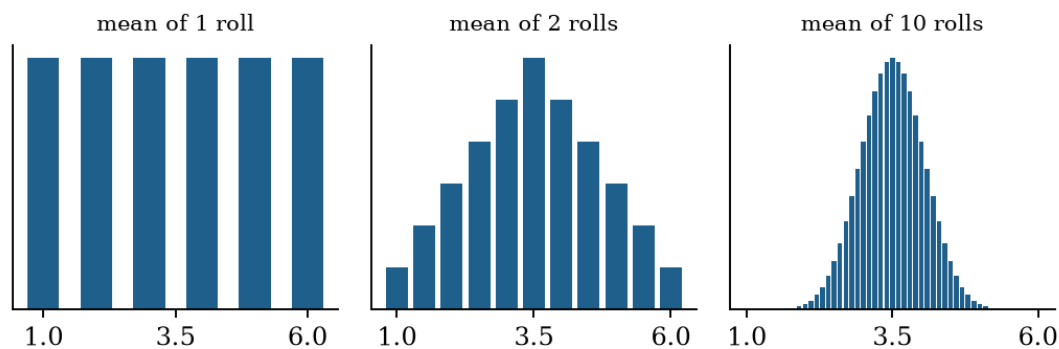
Concept: Linearity gives the mean and variance of $\bar{X}$ exactly; the CLT describes the SHAPE of $\bar{X}$'s distribution for large n .

(A) TRUE. $E[\bar{X}] = (1/n)\sum E[X_i] = \mu$ — unbiased for any n .

(B) TRUE. Independence $\Rightarrow \text{Var}(\bar{X}) = (1/n^2) \cdot n\sigma^2 = \sigma^2/n$.

(C) TRUE. Central limit theorem (finite variance).

(D) FALSE. The population does not change with n ; a die roll stays discrete uniform. Only averages (or sums) become bell-shaped:



Key idea: (A) and (B) are exact for all n and need no normality; (C) is an approximation for large n .

Topic: Logic of tests and intervals (concept check) · Medium

Q.30 [MSQ · 2 marks]

Correct answer: A, C, D

Concept: Two-sided tests and intervals are duals: H_0 is rejected at level α exactly when μ_0 lies outside the $(1 - \alpha)$ interval.

(A) TRUE. μ_0 outside $\bar{x} \pm 1.96\sigma/\sqrt{n} \Leftrightarrow |\bar{x} - \mu_0|/(\sigma/\sqrt{n}) > 1.96 \Leftrightarrow \text{reject}$.

(B) FALSE. The p-value is $P(\text{data at least this extreme} \mid H_0 \text{ true})$, not $P(H_0 \mid \text{data})$.

(C) TRUE. $2.576 > 1.96$, so the margin grows.

(D) TRUE. The k counts are constrained to sum to n , losing one df.

Trap: The p-value is a probability about the DATA computed assuming H_0 — never the probability that H_0 is true.

Mock Test 03 — Standard GATE-Level Test I (Data-Science Contexts)

Full-syllabus paper at actual GATE DA difficulty, set in spam filters, A/B tests, server logs, sensors and model evaluation.

Item	Details
Questions	30 (12 × 1-mark, 18 × 2-mark)
Maximum marks	48
Question types	MCQ: 14 · MSQ: 7 · NAT: 9
Suggested time	90 minutes
Negative marking	MCQ only: $-1/3$ (1-mark), $-2/3$ (2-mark). None for MSQ & NAT.
Calculator	Use only the GATE virtual scientific calculator

Topic focus of this set

Covers the entire Probability & Statistics syllabus at exam difficulty ($\approx 30\%$ easy, 50% medium, 20% hard): counting via hashing and cross-validation folds; axioms, independence and mutual exclusivity of filter rules; Bayes theorem for classifier precision and data-source attribution; joint/marginal/conditional PMFs and PDFs; law of total variance for bursty server traffic; summary statistics and skewness; covariance matrices of features; Bernoulli dropout, binomial click-through, Poisson/exponential request logs, uniform, normal sensor noise, t and χ^2 distributions; CDFs; the CLT for mini-batch means; confidence intervals for model accuracy and latency; z -test for an A/B experiment, one-sample t -test, and χ^2 tests of independence and goodness-of-fit.

Instructions

- Questions 1 onwards are arranged as in GATE: all 1-mark questions first, then 2-mark questions.
- **MCQ** – exactly one correct option. **MSQ** – ONE OR MORE options are correct; credit only if you select all correct options and no wrong option (no partial marks, no negative marks).
- **NAT** – enter a numerical value using the virtual keypad. The accepted range is given in the key; round as instructed in the question.
- Statistical tables (Z , t , χ^2) are provided in the Appendix; values in solutions use those tables.
- Attempt in one sitting, then check the answer key and read every solution — including those you got right — for the alternative methods and traps.

Mock Test 03 · Question Paper

1-mark questions

Q.1 MCQ · 1 mark · -1/3

A hash function maps each of 5 distinct keys independently and uniformly at random to one of 8 buckets. The probability that no two keys land in the same bucket is closest to

- (A) 0.410
- (B) 0.795
- (C) 0.205
- (D) 0.625

Q.2 MCQ · 1 mark · -1/3

A dense layer has 10 neurons. During training, dropout removes each neuron independently with probability 0.2. Let K be the number of neurons that are **kept**. Then $(E[K], \text{Var}(K))$ is

- (A) (2, 1.6)
- (B) (8, 1.6)
- (C) (8, 0.16)
- (D) (8, 2.0)

Q.3 NAT · 1 mark

The response latencies (in ms) of 7 API calls logged by a monitoring service are 12, 15, 15, 18, 20, 22, 95. The value of (sample mean – sample median) is _____ (round off to 2 decimal places).

Answer: _____

Q.4 MSQ · 1 mark

An email gateway applies two spam rules. Let $A = \{\text{rule 1 fires}\}$ and $B = \{\text{rule 2 fires}\}$ for a random incoming email. Logs show $P(A) = 0.3$, $P(B) = 0.2$ and $P(A \cup B) = 0.44$. Which of the following is/are TRUE?

- (A) A and B are independent
- (B) A and B are mutually exclusive
- (C) $P(A | B) = 0.3$
- (D) The probability that neither rule fires is 0.56

Q.5 MCQ · 1 mark · -1/3

The time T (in seconds) between consecutive requests to a micro-service is exponentially distributed with mean 2 s. Given that no request has arrived in the first 3 s after the last one, the probability that none arrives in the next 2 s is

- (A) e^{-1}
- (B) $e^{-5/2}$
- (C) e^{-2}
- (D) $1 - e^{-1}$

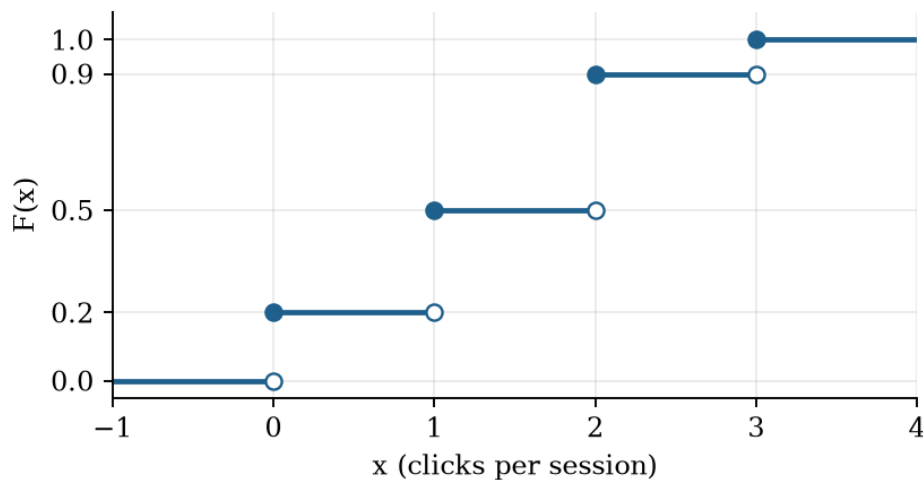
Q.6 NAT · 1 mark

A web server receives requests according to a Poisson process with an average of 3 requests per minute. The probability that it receives at most 1 request in a given minute is _____ (round off to 3 decimal places).

Answer: _____

Q.7 MCQ · 1 mark · -1/3

The number of ad clicks X in a user session has the CDF shown below ($F(x) = 0$ for $x < 0$; jumps occur only at $x = 0, 1, 2, 3$). The value of $E[X]$ is



- (A) 1.0
- (B) 2.6
- (C) 1.5
- (D) 1.4

Q.8 MCQ · 1 mark · -1/3

A temperature sensor's measurement error ε is normally distributed with mean 0 and standard deviation 0.5°C . The probability that the absolute error exceeds 1°C is closest to ($\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$)

- (A) 0.0228
- (B) 0.0455
- (C) 0.3173
- (D) 0.0500

Q.9 MSQ · 1 mark

Which of the following statements is/are TRUE?

- (A) The t-distribution with ν degrees of freedom has heavier tails than $N(0, 1)$
- (B) For $\nu > 2$, the variance of a t-distribution with ν degrees of freedom is $\nu/(\nu - 2)$
- (C) A χ^2 random variable with k degrees of freedom has mean $2k$
- (D) If $Z \sim N(0, 1)$, then Z^2 has a χ^2 distribution with 1 degree of freedom

Q.10 NAT · 1 mark

Two input features X and Y of a regression dataset have $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$ and $\text{Cov}(X, Y) = 3$. An engineered feature is defined as $U = 2X - Y$. The value of $\text{Var}(U)$ is _____ (answer as an integer).

Answer: _____

Q.11 MCQ · 1 mark · -1/3

Per-example losses on a training set have mean 2.0 and standard deviation 1.2. A mini-batch of 64 examples is drawn independently (with replacement). The standard deviation of the mini-batch average loss is

- (A) 0.0225
- (B) 1.2
- (C) 0.15
- (D) 0.019

Q.12 NAT · 1 mark

The processing time X (in seconds) of a batch job is uniformly distributed on $[2, 10]$. Given that a job has already run for more than 4 seconds, the probability that it runs for more than 7 seconds is _____ (round off to 2 decimal places).

Answer: _____

2-mark questions**Q.13 NAT · 2 marks**

In a company's inbox, 20% of emails are spam. A spam filter flags 90% of spam emails (recall = 0.90) and wrongly flags 5% of legitimate emails (false-positive rate = 0.05). The precision of the filter, i.e. $P(\text{spam} \mid \text{flagged})$, is _____ (round off to 3 decimal places).

Answer: _____

Q.14 MCQ · 2 marks · -2/3

A training corpus is assembled from three annotation vendors A, B and C, which contribute 50%, 30% and 20% of the records respectively. Their label-error rates are 2%, 4% and 10%. A record picked at random from the corpus is found to be mislabelled. The probability that it came from vendor C is closest to

- (A) 0.200
- (B) 0.476
- (C) 0.286
- (D) 0.238

Q.15 MSQ · 2 marks

A recommender logs, for each impression, the ad slot $Y \in \{1, 2, 3\}$ and whether it was clicked, $X \in \{0, 1\}$. The joint PMF is:

	Y = 1	Y = 2	Y = 3
X = 0	0.10	0.20	0.20
X = 1	0.20	0.20	0.10

Which of the following is/are TRUE?

- (A) X and Y are independent
- (B) $P(Y = 2 \mid X = 1) = 0.4$
- (C) $\text{Cov}(X, Y) = -0.1$
- (D) $E[X \mid Y = 1] = 1/3$

Q.16 NAT · 2 marks

The number of requests N hitting a cache in a one-second window is Poisson with rate Λ , where Λ itself is random: during peak traffic (probability 0.25) $\Lambda = 20$, otherwise (probability 0.75) $\Lambda = 4$. The variance of N is _____ (answer as an integer).

Answer: _____

Q.17 MCQ · 2 marks · -2/3

A classifier is evaluated on 400 independent test examples and classifies 340 of them correctly. Using the normal approximation, the 95% confidence interval for its true accuracy ($z_{0.025} = 1.96$) is closest to

- (A) (0.832, 0.868)
- (B) (0.804, 0.896)
- (C) (0.815, 0.885)
- (D) (0.780, 0.920)

Q.18 NAT · 2 marks

In an A/B test of a checkout page, variant A converted 200 of 2000 visitors and variant B converted 250 of 2000 visitors. Using the pooled two-proportion z-test of $H_0: p_A = p_B$, the value of the test statistic $z = (\hat{p}_B - \hat{p}_A)/SE$ is _____ (round off to 2 decimal places).

Answer: _____

Q.19 MSQ · 2 marks

The inference latency of a model is claimed to be 50 ms on average. On 16 random requests the sample mean is 52 ms and the sample standard deviation is 6 ms. Latencies are assumed normal. Consider testing $H_0: \mu = 50$ against $H_1: \mu > 50$. ($t_{0.05,15} = 1.753$, $t_{0.10,15} = 1.341$, $t_{0.05,35} = 1.690$.) Which of the following is/are TRUE?

- (A) The test statistic is approximately 1.33
- (B) H_0 is rejected at $\alpha = 0.10$
- (C) With the same sample mean and standard deviation but $n = 36$, H_0 would be rejected at $\alpha = 0.05$
- (D) The p-value of the test ($n = 16$) exceeds 0.05

Q.20 MCQ · 2 marks · -2/3

A survey records device type and whether the user clicked a promotional banner:

	Clicked	Not clicked	Total
Mobile	30	70	100
Desktop	20	80	100

At $\alpha = 0.05$ ($\chi^2_{0.05,1} = 3.841$), the Pearson χ^2 statistic (no continuity correction) and the decision about independence of device and clicking are

- (A) $\chi^2 \approx 2.67$; reject independence
- (B) $\chi^2 \approx 5.33$; reject independence
- (C) $\chi^2 \approx 1.33$; do not reject independence
- (D) $\chi^2 \approx 2.67$; do not reject independence

Q.21 NAT · 2 marks

Two normalised anomaly scores (X, Y) have joint density $f(x, y) = c \cdot x$ for $0 < y < x < 1$ and 0 elsewhere. The value of $P(X + Y < 1)$ is _____ (round off to 3 decimal places).

Answer: _____

Q.22 MCQ · 2 marks · -2/3

Random variables X and Y have joint density $f(x, y) = 3x$ for $0 < y < x < 1$ (0 elsewhere). For a fixed $0 < x < 1$, the conditional density $f_{Y|X}(y | x)$ and $E[Y | X = x]$ are

- (A) $f_{Y|X}(y | x) = 1/x$ on $0 < y < x$; $E[Y | X = x] = x/2$
- (B) $f_{Y|X}(y | x) = 3x$ on $0 < y < x$; $E[Y | X = x] = 3x/2$
- (C) $f_{Y|X}(y | x) = 2y/x^2$ on $0 < y < x$; $E[Y | X = x] = 2x/3$
- (D) $f_{Y|X}(y | x) = 1/(1 - x)$ on $x < y < 1$; $E[Y | X = x] = (1 + x)/2$

Q.23 MCQ · 2 marks · -2/3

A sensor's readings are i.i.d. $N(\mu, 4^2)$. The device reports the average of n readings. The smallest n for which the reported average is within ± 1 unit of μ with probability at least 0.95 is ($z_{0.025} = 1.96$)

- (A) 16
- (B) 31
- (C) 61
- (D) 62

Q.24 MSQ · 2 marks

Three input features X_1, X_2, X_3 have the covariance matrix

Σ	X_1	X_2	X_3
X_1	4	2	0
X_2	2	9	-1.5
X_3	0	-1.5	1

Which of the following is/are TRUE?

- (A) $\text{Corr}(X_1, X_2) = 1/3$
- (B) $\text{Var}(X_1 + X_2) = 17$
- (C) X_1 and X_3 must be independent
- (D) $\text{Var}(X_2 - 2X_3) = 19$

Q.25 MCQ · 2 marks · -2/3

A recommended item has a click-through rate of 5%, and clicks on different impressions are independent. If the item is shown 20 times, the probability that it receives at least 2 clicks is closest to

- (A) 0.358
- (B) 0.264
- (C) 0.736
- (D) 0.377

Q.26 MCQ · 2 marks · -2/3

Twelve distinct training examples are to be partitioned into 3 folds of 4 examples each for cross-validation. The folds are **unlabelled** (only the grouping matters, not which fold is called 'first'). The number of distinct partitions is

- (A) 34650
- (B) 495
- (C) 5775
- (D) 1925

Q.27 MSQ · 2 marks

Login requests arrive at an authentication server as a Poisson process with rate 4 per minute. Which of the following is/are TRUE?

- (A) The number of requests in a 30-second interval is Poisson with mean 2
- (B) The expected waiting time until the first request is 15 seconds
- (C) The probability of no request in a 30-second interval is e^{-2}
- (D) The waiting time until the second request is exponentially distributed with mean 30 seconds

Q.28 NAT · 2 marks

To check whether a hash function spreads keys uniformly, 200 keys are hashed into 4 buckets, giving observed counts 62, 45, 48, 45. The Pearson χ^2 goodness-of-fit statistic for the hypothesis of uniform bucket probabilities is _____ (round off to 2 decimal places).

Answer: _____

Q.29 MCQ · 2 marks · -2/3

The confidence score X output by a well-calibrated binary classifier on positive examples has density $f(x) = 2x$ for $0 \leq x \leq 1$ (0 elsewhere). Which ordering of its mean, median and mode is correct?

- (A) mode < median < mean
- (B) mean < median < mode
- (C) median < mean < mode
- (D) mean = median < mode

Q.30 MSQ · 2 marks

The page-load times (ms) of 25 randomly chosen sessions have sample mean 80 and sample standard deviation 10; load times are assumed normal. ($t_{0.025,24} = 2.064$, $z_{0.025} = 1.96$.) Which of the following is/are TRUE?

- (A) The 95% t-confidence interval for the mean is approximately (75.87, 84.13)
- (B) The interval computed with 1.96 in place of 2.064 is narrower than the t-interval
- (C) There is a 95% probability that the true mean lies in the computed interval (75.87, 84.13)
- (D) With $n = 100$ (same sample mean and s), the width of the 95% interval using $z = 1.96$ is 3.92

Mock Test 03 · Answer Key

Q	Type	Marks	Answer	Topic
1	MCQ	1	C	Counting (hash collisions)
2	MCQ	1	B	Bernoulli / binomial (dropout)
3	NAT	1	10.14 (range 10.13 to 10.15)	Mean, median (skewed data)
4	MSQ	1	A, C, D	Independence vs mutual exclusivity
5	MCQ	1	A	Exponential (memoryless)
6	NAT	1	0.199 (range 0.198 to 0.2)	Poisson PMF
7	MCQ	1	D	CDF of a discrete random variable
8	MCQ	1	B	Normal distribution (sensor noise)
9	MSQ	1	A, B, D	t and χ^2 distributions
10	NAT	1	13	Variance of linear combination
11	MCQ	1	C	CLT / standard error (mini-batch)
12	NAT	1	0.50 (range 0.49 to 0.51)	Continuous uniform, conditional probability
13	NAT	2	0.818 (range 0.817 to 0.819)	Bayes theorem (classifier precision)
14	MCQ	2	B	Bayes theorem (data provenance)
15	MSQ	2	B, C	Joint, marginal, conditional PMF; covariance
16	NAT	2	56	Conditional expectation & variance (law of total variance)
17	MCQ	2	C	Confidence interval for a proportion
18	NAT	2	2.50 (range 2.48 to 2.52)	z-test for two proportions (A/B test)
19	MSQ	2	A, C, D	One-sample t-test
20	MCQ	2	D	χ^2 test of independence
21	NAT	2	0.375 (range 0.374 to 0.376)	Joint PDF, region of integration
22	MCQ	2	A	Conditional PDF
23	MCQ	2	D	Normal / sample size via CLT
24	MSQ	2	A, B, D	Covariance matrix and correlation
25	MCQ	2	B	Binomial (click-through rate)
26	MCQ	2	C	Counting (cross-validation folds)
27	MSQ	2	A, B, C	Poisson process & exponential waiting times
28	NAT	2	3.96 (range 3.95 to 3.97)	χ^2 goodness-of-fit test
29	MCQ	2	B	Mean, median, mode of a continuous PDF
30	MSQ	2	A, B, D	Confidence interval (t-based)

Score tracker

	Attempted	Correct	Wrong (MCQ)	Marks
1-mark				
2-mark				
Total				/ 48

Mock Test 03 · Detailed Solutions

Q.1 [MCQ · 1 mark · -1/3]

Topic: Counting (hash collisions) · Easy

Correct answer: C

Concept: $P(\text{all distinct}) = (\text{ordered selections of distinct buckets}) / (\text{all assignments})$.

Each key has 8 choices, so the sample space has 8^5 equally likely assignments.

$$|S| = 8^5 = 32768$$

For no collision the first key may use any of 8 buckets, the second any of the remaining 7, and so on (a permutation of 5 out of 8):

$$|E| = {}^8P_5 = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 6720$$

$$P(\text{no collision}) = \frac{6720}{32768} \approx 0.2051$$

Option (B) 0.795 is the probability of at least one collision — the complement.

Key idea: This is the birthday problem: even with more buckets than keys, collisions are likely surprisingly early. $P(\text{collision}) = 1 - \prod (1 - i/m)$.

Q.2 [MCQ · 1 mark · -1/3]

Topic: Bernoulli / binomial (dropout) · Easy

Correct answer: B

Concept: a sum of n independent Bernoulli(p) indicators is Binomial(n, p) with mean np and variance $np(1 - p)$.

Each neuron is kept with probability $p = 1 - 0.2 = 0.8$, independently, so $K \sim \text{Bin}(10, 0.8)$.

$$E[K] = np = 10(0.8) = 8$$

$$\text{Var}(K) = np(1 - p) = 10(0.8)(0.2) = 1.6$$

(A) uses the dropped count's mean (2) — the variance is the same for dropped and kept counts, but the mean is not. (C) forgets the factor n . (D) reports $np \cdot (1 - p)/0.8$.

Shortcut: Var is symmetric in p and $1 - p$: the number dropped, $10 - K$, has the same variance 1.6.

Q.3 [NAT · 1 mark]

Topic: Mean, median (skewed data) · Easy

Correct answer: 10.14 (range 10.13 to 10.15)

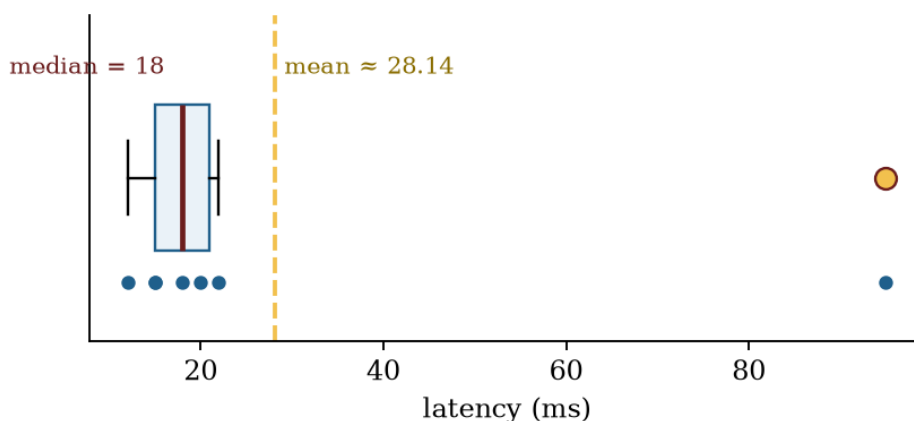
Concept: the mean uses every value (pulled by outliers); the median is the middle order statistic (robust).

$$\bar{x} = \frac{12 + 15 + 15 + 18 + 20 + 22 + 95}{7} = \frac{197}{7} \approx 28.1429$$

With $n = 7$ sorted values the median is the 4th value:

$$\text{median} = x_{(4)} = 18$$

$$\bar{x} - \text{median} = 28.1429 - 18 \approx 10.14$$



The single outlier 95 drags the mean far to the right of the bulk of the data (right skew), while the median stays at 18. The mode is 15.

Key idea: For right-skewed data (typical for latencies) $\text{mean} > \text{median} > \text{mode}$. This is why latency SLOs are quoted as medians/percentiles, not means.

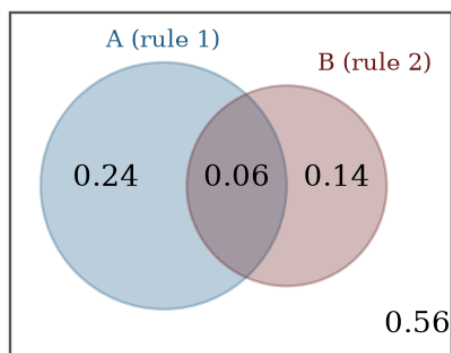
Q.4 [MSQ · 1 mark]

Topic: Independence vs mutual exclusivity · Easy

Correct answer: A, C, D

Concept: addition rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; independence means $P(A \cap B) = P(A)P(B)$.

$$P(A \cap B) = 0.3 + 0.2 - 0.44 = 0.06$$



(A) TRUE. $P(A)P(B) = 0.3 \times 0.2 = 0.06 = P(A \cap B)$.

(B) FALSE. Mutually exclusive needs $P(A \cap B) = 0$, but it is 0.06.

(C) TRUE. $P(A | B) = 0.06/0.2 = 0.3 = P(A)$, as expected under independence.

(D) TRUE. $P(\text{neither}) = 1 - P(A \cup B) = 1 - 0.44 = 0.56$ ($= 0.7 \times 0.8$, again consistent with independence).

Trap: Two events with positive probability cannot be both independent and mutually exclusive: exclusivity forces $P(A \cap B) = 0 \neq P(A)P(B)$.

Q.5 [MCQ · 1 mark · -1/3]**Topic: Exponential (memoryless) · Easy****Correct answer: A****Concept:** the exponential distribution is memoryless: $P(T > s + t \mid T > s) = P(T > t)$.Mean $2s \Rightarrow$ rate $\lambda = 1/2$ per second, and $P(T > t) = e^{-\lambda t}$.

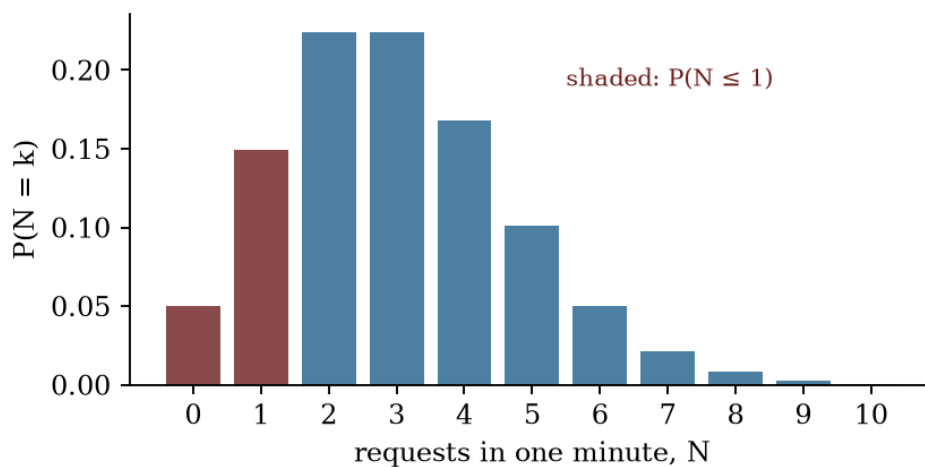
$$P(T > 5 \mid T > 3) = \frac{P(T > 5)}{P(T > 3)} = \frac{e^{-5/2}}{e^{-3/2}} = e^{-1}$$

$$e^{-1} \approx 0.368$$

(B) $e^{-5/2} = P(T > 5)$ ignores the conditioning. (C) uses $\lambda = 1$ (confusing mean with rate). (D) is the probability that a request does arrive.**Key idea:** Elapsed waiting time is irrelevant for exponential waits: the clock 'restarts'.**Q.6** [NAT · 1 mark]**Topic: Poisson PMF · Easy****Correct answer: 0.199 (range 0.198 to 0.2)****Concept:** $N \sim \text{Poisson}(\lambda)$ has $P(N = k) = e^{-\lambda} \lambda^k / k!$.Here $\lambda = 3$ per minute and the interval is one minute.

$$P(N \leq 1) = P(N = 0) + P(N = 1) = e^{-3} + 3e^{-3} = 4e^{-3}$$

$$4e^{-3} = 4(0.049787) \approx 0.1991$$

**Trap:** 'At most 1' includes 0. Forgetting the $k = 0$ term gives 0.149.

Q.7 [MCQ · 1 mark · -1/3]**Topic: CDF of a discrete random variable · Easy****Correct answer: D****Concept:** for a discrete r.v., $P(X = x)$ equals the jump of F at x .

x	0	1	2	3
$F(x)$	0.2	0.5	0.9	1.0
$P(X = x)$	0.2	0.3	0.4	0.1

$$E[X] = 0(0.2) + 1(0.3) + 2(0.4) + 3(0.1) = 0 + 0.3 + 0.8 + 0.3 = 1.4$$

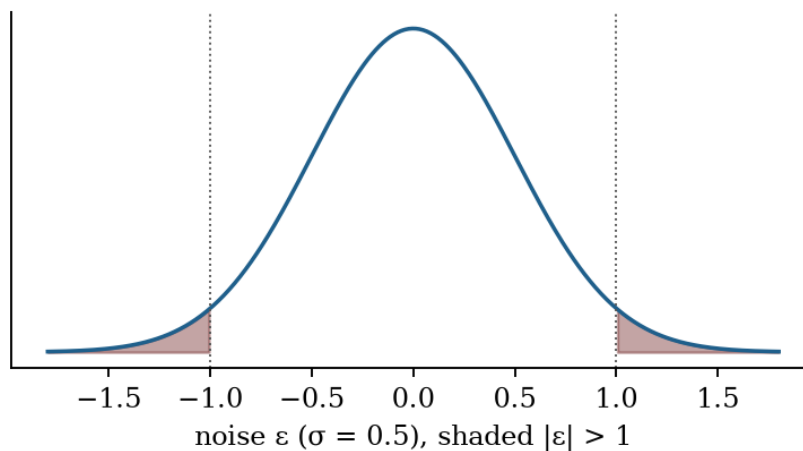
(B) $2.6 = 0.2 + 0.5 + 0.9 + 1.0$ adds the CDF values instead of weighting x by the PMF. (C) 1.5 is the mean if all four values were equally likely.

Shortcut: For a non-negative integer r.v., $E[X] = \sum_{k \geq 0} P(X > k) = 0.8 + 0.5 + 0.1 = 1.4$ — a quick cross-check straight from the CDF.

Q.8 [MCQ · 1 mark · -1/3]**Topic: Normal distribution (sensor noise) · Easy****Correct answer: B****Concept:** standardise, $Z = (\varepsilon - \mu)/\sigma \sim N(0, 1)$, and use symmetry.

$$P(|\varepsilon| > 1) = P\left(|Z| > \frac{1}{0.5}\right) = P(|Z| > 2)$$

$$P(|Z| > 2) = 2[1 - \Phi(2)] = 2(0.0228) = 0.0456$$



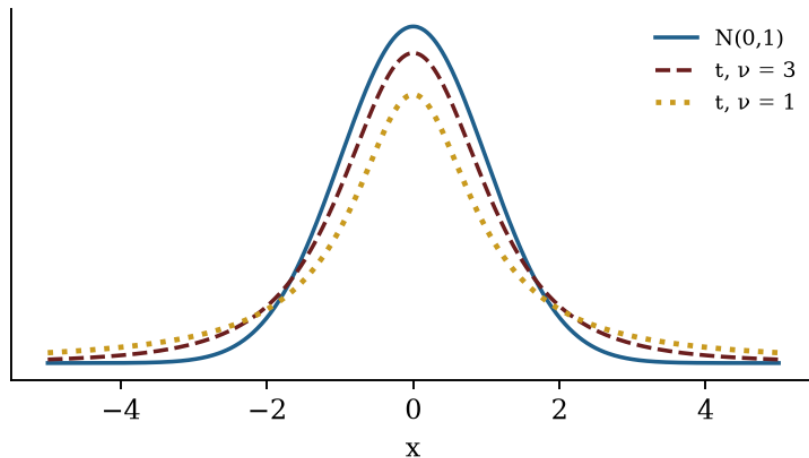
(A) is a single tail only. (C) is $P(|Z| > 1)$ — forgetting to divide by σ . (D) confuses 2σ with the 1.96σ two-sided 5% point.

Exact: $2(1 - \Phi(2)) = 0.0455$.

Trap: 'Absolute error exceeds' = both tails. Always ask: one tail or two?

Q.9 [MSQ · 1 mark]**Topic: t and χ^2 distributions · Medium****Correct answer: A, B, D**

Concept: $T = Z/\sqrt{V/\nu}$ with $V \sim \chi^2_\nu$; a χ^2_k variable is a sum of k squared independent standard normals.



(A) TRUE. Dividing by the random $\sqrt{V/\nu}$ inflates spread; t has polynomial (not Gaussian) tails and approaches $N(0, 1)$ as $\nu \rightarrow \infty$.

(B) TRUE. $\text{Var}(T) = \nu/(\nu - 2)$ for $\nu > 2$ (it is infinite for $\nu = 2$ and undefined for $\nu = 1$).

(C) FALSE. $E[\chi^2_k] = k$; it is the variance that equals $2k$.

$$E[Z^2] = 1 = E\left[\sum_{i=1}^k Z_i^2\right] = k, \quad \text{Var}(Z^2) = E[Z^4] - 1 = 2$$

(D) TRUE. By definition χ^2_1 is the distribution of Z^2 .

Trap: Mean k , variance $2k$ for χ^2_k — a favourite swap in options.

Q.10 [NAT · 1 mark]**Topic: Variance of linear combination · Easy****Correct answer: 13**

Concept: $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab \cdot \text{Cov}(X, Y)$.

With $a = 2$, $b = -1$:

$$\begin{aligned} \text{Var}(2X - Y) &= 4(4) + 1(9) + 2(2)(-1)(3) \\ &= 16 + 9 - 12 = 13 \end{aligned}$$

Trap: The cross-term carries the sign of ab . Forgetting it gives 25; using $+12$ gives 37.

Q.11 [MCQ · 1 mark · -1/3]

Topic: CLT / standard error (mini-batch) · Easy

Correct answer: C**Concept:** for an average of n i.i.d. values, $SD(\bar{X}) = \sigma/\sqrt{n}$ (the standard error).

$$SD(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{1.2}{\sqrt{64}} = \frac{1.2}{8} = 0.15$$

(A) 0.0225 is the variance $\sigma^2/n = 1.44/64$. (D) $0.019 = 1.2/64$ divides by n instead of $\sqrt{n}$. (B) ignores averaging.

By the CLT the batch mean is approximately $N(2.0, 0.15^2)$ — the basis of SGD noise analysis.

Key idea: To halve gradient noise you must quadruple the batch size ($\sqrt{n}$ law).

Q.12 [NAT · 1 mark]

Topic: Continuous uniform, conditional probability · Easy

Correct answer: 0.50 (range 0.49 to 0.51)**Concept:** $P(A | B) = P(A \cap B)/P(B)$; for a uniform r.v. probabilities are length ratios.

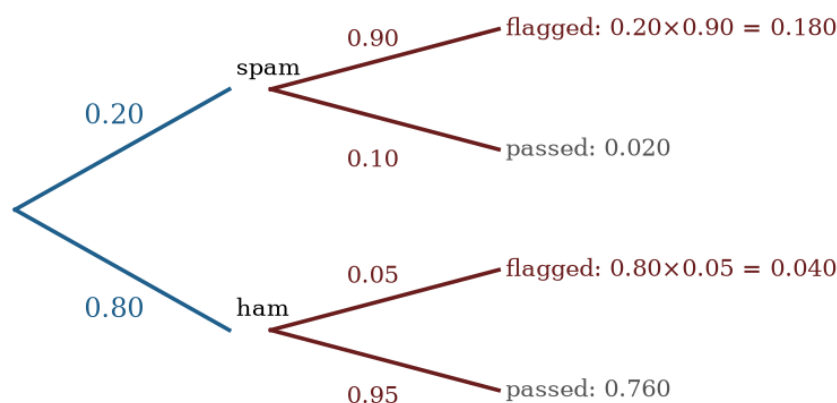
$$P(X > 7 | X > 4) = \frac{P(X > 7)}{P(X > 4)} = \frac{(10 - 7)/8}{(10 - 4)/8} = \frac{3}{6} = 0.5$$

Equivalently, given $X > 4$, X is uniform on $(4, 10]$, and $(7, 10]$ is half of that interval.

Trap: Unlike the exponential, the uniform is **not** memoryless: the unconditional $P(X > 3) = 7/8 \neq 0.5$.

Q.13 [NAT · 2 marks]

Topic: Bayes theorem (classifier precision) · Medium

Correct answer: 0.818 (range 0.817 to 0.819)**Concept:** Bayes theorem with the law of total probability in the denominator.

Total probability of a flag:

$$P(F) = P(F | S)P(S) + P(F | H)P(H) = 0.9(0.2) + 0.05(0.8)$$

$$P(F) = 0.18 + 0.04 = 0.22$$

Bayes theorem:

$$P(S | F) = \frac{0.18}{0.22} = \frac{9}{11} \approx 0.8182$$

In confusion-matrix language, per 100 emails: TP = 18, FP = 4, so precision = 18/22.

	Flagged	Not flagged	Total
Spam	18	2	20
Ham	4	76	80
Total	22	78	100

Key idea: Precision depends on the base rate; recall and FPR do not. If spam were only 2% of mail, precision would fall to $0.018/(0.018 + 0.049) \approx 0.27$.

Q.14 [MCQ · 2 marks · -2/3]

Topic: Bayes theorem (data provenance) · Medium

Correct answer: B

Concept: posterior $\propto$ prior $\times$ likelihood (Bayes theorem over a partition).

Let $M = \{\text{mislabelled}\}$. Joint probabilities:

Vendor	Prior	$P(M \text{vendor})$	Prior $\times$ likelihood
A	0.50	0.02	0.010
B	0.30	0.04	0.012
C	0.20	0.10	0.020
Total	1		0.042

$$P(M) = 0.010 + 0.012 + 0.020 = 0.042$$

$$P(C | M) = \frac{0.020}{0.042} = \frac{10}{21} \approx 0.476$$

(A) 0.200 is just the prior. (C) $0.286 = 0.012/0.042$ is the posterior for B. (D) $0.238 = 0.010/0.042$ is the posterior for A.

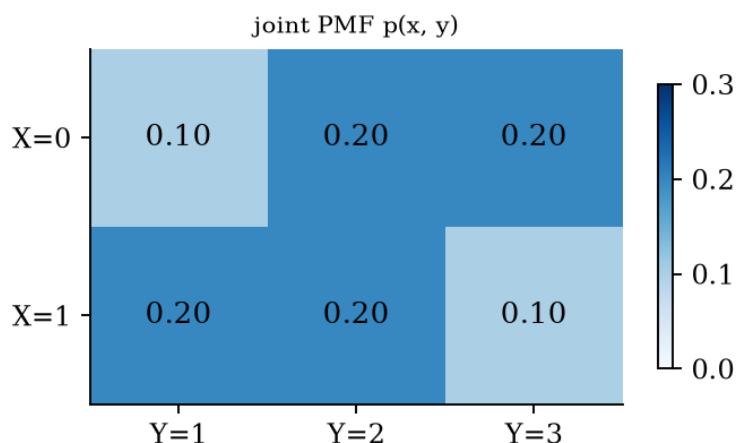
Key idea: A small vendor with a high error rate can dominate the error pool — C supplies 20% of data but $\approx 48\%$ of errors.

Q.15 [MSQ · 2 marks]

Topic: Joint, marginal, conditional PMF; covariance · Medium

Correct answer: B, C

Concept: marginals are row/column sums; conditional = joint / marginal; $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$.



Marginals: $P(X = 0) = 0.5$, $P(X = 1) = 0.5$; $P(Y = 1) = 0.3$, $P(Y = 2) = 0.4$, $P(Y = 3) = 0.3$.

(A) FALSE. $P(X = 0, Y = 1) = 0.10$ but $P(X = 0)P(Y = 1) = 0.5 \times 0.3 = 0.15$. One failing cell suffices.

(B) TRUE.

$$P(Y = 2 | X = 1) = \frac{0.20}{0.50} = 0.4$$

(C) TRUE.

$$\begin{aligned} E[X] &= 0.5, \quad E[Y] = 1(0.3) + 2(0.4) + 3(0.3) = 2.0 \\ E[XY] &= 1(1)(0.20) + 1(2)(0.20) + 1(3)(0.10) = 0.9 \\ \text{Cov}(X, Y) &= 0.9 - 0.5(2.0) = -0.1 \end{aligned}$$

(D) FALSE.

$$E[X | Y = 1] = P(X = 1 | Y = 1) = \frac{0.20}{0.30} = \frac{2}{3}$$

Key idea: Negative covariance: clicks are more likely in the top slot ($Y = 1$) and less likely in the bottom slot — position bias.

Topic: Conditional expectation & variance (law of total variance) · Hard

Q.16 [NAT · 2 marks]

Correct answer: 56

Concept: law of total variance, $\text{Var}(N) = E[\text{Var}(N | \Lambda)] + \text{Var}(E[N | \Lambda])$.

For a Poisson, $E[N | \Lambda] = \Lambda$ and $\text{Var}(N | \Lambda) = \Lambda$.

$$\begin{aligned} E[\Lambda] &= 0.25(20) + 0.75(4) = 5 + 3 = 8 \\ E[\Lambda^2] &= 0.25(400) + 0.75(16) = 100 + 12 = 112 \\ \text{Var}(\Lambda) &= 112 - 8^2 = 48 \end{aligned}$$

Now assemble:

$$\begin{aligned} E[\text{Var}(N | \Lambda)] &= E[\Lambda] = 8 \\ \text{Var}(E[N | \Lambda]) &= \text{Var}(\Lambda) = 48 \\ \text{Var}(N) &= 8 + 48 = 56 \end{aligned}$$

Also $E[N] = E[\Lambda] = 8$, so $\text{Var}(N)/E[N] = 7$, far above 1 — the mixture is over-dispersed.

Trap: Treating N as Poisson(8) would give $\text{Var} = 8$ — a big under-estimate. Rate heterogeneity adds $\text{Var}(\Lambda)$. This is why real request counts are often modelled as negative binomial.

Q.17 [MCQ · 2 marks · -2/3]

Topic: Confidence interval for a proportion · Medium

Correct answer: C

Concept: $\hat{p} \pm z_{\alpha/2} \sqrt{(\hat{p}(1 - \hat{p})/n)}$ (Wald interval).

$$\hat{p} = \frac{340}{400} = 0.85$$

$$SE = \sqrt{\frac{0.85(0.15)}{400}} = \sqrt{0.00031875} \approx 0.01785$$

$$\text{margin} = 1.96(0.01785) \approx 0.0350$$

$$0.85 \pm 0.035 \Rightarrow (0.815, 0.885)$$

(A) uses the margin $1 \times SE$ ($\approx 68\%$ interval). (B) uses $z = 2.576$ (a 99% interval, margin ≈ 0.046). (D) uses $n = 100$ instead of 400: $1.96\sqrt{(0.85 \cdot 0.15/100)} \approx 0.070$.

Shortcut: To halve the margin ($\approx 0.035 \rightarrow 0.0175$) you need $4\times$ as many test examples (1600).

Q.18 [NAT · 2 marks]

Topic: z-test for two proportions (A/B test) · Medium

Correct answer: 2.50 (range 2.48 to 2.52)

Concept: under H_0 both groups share one p , estimated by pooling; $SE = \sqrt{(\hat{p}(1 - \hat{p}))(1/n_A + 1/n_B))}$.

$$\hat{p}_A = \frac{200}{2000} = 0.100, \quad \hat{p}_B = \frac{250}{2000} = 0.125$$

$$\hat{p} = \frac{200 + 250}{2000 + 2000} = \frac{450}{4000} = 0.1125$$

$$SE = \sqrt{0.1125(0.8875) \left(\frac{1}{2000} + \frac{1}{2000} \right)}$$

$$SE = \sqrt{0.0998438 \times 0.001} \approx 0.009992$$

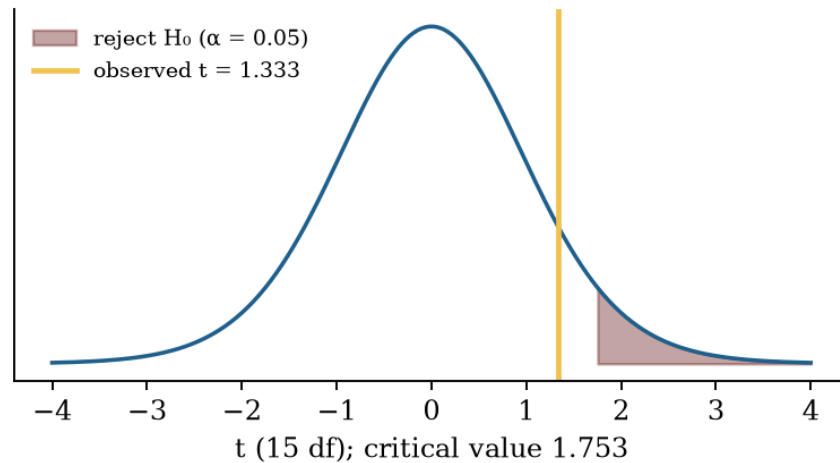
$$z = \frac{0.125 - 0.100}{0.009992} \approx 2.502$$

Since $|z| \approx 2.50 > 1.96$, the difference is significant at the 5% level (two-sided); the two-sided p-value is about 0.012.

Trap: Pooling is used for the test (H_0 says the p 's are equal). For a confidence interval of $p_B - p_A$ use unpooled SEs.

Q.19 [MSQ · 2 marks]**Topic: One-sample t-test · Medium****Correct answer: A, C, D****Concept:** $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$ with $n - 1$ df; reject for $t > t_{\alpha, n-1}$.

$$t = \frac{52 - 50}{6/\sqrt{16}} = \frac{2}{1.5} \approx 1.333$$

**(A) TRUE.** $t \approx 1.333$.**(B) FALSE.** $1.333 < 1.341 = t_{0.10, 15}$; even at the 10% level we narrowly fail to reject.**(C) TRUE.** With $n = 36$:

$$t = \frac{2}{6/\sqrt{36}} = \frac{2}{1} = 2.0 > 1.690$$

(D) TRUE. Since $t = 1.333 < 1.753$, the observed t lies outside the 5% rejection region, so $p > 0.05$ (in fact $p \approx 0.10$, just above it).

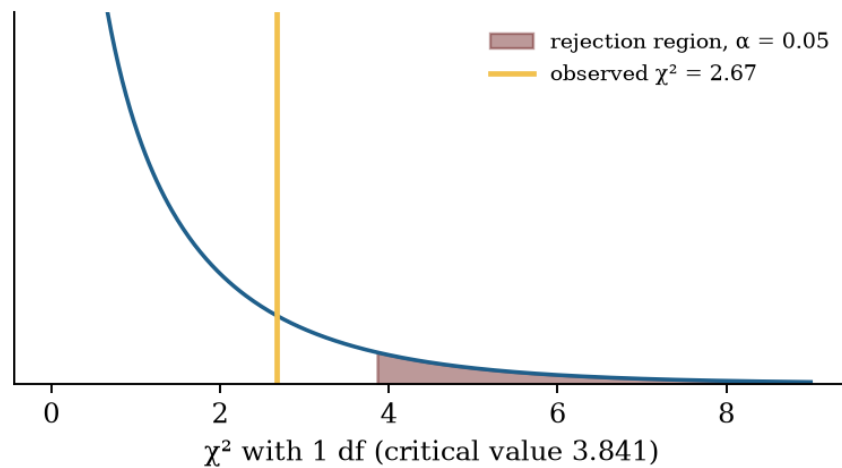
Key idea: The same 2 ms effect becomes 'significant' with more data: significance depends on effect size and n .

Q.20 [MCQ · 2 marks · -2/3]**Topic: χ^2 test of independence · Medium****Correct answer: D****Concept:** $E_{ij} = (\text{row total} \times \text{column total})/\text{grand total}$; $\chi^2 = \sum (O - E)^2/E$ with $(r - 1)(c - 1)$ df.
Column totals: clicked 50, not clicked 150; grand total 200.

Cell	O	E	$(O - E)^2/E$
Mobile, clicked	30	$100 \cdot 50 / 200 = 25$	$25/25 = 1.000$
Mobile, not	70	75	$25/75 = 0.333$
Desktop, clicked	20	25	1.000
Desktop, not	80	75	0.333

$$\chi^2 = 1 + 0.333 + 1 + 0.333 \approx 2.667$$

$$df = (2 - 1)(2 - 1) = 1$$



Since $2.667 < 3.841$ we do not reject H_0 : the data are consistent with device and clicking being independent.

(B) doubles the statistic; (C) uses only one row's contributions.

Key idea: A 30% vs 20% click rate looks large, but with only 100 users per group it is not statistically significant at 5%.

Q.21 [NAT · 2 marks]

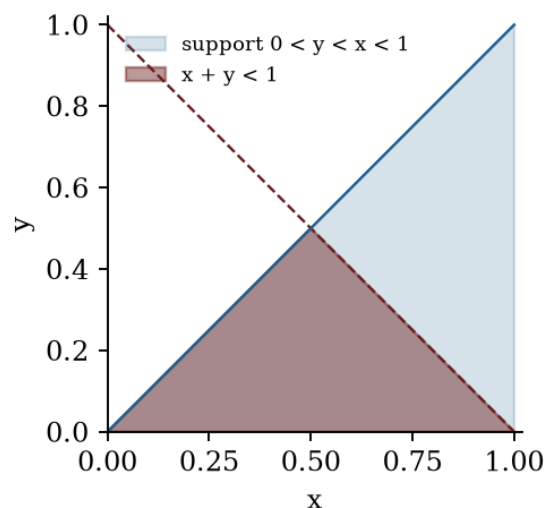
Topic: Joint PDF, region of integration · Hard

Correct answer: 0.375 (range 0.374 to 0.376)

Concept: find c from total probability = 1, then integrate f over the event region intersected with the support.

Normalisation:

$$\int_0^1 \int_0^x c x dy dx = c \int_0^1 x^2 dx = \frac{c}{3} = 1 \Rightarrow c = 3$$



The event $\{x + y < 1\}$ within $\{0 < y < x\}$ is the triangle with vertices $(0, 0)$, $(1, 0)$, $(\frac{1}{2}, \frac{1}{2})$. Integrate in y first on the outside: for $0 < y < \frac{1}{2}$, x runs from y to $1 - y$.

$$P = \int_0^{1/2} \int_y^{1-y} 3x dx dy = \int_0^{1/2} \frac{3}{2} [(1-y)^2 - y^2] dy$$

$$\begin{aligned}
&= \frac{3}{2} \int_0^{1/2} (1 - 2y) dy = \frac{3}{2} [y - y^2]_0^{1/2} \\
&= \frac{3}{2} \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{3}{8} = 0.375
\end{aligned}$$

(Monte Carlo check with 2×10^6 samples: ≈ 0.375 .)

Shortcut: Integrating x first from 0 to 1 would need a split at $x = 1/2$; choosing y as the outer variable avoids the split.

Q.22 [MCQ · 2 marks · -2/3]

Topic: Conditional PDF · Medium

Correct answer: A

Concept: $f_{Y|X}(y | x) = f(x, y)/f_X(x)$, where f_X integrates out y over the support.

$$\begin{aligned}
f_X(x) &= \int_0^x 3x dy = 3x^2, \quad 0 < x < 1 \\
f_{Y|X}(y | x) &= \frac{3x}{3x^2} = \frac{1}{x}, \quad 0 < y < x
\end{aligned}$$

This is Uniform(0, x), so

$$E[Y | X = x] = \int_0^x y \cdot \frac{1}{x} dy = \frac{x}{2}$$

(B) forgets to divide by $f_X(x)$ — it does not integrate to 1 over (0, x) unless $x = 1/\sqrt{3}$. (D) uses the wrong support ($y > x$). (C) is not proportional to the joint density in y .

Bonus: $\text{Var}(Y | X = x) = x^2/12$, so $E[\text{Var}(Y | X)] = E[X^2]/12 = (3/5)/12 = 1/20$.

Shortcut: If $f(x, y)$ does not depend on y over the slice, the conditional is uniform on that slice.

Q.23 [MCQ · 2 marks · -2/3]

Topic: Normal / sample size via CLT · Medium

Correct answer: D

Concept: $X \sim N(\mu, \sigma^2/n)$; require $z_{\alpha/2} \cdot \sigma/\sqrt{n} \leq E$.

$$\begin{aligned}
P(|\bar{X} - \mu| < 1) &= P\left(|Z| < \frac{1}{4/\sqrt{n}}\right) \geq 0.95 \\
\frac{\sqrt{n}}{4} &\geq 1.96 \Rightarrow \sqrt{n} \geq 7.84 \\
n &\geq 7.84^2 = 61.4656
\end{aligned}$$

n must be an integer, so round **up**: $n = 62$.

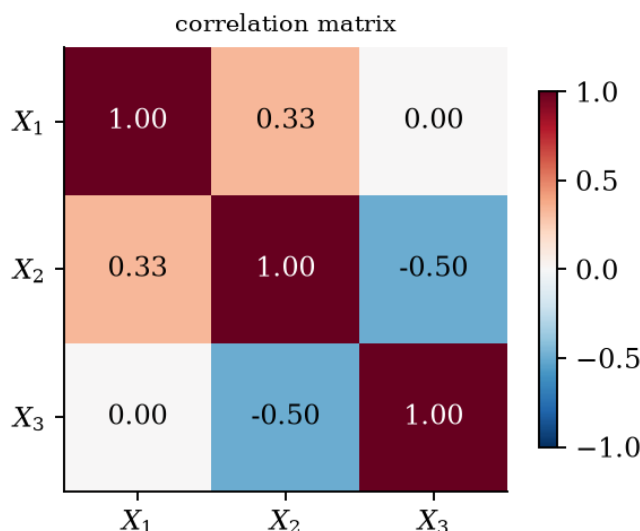
Check: $n = 61$ gives $\sqrt{61}/4 = 1.953 < 1.96$ (probability ≈ 0.949 , just short).

(C) rounds down; (B) forgets to square 1.96 ($1.96 \times 4^2 \approx 31$); (A) $= \sigma^2$, ignoring the confidence level.

Trap: Sample-size answers are always rounded up, never to the nearest integer.

Q.24 [MSQ · 2 marks]

Topic: Covariance matrix and correlation · Medium

Correct answer: A, B, D**Concept:** $\rho_{ij} = \sigma_{ij}/(\sigma_i\sigma_j)$; $\text{Var}(a^T X) = a^T \Sigma a$.**(A) TRUE.**

$$\rho_{12} = \frac{2}{\sqrt{4}\sqrt{9}} = \frac{2}{6} = \frac{1}{3}$$

(B) TRUE.

$$\text{Var}(X_1 + X_2) = 4 + 9 + 2(2) = 17$$

(C) FALSE. Zero covariance means zero linear association only; independence does not follow unless (X_1, X_3) is jointly normal, which is not given.**(D) TRUE.**

$$\text{Var}(X_2 - 2X_3) = 9 + 4(1) + 2(1)(-2)(-1.5) = 9 + 4 + 6 = 19$$

Trap: Uncorrelated $\neq$ independent. Classic counter-example: $X \sim N(0, 1)$, $Y = X^2$ has $\text{Cov}(X, Y) = E[X^3] = 0$ yet Y is a function of X .

Q.25 [MCQ · 2 marks · -2/3]

Topic: Binomial (click-through rate) · Medium

Correct answer: B**Concept:** $X \sim \text{Bin}(n, p)$; use the complement for 'at least'. $X \sim \text{Bin}(20, 0.05)$.

$$P(X = 0) = 0.95^{20} \approx 0.3585$$

$$P(X = 1) = \binom{20}{1}(0.05)(0.95)^{19} \approx 0.3774$$

$$P(X \geq 2) = 1 - 0.3585 - 0.3774 \approx 0.2642$$

(A) 0.358 is $P(X = 0)$; (D) 0.377 is $P(X = 1)$; (C) 0.736 is $P(X \leq 1)$.Poisson check ($\lambda = np = 1$): $1 - 2e^{-1} \approx 0.264$ — close, as n is large and p small.

Shortcut: 'At least k ' with small k : compute $1 - P(0) - \dots - P(k - 1)$.

Q.26 [MCQ · 2 marks · -2/3]**Topic: Counting (cross-validation folds) · Hard****Correct answer: C**

Concept: multinomial coefficient for labelled groups, then divide by the number of ways to permute equal-sized unlabelled groups.

Labelled folds (fold 1, fold 2, fold 3):

$$\binom{12}{4}\binom{8}{4}\binom{4}{4} = 495 \times 70 \times 1 = 34650 = \frac{12!}{4! 4! 4!}$$

Each unlabelled partition is counted $3! = 6$ times (once per labelling of the three folds):

$$\frac{34650}{3!} = \frac{34650}{6} = 5775$$

(A) is the labelled count — the trap. (B) $495 = C(12, 4)$ only chooses one fold. (D) $1925 = 34650/18$ over-divides.

Trap: Divide by $k!$ only when the k groups have the **same size** and are unlabelled. Folds of sizes 5, 4, 3 would need no division.

Q.27 [MSQ · 2 marks]**Topic: Poisson process & exponential waiting times · Medium****Correct answer: A, B, C**

Concept: in a Poisson process of rate λ , $N(t) \sim \text{Poisson}(\lambda t)$; inter-arrival times are i.i.d. $\text{Exp}(\lambda)$; the k -th arrival time is $\text{Gamma}(k, \lambda)$ (Erlang).

$\lambda = 4$ per minute $= 1/15$ per second.

(A) TRUE. $\lambda t = 4 \times 0.5 = 2$.

(B) TRUE. $E[T_1] = 1/\lambda = 1/4 \text{ min} = 15 \text{ s}$.

(C) TRUE.

$$P(N(0.5) = 0) = P(T_1 > 0.5) = e^{-4(0.5)} = e^{-2} \approx 0.135$$

(D) FALSE. $T_2 = T_1 + (T_2 - T_1)$ is a sum of two independent $\text{Exp}(\lambda)$ times: its mean is 30 s, but its distribution is $\text{Gamma}(2, \lambda)$ with density $\lambda^2 t e^{-\lambda t}$, not exponential (its density is 0 at $t = 0$).

Trap: Mean right, distribution wrong — MSQ options often hide the error in the word 'exponentially'.

Q.28 [NAT · 2 marks]Topic: χ^2 goodness-of-fit test · Medium**Correct answer: 3.96 (range 3.95 to 3.97)****Concept:** $\chi^2 = \sum (O_i - E_i)^2 / E_i$ with $k - 1$ df when no parameters are estimated.Under uniformity, $E_i = 200/4 = 50$ for every bucket.

Bucket	O	E	O - E	(O - E) ² /E
1	62	50	12	144/50 = 2.88
2	45	50	-5	0.50
3	48	50	-2	0.08
4	45	50	-5	0.50

$$\chi^2 = 2.88 + 0.50 + 0.08 + 0.50 = 3.96$$

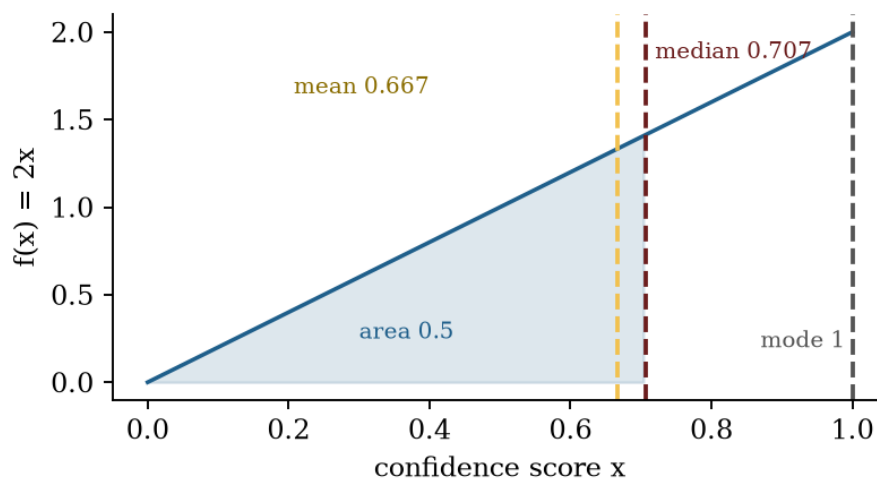
Degrees of freedom = $4 - 1 = 3$; $\chi^2_{0.05,3} = 7.815$. Since $3.96 < 7.815$, uniformity is not rejected.**Shortcut:** Check: $\sum (O - E)$ must be 0 ($12 - 5 - 2 - 5 = 0$) — a quick guard against arithmetic slips.**Q.29** [MCQ · 2 marks · -2/3]

Topic: Mean, median, mode of a continuous PDF · Medium

Correct answer: B**Concept:** mean = $\int x f(x) dx$; median m solves $F(m) = 1/2$; mode = $\operatorname{argmax} f$.

$$\text{mean} = \int_0^1 x \cdot 2x dx = \frac{2}{3} \approx 0.667$$

$$F(x) = x^2 = \frac{1}{2} \Rightarrow m = \frac{1}{\sqrt{2}} \approx 0.707$$

 $f(x) = 2x$ is increasing on $[0, 1]$, so the mode is at $x = 1$.

Hence mean (0.667) < median (0.707) < mode (1): the density is left-skewed.

Key idea: For a left-skewed distribution the long tail is on the left and pulls the mean below the median — the mirror image of Q.3.

Q.30 [MSQ · 2 marks]**Topic: Confidence interval (t-based) · Medium****Correct answer: A, B, D****Concept:** with σ unknown and normal data, $\bar{x} \pm t_{\alpha/2, n-1} \cdot s/\sqrt{n}$.

$$\frac{s}{\sqrt{n}} = \frac{10}{5} = 2, \quad \text{margin} = 2.064(2) = 4.128$$

(A) TRUE. $80 \pm 4.128 = (75.872, 84.128)$.**(B) TRUE.** $1.96 < 2.064$, so $80 \pm 3.92 = (76.08, 83.92)$ is narrower (and under-covers).**(C) FALSE.** μ is fixed; the computed interval either contains it or not. '95%' describes the long-run success rate of the procedure across repeated samples.**(D) TRUE.** $s/\sqrt{n} = 10/10 = 1$, width = $2 \times 1.96 \times 1 = 3.92$.**Trap:** Frequentist confidence $\neq$ probability about the fixed parameter. This is the single most common conceptual trap on interval questions.

Mock Test 04 — Standard GATE-Level Test II (Data-Science Contexts)

A second full-syllabus paper at actual GATE DA difficulty: fraud detection, naive Bayes, model comparison, dropout, logging pipelines and survey analytics.

Item	Details
Questions	30 (12 × 1-mark, 18 × 2-mark)
Maximum marks	48
Question types	MCQ: 14 · MSQ: 7 · NAT: 9
Suggested time	90 minutes
Negative marking	MCQ only: $-1/3$ (1-mark), $-2/3$ (2-mark). None for MSQ & NAT.
Calculator	Use only the GATE virtual scientific calculator

Topic focus of this set

Covers the whole syllabus at exam difficulty ($\approx 30\%$ easy, 50% medium, 20% hard): permutations with restrictions, hash-collision (birthday) probabilities and hypergeometric mini-batch sampling; mutually exclusive vs independent events; total probability, two-stage Bayes for fraud alerts and a naive Bayes spam posterior; joint/conditional PDFs (exponential pair, $x + y$ density), random-sum variance via conditional expectation; sample and grouped-data summaries, correlation invariance and the regression slope; Bernoulli/inverted dropout, discrete uniform, Poisson, exponential, Gamma-type CDFs, normal comparisons, χ^2 of the sample variance; CLT for dropout counts; sample size for a CTR interval; two-sample z-test for an A/B experiment, paired t-test comparing models across datasets, and a 2×3 χ^2 test of independence.

Instructions

- Questions 1 onwards are arranged as in GATE: all 1-mark questions first, then 2-mark questions.
- **MCQ** – exactly one correct option. **MSQ** – ONE OR MORE options are correct; credit only if you select all correct options and no wrong option (no partial marks, no negative marks).
- **NAT** – enter a numerical value using the virtual keypad. The accepted range is given in the key; round as instructed in the question.
- Statistical tables (Z , t , χ^2) are provided in the Appendix; values in solutions use those tables.
- Attempt in one sitting, then check the answer key and read every solution — including those you got right — for the alternative methods and traps.

Mock Test 04 · Question Paper

1-mark questions

Q.1 MCQ · 1 mark · -1/3

Six model variants (one baseline and five candidates) are to be deployed in six consecutive one-hour traffic slots, one variant per slot. If the baseline must **not** occupy the first slot, the number of possible schedules is

- (A) 600
- (B) 720
- (C) 120
- (D) 480

Q.2 MCQ · 1 mark · -1/3

In inverted dropout, an activation x is replaced by $Y = x \cdot M/q$, where $M \sim \text{Bernoulli}(q)$ is the keep-mask and q is the keep probability. For $x = 2$ and $q = 0.8$, the pair $(E[Y], \text{Var}(Y))$ is

- (A) (2, 0.64)
- (B) (1.6, 0.64)
- (C) (2, 1)
- (D) (2.5, 1)

Q.3 NAT · 1 mark

The numbers of GPU hours used by five training runs are 4, 8, 6, 5, 7. Their sample standard deviation (with divisor $n - 1$) is _____ (round off to 2 decimal places).

Answer: _____

Q.4 MCQ · 1 mark · -1/3

For a read request, let $A = \{\text{served from the in-memory cache}\}$ and $B = \{\text{served from the disk database}\}$. A request is served from exactly one source, and $P(A) = 0.7$, $P(B) = 0.2$. Which statement is correct?

- (A) A and B are independent
- (B) $P(A | B) = 0$, so A and B are dependent
- (C) $P(A \cup B) = 0.76$
- (D) $P(A \cap B) = 0.14$

Q.5 NAT · 1 mark

A log-parsing job encounters malformed lines at an average rate of 0.5 per 1000 lines, independently, so that counts follow a Poisson distribution. For a file of 4000 lines, the probability that no malformed line is encountered is _____ (round off to 3 decimal places).

Answer: _____

Q.6 MSQ · 1 mark

A feature X is normally distributed with mean 10 and variance 4. Which of the following is/are TRUE?

- (A) $P(X < 10) = 0.5$
- (B) $(X - 10)/2 \sim N(0, 1)$
- (C) $2X \sim N(20, 8)$
- (D) $P(8 < X < 12) \approx 0.683$

Q.7 MCQ · 1 mark · -1/3

A data-augmentation routine picks a random crop offset X uniformly from the integers $\{0, 1, 2, \dots, 9\}$. The variance of X is

- (A) 8.33
- (B) 9.17
- (C) 6.75
- (D) 8.25

Q.8 NAT · 1 mark

A dataset merges records from two sources: 40% come from source S and the rest from source T . Among S -records 25% contain missing values; among T -records 10% do. The probability that a randomly chosen record contains missing values is _____ (round off to 2 decimal places).

Answer: _____

Q.9 MCQ · 1 mark · -1/3

The time to failure of a GPU node is exponentially distributed with mean 100 hours. The median time to failure is approximately

- (A) 100 hours
- (B) 69.3 hours
- (C) 50 hours
- (D) 144.3 hours

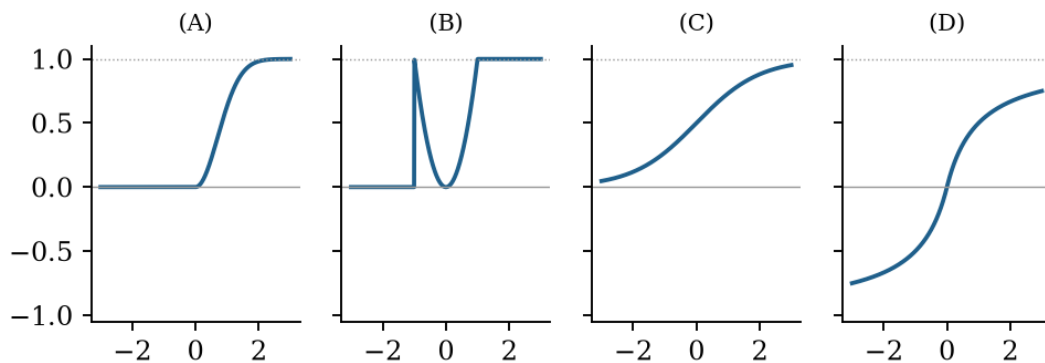
Q.10 MCQ · 1 mark · -1/3

Two features have $\text{Corr}(X, Y) = 0.6$. New features are defined as $U = 3 - 2X$ and $V = 5Y + 1$. Then $\text{Corr}(U, V)$ equals

- (A) -0.6
- (B) 0.6
- (C) -6.0
- (D) -0.06

Q.11 MSQ · 1 mark

Which of the following functions is/are valid cumulative distribution functions on the real line? (Each is plotted below on $[-3, 3]$.)



- (A) $F(x) = 1 - e^{-x^2}$ for $x \geq 0$, and $F(x) = 0$ for $x < 0$
 (B) $F(x) = 0$ for $x < -1$, $F(x) = x^2$ for $-1 \leq x \leq 1$, $F(x) = 1$ for $x > 1$
 (C) $F(x) = 1/(1 + e^{-x})$ for all x
 (D) $F(x) = x/(1 + |x|)$ for all x

Q.12 NAT · 1 mark

Let $Z_1, \dots, Z_5$ be independent standard normal random variables (e.g. standardised residuals) and $W = Z_1^2 + \dots + Z_5^2$. The value of $E[W] + \text{Var}(W)$ is _____ (answer as an integer).

Answer: _____

2-mark questions**Q.13 NAT · 2 marks**

0.5% of transactions are fraudulent. Two fraud models are run independently on every transaction; conditional on the true class, their decisions are independent. Each model flags a fraudulent transaction with probability 0.98 and a legitimate one with probability 0.03. Given that **both** models flag a transaction, the probability that it is fraudulent is _____ (round off to 3 decimal places).

Answer: _____

Q.14 MCQ · 2 marks · -2/3

A naive Bayes spam classifier uses two binary word features. Priors: $P(\text{spam}) = 0.4$, $P(\text{ham}) = 0.6$. Likelihoods: $P(\text{"free"} | \text{spam}) = 0.5$, $P(\text{"free"} | \text{ham}) = 0.1$, $P(\text{"meeting"} | \text{spam}) = 0.1$, $P(\text{"meeting"} | \text{ham}) = 0.4$. Assuming the words are conditionally independent given the class, the posterior probability that an email containing both words is spam is closest to

- (A) 0.400
 (B) 0.545
 (C) 0.455
 (D) 0.833

Q.15 MSQ · 2 marks

The latencies X and Y (in seconds) of two independent micro-services have joint density $f(x, y) = 6e^{-2x-3y}$ for $x > 0, y > 0$ (0 elsewhere). Which of the following is/are TRUE?

- (A) X and Y are independent
- (B) X is exponential with rate 2 (mean 0.5 s)
- (C) $P(X < Y) = 0.6$
- (D) $E[XY] = 1/6$

Q.16 NAT · 2 marks

A data loader produces mini-batches whose size N is 32 or 64 with equal probability. Each example in a batch is independently a positive with probability 0.25. Let S be the number of positives in a batch. The value of $\text{Var}(S)$ is _____ (answer as an integer).

Answer: _____

Q.17 MCQ · 2 marks · -2/3

An A/B test compares mean session duration (minutes). Variant A: $n = 400$, mean 5.2, standard deviation 2.0. Variant B: $n = 400$, mean 5.5, standard deviation 2.4. Using a large-sample z-test ($z_{0.025} = 1.96, z_{0.05} = 1.645$), which conclusion is correct?

- (A) $z \approx 1.92$; reject $H_0: \mu_A = \mu_B$ in a two-sided test at 5%
- (B) $z \approx 1.92$; do not reject H_0 in a two-sided test at 5%, but reject it in favour of $\mu_B > \mu_A$ in a one-sided test at 5%
- (C) $z \approx 0.10$; do not reject H_0 in either test
- (D) $z \approx 2.50$; reject H_0 in both tests

Q.18 NAT · 2 marks

Two models are evaluated on the same 6 benchmark datasets. The accuracy differences (model 2 – model 1, in percentage points) are:

Dataset	1	2	3	4	5	6
Difference	1.2	0.8	1.5	0.3	1.0	0.6

The paired t-statistic for H_0 : mean difference = 0 is _____ (round off to 2 decimal places).

Answer: _____

Q.19 MSQ · 2 marks

A product survey cross-tabulates user plan against the most-used feature:

	Feature A	Feature B	Feature C	Total
Free	50	30	20	100
Paid	30	40	30	100
Total	80	70	50	200

A χ^2 test of independence is performed ($\chi^2_{0.05,2} = 5.991$, $\chi^2_{0.01,2} = 9.210$). Which of the following is/are TRUE?

- (A) The test has 2 degrees of freedom
- (B) The expected count for (Free, Feature A) is 40
- (C) The χ^2 statistic is about 8.43, so independence is rejected at $\alpha = 0.05$
- (D) Independence is also rejected at $\alpha = 0.01$

Q.20 MCQ · 2 marks · -2/3

A team wants to estimate the click-through rate of a new widget to within ± 0.02 with 95% confidence, with no prior idea of its value. Using the normal approximation and the conservative choice for p ($z_{0.025} = 1.96$), the minimum number of impressions required is

- (A) 2401
- (B) 2400
- (C) 1692
- (D) 4802

Q.21 NAT · 2 marks

The response time X (in seconds) of a search service has CDF $F(x) = 1 - (1 + x)e^{-x}$ for $x \geq 0$ and $F(x) = 0$ for $x < 0$. The value of $P(1 < X \leq 2)$ is _____ (round off to 3 decimal places).

Answer: _____

Q.22 MCQ · 2 marks · -2/3

A query is fanned out to 3 replicas and the response is returned only when **all three** have replied. The replicas' latencies are independent and uniformly distributed on (0, 1) second. The expected response time (in seconds) is

- (A) 1/2
- (B) 2/3
- (C) 1/4
- (D) 3/4

Q.23 MSQ · 2 marks

A layer has 400 neurons and dropout keeps each neuron independently with probability 0.9. Let S be the number of kept neurons. Which of the following is/are TRUE? ($\Phi(2) = 0.9772$.)

- (A) $E[S] = 360$
- (B) The standard deviation of S is 6
- (C) Using the normal approximation without continuity correction, $P(S \geq 372) \approx 0.023$
- (D) The standard deviation of the kept fraction $S/400$ is 0.15

Q.24 MCQ · 2 marks · -2/3

For users of an app, weekly sessions X and weekly spend Y have $E[X] = 10$, $E[Y] = 50$, $SD(X) = 2$, $SD(Y) = 5$ and $\text{Corr}(X, Y) = 0.6$. The best linear (least-squares) prediction of Y for a user with $X = 12$ is

- (A) 51.2
- (B) 55.0
- (C) 53.0
- (D) 50.48

Q.25 MCQ · 2 marks · -2/3

Ten user IDs are hashed independently and uniformly into a table of 100 slots. The probability that at least two IDs collide (share a slot) is closest to

- (A) 0.628
- (B) 0.372
- (C) 0.100
- (D) 0.450

Q.26 MCQ · 2 marks · -2/3

A small dataset of 20 examples contains 4 mislabelled ones. A mini-batch of 5 examples is drawn uniformly at random **without replacement**. The probability that the batch contains at least one mislabelled example is closest to

- (A) 0.718
- (B) 0.672
- (C) 0.282
- (D) 0.470

Q.27 MSQ · 2 marks

Ten readings $X_1, \dots, X_{10}$ are taken from a sensor whose noise is $N(\mu, \sigma^2)$ with $\sigma^2 = 4$. Let $\bar{X}$ and S^2 be the sample mean and sample variance (divisor $n - 1$). Which of the following is/are TRUE?

- (A) $9S^2/4$ follows a χ^2 distribution with 9 degrees of freedom
- (B) $E[S^2] = 4$
- (C) $\text{Var}(S^2) = 3.2$
- (D) $\bar{X}$ and S^2 are dependent random variables

Q.28 NAT · 2 marks

Two temperature sensors give independent readings $X \sim N(20, 9)$ and $Y \sim N(18, 16)$ (second parameter = variance). The probability that sensor X reads higher than sensor Y is _____ (round off to 3 decimal places). ($\Phi(0.4) = 0.6554$.)

Answer: _____

Q.29 MCQ · 2 marks · -2/3

Two normalised scores (X, Y) have joint density $f(x, y) = x + y$ on the unit square $0 < x < 1, 0 < y < 1$ (0 elsewhere). The value of $E[Y | X = 0.5]$ is

- (A) $1/2$
- (B) $2/3$
- (C) $5/12$
- (D) $7/12$

Q.30 MSQ · 2 marks

Session lengths (minutes) of 50 users are summarised below.

Class (min)	0-10	10-20	20-30	30-40
Frequency	5	15	20	10

Using class mid-points for the mean and SD (divisor N) and linear interpolation within the median class for the median, which of the following is/are TRUE?

- (A) The median is 22.5 minutes
- (B) The mean is 22 minutes
- (C) The modal class is 20-30
- (D) The standard deviation exceeds 10 minutes

Mock Test 04 · Answer Key

Q	Type	Marks	Answer	Topic
1	MCQ	1	A	Permutations with a restriction
2	MCQ	1	C	Bernoulli r.v. (inverted dropout)
3	NAT	1	1.58 (range 1.57 to 1.59)	Sample standard deviation
4	MCQ	1	B	Mutually exclusive vs independent
5	NAT	1	0.135 (range 0.134 to 0.136)	Poisson distribution
6	MSQ	1	A, B, D	Normal distribution properties
7	MCQ	1	D	Discrete uniform distribution
8	NAT	1	0.16	Total probability
9	MCQ	1	B	Exponential distribution (median vs mean)
10	MCQ	1	A	Correlation under linear transformations
11	MSQ	1	A, C	Valid CDFs
12	NAT	1	15	Chi-squared distribution
13	NAT	2	0.843 (range 0.842 to 0.844)	Bayes theorem (two-stage screening)
14	MCQ	2	C	Bayes theorem (naive Bayes classifier)
15	MSQ	2	A, B, D	Joint PDF, independence, $P(X < Y)$
16	NAT	2	25	Conditional expectation & variance (random sum)
17	MCQ	2	B	Two-sample z-test (A/B test of means)
18	NAT	2	5.14 (range 5.12 to 5.16)	Paired t-test
19	MSQ	2	A, B, C	χ^2 test of independence (2×3)
20	MCQ	2	A	Confidence interval — sample size for a proportion
21	NAT	2	0.330 (range 0.329 to 0.331)	CDF and PDF of a continuous r.v.
22	MCQ	2	D	CDF of the maximum (order statistics)
23	MSQ	2	A, B, C	CLT / normal approximation (dropout count)
24	MCQ	2	C	Covariance, correlation, regression slope
25	MCQ	2	B	Counting (birthday problem for hashing)
26	MCQ	2	A	Hypergeometric sampling (mini-batches)
27	MSQ	2	A, B	Sampling distribution of S^2 (χ^2)
28	NAT	2	0.655 (range 0.654 to 0.657)	Normal distribution (difference of normals)
29	MCQ	2	D	Conditional PDF and conditional expectation
30	MSQ	2	A, B, C	Grouped data: mean, median, mode, SD

Score tracker

	Attempted	Correct	Wrong (MCQ)	Marks
1-mark				
2-mark				
Total				/ 48

Mock Test 04 · Detailed Solutions

Q.1 [MCQ · 1 mark · -1/3]

Topic: Permutations with a restriction · Easy

Correct answer: A

Concept: count the complement — total arrangements minus those violating the restriction.

$$\text{total} = 6! = 720$$

Schedules with the baseline in slot 1: the remaining 5 variants fill 5 slots in $5!$ ways.

$$\text{bad} = 5! = 120$$

$$\text{answer} = 720 - 120 = 600$$

Direct check: slot 1 can take any of the 5 candidates, then the other 5 variants (including the baseline) fill the rest in $5!$ ways: $5 \times 120 = 600$.

(D) $480 = 4 \times 5!$ would forbid the baseline from both the first and the last slot.

Shortcut: For 'X not in position k' restrictions, $(n-1) \cdot (n-1)!$ is a fast direct count.

Q.2 [MCQ · 1 mark · -1/3]

Topic: Bernoulli r.v. (inverted dropout) · Medium

Correct answer: C

Concept: for $M \sim \text{Bernoulli}(q)$: $E[M] = q$, $\text{Var}(M) = q(1-q)$; and $\text{Var}(cM) = c^2 \text{Var}(M)$.

Here $Y = (2/0.8)M = 2.5M$.

$$E[Y] = 2.5 E[M] = 2.5(0.8) = 2$$

$$\text{Var}(Y) = 2.5^2 \text{Var}(M) = 6.25(0.8)(0.2) = 6.25(0.16) = 1$$

The $1/q$ scaling makes $E[Y] = x$, so no rescaling is needed at test time — that is the point of inverted dropout.

(A) uses $\text{Var}(2M) = 0.64$ (forgets the $1/q$ scaling in the variance). (B) has no scaling at all. (D) reports the value of Y when kept as its mean.

Key idea: Dropout keeps the mean but injects variance $x^2(1-q)/q$ — here $4(0.2)/0.8 = 1$.

Q.3 [NAT · 1 mark]

Topic: Sample standard deviation · Easy

Correct answer: 1.58 (range 1.57 to 1.59)

Concept: $s^2 = \sum(x_i - \bar{x})^2 / (n-1)$.

$$\bar{x} = \frac{4 + 8 + 6 + 5 + 7}{5} = \frac{30}{5} = 6$$

x	4	8	6	5	7
$x - \bar{x}$	-2	2	0	-1	1
$(x - \bar{x})^2$	4	4	0	1	1

$$\sum(x_i - \bar{x})^2 = 10, \quad s^2 = \frac{10}{5-1} = 2.5$$

$$s = \sqrt{2.5} \approx 1.5811$$

Trap: With divisor n you would get $\sqrt{2} \approx 1.41$ (the population formula). Read which divisor the question asks for.

Q.4 [MCQ · 1 mark · -1/3]**Topic: Mutually exclusive vs independent · Easy****Correct answer: B****Concept:** mutually exclusive means $P(A \cap B) = 0$; independent means $P(A \cap B) = P(A)P(B)$.'Exactly one source' $\Rightarrow$ A and B cannot both occur $\Rightarrow P(A \cap B) = 0$.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{0.2} = 0 \neq P(A) = 0.7$$

So knowing B changes the probability of A: they are **dependent**. (B) is correct.(A) would need $P(A \cap B) = 0.14 \neq 0$. (C) $P(A \cup B) = 0.7 + 0.2 - 0 = 0.9$, not 0.76 (0.76 is what independence would give). (D) is the independence value, not the actual one.

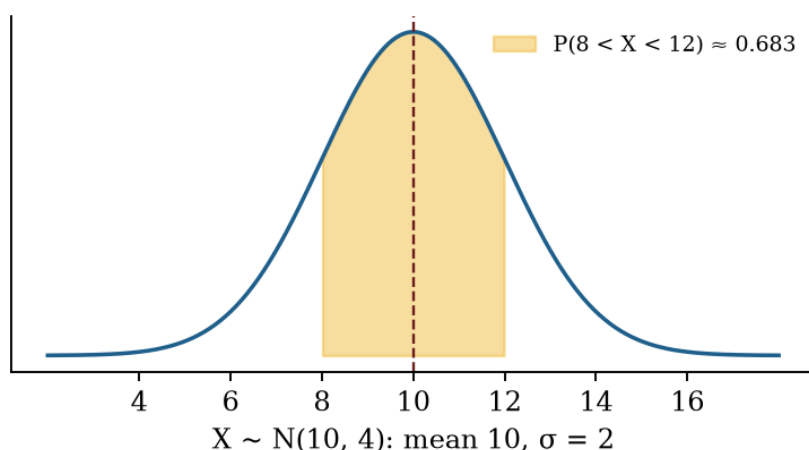
Trap: Disjoint events with non-zero probabilities are always dependent — the strongest form of dependence: one rules the other out.

Q.5 [NAT · 1 mark]**Topic: Poisson distribution · Easy****Correct answer: 0.135 (range 0.134 to 0.136)****Concept:** scale the Poisson rate to the interval length: $\lambda = \text{rate} \times \text{length}$.

$$\lambda = 0.5 \times \frac{4000}{1000} = 2$$

$$P(N = 0) = e^{-\lambda} = e^{-2} \approx 0.1353$$

Trap: Forgetting to rescale (using $\lambda = 0.5$) gives $e^{-0.5} \approx 0.607$.

Q.6 [MSQ · 1 mark]**Topic: Normal distribution properties · Easy****Correct answer: A, B, D****Concept:** $X \sim N(\mu, \sigma^2) = aX + b \sim N(a\mu + b, a^2\sigma^2)$; σ here is $\sqrt{4} = 2$.**(A) TRUE.** The normal is symmetric about its mean, so half the mass lies below 10.**(B) TRUE.** Standardising with $\sigma = 2$ (not the variance 4) gives $N(0, 1)$.**(C) FALSE.**

$$\text{Var}(2X) = 2^2(4) = 16 \Rightarrow 2X \sim N(20, 16)$$

(D) TRUE. 8 and 12 are $\mu \mp \sigma$, and $P(|Z| < 1) = 2\Phi(1) - 1 = 0.6827$.

Trap: Scaling multiplies the variance by a^2 , not a . $N(\mu, \sigma^2)$ notation uses the variance — divide by its square root when standardising.

Q.7 [MCQ · 1 mark · -1/3]

Topic: Discrete uniform distribution · Medium

Correct answer: D

Concept: for X uniform on n consecutive integers, $\text{Var}(X) = (n^2 - 1)/12$.

Here $n = 10$:

$$\text{Var}(X) = \frac{10^2 - 1}{12} = \frac{99}{12} = 8.25$$

Direct check: $E[X] = 4.5$ and $E[X^2] = (0 + 1 + 4 + \dots + 81)/10 = 285/10 = 28.5$, so

$$\text{Var}(X) = 28.5 - 4.5^2 = 28.5 - 20.25 = 8.25$$

(A) $100/12$ uses $n^2/12$; (C) $81/12$ uses the continuous formula on $[0, 9]$; (B) $99/12 \times 10/9$ applies an $n - 1$ correction that does not belong to a population variance.

Trap: Discrete uniform: $(n^2 - 1)/12$. Continuous uniform on $[a, b]$: $(b - a)^2/12$. Do not mix them.

Q.8 [NAT · 1 mark]

Topic: Total probability · Easy

Correct answer: 0.16

Concept: law of total probability over the partition $\{S, T\}$.

$$P(M) = P(M|S)P(S) + P(M|T)P(T)$$

$$P(M) = 0.25(0.4) + 0.10(0.6) = 0.10 + 0.06 = 0.16$$

Bonus (Bayes): $P(S|M) = 0.10/0.16 = 0.625$ — a record with missing values is more likely from S.

Trap: Averaging the two rates (0.175) ignores the unequal source sizes.

Q.9 [MCQ · 1 mark · -1/3]

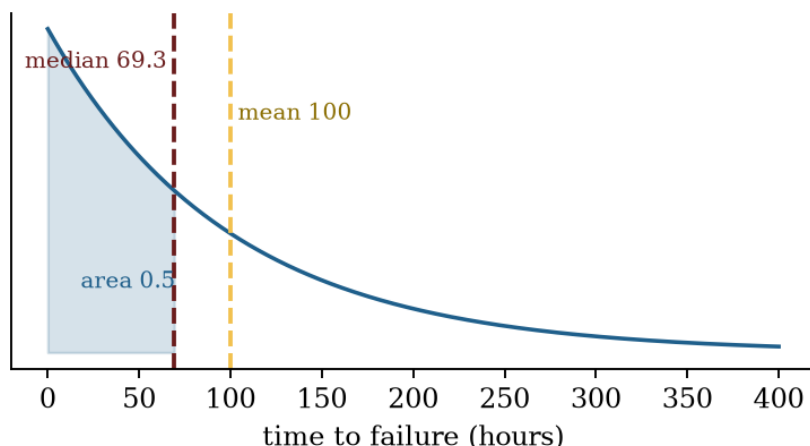
Topic: Exponential distribution (median vs mean) · Easy

Correct answer: B

Concept: the median m solves $F(m) = 1 - e^{-\lambda m} = 1/2$, so $m = \ln 2/\lambda$.

$$\lambda = \frac{1}{100}, \quad 1 - e^{-m/100} = \frac{1}{2} \Rightarrow m = 100 \ln 2$$

$$m = 100(0.6931) \approx 69.3 \text{ hours}$$



(A) is the mean; (C) assumes symmetry; (D) = $100/\ln 2$ inverts the factor.

Key idea: The exponential is right-skewed: median $\approx 0.693 \times$ mean. Half of all nodes fail before 70 hours even though the mean is 100.

Q.10 [MCQ · 1 mark · -1/3]

Topic: Correlation under linear transformations · Easy

Correct answer: A

Concept: $\text{Corr}(aX + b, cY + d) = \text{sign}(ac) \cdot \text{Corr}(X, Y)$: shifts and positive scalings do not change correlation; a negative factor flips its sign.

$$\begin{aligned}\text{Cov}(U, V) &= (-2)(5) \text{Cov}(X, Y) = -10 \text{Cov}(X, Y) \\ \sigma_U \sigma_V &= |-2| \cdot |5| \sigma_X \sigma_Y = 10 \sigma_X \sigma_Y \\ \text{Corr}(U, V) &= \frac{-10 \text{Cov}(X, Y)}{10 \sigma_X \sigma_Y} = -0.6\end{aligned}$$

(C) multiplies the correlation by ac as if it were a covariance; correlation always lies in $[-1, 1]$.

Key idea: Standardising features (z-scores) never changes their correlations.

Q.11 [MSQ · 1 mark]

Topic: Valid CDFs · Medium

Correct answer: A, C

Concept: F is a CDF iff it is non-decreasing, right-continuous, $F(-\infty) = 0$ and $F(\infty) = 1$.

(A) VALID. 0 for $x < 0$, continuous at 0 ($F(0) = 0$), increasing for $x > 0$, $\rightarrow 1$ as $x \rightarrow \infty$. (This is the Rayleigh distribution.)

(B) INVALID. It jumps from 0 to $F(-1) = 1$ at $x = -1$ and then decreases to 0 at $x = 0$ — a CDF can never decrease.

(C) VALID. The logistic sigmoid is continuous, strictly increasing, with limits 0 and 1.

(D) INVALID. It is increasing, but

$$\lim_{x \rightarrow -\infty} \frac{x}{1 + |x|} = -1 \neq 0$$

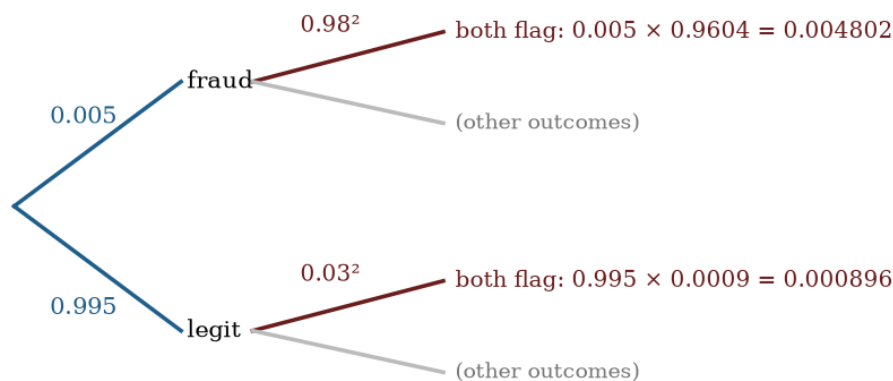
Trap: Check all four conditions — (D) passes monotonicity and the upper limit but fails the lower limit. A CDF takes values only in $[0, 1]$.

Q.12 [NAT · 1 mark]**Topic: Chi-squared distribution · Easy****Correct answer: 15****Concept:** $W \sim \chi^2_k$ with $k = 5$; a χ^2_k variable has mean k and variance $2k$.Derivation for one term: $E[Z^2] = 1$ and $E[Z^4] = 3$, so $\text{Var}(Z^2) = 3 - 1 = 2$. By independence:

$$E[W] = 5(1) = 5, \quad \text{Var}(W) = 5(2) = 10$$

$$E[W] + \text{Var}(W) = 5 + 10 = 15$$

Key idea: Sum of squared residuals of standardised errors is χ^2 — the basis of the χ^2 tests and of the sample-variance distribution.

Q.13 [NAT · 2 marks]**Topic: Bayes theorem (two-stage screening) · Hard****Correct answer: 0.843 (range 0.842 to 0.844)****Concept:** Bayes theorem; conditional independence lets the likelihoods multiply.

Likelihood of 'both flag' under each class:

$$P(++ | F) = 0.98^2 = 0.9604, \quad P(++ | L) = 0.03^2 = 0.0009$$

Joint probabilities:

$$P(F)P(++ | F) = 0.005(0.9604) = 0.004802$$

$$P(L)P(++ | L) = 0.995(0.0009) = 0.0008955$$

$$P(F | ++) = \frac{0.004802}{0.004802 + 0.0008955} \approx 0.8428$$

For comparison, a single flag gives only $P(F | +) = 0.0049/(0.0049 + 0.02985) \approx 0.141$.

Equivalent sequential view: the posterior after the first flag (0.141) becomes the prior for the second flag:

$$\frac{0.141(0.98)}{0.141(0.98) + 0.859(0.03)} \approx 0.843$$

Key idea: With rare positives, a single good test still gives low precision; a second, conditionally independent test multiplies the likelihood ratio (≈ 32.7 each) and boosts precision dramatically.

Q.14 [MCQ · 2 marks · -2/3]**Topic: Bayes theorem (naive Bayes classifier) · Medium****Correct answer: C**

Concept: posterior $\propto$ prior $\times$ [] likelihoods (naive conditional independence), then normalise.
Unnormalised scores:

$$\text{spam: } 0.4(0.5)(0.1) = 0.020$$

$$\text{ham: } 0.6(0.1)(0.4) = 0.024$$

$$P(\text{spam} \mid \text{free, meeting}) = \frac{0.020}{0.020 + 0.024} = \frac{5}{11} \approx 0.455$$

So the classifier labels the email ham ($0.455 < 0.5$).

(A) is the prior. (B) is the ham posterior. (D) $0.833 = 0.020/0.024$ is the ratio of the two scores, not a normalised probability.

Key idea: The word 'free' alone gives $P(\text{spam} \mid \text{free}) = 0.2/0.26 \approx 0.77$; the counter-evidence of 'meeting' (likelihood ratio $1/4$) flips the decision.

Q.15 [MSQ · 2 marks]**Topic: Joint PDF, independence, $P(X < Y)$ · Medium****Correct answer: A, B, D**

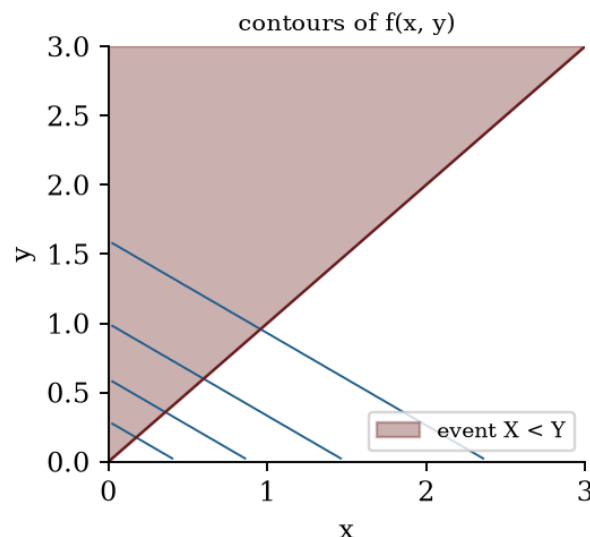
Concept: if a joint density factorises as $g(x)h(y)$ on a product support, the variables are independent.

$$6e^{-2x-3y} = (2e^{-2x})(3e^{-3y}), \quad x > 0, y > 0$$

(A) **TRUE.** Factorisation on the rectangle $(0, \infty) \times (0, \infty)$.

(B) **TRUE.** $f_X(x) = 2e^{-2x}$: Exp(rate 2), mean $1/2$.

(C) **FALSE.**



$$\begin{aligned} P(X < Y) &= \int_0^\infty 2e^{-2x} P(Y > x) dx = \int_0^\infty 2e^{-2x} e^{-3x} dx \\ &= \frac{2}{2+3} = 0.4 \end{aligned}$$

0.6 is $P(Y < X)$ — the faster service (rate 3) usually finishes first.

(D) **TRUE.** By independence:

$$E[XY] = E[X]E[Y] = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$$

Shortcut: For independent exponentials, $P(X < Y) = \lambda_X / (\lambda_X + \lambda_Y)$: the one with the larger rate wins more often.

Topic: Conditional expectation & variance (random sum)
· Hard

Q.16 [NAT · 2 marks]

Correct answer: 25

Concept: law of total variance, $\text{Var}(S) = E[\text{Var}(S | N)] + \text{Var}(E[S | N])$.

Given $N, S \sim \text{Bin}(N, 0.25)$:

$$E[S | N] = 0.25N, \quad \text{Var}(S | N) = N(0.25)(0.75) = 0.1875N$$

Moments of N : $E[N] = 48$ and, since $N = 48 \pm 16$ with equal probability, $\text{Var}(N) = 16^2 = 256$.

$$E[\text{Var}(S | N)] = 0.1875 E[N] = 0.1875(48) = 9$$

$$\text{Var}(E[S | N]) = 0.25^2 \text{Var}(N) = 0.0625(256) = 16$$

$$\text{Var}(S) = 9 + 16 = 25$$

Monte Carlo check (2×10^6 batches): ≈ 24.98 .

Trap: Treating S as $\text{Bin}(48, 0.25)$ gives only 9 — it misses the variability that comes from the random batch size.

Q.17 [MCQ · 2 marks · -2/3]

Topic: Two-sample z-test (A/B test of means) · Medium

Correct answer: B

Concept: $z = (\bar{x}_B - \bar{x}_A) / \sqrt{(s_A^2/n_A + s_B^2/n_B)}$.

$$SE = \sqrt{\frac{2.0^2}{400} + \frac{2.4^2}{400}} = \sqrt{0.01 + 0.0144} = \sqrt{0.0244}$$

$$SE \approx 0.1562, \quad z = \frac{5.5 - 5.2}{0.1562} \approx 1.921$$

Two-sided at 5%: $|z| = 1.92 < 1.96 \Rightarrow$ do not reject.

One-sided ($H_1: \mu_B > \mu_A$) at 5%: $1.92 > 1.645 \Rightarrow$ reject.

(C) forgets to divide the variances by n : $0.3/\sqrt{4 + 5.76} \approx 0.10$. (D) uses only group B's standard error, $0.3/\sqrt{5.76/400} = 0.3/0.12 = 2.50$, ignoring the uncertainty in $\bar{x}_A$.

Trap: The choice of one- vs two-sided alternative must be fixed **before** seeing the data; switching after the fact inflates the false-positive rate.

Q.18 [NAT · 2 marks]**Topic: Paired t-test · Medium****Correct answer: 5.14 (range 5.12 to 5.16)****Concept:** a paired test is a one-sample t-test on the differences: $t = \bar{d}/(s_d/\sqrt{n})$, $df = n - 1$.

$$\bar{d} = \frac{1.2 + 0.8 + 1.5 + 0.3 + 1.0 + 0.6}{6} = \frac{5.4}{6} = 0.9$$

d	1.2	0.8	1.5	0.3	1.0	0.6
$d - \bar{d}$	0.3	-0.1	0.6	-0.6	0.1	-0.3
$(d - \bar{d})^2$	0.09	0.01	0.36	0.36	0.01	0.09

$$s_d^2 = \frac{0.92}{5} = 0.184, \quad s_d \approx 0.4290$$

$$t = \frac{0.9}{0.4290/\sqrt{6}} = \frac{0.9}{0.1751} \approx 5.139$$

With 5 df, $t_{0.025,5} = 2.571$, so model 2 is significantly better at the 5% level.

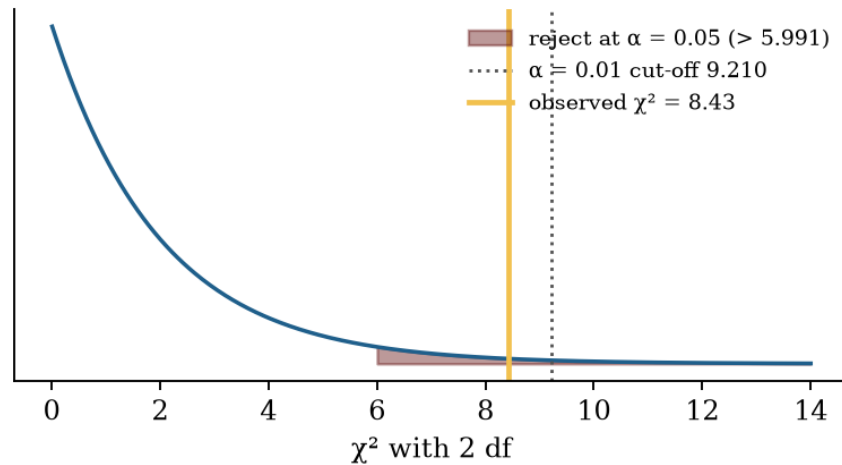
Key idea: Pairing removes dataset-to-dataset difficulty variation. An unpaired two-sample test on the raw accuracies would be far less powerful.

Q.19 [MSQ · 2 marks]**Topic: χ^2 test of independence (2×3) · Medium****Correct answer: A, B, C****Concept:** $E_{ij} = R_i C_j / N$; $\chi^2 = \sum (O - E)^2 / E$; $df = (r - 1)(c - 1)$.**(A) TRUE.** $(2 - 1)(3 - 1) = 2$.**(B) TRUE.** $E = 100 \times 80/200 = 40$. Likewise the expected counts in each row are 40, 35, 25.

Cell	O	E	$(O - E)^2/E$
Free, A	50	40	2.500
Free, B	30	35	0.714
Free, C	20	25	1.000
Paid, A	30	40	2.500
Paid, B	40	35	0.714
Paid, C	30	25	1.000

$$\chi^2 = 2(2.5 + 0.714 + 1.0) \approx 8.429$$

(C) TRUE. $8.43 > 5.991$.**(D) FALSE.** $8.43 < 9.210$, so not rejected at 1% ($p \approx 0.015$).



Key idea: Significance depends on α : report the p-value (≈ 0.015 here) rather than only a reject/accept verdict.

Topic: Confidence interval — sample size for a proportion · Medium

Q.20 [MCQ · 2 marks · -2/3]

Correct answer: A

Concept: margin $E = z\sqrt{p(1-p)/n} \Rightarrow n = z^2 p(1-p)/E^2$; with no prior information use $p = 0.5$, which maximises $p(1-p)$.

$$n \geq \frac{1.96^2(0.5)(0.5)}{0.02^2} = \frac{3.8416(0.25)}{0.0004}$$

$$n \geq \frac{0.9604}{0.0004} = 2401$$

The bound is met exactly at 2401 ($98^2 \times 0.25$), so $n = 2401$. 2400 would give a margin slightly above 0.02.

(C) uses $z = 1.645$ (a 90% interval: $1691.3 \rightarrow 1692$). (D) 4802 uses $p(1-p) = 0.5$ instead of its maximum 0.25, doubling the requirement.

Key idea: $p(1-p) \leq 1/4$ — the conservative choice $p = 0.5$ guarantees the margin whatever the true CTR.

Q.21 [NAT · 2 marks]

Topic: CDF and PDF of a continuous r.v. · Medium

Correct answer: 0.330 (range 0.329 to 0.331)**Concept:** $P(a < X \leq b) = F(b) - F(a)$; the PDF is $F'(x)$.

$$F(2) = 1 - 3e^{-2}, \quad F(1) = 1 - 2e^{-1}$$

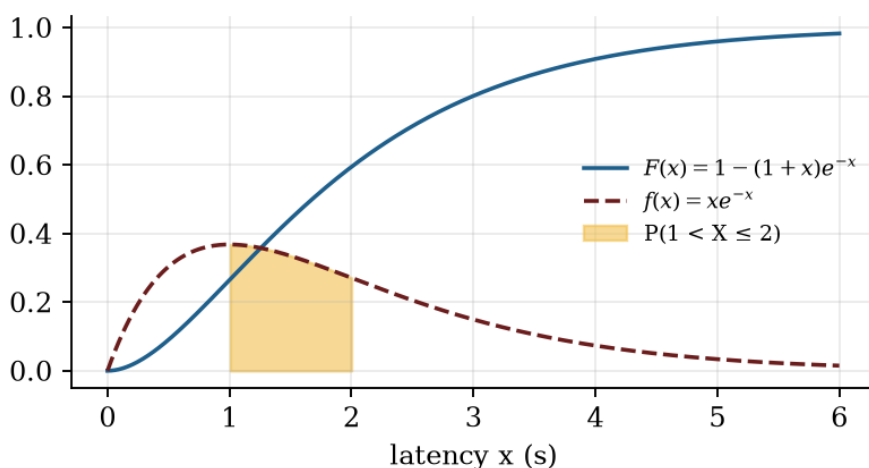
$$P(1 < X \leq 2) = (1 - 3e^{-2}) - (1 - 2e^{-1}) = 2e^{-1} - 3e^{-2}$$

$$= 0.735759 - 0.406006 \approx 0.3298$$

Check via the density:

$$f(x) = F'(x) = -e^{-x} + (1+x)e^{-x} = xe^{-x}, \quad x \geq 0$$

This is a Gamma(2, 1) density — the time until the second event of a rate-1 Poisson process.

**Key idea:** For a continuous r.v. the endpoints don't matter: $P(1 < X \leq 2) = P(1 \leq X \leq 2)$.**Q.22** [MCQ · 2 marks · -2/3]

Topic: CDF of the maximum (order statistics) · Hard

Correct answer: D**Concept:** for independent X_i , $P(\max \leq t) = \prod P(X_i \leq t)$; differentiate to get the density.Let $M = \max(X_1, X_2, X_3)$.

$$F_M(t) = P(X_1 \leq t)P(X_2 \leq t)P(X_3 \leq t) = t^3, \quad 0 < t < 1$$

$$f_M(t) = 3t^2$$

$$E[M] = \int_0^1 t \cdot 3t^2 dt = \frac{3}{4}$$

Alternatively $E[M] = \int_0^1 (1 - t^3) dt = 1 - 1/4 = 3/4$.(A) is a single replica's mean. (C) 1/4 is the expected minimum (first reply). (B) 2/3 is $E[\max]$ for only 2 replicas.**Key idea:** General result: max of n i.i.d. $U(0, 1)$ has mean $n/(n+1)$ — tail latency grows with fan-out.

Topic: CLT / normal approximation (dropout count) ·
Medium

Q.23 [MSQ · 2 marks]

Correct answer: A, B, C

Concept: $S \sim \text{Bin}(n, p) \approx N(np, np(1 - p))$ for large n (CLT).

(A) TRUE. $np = 400(0.9) = 360$.

(B) TRUE.

$$\text{Var}(S) = 400(0.9)(0.1) = 36, \quad \text{SD}(S) = 6$$

(C) TRUE.

$$P(S \geq 372) \approx P\left(Z \geq \frac{372 - 360}{6}\right) = P(Z \geq 2) = 0.0228$$

(D) FALSE.

$$\text{SD}(S/400) = \frac{6}{400} = 0.015$$

0.15 is off by a factor of 10 (it would be $\sqrt{(0.09/4)}$).

With continuity correction one would use 371.5: $z = 1.917$, $P \approx 0.028$ — the question fixes the convention, so (C) is decidable.

Key idea: Proportions shrink like $1/\sqrt{n}$: $\text{SD}(S/n) = \sqrt{p(1 - p)/n}$.

Topic: Covariance, correlation, regression slope ·
Medium

Q.24 [MCQ · 2 marks · -2/3]

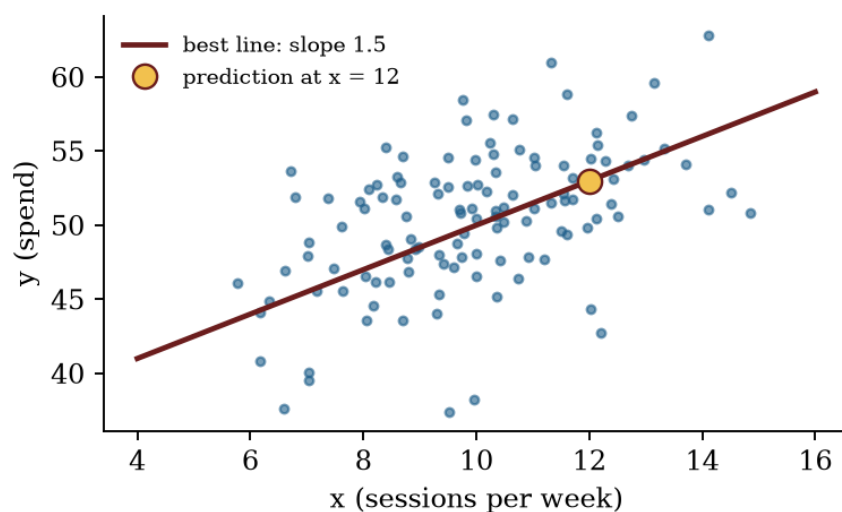
Correct answer: C

Concept: the least-squares line is $\hat{Y} = E[Y] + \beta(X - E[X])$ with $\beta = \text{Cov}(X, Y)/\text{Var}(X) = \rho\sigma_Y/\sigma_X$.

$$\text{Cov}(X, Y) = \rho\sigma_X\sigma_Y = 0.6(2)(5) = 6$$

$$\beta = \frac{6}{2^2} = 1.5$$

$$\hat{Y} = 50 + 1.5(12 - 10) = 53$$



(A) uses $\beta = \rho = 0.6$. (B) uses $\beta = \sigma_Y/\sigma_X = 2.5$, ignoring ρ . (D) uses $\text{Cov}/\text{Var}(Y) = 0.24$, the slope for regressing X on Y .

Key idea: Regression to the mean: X is 1 SD above average, but the prediction is only $\rho = 0.6$ SD of Y above average ($0.6 \times 5 = 3$).

Topic: Counting (birthday problem for hashing) ·
Medium

Q.25 [MCQ · 2 marks · -2/3]

Correct answer: B

Concept: $P(\text{at least one collision}) = 1 - P(\text{all distinct})$, with $P(\text{all distinct}) = {}^m P_k / m^k$.

$$\begin{aligned} P(\text{all distinct}) &= \frac{100 \cdot 99 \dots 91}{100^{10}} = \prod_{i=0}^9 \left(1 - \frac{i}{100}\right) \\ &= 1(0.99)(0.98) \dots (0.91) \approx 0.6282 \\ P(\text{collision}) &\approx 1 - 0.6282 = 0.3718 \end{aligned}$$

Approximation check: $1 - \exp(-k(k-1)/(2m)) = 1 - e^{-0.45} \approx 0.362$ — close.

(A) is $P(\text{no collision})$. (D) $0.45 = C(10, 2)/100$ is the union bound — an upper bound that double-counts overlapping pairs. (C) is k/m .

Trap: Expected number of colliding pairs = $C(k, 2)/m = 0.45$, but probability $\neq$ expectation; use the complement product for the exact value.

Topic: Hypergeometric sampling (mini-batches) ·
Medium

Q.26 [MCQ · 2 marks · -2/3]

Correct answer: A

Concept: sampling without replacement = hypergeometric counts; use the complement.

$$P(\text{none}) = \frac{\binom{16}{5}}{\binom{20}{5}} = \frac{4368}{15504} \approx 0.2817$$

$$P(\text{at least one}) = 1 - 0.2817 \approx 0.7183$$

(B) $0.672 = 1 - 0.8^5$ treats draws as with replacement (binomial). (C) is $P(\text{none})$. (D) is $P(\text{exactly one})$:

$$P(\text{exactly one}) = \frac{\binom{4}{1} \binom{16}{4}}{\binom{20}{5}} = \frac{7280}{15504} \approx 0.470$$

Key idea: Without replacement, each clean draw makes the next draw slightly more likely to be mislabelled, so $P(\text{none})$ is smaller than the binomial 0.328.

Q.27 [MSQ · 2 marks]**Topic: Sampling distribution of S^2 (χ^2) · Hard****Correct answer: A, B****Concept:** for normal samples, $(n - 1)S^2/\sigma^2 \sim \chi^2_{n-1}$, and $\bar{X}$ is independent of S^2 .**(A) TRUE.** $(n - 1)S^2/\sigma^2 = 9S^2/4 \sim \chi^2_9$ (one df is lost by estimating μ with $\bar{X}$).**(B) TRUE.** $E[\chi^2_9] = 9 \Rightarrow E[9S^2/4] = 9 \Rightarrow E[S^2] = 4$: S^2 is unbiased.**(C) FALSE.** $\text{Var}(\chi^2_9) = 18$, so

$$\text{Var}(S^2) = \left(\frac{4}{9}\right)^2 (18) = \frac{2\sigma^4}{n-1} = \frac{32}{9} \approx 3.56$$

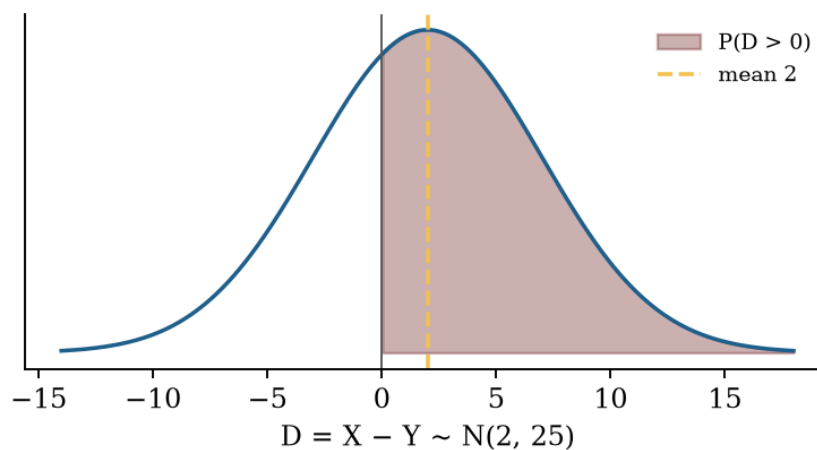
 $3.2 = 2\sigma^4/n$ uses n instead of $n - 1$.**(D) FALSE.** For normal data $\bar{X}$ and S^2 are independent (a characterising property of the normal distribution).

Key idea: Remember both facts together: χ^2_{n-1} for the scaled variance and independence of $\bar{X}$ and S^2 — together they give the t-statistic its t_{n-1} distribution.

Q.28 [NAT · 2 marks]**Topic: Normal distribution (difference of normals) · Medium****Correct answer: 0.655 (range 0.654 to 0.657)****Concept:** a difference of independent normals is normal; variances add.

$$D = X - Y \sim N(20 - 18, 9 + 16) = N(2, 25)$$

$$P(X > Y) = P(D > 0) = P\left(Z > \frac{0 - 2}{5}\right) = P(Z > -0.4) \\ = \Phi(0.4) \approx 0.6554$$



Trap: $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$ for independent variables — never subtract variances ($9 - 16$ would be negative!).

Topic: Conditional PDF and conditional expectation · Hard

Q.29 [MCQ · 2 marks · -2/3]

Correct answer: D

Concept: $f_{Y|X}(y | x) = f(x, y)/f_X(x)$; then integrate y against it.

$$f_X(x) = \int_0^1 (x + y) dy = x + \frac{1}{2}$$

$$f_{Y|X}(y | x) = \frac{x + y}{x + 1/2}, \quad 0 < y < 1$$

At $x = 0.5$ the denominator is 1, so $f_{Y|X}(y | 0.5) = 0.5 + y$.

$$E[Y | X = 0.5] = \int_0^1 y(0.5 + y) dy = \frac{0.5}{2} + \frac{1}{3} = \frac{1}{4} + \frac{1}{3} = \frac{7}{12}$$

General formula:

$$E[Y | X = x] = \frac{x/2 + 1/3}{x + 1/2}$$

(A) assumes Y is uniform given X . (B) $2/3$ is $E[Y | X = 0] = (1/3)/(1/2)$. (C) is $1 - 7/12$.

Monte Carlo check (rejection sampling, X near 0.5): ≈ 0.582 .

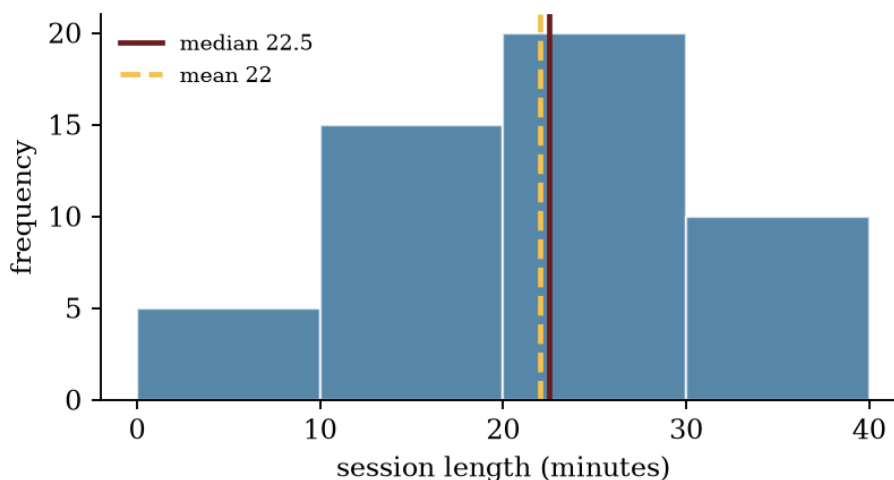
Key idea: The conditional mean decreases from $2/3$ ($x = 0$) to $5/9$ ($x = 1$): X and Y are slightly negatively correlated under this density.

Q.30 [MSQ · 2 marks]

Topic: Grouped data: mean, median, mode, SD · Medium

Correct answer: A, B, C

Concept: grouped median = $L + ((N/2 - CF)/f) \cdot h$; grouped mean = $\sum f \cdot m / N$.



(A) TRUE. $N/2 = 25$. Cumulative frequencies: 5, 20, 40, 50 $\Rightarrow$ the median class is 20-30 ($L = 20$, $CF = 20$, $f = 20$, $h = 10$).

$$\text{median} = 20 + \frac{25 - 20}{20}(10) = 22.5$$

(B) TRUE.

$$\bar{x} = \frac{5(5) + 15(15) + 20(25) + 10(35)}{50} = \frac{1100}{50} = 22$$

(C) TRUE. 20-30 has the highest frequency (20).

(D) FALSE.

$$\frac{\sum fm^2}{N} = \frac{125 + 3375 + 12500 + 12250}{50} = 565$$
$$\sigma^2 = 565 - 22^2 = 81, \quad \sigma = 9$$

Key idea: Here the mean (22) is slightly below the median (22.5): the long tail is on the left (few short sessions), a mild left skew.

Mock Test 05 — Topic Test: Probability, Counting, Bayes & Conditional Expectation

A topic-intensive set on counting, axioms, independence, Bayes' theorem, joint PMFs and conditional expectation/variance.

Item	Details
Questions	30 (12 × 1-mark, 18 × 2-mark)
Maximum marks	48
Question types	MCQ: 14 · MSQ: 7 · NAT: 9
Suggested time	90 minutes
Negative marking	MCQ only: $-1/3$ (1-mark), $-2/3$ (2-mark). None for MSQ & NAT.
Calculator	Use only the GATE virtual scientific calculator

Topic focus of this set

Counting (permutations, combinations, multinomial, stars and bars, derangements and inclusion–exclusion), probability axioms and set algebra, mutual exclusivity vs independence (pairwise vs mutual), joint/marginal/conditional probability, total probability and multi-stage Bayes, random variables and indicator methods, conditional expectation $E[X | Y]$, the laws of total expectation and total variance, and joint PMF tables.

Instructions

- Questions 1 onwards are arranged as in GATE: all 1-mark questions first, then 2-mark questions.
- **MCQ** – exactly one correct option. **MSQ** – ONE OR MORE options are correct; credit only if you select all correct options and no wrong option (no partial marks, no negative marks).
- **NAT** – enter a numerical value using the virtual keypad. The accepted range is given in the key; round as instructed in the question.
- Statistical tables (Z , t , χ^2) are provided in the Appendix; values in solutions use those tables.
- Attempt in one sitting, then check the answer key and read every solution — including those you got right — for the alternative methods and traps.

Mock Test 05 · Question Paper

1-mark questions

Q.1 MCQ · 1 mark · -1/3

A teacher distributes 10 identical reward stickers among 4 students so that every student receives at least one sticker. The number of possible distributions is

- (A) 286
- (B) 120
- (C) 84
- (D) 210

Q.2 MCQ · 1 mark · -1/3

Events A and B are mutually exclusive with $P(A) = 0.3$ and $P(B) = 0.4$. Which statement is TRUE?

- (A) A and B are independent
- (B) A and B are dependent, and $P(A | B) = 0$
- (C) $P(A \cup B) = P(A) + P(B) - P(A)P(B) = 0.58$
- (D) $P(A | B) = P(A) = 0.3$

Q.3 NAT · 1 mark

For events A and B, $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cup B) = 0.75$. The value of $P(A | B^c)$ is _____ (round off to 2 decimal places).

Answer: _____

Q.4 MSQ · 1 mark

For ANY two events A and B in a probability space, which of the following is/are ALWAYS TRUE? ($A \Delta B$ denotes the symmetric difference $(A \cap B^c) \cup (A^c \cap B)$.)

- (A) $P(A \cap B) \geq P(A) + P(B) - 1$
- (B) $P(A \cup B) \leq P(A) + P(B)$
- (C) $P(A \cap B^c) = P(A) - P(B)$
- (D) $P(A \Delta B) = P(A) + P(B) - 2P(A \cap B)$

Q.5 MCQ · 1 mark · -1/3

Five letters are written to five different people and the five addressed envelopes are filled at random, one letter per envelope. The probability that NO letter goes into its correct envelope is

- (A) 11/30
- (B) 53/144
- (C) 3/8
- (D) 1/5

Q.6 NAT · 1 mark

A fair six-sided die is rolled once; if it shows k , a fair coin is tossed k times. Let X be the number of heads obtained. Then $E[X] = \underline{\hspace{1cm}}$ (round off to 2 decimal places).

Answer:

Q.7 MCQ · 1 mark · -1/3

Five people are selected at random. Assuming each person's birth month is equally likely to be any of the 12 months, independently of the others, the probability that at least two of them share a birth month is closest to

- (A) 0.382
- (B) 0.417
- (C) 0.583
- (D) 0.618

Q.8 MCQ · 1 mark · -1/3

Nine students are to be split into three study circles of three students each. The circles are NOT named or distinguished in any way. The number of ways of doing this is

- (A) 1680
- (B) 280
- (C) 84
- (D) 560

Q.9 MSQ · 1 mark

The joint PMF of discrete random variables X and Y is shown below.

Joint PMF $P(X = x, Y = y)$

		$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.10	0.20	0.10	
$x = 1$	0.15	0.30	0.15	
		$y = 0$	$y = 1$	$y = 2$

Y

Which of the following is/are TRUE?

- (A) X and Y are independent
- (B) $P(X + Y = 2) = 0.30$
- (C) $E[Y] = 1$
- (D) $P(X = 1 | Y = 2) = 0.4$

Q.10 NAT · 1 mark

Two cards are drawn at random without replacement from a well-shuffled standard deck of 52 cards. Given that at least one of them is an ace, the probability that both are aces is _____ (round off to 4 decimal places).

Answer: _____

Q.11 MCQ · 1 mark · -1/3

An annotation team has three labellers who label 50%, 30% and 20% of all images. Their error rates are 1%, 2% and 3% respectively. The probability that a randomly chosen image is mislabelled is

- (A) 0.020
- (B) 0.060
- (C) 0.017
- (D) 0.006

Q.12 MSQ · 1 mark

Let X and Y be random variables with finite variances. Which of the following is/are ALWAYS TRUE?

- (A) If $E[X | Y] = E[X]$ (a constant), then X and Y are independent
- (B) $E[E[X | Y]] = E[X]$
- (C) $E[X | Y]$ is a constant (non-random) number
- (D) $\text{Var}(X) \geq \text{Var}(E[X | Y])$

2-mark questions**Q.13 NAT · 2 marks**

Urn I contains 3 white and 2 black balls; Urn II contains 2 white and 4 black balls. One ball is drawn at random from Urn I and placed (unseen) into Urn II. A ball is then drawn at random from Urn II and is found to be white. The probability that the transferred ball was white is _____ (round off to 3 decimal places).

Answer: _____

Q.14 MCQ · 2 marks · -2/3

A server allocates 12 identical jobs to 4 distinct worker machines; each machine can hold at most 5 jobs (a machine may receive 0 jobs). The number of possible allocations is

- (A) 125
- (B) 455
- (C) 119
- (D) 131

Q.15 NAT · 2 marks

The number N of customers entering a shop in an hour is Poisson with mean 4. Customer i spends an amount X_i (in ₹) with mean 50 and standard deviation 10; the X_i are i.i.d. and independent of N . Let $S = X_1 + \dots + X_N$ ($S = 0$ if $N = 0$). The standard deviation of S is _____. (round off to 2 decimal places).

Answer: _____

Q.16 MCQ · 2 marks · -2/3

The joint PMF of $X \in \{1, 2, 3\}$ and $Y \in \{0, 1\}$ is given below.

	$X = 1$	$X = 2$	$X = 3$
$Y = 0$	0.10	0.20	0.10
$Y = 1$	0.25	0.05	0.30

The value of $E[X | Y = 1]$ is

- (A) 2.05
- (B) 2
- (C) 1.25
- (D) 25/12

Q.17 MSQ · 2 marks

Two fair dice are rolled. Define $A = \{\text{first die shows an odd number}\}$, $B = \{\text{second die shows an odd number}\}$, $C = \{\text{the sum of the two dice is odd}\}$. Which of the following is/are TRUE?

- (A) A and C are independent
- (B) A , B and C are mutually independent
- (C) $P(A \cap B \cap C) = 0$
- (D) B and C are independent

Q.18 NAT · 2 marks

The number of integers in $\{1, 2, \dots, 1000\}$ that are divisible by NONE of 2, 3 and 5 is _____.

Answer: _____

Q.19 MCQ · 2 marks · -2/3

Let $X \sim \text{Uniform}(0, 1)$. Given $X = x$, let $Y \sim \text{Uniform}(0, x)$. Then $\text{Var}(Y)$ equals

- (A) 1/48
- (B) 7/144
- (C) 1/36
- (D) 1/12

Q.20 MSQ · 2 marks

A disease has prevalence 2%. A screening test has sensitivity 0.90 and false-positive rate 0.05. A person takes the test twice; given the disease status, the two results are independent. Which of the following is/are TRUE?

- (A) The probability that a single test is positive is 0.067
- (B) $P(\text{disease} \mid \text{one positive test})$ lies between 0.26 and 0.28
- (C) $P(\text{disease} \mid \text{two positive tests})$ is greater than 0.85
- (D) The two test results are (unconditionally) independent events

Q.21 NAT · 2 marks

A fair die is rolled 10 times. The expected number of DISTINCT faces that appear is _____ (round off to 2 decimal places).

Answer: _____

Q.22 MCQ · 2 marks · -2/3

The number of distinct arrangements of the letters of the word BALLOON in which the two L's are NOT adjacent is

- (A) 1260
- (B) 360
- (C) 900
- (D) 1080

Q.23 NAT · 2 marks

The joint PMF of X and Y is $p(x, y) = c(x + 2y)$ for $x \in \{0, 1, 2\}$ and $y \in \{0, 1\}$, and 0 otherwise. The covariance $\text{Cov}(X, Y)$ is _____ (round off to 3 decimal places).

Answer: _____

Q.24 MCQ · 2 marks · -2/3

Players A and B roll a fair die alternately, A first. A wins as soon as A rolls a 6; B wins as soon as B rolls a 5 or a 6. The game continues until someone wins. The probability that A wins is

- (A) $3/8$
- (B) $6/11$
- (C) $1/3$
- (D) $5/8$

Q.25 MSQ · 2 marks

X is uniformly distributed on $\{-2, -1, 0, 1, 2\}$ and $Y = X^2$. Which of the following is/are TRUE?

- (A) X and Y are independent
- (B) $\text{Cov}(X, Y) = 0$
- (C) $E[Y] = 2$
- (D) $P(Y = 1) = 1/5$

Q.26 NAT · 2 marks

A biased coin shows heads with probability 0.6. It is tossed repeatedly until two CONSECUTIVE heads appear. The expected number of tosses is _____ (round off to 2 decimal places).

Answer: _____

Q.27 MCQ · 2 marks · -2/3

A committee of 5 is to be chosen from 6 men and 5 women. It must contain at least 2 women, and a particular man M and a particular woman W refuse to serve together. The number of possible committees is

- (A) 381
- (B) 74
- (C) 462
- (D) 307

Q.28 MCQ · 2 marks · -2/3

A point (X, Y) is chosen uniformly at random from the unit square $[0, 1] \times [0, 1]$. The value of $P(XY < 1/4 \mid X > 1/2)$ is

- (A) $(\ln 2)/4$
- (B) $(\ln 2)/2$
- (C) $1/4$
- (D) $1 - (\ln 2)/2$

Q.29 MCQ · 2 marks · -2/3

A source sends bit 1 with probability 0.7 and bit 0 with probability 0.3. The bit passes through two independent noisy relays in series; each relay flips the bit it receives with probability 0.1. Given that the final received bit is 1, the probability that a 1 was sent is closest to

- (A) 0.955
- (B) 0.820
- (C) 0.914
- (D) 0.574

Q.30 MSQ · 2 marks

X and Y are independent Poisson random variables, each with mean $\lambda > 0$. Fix a positive integer n. Which of the following is/are TRUE?

- (A) $X + Y$ is Poisson with mean λ
- (B) Conditional on $X + Y = n$, X is Binomial(n, 1/2)
- (C) $E[X \mid X + Y = n] = n/2$
- (D) $P(X = 0 \mid X + Y = n) = 2^{-n}$

Mock Test 05 · Answer Key

Q	Type	Marks	Answer	Topic
1	MCQ	1	C	Counting: stars and bars
2	MCQ	1	B	Mutually exclusive vs independent
3	NAT	1	0.50	Conditional probability, set algebra
4	MSQ	1	A, B, D	Probability axioms & set algebra
5	MCQ	1	A	Derangements (inclusion-exclusion)
6	NAT	1	1.75	Law of total expectation
7	MCQ	1	D	Counting: complementary probability
8	MCQ	1	B	Multinomial counting (unlabelled groups)
9	MSQ	1	A, C	Joint PMF, marginals, independence
10	NAT	1	0.0303 (range 0.03 to 0.0306)	Conditional probability (counting)
11	MCQ	1	C	Total probability
12	MSQ	1	B, D	Conditional expectation properties
13	NAT	2	0.692 (range 0.691 to 0.693)	Bayes theorem (two-stage experiment)
14	MCQ	2	A	Bounded stars and bars via inclusion-exclusion
15	NAT	2	101.98 (range 101.97 to 101.99)	Law of total variance (random sum)
16	MCQ	2	D	Conditional expectation from a joint PMF
17	MSQ	2	A, C, D	Pairwise vs mutual independence
18	NAT	2	266	Inclusion-exclusion counting
19	MCQ	2	B	Conditional expectation & total variance (continuous)
20	MSQ	2	A, B, C	Bayes with repeated tests
21	NAT	2	5.03 (range 5.02 to 5.04)	Expectation via indicator random variables
22	MCQ	2	C	Permutations with repetition & restriction
23	NAT	2	-0.083 (range -0.084 to -0.083)	Joint PMF: normalisation & covariance
24	MCQ	2	A	Total probability by first-step conditioning
25	MSQ	2	B, C	Functions of a random variable; uncorrelated vs independent
26	NAT	2	4.44 (range 4.43 to 4.46)	Expected waiting time by conditioning
27	MCQ	2	D	Combinations with constraints
28	MCQ	2	B	Conditional probability on a continuous region
29	MCQ	2	C	Multi-stage Bayes (cascaded channels)
30	MSQ	2	B, C, D	Conditional distribution of Poisson given the sum

Score tracker

	Attempted	Correct	Wrong (MCQ)	Marks
1-mark				
2-mark				
Total				/ 48

Mock Test 05 · Detailed Solutions

Q.1 [MCQ · 1 mark · -1/3]

Topic: Counting: stars and bars · Easy

Correct answer: C

Concept: the number of positive integer solutions of $x_1 + \dots + x_k = n$ is $C(n-1, k-1)$ (stars and bars with every bin non-empty).

Let $x_i \geq 1$ be the stickers of student i . Then

$$x_1 + x_2 + x_3 + x_4 = 10, \quad x_i \geq 1$$

Substitute $y_i = x_i - 1 \geq 0$, giving $y_1 + \dots + y_4 = 6$ with non-negative integers:

$$\binom{6+4-1}{4-1} = \binom{9}{3} = 84$$

Why the others are wrong: $286 = C(13, 3)$ counts distributions where a student may get zero; $120 = C(10, 3)$ and $210 = C(10, 4)$ come from mis-placing the -1 .

Trap: Stickers identical, students distinct $\rightarrow$ stars and bars. If the stickers were distinct the answer would be a surjection count (inclusion-exclusion), not a binomial.

Q.2 [MCQ · 1 mark · -1/3]

Topic: Mutually exclusive vs independent · Easy

Correct answer: B

Concept: mutually exclusive means $P(A \cap B) = 0$; independent means $P(A \cap B) = P(A)P(B)$. Both can hold only if $P(A) = 0$ or $P(B) = 0$.

$$P(A \cap B) = 0 \neq P(A)P(B) = 0.3 \times 0.4 = 0.12$$

Hence A and B are **dependent**. Moreover

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{0.4} = 0$$

(A) false (shown above). (C) false: for disjoint events $P(A \cup B) = 0.3 + 0.4 = 0.7$, the product formula is for independent events. (D) false: $P(A | B) = 0 \neq 0.3$.

Key idea: Learning that B occurred tells you A certainly did NOT occur — the strongest possible dependence. Disjoint events with positive probabilities are never independent.

Q.3 [NAT · 1 mark]

Topic: Conditional probability, set algebra · Easy

Correct answer: 0.50

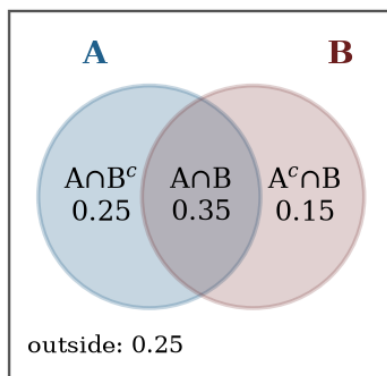
Concept: $P(A | B^c) = P(A \cap B^c) / P(B^c)$, with $P(A \cap B^c) = P(A) - P(A \cap B)$.

Step 1 (addition rule):

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.6 + 0.5 - 0.75 = 0.35$$

Step 2: $P(A \cap B^c) = 0.6 - 0.35 = 0.25$ and $P(B^c) = 1 - 0.5 = 0.5$.

$$P(A | B^c) = \frac{0.25}{0.5} = 0.50$$



Key idea: Check: $P(A \cap B) = 0.35 \neq P(A)P(B) = 0.30$, so A and B are dependent; hence $P(A | B^c)$ differs from $P(A) = 0.6$.

Q.4 [MSQ · 1 mark]

Topic: Probability axioms & set algebra · Medium

Correct answer: A, B, D

Concept: everything follows from the axioms plus the inclusion-exclusion identity $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

(A) TRUE (Bonferroni): since $P(A \cup B) \leq 1$,

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1$$

(B) TRUE (Boole): $P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B)$ because $P(A \cap B) \geq 0$.

(C) FALSE: the correct identity is $P(A \cap B^c) = P(A) - P(A \cap B)$. It equals $P(A) - P(B)$ only when $B \subset A$. Counter-example: A, B disjoint with $P(A) = 0.2$, $P(B) = 0.5$ gives 0.2 on the left but -0.3 on the right.

(D) TRUE: $A \Delta B$ is the disjoint union of $A \cap B^c$ and $A^c \cap B$:

$$[P(A) - P(A \cap B)] + [P(B) - P(A \cap B)] = P(A) + P(B) - 2P(A \cap B)$$

Shortcut: A probability can never be negative — any 'identity' that can produce a negative value for some valid inputs (like option C) must be false in general.

Q.5 [MCQ · 1 mark · -1/3]

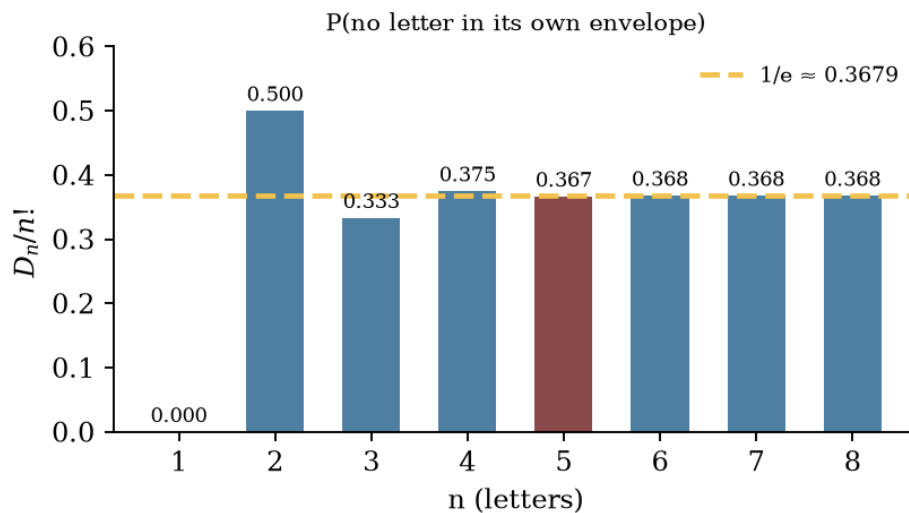
Topic: Derangements (inclusion-exclusion) · Medium

Correct answer: A

Concept: the number of derangements of n objects is $D_n = n! \sum_{k=0}^n (-1)^k / k!$, obtained by inclusion-exclusion on the events 'letter i is correct'.

$$\begin{aligned} D_5 &= 5! \left(1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right) \\ D_5 &= 120 \left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} \right) = 60 - 20 + 5 - 1 = 44 \\ P(\text{no match}) &= \frac{44}{120} = \frac{11}{30} \approx 0.3667 \end{aligned}$$

Distractors: $3/8 = D_4/4!$ and $53/144 = D_6/6!$ (wrong n); $1/5$ is the chance a particular letter is correct.



Shortcut: $D_n/n!$ converges to $1/e \approx 0.3679$ extremely fast; for $n \geq 5$ it is within 0.002. Recursion $D_n = (n-1)(D_{n-1} + D_{n-2})$ is handy for small n .

Q.6 [NAT · 1 mark]

Topic: Law of total expectation · Easy

Correct answer: 1.75

Concept: law of total (iterated) expectation $E[X] = E[E[X | K]]$.

Given $K = k$, $X \sim \text{Binomial}(k, 1/2)$, so $E[X | K] = K/2$.

$$E[X] = E\left[\frac{K}{2}\right] = \frac{1}{2} \cdot \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{3.5}{2} = 1.75$$

Key idea: There is no need for the PMF of X ; condition on the random number of trials, then average.

Q.7 [MCQ · 1 mark · -1/3]

Topic: Counting: complementary probability · Easy

Correct answer: D

Concept: $P(\text{at least one repeat}) = 1 - P(\text{all different})$; count ordered assignments.

All-distinct assignments: $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$ out of $12^5 = 248832$.

$$P(\text{all different}) = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{12^5} = \frac{95040}{248832} \approx 0.3819$$

$$P(\text{at least two share}) = 1 - 0.3819 = 0.6181$$

0.382 is the complement (all different). 0.417 = 5/12 and 0.583 = 7/12 arise from wrongly treating the problem as a single comparison.

Shortcut: 'At least one coincidence' → always go through the complement 'all distinct'.

Topic: Multinomial counting (unlabelled groups) · Medium

Q.8 [MCQ · 1 mark · -1/3]

Correct answer: B

Concept: multinomial coefficient $n!/(n_1! n_2! \dots)$ counts splits into LABELLED groups; divide by (number of equal-size groups)! when groups are unlabelled.

Labelled circles (Circle 1, 2, 3):

$$\frac{9!}{3! 3! 3!} = \frac{362880}{216} = 1680$$

The 3! orderings of the same three circles give the same partition, so

$$\frac{1680}{3!} = 280$$

1680 forgets the over-counting; $84 = C(9, 3)$ chooses only one circle; $560 = 1680/3$ divides by the wrong factor.

Trap: Divide by $k!$ only for groups of EQUAL size that are indistinguishable. Groups of sizes 2, 3, 4 would need no division.

Q.9 [MSQ · 1 mark]

Topic: Joint PMF, marginals, independence · Easy

Correct answer: A, C

Concept: X, Y independent $\Leftrightarrow p(x, y) = p_X(x) p_Y(y)$ for EVERY cell.

	$y = 0$	$y = 1$	$y = 2$	$p_X(x)$
$x = 0$	0.10	0.20	0.10	0.40
$x = 1$	0.15	0.30	0.15	0.60
$p_Y(y)$	0.25	0.50	0.25	1

(A) TRUE: $0.4 \times 0.25 = 0.10$, $0.4 \times 0.5 = 0.20$, $0.6 \times 0.25 = 0.15$, $0.6 \times 0.5 = 0.30$ — every cell factorises.

(B) FALSE: $X + Y = 2$ at $(0, 2)$ and $(1, 1)$: $0.10 + 0.30 = 0.40$.

(C) TRUE: $E[Y] = 0(0.25) + 1(0.5) + 2(0.25) = 1$.

(D) FALSE: $P(X = 1 | Y = 2) = 0.15/0.25 = 0.6$ ($= P(X = 1)$, as independence demands).

Shortcut: Rows proportional to each other ($0.10 : 0.20 : 0.10$ and $0.15 : 0.30 : 0.15$) is a quick visual test of independence.

Q.10 [NAT · 1 mark]**Topic: Conditional probability (counting) · Medium****Correct answer: 0.0303 (range 0.03 to 0.0306)****Concept:** $P(\text{both} \mid \text{at least one}) = P(\text{both})/P(\text{at least one})$, since 'both' $\subset$ 'at least one'.

$$P(\text{both}) = \frac{\binom{4}{2}}{\binom{52}{2}} = \frac{6}{1326}$$

$$P(\text{at least one}) = 1 - \frac{\binom{48}{2}}{\binom{52}{2}} = 1 - \frac{1128}{1326} = \frac{198}{1326}$$

$$P(\text{both} \mid \geq 1) = \frac{6}{198} = \frac{1}{33} \approx 0.0303$$

Trap: Compare with $P(\text{both} \mid \text{the FIRST card is an ace}) = 3/51 \approx 0.0588$. 'At least one' is a weaker condition and gives a smaller answer.

Q.11 [MCQ · 1 mark · -1/3]**Topic: Total probability · Easy****Correct answer: C****Concept:** law of total probability $P(E) = \sum P(E \mid L_i) P(L_i)$ over a partition.

$$P(E) = 0.5(0.01) + 0.3(0.02) + 0.2(0.03) = 0.005 + 0.006 + 0.006 = 0.017$$

0.020 is the unweighted mean of the error rates; 0.060 is their sum.

Shortcut: The overall rate must lie between the smallest and largest conditional rates (1% and 3%).

Q.12 [MSQ · 1 mark]**Topic: Conditional expectation properties · Medium****Correct answer: B, D****Concept:** $E[X \mid Y] = g(Y)$ is a random variable; tower property and the law of total variance $\text{Var}(X) = E[\text{Var}(X \mid Y)] + \text{Var}(E[X \mid Y])$.

(A) FALSE: mean-independence is weaker than independence. Example: $Y \in \{1, 2\}$ equally likely, $X \mid Y=1 \sim \pm 1$ and $X \mid Y=2 \sim \pm 2$ (equally likely signs). $E[X \mid Y] = 0$ always, but $|X| = Y$, so they are dependent.

(B) TRUE: tower property (law of total expectation).

(C) FALSE: $E[X \mid Y] = g(Y)$ changes with Y ; only $E[X \mid Y = y]$ for a fixed y is a number.

(D) TRUE: from the law of total variance, since $E[\text{Var}(X \mid Y)] \geq 0$,

$$\text{Var}(X) = E[\text{Var}(X \mid Y)] + \text{Var}(E[X \mid Y]) \geq \text{Var}(E[X \mid Y])$$

Key idea: Independence $\Rightarrow E[X \mid Y] = E[X] \Rightarrow \text{Cov}(X, Y) = 0$, but neither arrow reverses.

Q.13 [NAT · 2 marks]

Topic: Bayes theorem (two-stage experiment) · Medium

Correct answer: 0.692 (range 0.691 to 0.693)

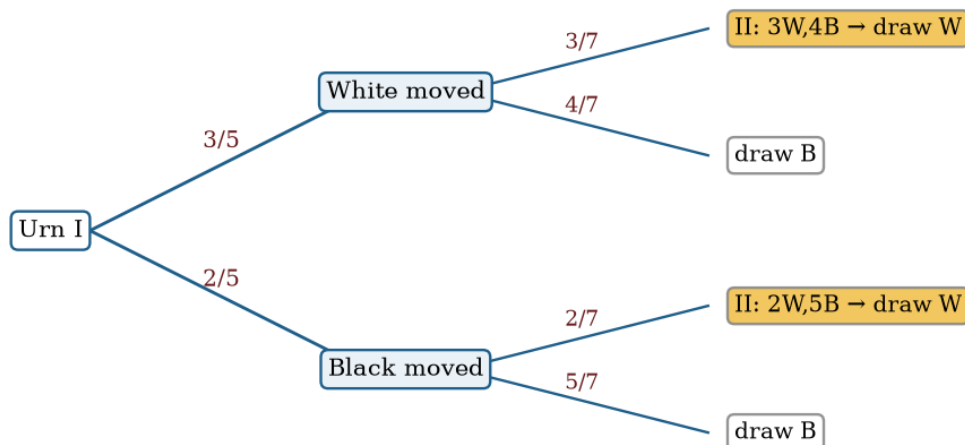
Concept: Bayes' theorem over the partition {transferred W, transferred B}, with likelihoods computed from the updated composition of Urn II.

Let T_W, T_B be the colour transferred, and W the event that the second draw is white.

Step 1 — priors: $P(T_W) = 3/5$, $P(T_B) = 2/5$.

Step 2 — likelihoods: after T_W , Urn II has 3W, 4B (7 balls); after T_B , it has 2W, 5B.

$$P(W | T_W) = \frac{3}{7}, \quad P(W | T_B) = \frac{2}{7}$$



Step 3 — total probability:

$$P(W) = \frac{3}{5} \cdot \frac{3}{7} + \frac{2}{5} \cdot \frac{2}{7} = \frac{9+4}{35} = \frac{13}{35}$$

Step 4 — Bayes:

$$P(T_W | W) = \frac{9/35}{13/35} = \frac{9}{13} \approx 0.692$$

Key idea: Observing white raises the probability that white was transferred from the prior 0.600 to 0.692 — evidence pushes belief toward the hypothesis that makes it more likely.

Q.14 [MCQ · 2 marks · -2/3]

Topic: Bounded stars and bars via inclusion-exclusion · Hard

Correct answer: A

Concept: count non-negative solutions of $x_1 + \dots + x_4 = 12$ with $x_i \leq 5$ by inclusion-exclusion on the 'bad' events $B_i = \{x_i \geq 6\}$.

Step 1 — no upper bound:

$$N_0 = \binom{12+3}{3} = \binom{15}{3} = 455$$

Step 2 — one machine overloaded: set $x_i = 6 + z_i$; remaining sum 6:

$$|B_i| = \binom{6+3}{3} = 84, \quad \sum |B_i| = 4 \times 84 = 336$$

Step 3 — two machines overloaded: remaining sum 0, exactly one way:

$$|B_i \cap B_j| = \binom{0+3}{3} = 1, \quad \sum |B_i \cap B_j| = \binom{4}{2} \times 1 = 6$$

Step 4 — three overloaded needs $\geq 18 > 12$ jobs: impossible.

$$N = 455 - 336 + 6 = 125$$

455 ignores the cap; 119 = 455 - 336 forgets to add back the double overlaps; 131 adds them twice.

Trap: Alternating signs: subtract singles, ADD BACK pairs, subtract triples... Forgetting the pair term is the commonest error.

Q.15 [NAT · 2 marks]

Topic: Law of total variance (random sum) · Medium

Correct answer: 101.98 (range 101.97 to 101.99)

Concept: for a random sum, condition on N and use $\text{Var}(S) = E[\text{Var}(S | N)] + \text{Var}(E[S | N])$.

Given $N = n$, S is a sum of n i.i.d. terms:

$$E[S | N] = 50N, \quad \text{Var}(S | N) = 100N$$

First term:

$$E[\text{Var}(S | N)] = 100 E[N] = 100 \times 4 = 400$$

Second term (for Poisson, $\text{Var}(N) = E[N] = 4$):

$$\text{Var}(E[S | N]) = 50^2 \text{Var}(N) = 2500 \times 4 = 10000$$

$$\text{Var}(S) = 400 + 10000 = 10400, \quad \text{SD}(S) = \sqrt{10400} \approx 101.98$$

Also $E[S] = E[N]E[X] = 200$.

Trap: Forgetting the second term (variability in the NUMBER of customers) gives $\text{SD} = 20$ — a huge underestimate. The randomness of N dominates here.

Q.16 [MCQ · 2 marks · -2/3]

Topic: Conditional expectation from a joint PMF · Medium

Correct answer: D

Concept: $E[X | Y = y] = \sum_x x p(x, y) / p_Y(y)$: renormalise the row.

Step 1: $P(Y = 1) = 0.25 + 0.05 + 0.30 = 0.60$.

Step 2: conditional PMF of X given $Y = 1$:

x	1	2	3
$P(X = x Y = 1)$	$0.25/0.6 = 5/12$	$0.05/0.6 = 1/12$	$0.30/0.6 = 6/12$

$$E[X | Y = 1] = 1 \cdot \frac{5}{12} + 2 \cdot \frac{1}{12} + 3 \cdot \frac{6}{12} = \frac{25}{12} \approx 2.083$$

Distractors: 1.25 is $\sum x p(x, 1)$ without dividing by 0.6; 2.05 is the unconditional $E[X]$; 2 is $E[X | Y = 0] = 0.8/0.4$.

Check of the tower property: $E[X] = 0.4(2) + 0.6(25/12) = 0.8 + 1.25 = 2.05$ (verified).

Key idea: A conditional PMF is just a row (or column) rescaled to sum to 1.

Q.17 [MSQ · 2 marks]**Topic: Pairwise vs mutual independence · Hard****Correct answer: A, C, D**

Concept: mutual independence of three events needs all three pairwise product rules AND $P(A \cap B \cap C) = P(A)P(B)P(C)$.

Each die is odd with probability $1/2$, so $P(A) = P(B) = 1/2$. The sum is odd iff exactly one die is odd:

$$P(C) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

(A) TRUE: $A \cap C = \{\text{first odd, second even}\}$:

$$P(A \cap C) = \frac{1}{4} = P(A)P(C)$$

(D) TRUE: by symmetry, $P(B \cap C) = P(\text{second odd, first even}) = 1/4 = P(B)P(C)$. (A and B are also independent, as the dice are independent.)

(C) TRUE: if both dice are odd, the sum is even, so $A \cap B \cap C = \emptyset$.

(B) FALSE: $P(A \cap B \cap C) = 0 \neq P(A)P(B)P(C) = 1/8$. So the events are pairwise independent but not mutually independent.

Key idea: Knowing A and B together determines C completely, though any single one tells nothing about another. Pairwise independence does NOT imply mutual independence.

Q.18 [NAT · 2 marks]**Topic: Inclusion-exclusion counting · Medium****Correct answer: 266**

Concept: $|A_2 \cup A_3 \cup A_5|$ by inclusion-exclusion, with $|A_d| = \lfloor 1000/d \rfloor$ and $A_d \cap A_e = A_{\text{lcm}(d,e)}$.

Set	Count	Set	Count
A_2	500	A_6	166
A_3	333	A_{10}	100
A_5	200	A_{15}	66
		A_{30}	33

$$|A_2 \cup A_3 \cup A_5| = (500 + 333 + 200) - (166 + 100 + 66) + 33$$

$$= 1033 - 332 + 33 = 734$$

$$\text{None} = 1000 - 734 = 266$$

Sanity check: the density of such numbers is $(1 - 1/2)(1 - 1/3)(1 - 1/5) = 4/15$, and $1000 \times 4/15 \approx 266.7$.

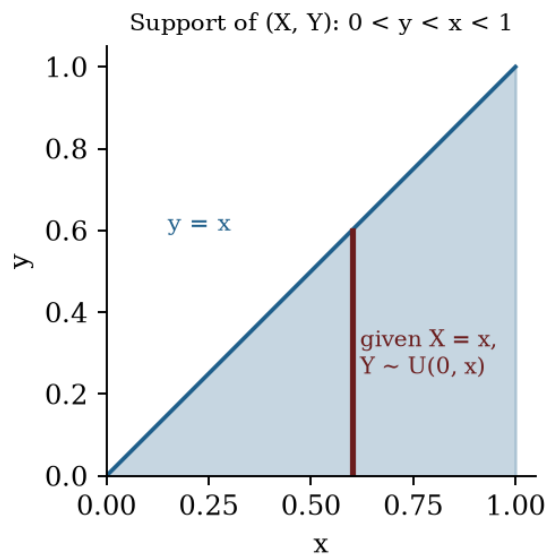
Trap: Use the floor at every step ($\lfloor 1000/3 \rfloor = 333$, $\lfloor 1000/15 \rfloor = 66$); using exact fractions gives 266.7, which is not an integer count.

Topic: Conditional expectation & total variance
(continuous) · Hard

Q.19 [MCQ · 2 marks · -2/3]

Correct answer: B

Concept: law of total variance $\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E[Y | X])$.



Given X : $Y \sim U(0, X)$, so

$$E[Y|X] = \frac{X}{2}, \quad \text{Var}(Y|X) = \frac{X^2}{12}$$

Term 1 ($E[X^2] = 1/3$ for $U(0, 1)$):

$$E\left[\frac{X^2}{12}\right] = \frac{1}{12} \cdot \frac{1}{3} = \frac{1}{36}$$

Term 2 ($\text{Var}(X) = 1/12$):

$$\begin{aligned} \text{Var}\left(\frac{X}{2}\right) &= \frac{1}{4} \cdot \frac{1}{12} = \frac{1}{48} \\ \text{Var}(Y) &= \frac{1}{36} + \frac{1}{48} = \frac{4+3}{144} = \frac{7}{144} \end{aligned}$$

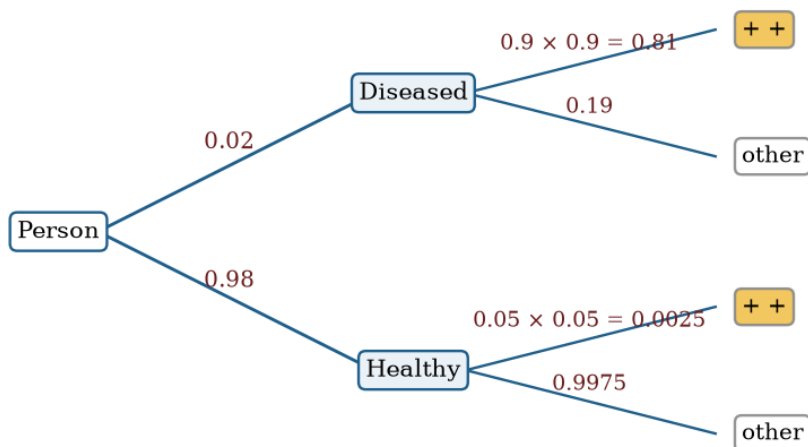
Check directly: $E[Y] = E[X/2] = 1/4$, $E[Y^2] = E[X^2/3] = 1/9$, so $\text{Var}(Y) = 1/9 - 1/16 = 7/144$ (verified).

$1/48$ and $1/36$ are the two components alone; $1/12$ is the variance of a $U(0, 1)$.

Key idea: Both pieces are needed: within-group spread $E[\text{Var}(Y | X)]$ plus between-group spread $\text{Var}(E[Y | X])$.

Q.20 [MSQ · 2 marks]

Topic: Bayes with repeated tests · Hard

Correct answer: A, B, C**Concept:** Bayes' theorem; conditional independence given D does not give unconditional independence.**(A) TRUE:** $P(+) = 0.02(0.90) + 0.98(0.05) = 0.018 + 0.049 = 0.067$.**(B) TRUE:** $P(D | +) = 0.018/0.067 = 0.2687$.**(C) TRUE:** for two positives, multiply likelihoods (conditional independence):

$$P(D | ++) = \frac{0.02(0.81)}{0.02(0.81) + 0.98(0.0025)} = \frac{0.0162}{0.01865}$$

$$P(D | ++) \approx 0.8686$$

(D) FALSE: $P(++) = 0.01865$ but $P(+)^2 = 0.067^2 \approx 0.00449$. A first positive raises the chance of disease, which raises the chance of a second positive.Equivalently, sequential updating: the posterior after test 1 (0.2687) becomes the prior for test 2: $0.2687(0.9)/[0.2687(0.9) + 0.7313(0.05)] \approx 0.8686$.**Key idea:** A single positive from a rare-disease screen is usually a false alarm ($\approx 73\%$ here); a confirmatory second test is what makes the diagnosis convincing.

Q.21 [NAT · 2 marks]

Topic: Expectation via indicator random variables · Medium

Correct answer: 5.03 (range 5.02 to 5.04)**Concept:** write the count as a sum of indicators and use linearity of expectation (no independence needed).Let $I_j = 1$ if face j appears at least once, $j = 1, \dots, 6$. The number of distinct faces is $D = I_1 + \dots + I_6$.

$$E[I_j] = P(\text{face } j \text{ appears}) = 1 - \left(\frac{5}{6}\right)^{10}$$

$$\left(\frac{5}{6}\right)^{10} \approx 0.16151$$

$$E[D] = 6 \left[1 - \left(\frac{5}{6}\right)^{10} \right] = 6(0.83849) \approx 5.03$$

Shortcut: The indicators are dependent (they cannot all be 0), yet linearity still holds. This trick turns hard PMFs into one-line expectations.

Q.22 [MCQ · 2 marks · -2/3]

Topic: Permutations with repetition & restriction · **Medium**

Correct answer: C

Concept: permutations of a multiset: $n!/(n_1! n_2! \dots)$; restriction via complement (glue the L's together).

BALLOON has 7 letters: B, A, L $\times$ 2, O $\times$ 2, N.

$$\text{Total} = \frac{7!}{2! 2!} = \frac{5040}{4} = 1260$$

Treat 'LL' as one block: 6 objects (B, A, LL, O, O, N) with O repeated:

$$\text{L's together} = \frac{6!}{2!} = 360$$

$$\text{L's apart} = 1260 - 360 = 900$$

1260 ignores the restriction; 360 is the complement; $1080 = 1260 - 180$ comes from wrongly dividing $6!$ by $2! \cdot 2!$ after gluing (once glued, LL is a single object and is no longer a repeated letter).

Shortcut: Gap method check: arrange B, A, O, O, N in $5!/2! = 60$ ways, choose 2 of the 6 gaps for the L's: $60 \times C(6, 2) = 900$ (verified).

Q.23 [NAT · 2 marks]

Topic: Joint PMF: normalisation & covariance · **Medium**

Correct answer: -0.083 (range -0.084 to -0.083)

Concept: normalise to find c , then $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$.

Step 1 — normalisation: sum of $(x + 2y)$ over the six cells = $(0+1+2) + (2+3+4) = 12$, so $c = 1/12$.

$p(x, y) = (x + 2y)/12$

Y	$y = 0$	0/12	1/12	2/12
	$y = 1$	2/12	3/12	4/12
		$x = 0$	$x = 1$	$x = 2$
		X		

Step 2 — marginals: $p_X = (2, 4, 6)/12$ for $x = 0, 1, 2$; $p_Y = (3, 9)/12$ for $y = 0, 1$.

$$E[X] = \frac{0 \cdot 2 + 1 \cdot 4 + 2 \cdot 6}{12} = \frac{16}{12} = \frac{4}{3}, \quad E[Y] = \frac{9}{12} = \frac{3}{4}$$

Step 3 — only $y = 1$ cells contribute to $E[XY]$:

$$E[XY] = \frac{1 \cdot 3 + 2 \cdot 4}{12} = \frac{11}{12}$$

$$\text{Cov}(X, Y) = \frac{11}{12} - \frac{4}{3} \cdot \frac{3}{4} = \frac{11}{12} - 1 = -\frac{1}{12} \approx -0.083$$

Key idea: Small negative covariance: when $y = 1$ the weights (2, 3, 4) are flatter across x than when $y = 0$ (0, 1, 2), so $E[X | Y = 1] = 11/9 < E[X | Y = 0] = 5/3$.

Q.24 [MCQ · 2 marks · -2/3]

Topic: Total probability by first-step conditioning · Hard

Correct answer: A

Concept: condition on the first round; if nobody wins, the game restarts in the same state (renewal argument).

Let $p = P(\text{A wins})$. In round 1: A wins at once with probability $1/6$. Otherwise $(5/6)$, B fails with probability $4/6 = 2/3$, and then the game restarts.

$$p = \frac{1}{6} + \frac{5}{6} \cdot \frac{2}{3} p$$

$$p \left(1 - \frac{10}{18} \right) = \frac{1}{6} \Rightarrow p = \frac{1}{6} \cdot \frac{18}{8} = \frac{3}{8}$$

Equivalently, a geometric series: $p = (1/6)[1 + (10/18) + (10/18)^2 + \dots]$.

$6/11$ is the answer when both need a 6; $1/3$ and $5/8 = 1 - 3/8$ (B's chance) are the other traps.

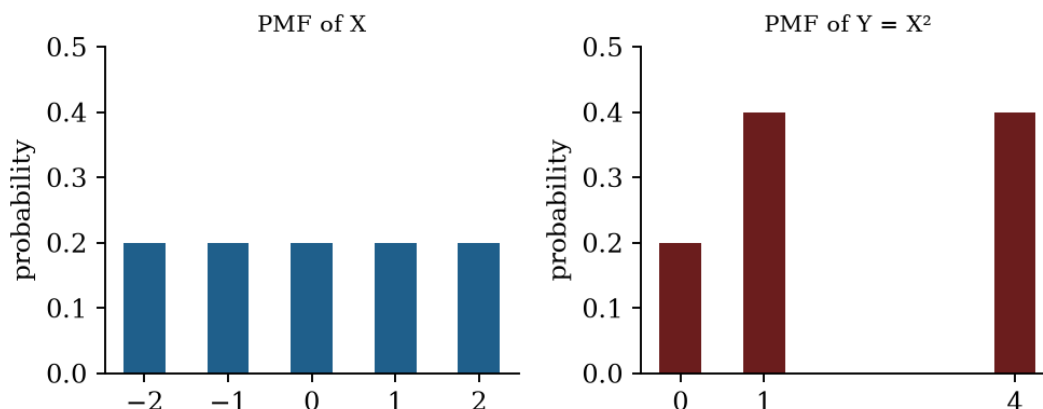
Key idea: Despite moving first, A is the underdog because B's success chance per turn ($1/3$) is double A's ($1/6$).

Q.25 [MSQ · 2 marks]

Topic: Functions of a random variable; uncorrelated vs independent · Medium

Correct answer: B, C

Concept: zero covariance only rules out LINEAR dependence; $Y = g(X)$ is fully dependent on X .



(A) FALSE: $P(X = 0, Y = 4) = 0$ but $P(X = 0)P(Y = 4) = (1/5)(2/5) = 2/25 \neq 0$.

(B) TRUE: $E[X] = 0$ and $E[XY] = E[X^3] = (-8 - 1 + 0 + 1 + 8)/5 = 0$, so

$$\text{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0 - 0 = 0$$

(C) TRUE: $E[Y] = (4 + 1 + 0 + 1 + 4)/5 = 10/5 = 2$.

(D) FALSE: $Y = 1$ when $X = \pm 1$, so $P(Y = 1) = 2/5$.

Trap: Symmetric X with an even function $Y = g(X)$ always gives $\text{Cov} = 0$ — the classic 'uncorrelated but dependent' example.

Q.26 [NAT · 2 marks]

Topic: Expected waiting time by conditioning · Hard

Correct answer: 4.44 (range 4.43 to 4.46)

Concept: law of total expectation applied to states: let E_0 = expected tosses from scratch, E_1 = expected tosses when the last toss was H.

From state 1: next toss H (0.6) finishes; T (0.4) sends us back to scratch:

$$E_1 = 1 + 0.4E_0$$

From scratch: H (0.6) moves to state 1; T (0.4) leaves us at scratch:

$$E_0 = 1 + 0.6E_1 + 0.4E_0$$

Substitute E_1 :

$$0.6E_0 = 1 + 0.6(1 + 0.4E_0) = 1.6 + 0.24E_0 \Rightarrow 0.36E_0 = 1.6$$

$$E_0 = \frac{1.6}{0.36} \approx 4.44$$

General formula: $E = (1 + p)/p^2$; for a fair coin it gives 6.

Trap: $1/p^2 = 2.78$ (treating the pair as a single trial of probability p^2) is wrong because overlapping attempts are not independent trials.

Q.27 [MCQ · 2 marks · -2/3]

Topic: Combinations with constraints · Hard

Correct answer: D

Concept: count all committees meeting the 'at least 2 women' rule, then subtract those that contain both M and W (and still meet the rule).

Step 1 — at least 2 women (k women, $5 - k$ men):

$$\begin{aligned} & \binom{5}{2}\binom{6}{3} + \binom{5}{3}\binom{6}{2} + \binom{5}{4}\binom{6}{1} + \binom{5}{5} \\ &= 10(20) + 10(15) + 5(6) + 1 = 200 + 150 + 30 + 1 = 381 \end{aligned}$$

Step 2 — committees with both M and W: choose 3 more from the remaining 5 men and 4 women, with at least 1 more woman (W already counts as one):

$$\binom{9}{3} - \binom{5}{3} = 84 - 10 = 74$$

$$\text{Answer} = 381 - 74 = 307$$

381 ignores the conflict; 74 is only the subtracted set; $462 = C(11, 5)$ ignores both constraints.

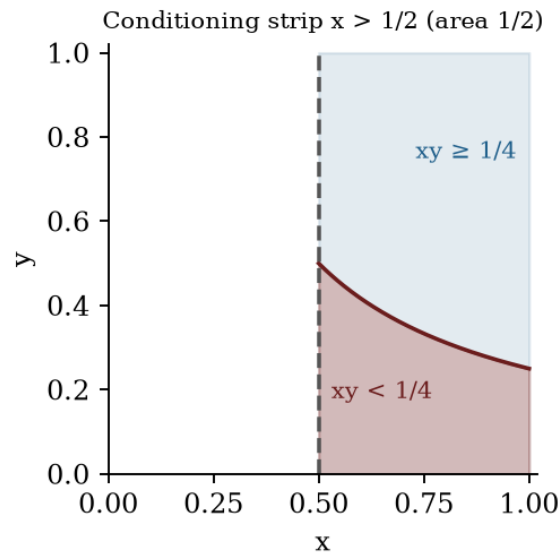
Trap: When subtracting the 'bad' committees, remember they must ALSO satisfy every other constraint (here: ≥ 2 women in total).

Topic: Conditional probability on a continuous region · Hard

Q.28 [MCQ · 2 marks · -2/3]

Correct answer: B

Concept: for a uniform point, conditional probability = (area of $A \cap B$)/(area of B).



Conditioning event $B = \{X > 1/2\}$ has area $1/2$. On B, the curve $y = 1/(4x)$ lies in $(1/4, 1/2]$, so it stays inside the square.

$$\text{Area}(A \cap B) = \int_{1/2}^1 \frac{1}{4x} dx = \frac{1}{4} \ln 2$$

$$P(XY < 1/4 | X > 1/2) = \frac{(\ln 2)/4}{1/2} = \frac{\ln 2}{2} \approx 0.3466$$

$(\ln 2)/4$ is the unconditional area — forgetting to divide by $P(B)$. $1 - (\ln 2)/2$ is the complement.

Equivalent view: given $X = x \in (1/2, 1)$, X is $U(1/2, 1)$ with density 2 and $P(Y < 1/(4x)) = 1/(4x)$, so the answer is $\int 2 \cdot 1/(4x) dx = (\ln 2)/2$.

Trap: Always check whether the boundary curve leaves the square on the conditioning region; if it did, the integrand would need to be capped at 1.

Q.29 [MCQ · 2 marks · -2/3]

Topic: Multi-stage Bayes (cascaded channels) · Hard

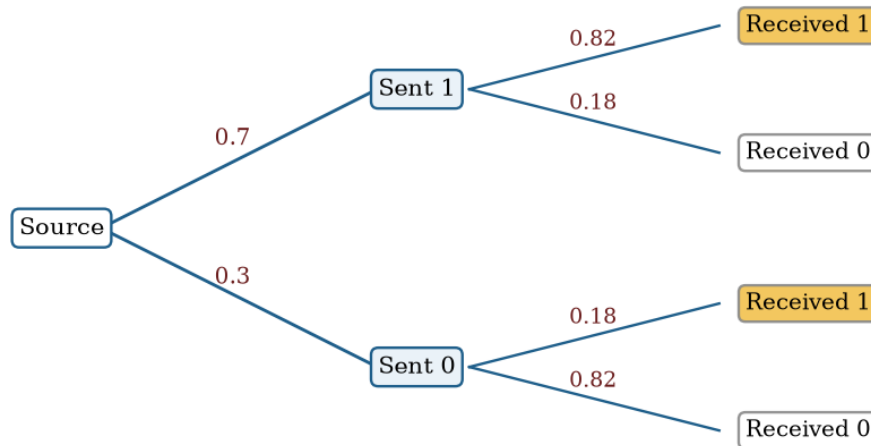
Correct answer: C

Concept: compose the stages first (total probability over the intermediate bit), then apply Bayes' theorem.

Step 1 — end-to-end flip probability: the bit is wrong at the end iff exactly one relay flips:

$$q = 2(0.1)(0.9) = 0.18, \quad 1 - q = 0.82$$

(Two flips restore the original bit.)



Step 2 — total probability of receiving 1:

$$P(R = 1) = 0.7(0.82) + 0.3(0.18) = 0.574 + 0.054 = 0.628$$

Step 3 — Bayes:

$$P(S = 1 | R = 1) = \frac{0.574}{0.628} \approx 0.914$$

0.955 is the answer for a SINGLE relay (0.63/0.66); 0.82 is $P(R = 1 | S = 1)$, the reverse conditional; 0.574 is the joint probability.

Key idea: Each extra noisy stage reduces how much the output tells us about the input: posterior drops from 0.955 (one relay) to 0.914 (two relays).

Topic: Conditional distribution of Poisson given the sum
· Hard

Q.30 [MSQ · 2 marks]

Correct answer: B, C, D

Concept: sums of independent Poissons are Poisson (means add); conditioning on the total splits it binomially in proportion to the means.

(A) FALSE: $X + Y \sim \text{Poisson}(2\lambda)$.

(B) TRUE: for $0 \leq k \leq n$,

$$\begin{aligned}
 P(X = k | X + Y = n) &= \frac{e^{-\lambda} \frac{\lambda^k}{k!} e^{-\lambda} \frac{\lambda^{n-k}}{(n-k)!}}{e^{-2\lambda} \frac{(2\lambda)^n}{n!}} \\
 &= \binom{n}{k} \left(\frac{1}{2}\right)^n
 \end{aligned}$$

which is the Binomial(n , $1/2$) PMF — notice λ cancels.

(C) TRUE: mean of Binomial(n , $1/2$) is $n/2$ (also follows from symmetry: $E[X | S] = E[Y | S]$ and they add to S).

(D) TRUE: Binomial(n , $1/2$) at $k = 0$ gives $(1/2)^n = 2^{-n}$.

For the record, $\text{Var}(X | X + Y = n) = n/4$.

Key idea: With means λ_1, λ_2 , $X | X + Y = n \sim \text{Binomial}(n, \lambda_1/(\lambda_1 + \lambda_2))$.

Mock Test 06 — Topic Test: Random Variables, Distributions, PDFs & CDFs

A topic-intensive set on discrete and continuous distributions, CDFs, transformations, joint and conditional densities.

Item	Details
Questions	30 (12 × 1-mark, 18 × 2-mark)
Maximum marks	48
Question types	MCQ: 14 · MSQ: 7 · NAT: 9
Suggested time	90 minutes
Negative marking	MCQ only: $-1/3$ (1-mark), $-2/3$ (2-mark). None for MSQ & NAT.
Calculator	Use only the GATE virtual scientific calculator

Topic focus of this set

Discrete distributions (uniform, Bernoulli, binomial, Poisson and the Poisson approximation), continuous distributions (uniform, exponential and memorylessness, normal and standardisation, t and χ^2 properties), CDF properties and mixed CDFs, pdf from CDF, transformations of random variables, joint/marginal/conditional pdfs, moments and covariance from densities, and quantiles, medians and modes.

Instructions

- Questions 1 onwards are arranged as in GATE: all 1-mark questions first, then 2-mark questions.
- **MCQ** – exactly one correct option. **MSQ** – ONE OR MORE options are correct; credit only if you select all correct options and no wrong option (no partial marks, no negative marks).
- **NAT** – enter a numerical value using the virtual keypad. The accepted range is given in the key; round as instructed in the question.
- Statistical tables (Z , t , χ^2) are provided in the Appendix; values in solutions use those tables.
- Attempt in one sitting, then check the answer key and read every solution — including those you got right — for the alternative methods and traps.

Mock Test 06 · Question Paper

1-mark questions

Q.1 MCQ · 1 mark · -1/3

A binomial random variable X has mean 6 and variance 4.2. The parameters (n , p) of X are

- (A) $n = 20$, $p = 0.3$
- (B) $n = 30$, $p = 0.2$
- (C) $n = 15$, $p = 0.4$
- (D) $n = 20$, $p = 0.7$

Q.2 MCQ · 1 mark · -1/3

The lifetime of an SSD is exponentially distributed with mean 5 years. A drive has already worked for 3 years. The probability that it works for at least 4 MORE years is

- (A) $e^{-1.4}$
- (B) $1 - e^{-0.8}$
- (C) $e^{-0.8}$
- (D) $e^{-0.6}$

Q.3 NAT · 1 mark

$X \sim N(50, 8^2)$. Using $\Phi(1) = 0.8413$ and $\Phi(2) = 0.9772$, the value of $P(42 < X < 66)$ is _____ (round off to 3 decimal places).

Answer: _____

Q.4 MSQ · 1 mark

Which of the following functions is/are valid cumulative distribution functions?

- (A) $F(x) = 1 - e^{-x}$ for all real x
- (B) $F(x) = \exp(-e^{-x})$ for all real x
- (C) $F(x) = 0$ for $x < 0$; $F(x) = (x + 1)/2$ for $0 \leq x < 1$; $F(x) = 1$ for $x \geq 1$
- (D) $F(x) = 0$ for $x < 0$; $F(x) = \sin x$ for $0 \leq x < \pi$; $F(x) = 1$ for $x \geq \pi$

Q.5 MCQ · 1 mark · -1/3

$Z_1, \dots, Z_5$ are i.i.d. $N(0, 1)$ and $W = Z_1^2 + \dots + Z_5^2$. The mean and variance of W are respectively

- (A) 5 and 5
- (B) 0 and 5
- (C) 5 and 25
- (D) 5 and 10

Q.6 MSQ · 1 mark

Help-desk calls arrive as a Poisson process at 3 calls per hour. Let X be the number of calls in a 40-minute window. Which of the following is/are TRUE?

- (A) $P(X = 1) = P(X = 2)$
- (B) $E[X] = \text{Var}(X)$
- (C) X has a unique mode
- (D) $P(X \geq 1) = 1 - e^{-2}$

Q.7 MCQ · 1 mark · -1/3

X is uniformly distributed on (a, b) with $E[X] = 5$ and $\text{Var}(X) = 3$. Then $P(X > 6)$ equals

- (A) $1/4$
- (B) $1/3$
- (C) $1/2$
- (D) $1/6$

Q.8 MSQ · 1 mark

Let T_ν denote a random variable having Student's t -distribution with ν degrees of freedom. Which of the following is/are TRUE?

- (A) The density of T_ν is symmetric about 0 for every ν
- (B) $\text{Var}(T_5) = 5/3$
- (C) T_ν has heavier tails than $N(0, 1)$, e.g. $P(|T_5| > 2) > P(|Z| > 2)$
- (D) T_1 has a finite mean equal to 0

Q.9 MCQ · 1 mark · -1/3

A continuous random variable has CDF $F(x) = 1 - (1 + x)e^{-x}$ for $x \geq 0$ (and $F(x) = 0$ for $x < 0$). Its probability density function for $x > 0$ is

- (A) $x e^{-x}$
- (B) $(1 + x) e^{-x}$
- (C) e^{-x}
- (D) $(2 + x) e^{-x}$

Q.10 NAT · 1 mark

The waiting time T (in minutes) for a ride-share has pdf $f(t) = 0.5 e^{-0.5t}$ for $t \geq 0$. The median of T is _____ minutes (round off to 2 decimal places).

Answer: _____

Q.11 MCQ · 1 mark · -1/3

A Bernoulli(p) random variable X has $\text{Var}(X) = 0.21$ and $p > 0.5$. The value of p is

- (A) 0.3
- (B) 0.79
- (C) 0.7
- (D) 0.21

Q.12 NAT · 1 mark

X is uniformly distributed on $\{1, 2, \dots, n\}$. If $\text{Var}(X) = 10$, then $n = \underline{\hspace{2cm}}$.

Answer:

2-mark questions**Q.13 NAT · 2 marks**

The joint pdf of (X, Y) is $f(x, y) = kxy$ for $0 < y < x < 1$, and 0 otherwise. The value of $P(X + Y < 1)$ is (round off to 3 decimal places).

Answer:

Q.14 MSQ · 2 marks

(X, Y) has joint pdf $f(x, y) = e^{-x}$ for $0 < y < x < \infty$, and 0 otherwise. Which of the following is/are TRUE?

- (A) The marginal pdf of Y is $f_Y(y) = e^{-y}$, $y > 0$
- (B) X and Y are independent
- (C) Given $X = x$, Y is uniformly distributed on $(0, x)$
- (D) $\text{Cov}(X, Y) = 1$

Q.15 MCQ · 2 marks · -2/3

For the joint pdf $f(x, y) = 8xy$ on $0 < y < x < 1$ (0 elsewhere), the conditional expectation $E[Y | X = x]$ for $0 < x < 1$ is

- (A) $x/2$
- (B) $2x/3$
- (C) $x/3$
- (D) $3x/4$

Q.16 MCQ · 2 marks · -2/3

$X \sim \text{Uniform}(-1, 2)$ and $Y = X^2$. The pdf of Y is

- (A) $f_Y(y) = 1/(6\sqrt{y})$ for $0 < y < 4$
- (B) $f_Y(y) = 1/(3\sqrt{y})$ for $0 < y < 4$
- (C) $f_Y(y) = 1/(6\sqrt{y})$ for $0 < y < 1$, and $1/(3\sqrt{y})$ for $1 < y < 4$
- (D) $f_Y(y) = 1/(3\sqrt{y})$ for $0 < y < 1$, and $1/(6\sqrt{y})$ for $1 < y < 4$

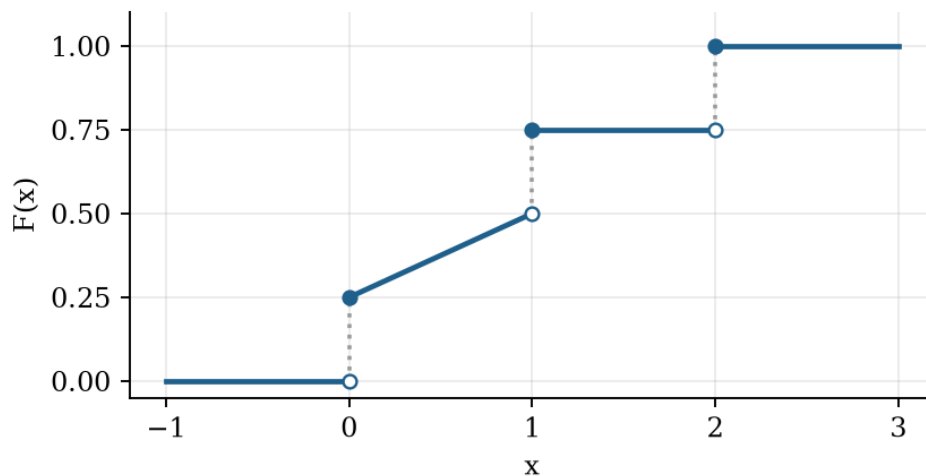
Q.17 NAT · 2 marks

$U \sim \text{Uniform}(0, 1)$ and $Y = -2 \ln U$. The value of $P(Y > 3)$ is (round off to 3 decimal places).

Answer:

Q.18 MSQ · 2 marks

The CDF of a random variable X is plotted below: $F(x) = 0$ for $x < 0$; $F(x) = (1 + x)/4$ for $0 \leq x < 1$; $F(x) = 3/4$ for $1 \leq x < 2$; $F(x) = 1$ for $x \geq 2$.



Which of the following is/are TRUE?

- (A) $P(0 \leq X < 1) = 1/4$
- (B) $P(X = 1) = 1/4$
- (C) $P(0 < X < 1) = 1/4$
- (D) $E[X] = 7/8$

Q.19 NAT · 2 marks

The response time of a web service is normally distributed. 10% of requests take longer than 72 ms and 25% take less than 60 ms. Use $\Phi(1.2816) = 0.90$ and $\Phi(0.6745) = 0.75$. The mean response time is _____ ms (round off to 2 decimal places).

Answer: _____

Q.20 MCQ · 2 marks · -2/3

A chip has a defect with probability 0.002, independently of other chips. Using the Poisson approximation, the probability that a batch of 1500 chips contains at least 2 defective chips is approximately

- (A) 0.801
- (B) 0.199
- (C) 0.950
- (D) 0.577

Q.21 NAT · 2 marks

(X, Y) has joint pdf $f(x, y) = x + y$ for $0 < x < 1$, $0 < y < 1$. The correlation coefficient $\rho(X, Y)$ is _____ (round off to 3 decimal places).

Answer: _____

Q.22 MSQ · 2 marks

$Z \sim N(0, 1)$ and $V \sim \chi^2_n$ are independent. Which of the following is/are TRUE?

- (A) $Z/\sqrt{V/n}$ has the t-distribution with n degrees of freedom
- (B) $E[V] = n$ and $\text{Var}(V) = 2n$
- (C) $Z + V$ has the χ^2_{n+1} distribution
- (D) For $n = 10$, $P(V > 10) = 0.5$

Q.23 MCQ · 2 marks · -2/3

A system has two independent servers with exponentially distributed lifetimes of mean 2 hours and 3 hours. The expected time until BOTH servers have failed is

- (A) 5 hours
- (B) 1.2 hours
- (C) 3.8 hours
- (D) 3.0 hours

Q.24 NAT · 2 marks

X has pdf $f(x) = 12x^2(1 - x)$ for $0 < x < 1$. The median of X is _____ (round off to 3 decimal places).

Answer: _____

Q.25 MCQ · 2 marks · -2/3

$X \sim \text{Uniform}(0, 1)$. Given $X = x$, Y has the exponential distribution with rate x (pdf $x e^{-xy}$, $y > 0$). The value of $P(Y > 1)$ is

- (A) $1/e$
- (B) $1 - 1/e$
- (C) $1/2$
- (D) $e - 2$

Q.26 MCQ · 2 marks · -2/3

A storage array has 10 disks, each failing within a year with probability 0.2 independently. The array survives the year if at most 2 disks fail. A data centre runs 5 such arrays independently. The expected number of arrays that survive the year is closest to

- (A) 1.61
- (B) 1.88
- (C) 0.68
- (D) 3.39

Q.27 MCQ · 2 marks · -2/3

Daily demand $X \sim N(10, 9)$ units at store A and $Y \sim N(6, 16)$ units at store B, independently (second parameter = variance). Using $\Phi(0.8) = 0.7881$, $\Phi(1) = 0.8413$ and $\Phi(0.57) = 0.7157$, $P(X > Y)$ is

- (A) 0.7881
- (B) 0.2119
- (C) 0.8413
- (D) 0.7157

Q.28 MSQ · 2 marks

X_1, X_2, X_3 are i.i.d. Uniform(0, 1). Let $M = \max(X_1, X_2, X_3)$ and $L = \min(X_1, X_2, X_3)$. Which of the following is/are TRUE?

- (A) The pdf of M is $3(1 - m)^2$ for $0 < m < 1$
- (B) $E[M] = 3/4$
- (C) $P(L > 0.5) = 1/2$
- (D) The median of M is $2^{-1/3} \approx 0.794$

Q.29 NAT · 2 marks

$X \sim$ Exponential with rate 1. The value of $E[X | X < 2]$ is _____ (round off to 3 decimal places).

Answer: _____

Q.30 MCQ · 2 marks · -2/3

X and Y are i.i.d. Uniform(0, 1) and $Z = X + Y$. The value of the pdf $f_Z(1.5)$ and of $P(Z > 1.5)$ are respectively

- (A) 0.5 and 0.25
- (B) 1 and 0.125
- (C) 0.5 and 0.125
- (D) 0.75 and 0.125

Mock Test 06 · Answer Key

Q	Type	Marks	Answer	Topic
1	MCQ	1	A	Binomial: mean and variance
2	MCQ	1	C	Exponential: memorylessness
3	NAT	1	0.819 (range 0.817 to 0.82)	Normal: standardisation
4	MSQ	1	B, C	Properties of a CDF
5	MCQ	1	D	Chi-squared as sum of squared normals
6	MSQ	1	A, B, D	Poisson PMF & mode
7	MCQ	1	B	Continuous uniform
8	MSQ	1	A, B, C	Student t-distribution properties
9	MCQ	1	A	PDF from CDF
10	NAT	1	1.39 (range 1.38 to 1.39)	Median (quantile) of exponential
11	MCQ	1	C	Bernoulli variance
12	NAT	1	11	Discrete uniform variance
13	NAT	2	0.167 (range 0.166 to 0.168)	Joint pdf: normalisation & region probability
14	MSQ	2	A, C, D	Marginal & conditional pdfs, covariance
15	MCQ	2	B	Conditional pdf & conditional expectation
16	MCQ	2	D	Transformation of a random variable (non-monotone)
17	NAT	2	0.223 (range 0.222 to 0.224)	Transformation: $-2 \ln U$ and chi-squared(2)
18	MSQ	2	B, C, D	Mixed random variable from its CDF
19	NAT	2	64.14 (range 64.05 to 64.25)	Normal: solving for μ and σ from quantiles
20	MCQ	2	A	Poisson approximation to binomial
21	NAT	2	-0.091 (range -0.092 to -0.09)	Correlation from a joint pdf
22	MSQ	2	A, B	Chi-squared and t relationships
23	MCQ	2	C	Exponential: minimum and maximum
24	NAT	2	0.614 (range 0.612 to 0.616)	Median, mean and mode from a pdf
25	MCQ	2	B	Hierarchical model: marginal probability
26	MCQ	2	D	Binomial in two stages
27	MCQ	2	A	Linear combinations of normals
28	MSQ	2	B, D	Distribution of max/min of uniforms (order statistics)
29	NAT	2	0.687 (range 0.685 to 0.689)	Conditional expectation of a truncated exponential
30	MCQ	2	C	Sum of independent uniforms (convolution)

Score tracker

	Attempted	Correct	Wrong (MCQ)	Marks
1-mark				
2-mark				
Total				/ 48

Mock Test 06 · Detailed Solutions

Q.1 [MCQ · 1 mark · -1/3]

Topic: Binomial: mean and variance · Easy

Correct answer: A

Concept: for $X \sim \text{Binomial}(n, p)$: $E[X] = np$ and $\text{Var}(X) = np(1 - p)$.

Divide variance by mean:

$$1 - p = \frac{np(1 - p)}{np} = \frac{4.2}{6} = 0.7 \Rightarrow p = 0.3$$

$$n = \frac{6}{0.3} = 20$$

Check the distractors: $(n, p) = (30, 0.2)$ gives variance 4.8; $(15, 0.4)$ gives 3.6; $(20, 0.7)$ has mean 14 — it swaps p and $1 - p$.

Key idea: For a binomial, variance $<$ mean always (since $1 - p < 1$). If a problem quotes variance $\geq$ mean, no binomial fits.

Q.2 [MCQ · 1 mark · -1/3]

Topic: Exponential: memorylessness · Easy

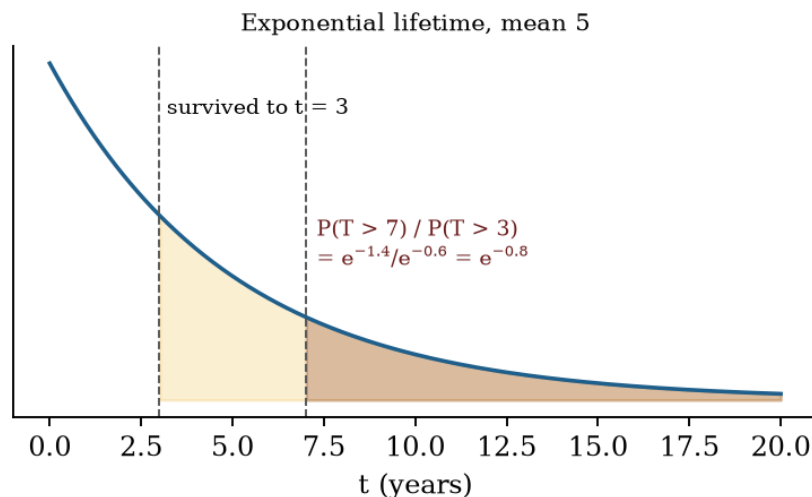
Correct answer: C

Concept: memoryless property $P(T > s + t \mid T > s) = P(T > t)$; for rate λ , $P(T > t) = e^{-\lambda t}$.

Rate $\lambda = 1/\text{mean} = 0.2$ per year.

$$P(T > 7 \mid T > 3) = \frac{e^{-0.2(7)}}{e^{-0.2(3)}} = e^{-0.2(4)} = e^{-0.8}$$

Numerically $e^{-0.8} \approx 0.4493$.



$e^{-1.4} = P(T > 7)$ ignores the conditioning; $e^{-0.6} = P(T > 3)$; $1 - e^{-0.8}$ is the probability of failing within the next 4 years.

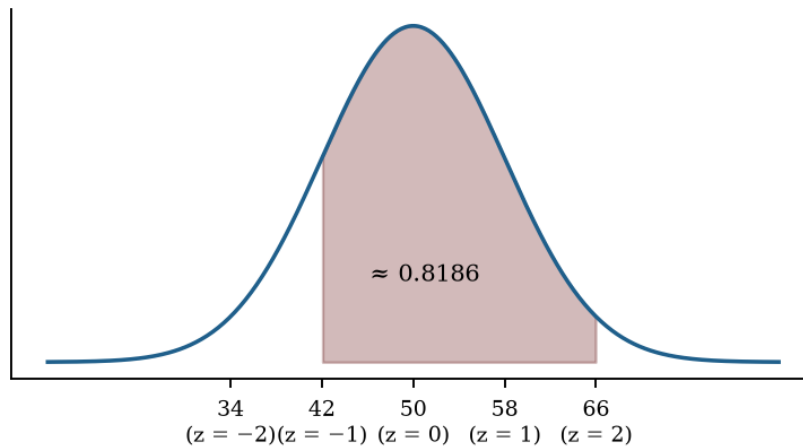
Key idea: The exponential is the only continuous distribution without memory: a used drive is as good as new.

Q.3 [NAT · 1 mark]**Topic: Normal: standardisation · Easy****Correct answer: 0.819 (range 0.817 to 0.82)****Concept:** standardise $Z = (X - \mu)/\sigma$ and use $\Phi(-a) = 1 - \Phi(a)$.

$$z_1 = \frac{42 - 50}{8} = -1, \quad z_2 = \frac{66 - 50}{8} = 2$$

$$P(-1 < Z < 2) = \Phi(2) - \Phi(-1) = \Phi(2) - [1 - \Phi(1)]$$

$$= 0.9772 - 0.1587 = 0.8185 \approx 0.819$$



Shortcut: Asymmetric intervals must be split: 0.3413 (from -1 to 0) + 0.4772 (from 0 to 2) = 0.8185 .

Q.4 [MSQ · 1 mark]**Topic: Properties of a CDF · Medium****Correct answer: B, C****Concept:** F is a CDF iff it is non-decreasing, right-continuous, $F(-\infty) = 0$ and $F(+\infty) = 1$.**(A) INVALID:** as $x \rightarrow -\infty$, $e^{-x} \rightarrow \infty$ so $F(x) \rightarrow -\infty$; also $F(-1) = 1 - e \approx -1.72 < 0$.**(B) VALID** (Gumbel CDF): $-e^{-x}$ increases with x , so F is increasing and continuous; $F \rightarrow e^{-\infty} = 0$ as $x \rightarrow -\infty$ and $\rightarrow e^0 = 1$ as $x \rightarrow \infty$.**(C) VALID** (mixed): F jumps from 0 to $1/2$ at $x = 0$ (a point mass of $1/2$, right-continuous), then rises linearly to 1 at $x = 1$. Non-decreasing with correct limits.**(D) INVALID:** $\sin x$ decreases on $(\pi/2, \pi)$, e.g. $F(\pi/2) = 1 > F(3\pi/4) \approx 0.707$.

Key idea: A CDF may jump (discrete part) but may never go down. Jumps must be closed at the top (right-continuity).

Q.5 [MCQ · 1 mark · -1/3]**Topic: Chi-squared as sum of squared normals · Easy****Correct answer: D**

Concept: a sum of k squared independent standard normals is χ^2_k , with mean k and variance $2k$.

For one term: $E[Z^2] = \text{Var}(Z) = 1$ and $E[Z^4] = 3$, so

$$\text{Var}(Z^2) = E[Z^4] - (E[Z^2])^2 = 3 - 1 = 2$$

$$E[W] = 5(1) = 5, \quad \text{Var}(W) = 5(2) = 10$$

'5 and 5' assumes $\text{Var}(Z^2) = 1$; '0 and 5' confuses W with $\sum Z_i$; '5 and 25' squares the mean.

Shortcut: χ^2_k : mean = df, variance = $2 \cdot \text{df}$. Memorise $E[Z^4] = 3$.

Q.6 [MSQ · 1 mark]**Topic: Poisson PMF & mode · Medium****Correct answer: A, B, D**

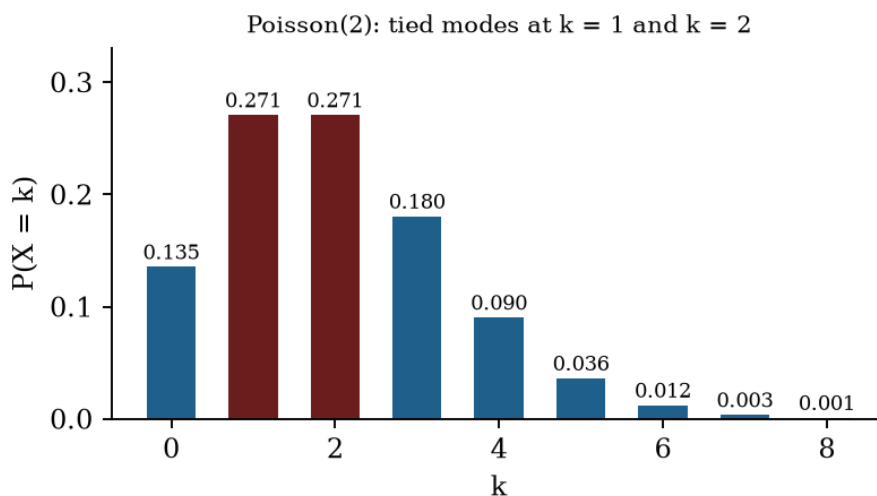
Concept: counts in a window of length t have $\text{Poisson}(\lambda t)$ distribution, $P(X = k) = e^{-\mu} \mu^k / k!$.

$$\mu = \lambda t = 3 \times \frac{40}{60} = 2$$

(A) **TRUE:** $P(X = 1) = 2e^{-2}$ and $P(X = 2) = 2^2 e^{-2} / 2 = 2e^{-2}$.

(B) **TRUE:** for a Poisson, mean = variance = $\mu = 2$.

(C) **FALSE:** since $\mu = 2$ is an integer, the PMF ratio $P(k)/P(k-1) = \mu/k$ equals 1 at $k = 2$, so $k = 1$ and $k = 2$ are tied modes.



(D) **TRUE:** $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-2} \approx 0.8647$.

Trap: Poisson mode = $\lfloor \mu \rfloor$, with a tie between $\mu - 1$ and μ whenever μ is an integer.

Q.7 [MCQ · 1 mark · -1/3]**Topic: Continuous uniform · Easy****Correct answer: B****Concept:** for $U(a, b)$, mean = $(a + b)/2$, variance = $(b - a)^2/12$.

$$\frac{(b - a)^2}{12} = 3 \Rightarrow b - a = 6, \quad a + b = 10$$

So $a = 2$, $b = 8$ and the density is $1/6$ on $(2, 8)$.

$$P(X > 6) = \frac{8 - 6}{8 - 2} = \frac{2}{6} = \frac{1}{3}$$

 $1/6$ confuses the density height with a probability; $1/2$ is $P(X > \text{mean})$.**Shortcut:** Uniform probabilities are length ratios — find the endpoints first.**Q.8** [MSQ · 1 mark]**Topic: Student t-distribution properties · Medium****Correct answer: A, B, C****Concept:** $T_\nu = Z/\sqrt{V/\nu}$ with $Z \sim N(0,1)$, $V \sim \chi^2_\nu$ independent; $E[T] = 0$ for $\nu > 1$ and $\text{Var}(T) = \nu/(\nu - 2)$ for $\nu > 2$.**(A) TRUE:** Z is symmetric and independent of V , so $-T$ has the same distribution as T .**(B) TRUE:**

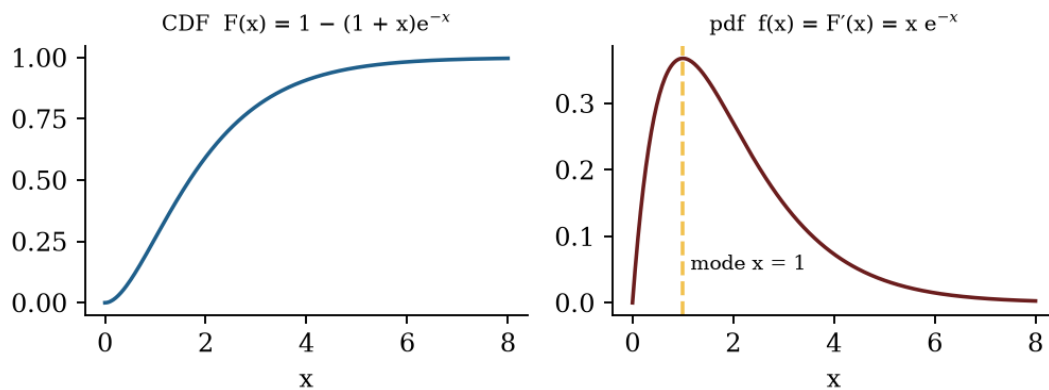
$$\text{Var}(T_5) = \frac{5}{5 - 2} = \frac{5}{3} \approx 1.667$$

(C) TRUE: $P(|T_5| > 2) = 0.1019$ versus $P(|Z| > 2) = 0.0455$. The extra randomness in the denominator fattens tails.**(D) FALSE:** T_1 is the standard Cauchy distribution; $\int |t|/(\pi(1 + t^2)) dt$ diverges, so its mean does not exist (even though the density is symmetric about 0).**Key idea:** As $\nu \rightarrow \infty$, $T_\nu \rightarrow N(0, 1)$ and the variance $\nu/(\nu - 2) \rightarrow 1$.**Q.9** [MCQ · 1 mark · -1/3]**Topic: PDF from CDF · Easy****Correct answer: A****Concept:** $f(x) = F'(x)$ wherever F is differentiable.

$$f(x) = -\frac{d}{dx}[(1 + x)e^{-x}] = -[e^{-x} - (1 + x)e^{-x}]$$

$$f(x) = -e^{-x} + (1 + x)e^{-x} = xe^{-x}, \quad x > 0$$

This is the Gamma(2, 1) density (sum of two independent Exp(1) variables), with mean 2 and mode 1.



$(1+x)e^{-x}$ is $1 - F(x)$, the survival function — not the pdf.

Shortcut: Quick check: the candidate must integrate to 1 and be ≥ 0 ; $(1+x)e^{-x}$ integrates to 2, so it cannot be a pdf.

Q.10 [NAT · 1 mark]

Topic: Median (quantile) of exponential · Easy

Correct answer: 1.39 (range 1.38 to 1.39)

Concept: the median m solves $F(m) = 1/2$; for $\text{Exp}(\lambda)$, $F(t) = 1 - e^{-\lambda t}$.

$$1 - e^{-0.5m} = \frac{1}{2} \Rightarrow e^{-0.5m} = \frac{1}{2} \Rightarrow m = \frac{\ln 2}{0.5}$$

$$m = 2 \ln 2 \approx 1.386 \approx 1.39$$

Compare: the mean is $1/\lambda = 2$ minutes; the mode is 0.

Key idea: Exponential: mode (0) < median ($0.693/\lambda$) < mean ($1/\lambda$), as for every right-skewed density of this shape.

Q.11 [MCQ · 1 mark · -1/3]

Topic: Bernoulli variance · Easy

Correct answer: C

Concept: $\text{Var}(\text{Bernoulli}(p)) = p(1-p)$, which is symmetric about $p = 1/2$.

$$p(1-p) = 0.21 \Rightarrow p^2 - p + 0.21 = 0 \Rightarrow p = \frac{1 \pm \sqrt{1 - 0.84}}{2} = \frac{1 \pm 0.4}{2}$$

So $p = 0.7$ or $p = 0.3$; the condition $p > 0.5$ selects $p = 0.7$.

Trap: $\text{Var}(X) \leq 1/4$ for any Bernoulli; both p and $1-p$ give the same variance, so an extra condition is always needed to pin p down.

Q.12 [NAT · 1 mark]

Topic: Discrete uniform variance · Easy

Correct answer: 11**Concept:** for X uniform on $\{1, \dots, n\}$: $E[X] = (n + 1)/2$ and $\text{Var}(X) = (n^2 - 1)/12$.

$$\frac{n^2 - 1}{12} = 10 \Rightarrow n^2 = 121 \Rightarrow n = 11$$

Derivation: $E[X^2] = (n + 1)(2n + 1)/6$, so

$$\text{Var}(X) = \frac{(n + 1)(2n + 1)}{6} - \frac{(n + 1)^2}{4} = \frac{(n + 1)(n - 1)}{12}$$

Trap: Do not confuse with the continuous $U(a, b)$ variance $(b - a)^2/12$, which would give $n - 1 = \sqrt{120}$ — not an integer.

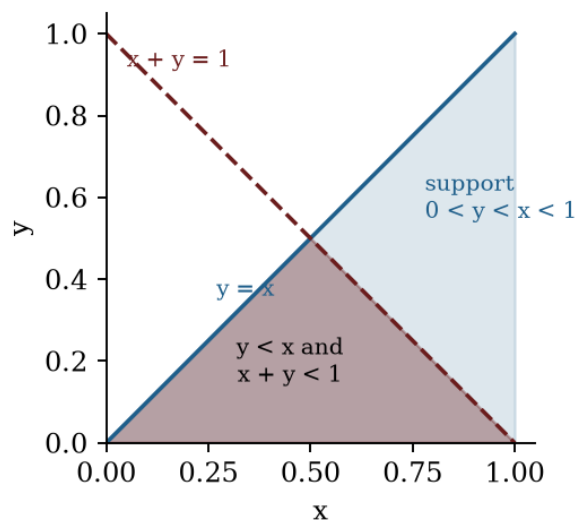
Q.13 [NAT · 2 marks]

Topic: Joint pdf: normalisation & region probability · Medium

Correct answer: 0.167 (range 0.166 to 0.168)**Concept:** find k from $\iint f = 1$, then integrate f over the intersection of the support with the event.

Step 1 — normalisation:

$$\int_0^1 \int_0^x kxy \, dy \, dx = k \int_0^1 \frac{x^3}{2} \, dx = \frac{k}{8} = 1 \Rightarrow k = 8$$



Step 2 — region: $y < x$ and $x + y < 1$ means $y < x < 1 - y$, which requires $y < 1/2$. Integrate x first:

$$\begin{aligned} P &= \int_0^{1/2} \int_y^{1-y} 8xy \, dx \, dy = \int_0^{1/2} 4y[(1-y)^2 - y^2] \, dy \\ &= \int_0^{1/2} 4y(1 - 2y) \, dy = 4 \left[\frac{y^2}{2} - \frac{2y^3}{3} \right]_0^{1/2} \\ &= 4 \left(\frac{1}{8} - \frac{1}{12} \right) = \frac{4}{24} = \frac{1}{6} \approx 0.167 \end{aligned}$$

Shortcut: Integrating y first would need a split at $x = 1/2$ (upper limit $\min(x, 1 - x)$); choosing the order with a single expression for the limits saves work.

Q.14 [MSQ · 2 marks]

Topic: Marginal & conditional pdfs, covariance · Hard

Correct answer: A, C, D

Concept: marginals by integrating out the other variable over the support; $f_{Y|X} = f/f_X$.

(A) TRUE:

$$f_Y(y) = \int_y^\infty e^{-x} dx = e^{-y}, \quad y > 0$$

(B) FALSE: the support $0 < y < x$ is not a rectangle, so independence is impossible. Indeed $f_X(x) = \int_0^x e^{-x} dy = x e^{-x}$ and $f_X f_Y = x e^{-x-y} \neq e^{-x}$.

(C) TRUE:

$$f_{Y|X}(y|x) = \frac{e^{-x}}{x e^{-x}} = \frac{1}{x}, \quad 0 < y < x$$

(D) TRUE: write $W = X - Y$. The joint pdf of (Y, W) is $e^{-y}e^{-w}$ (Jacobian 1), so Y and W are independent $\text{Exp}(1)$. Then $X = Y + W$ and

$$\text{Cov}(X, Y) = \text{Cov}(Y + W, Y) = \text{Var}(Y) = 1$$

Check by moments: $E[X] = 2$, $E[Y] = 1$, $E[XY] = E[X \cdot E[Y|X]] = E[X^2/2] = 6/2 = 3$, so $\text{Cov} = 3 - 2 = 1$.

Trap: Non-rectangular support $\Rightarrow$ dependent, regardless of whether the formula factorises.

Q.15 [MCQ · 2 marks · -2/3]

Topic: Conditional pdf & conditional expectation · Medium

Correct answer: B

Concept: $f_{Y|X}(y|x) = f(x, y)/f_X(x)$; then $E[Y|X=x] = \int y f_{Y|X}(y|x) dy$.

Step 1 — marginal of X :

$$f_X(x) = \int_0^x 8xy dy = 4x^3, \quad 0 < x < 1$$

Step 2 — conditional pdf:

$$f_{Y|X}(y|x) = \frac{8xy}{4x^3} = \frac{2y}{x^2}, \quad 0 < y < x$$

Step 3 — conditional mean:

$$E[Y|X=x] = \int_0^x y \cdot \frac{2y}{x^2} dy = \frac{2}{x^2} \cdot \frac{x^3}{3} = \frac{2x}{3}$$

$x/2$ would be right only if $Y|X$ were uniform on $(0, x)$; here the conditional density $2y/x^2$ leans toward larger y .

Key idea: Tower check: $E[Y] = E[2X/3] = (2/3)(4/5) = 8/15$, which matches $\int y \cdot 4y(1 - y^2) dy = 8/15$ computed from $f_Y(y) = 4y(1 - y^2)$.

**Topic: Transformation of a random variable
(non-monotone) · Hard**

Q.16 [MCQ · 2 marks · -2/3]

Correct answer: D

Concept: for a non-monotone g , sum over all roots: $f_Y(y) = \sum f_X(x_i)/|g'(x_i)|$.

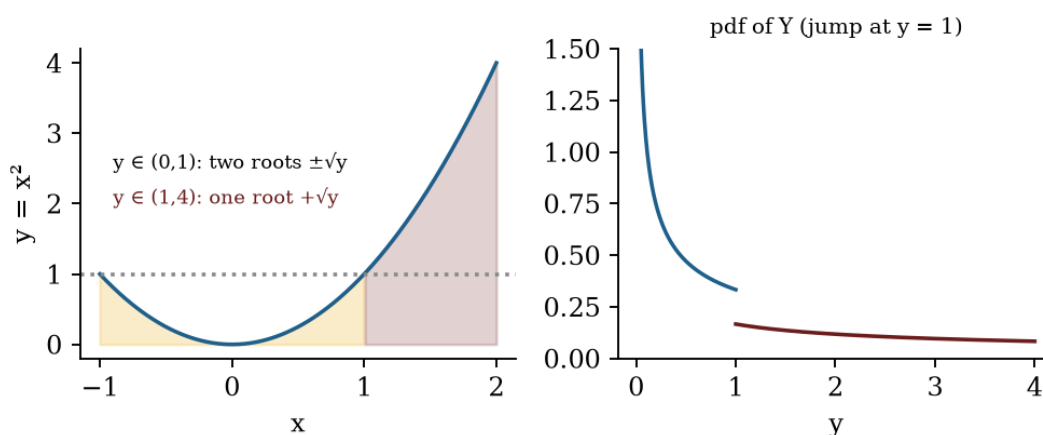
$f_X = 1/3$ on $(-1, 2)$ and $|g'(x)| = 2|x| = 2\sqrt{y}$ at $x = \pm\sqrt{y}$.

Case $0 < y < 1$: both roots $\pm\sqrt{y}$ lie in $(-1, 2)$:

$$f_Y(y) = \frac{1/3}{2\sqrt{y}} + \frac{1/3}{2\sqrt{y}} = \frac{1}{3\sqrt{y}}$$

Case $1 < y < 4$: only $+\sqrt{y} \in (1, 2)$ is in the support ($-\sqrt{y} < -1$):

$$f_Y(y) = \frac{1/3}{2\sqrt{y}} = \frac{1}{6\sqrt{y}}$$



Check: $\int_0^1 dy/(3\sqrt{y}) = 2/3$ and $\int_1^4 dy/(6\sqrt{y}) = 1/3$; total 1 (verified). (Option B integrates to $4/3$, option A to $2/3$ — neither is a pdf.)

Shortcut: CDF method as cross-check: for $0 < y < 1$, $F_Y(y) = P(-\sqrt{y} < X < \sqrt{y}) = 2\sqrt{y}/3$; for $1 < y < 4$, $F_Y(y) = P(-1 < X < \sqrt{y}) = (\sqrt{y} + 1)/3$.

Topic: Transformation: $-2 \ln U$ and chi-squared(2) · Medium

Q.17 [NAT · 2 marks]

Correct answer: 0.223 (range 0.222 to 0.224)

Concept: CDF method for a monotone transformation; $-\ln U \sim \text{Exp}(1)$.

$Y > y \Leftrightarrow -2 \ln U > y \Leftrightarrow U < e^{-y/2}$, so

$$P(Y > y) = P(U < e^{-y/2}) = e^{-y/2}, \quad y > 0$$

Hence $Y \sim \text{Exponential}$ with rate $1/2$ (mean 2) — which is exactly the χ^2_2 distribution.

$$f_Y(y) = \frac{1}{2}e^{-y/2}, \quad P(Y > 3) = e^{-1.5} \approx 0.223$$

Key idea: $\chi^2_2 = \text{Exp}(\text{mean } 2)$. This is the basis of the Box-Muller method: $Z_1^2 + Z_2^2 = -2 \ln U$.

Q.18 [MSQ · 2 marks]**Topic: Mixed random variable from its CDF · Hard****Correct answer: B, C, D**

Concept: $P(X = a) = F(a) - F(a^-)$ (jump size); $P(a < X < b) = F(b^-) - F(a)$. $E[X] = \sum (\text{atoms}) + \int x f(x) dx$ over the continuous part.

Piece	Probability
Atom at $x = 0$	$F(0) - F(0^-) = 1/4 - 0 = 1/4$
Density $1/4$ on $(0, 1)$	$1/2 - 1/4 = 1/4$
Atom at $x = 1$	$3/4 - 1/2 = 1/4$
Atom at $x = 2$	$1 - 3/4 = 1/4$

(A) FALSE: $P(0 \leq X < 1) = F(1^-) - F(0^-) = 1/2 - 0 = 1/2$ (it includes the atom at 0).

(B) TRUE: jump at 1 is $3/4 - 1/2 = 1/4$.

(C) TRUE: $P(0 < X < 1) = F(1^-) - F(0) = 1/2 - 1/4 = 1/4$.

(D) TRUE:

$$\begin{aligned}
 E[X] &= 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{4} + \int_0^1 x \cdot \frac{1}{4} dx \\
 &= \frac{3}{4} + \frac{1}{8} = \frac{7}{8}
 \end{aligned}$$

Trap: Whether an endpoint is included matters ONLY at jump points; on the continuous part $P(X = a) = 0$.

Q.19 [NAT · 2 marks]**Topic: Normal: solving for μ and σ from quantiles · Medium****Correct answer: 64.14 (range 64.05 to 64.25)**

Concept: each percentile gives a linear equation $x = \mu + z\sigma$.

$P(X > 72) = 0.10 \Rightarrow 72$ is the 90th percentile; $P(X < 60) = 0.25 \Rightarrow 60$ is the 25th percentile ($z = -0.6745$).

$$\mu + 1.2816\sigma = 72, \quad \mu - 0.6745\sigma = 60$$

Subtract:

$$\begin{aligned}
 1.9561\sigma &= 12 \Rightarrow \sigma = 6.1347 \\
 \mu &= 60 + 0.6745(6.1347) = 64.1378 \approx 64.14
 \end{aligned}$$

Check: $72 - 1.2816\sigma = 72 - 7.862 = 64.138$ (verified).

Trap: Mind the sign: the lower quartile has $z = -0.6745$, not $+0.6745$. Using $+0.6745$ gives a negative σ — a red flag.

Q.20 [MCQ · 2 marks · -2/3]**Topic: Poisson approximation to binomial · Medium****Correct answer: A****Concept:** Binomial(n, p) $\approx$ Poisson(np) for large n , small p .

$$\lambda = np = 1500 \times 0.002 = 3$$

$$P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-3}(1 + 3) = 1 - 4e^{-3} \\ = 1 - 4(0.049787) = 0.8009$$

The exact binomial value is 0.8012 — the approximation is excellent.

0.199 is $P(X \leq 1)$ (the complement); $0.950 = 1 - e^{-3}$ is $P(X \geq 1)$; $0.577 = 1 - e^{-3}(1 + 3 + 4.5)$ is $P(X \geq 3)$ (subtracting $P(2)$ as well).**Trap:** 'At least 2' = $1 - P(0) - P(1)$: forgetting either term is the standard slip.**Q.21** [NAT · 2 marks]**Topic: Correlation from a joint pdf · Hard****Correct answer: -0.091 (range -0.092 to -0.09)****Concept:** $\rho = \text{Cov}(X, Y)/(\sigma_X \sigma_Y)$, with all moments from the joint pdf.Marginal: $f_X(x) = \int_0^1 (x + y) dy = x + 1/2$ (same for Y by symmetry).

$$E[X] = \int_0^1 x \left(x + \frac{1}{2}\right) dx = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$E[X^2] = \int_0^1 x^2 \left(x + \frac{1}{2}\right) dx = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$

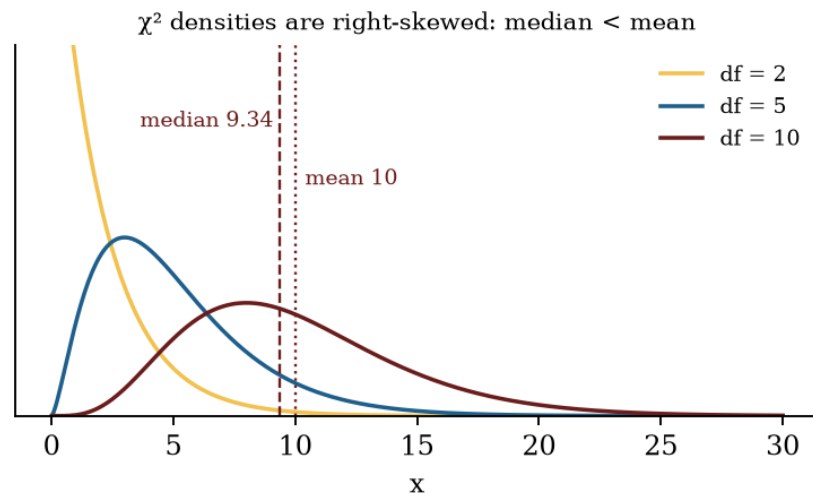
$$\text{Var}(X) = \frac{5}{12} - \frac{49}{144} = \frac{11}{144} = \text{Var}(Y)$$

$$E[XY] = \int_0^1 \int_0^1 xy(x + y) dx dy = \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{3}$$

$$\text{Cov}(X, Y) = \frac{1}{3} - \frac{49}{144} = -\frac{1}{144}$$

$$\rho = \frac{-1/144}{11/144} = -\frac{1}{11} \approx -0.091$$

Key idea: Because $\sigma_X = \sigma_Y$, $\rho = \text{Cov}/\text{Var}$. The weak negative correlation arises because the density $x + y$ piles mass in the corners where one variable is large.

Q.22 [MSQ · 2 marks]**Topic: Chi-squared and t relationships · Medium****Correct answer: A, B****Concept:** definitions — χ^2_n is a sum of n squared independent $N(0,1)$; $t_n = Z/\sqrt{V/n}$.**(A) TRUE:** this is the definition of Student's t with n degrees of freedom.**(B) TRUE:** each squared normal has mean 1 and variance 2; add n of them.**(C) FALSE:** it is $Z^2 + V$ that is χ^2_{n+1} . $Z + V$ can be negative, but a chi-squared variable never is.**(D) FALSE:** χ^2 is right-skewed, so its median is below its mean. For $n = 10$ the median is 9.34 and $P(V > 10) = 0.4405 < 0.5$.**Key idea:** Useful facts: median of $\chi^2_k \approx k(1 - 2/(9k))^3$; for large k , $\chi^2_k \approx N(k, 2k)$.**Q.23** [MCQ · 2 marks · -2/3]**Topic: Exponential: minimum and maximum · Hard****Correct answer: C****Concept:** min of independent exponentials is exponential with summed rates; $\max = X + Y - \min$.Rates: $\lambda_1 = 1/2$, $\lambda_2 = 1/3$. The first failure time is

$$\min(X, Y) \sim \text{Exp}\left(\frac{1}{2} + \frac{1}{3} = \frac{5}{6}\right), \quad E[\min] = \frac{6}{5} = 1.2$$

Since $\max + \min = X + Y$,

$$E[\max] = E[X] + E[Y] - E[\min] = 2 + 3 - 1.2 = 3.8$$

Direct check: $E[\max] = \int_0^\infty [1 - (1 - e^{-t/2})(1 - e^{-t/3})] dt = 2 + 3 - 6/5 = 3.8$.5 hours = $E[X + Y]$ (a standby/backup system, not parallel); 1.2 is the first failure; 3.0 is the larger mean.**Shortcut:** $E[\max]$ always exceeds the larger individual mean (3) but is less than the sum (5).

Q.24 [NAT · 2 marks]

Topic: Median, mean and mode from a pdf · Medium

Correct answer: 0.614 (range 0.612 to 0.616)**Concept:** median m solves $F(m) = 1/2$; mode maximises f ; mean $= \int x f(x) dx$.

Step 1 — CDF:

$$F(x) = \int_0^x 12t^2(1-t) dt = 4x^3 - 3x^4$$

Step 2 — solve $4m^3 - 3m^4 = 0.5$ numerically (bisection / Newton):

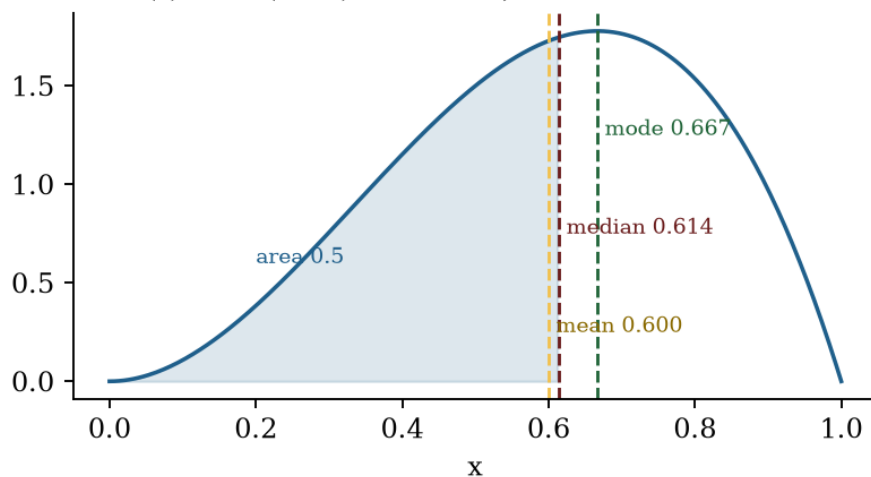
m	0.60	0.61	0.614	0.615	0.62
F(m)	0.4752	0.4925	0.4995	0.5013	0.5100

$$m \approx 0.6143$$

For comparison:

$$E[X] = \int_0^1 12x^3(1-x) dx = 12 \left(\frac{1}{4} - \frac{1}{5} \right) = \frac{3}{5}$$

$$f'(x) = 24x - 36x^2 = 0 \Rightarrow \text{mode} = \frac{2}{3}$$

 $f(x) = 12x^2(1-x)$: left-skewed, mean < median < mode

Key idea: For a left-skewed density, mean < median < mode ($0.600 < 0.614 < 0.667$) — the reverse of the right-skewed order.

Q.25 [MCQ · 2 marks · -2/3]**Topic: Hierarchical model: marginal probability · Hard****Correct answer: B**

Concept: continuous law of total probability: $P(A) = \int P(A | X = x) f_X(x) dx$.

Given $X = x$, $P(Y > 1 | X = x) = e^{-x}$.

$$P(Y > 1) = \int_0^1 e^{-x} \cdot 1 dx = 1 - e^{-1} \approx 0.632$$

Note that $1/e = P(Y > 1 | X = 1)$ would be the answer only if the rate were fixed at 1.

For completeness, the marginal pdf of Y is

$$f_Y(y) = \int_0^1 x e^{-xy} dx = \frac{1 - (1+y)e^{-y}}{y^2}, \quad y > 0$$

and $E[Y] = E[1/X] = \int_0^1 dx/x = \infty$ — small rates make Y huge.

Key idea: Mixing over the rate gives a heavier tail than any single exponential: $P(Y > 1) = 0.632$ exceeds $e^{-0.5} = 0.607$ (the exponential with the average rate 0.5).

Q.26 [MCQ · 2 marks · -2/3]**Topic: Binomial in two stages · Medium****Correct answer: D**

Concept: stage 1 — Binomial(10, 0.2) for failures in one array; stage 2 — the number of surviving arrays is Binomial(5, p_s) with mean $5p_s$.

$$P(X = 0) = 0.8^{10} \approx 0.10737$$

$$P(X = 1) = 10(0.2)(0.8)^9 \approx 0.26844$$

$$P(X = 2) = \binom{10}{2}(0.2)^2(0.8)^8 = 45(0.04)(0.16777) \approx 0.30199$$

$$p_s = P(X \leq 2) \approx 0.67780, \quad E[\text{survivors}] = 5p_s \approx 3.389$$

$1.61 = 5 \times P(X \geq 3)$ counts failing arrays; 1.88 uses $P(X < 2)$ (forgets $X = 2$); 0.68 is p_s itself.

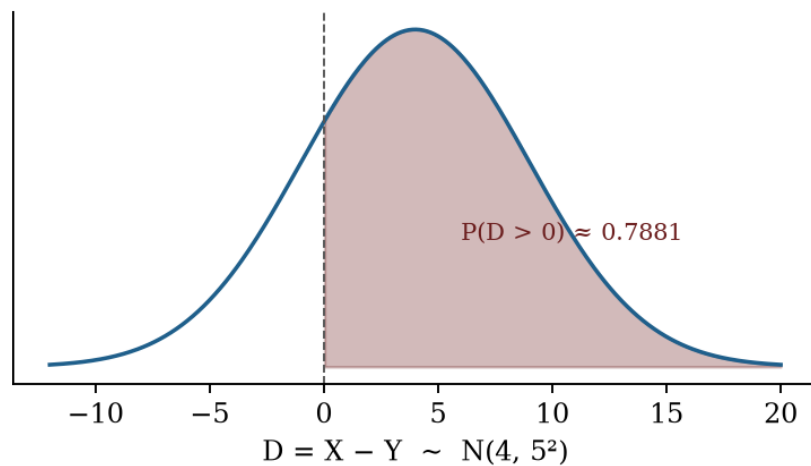
Shortcut: Linearity: $E[\sum I_j] = 5 P(\text{array survives})$ — no need for the full distribution of the number of survivors.

Q.27 [MCQ · 2 marks · -2/3]**Topic: Linear combinations of normals · Medium****Correct answer: A**

Concept: independent normals: $X - Y \sim N(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2)$ — variances ADD even for a difference.

$$D = X - Y \sim N(10 - 6, 9 + 16) = N(4, 25), \quad \sigma_D = 5$$

$$P(X > Y) = P(D > 0) = P\left(Z > \frac{0 - 4}{5}\right) = P(Z > -0.8) = \Phi(0.8) = 0.7881$$



0.2119 is $P(X < Y)$; $0.8413 = \Phi(1)$ uses $\sigma_D = 4$ ($\sqrt{16}$); $0.7157 = \Phi(4/7)$ adds the SDs ($3 + 4 = 7$) instead of the variances.

Trap: $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$ for independent X, Y — never subtract variances.

Topic: Distribution of max/min of uniforms (order statistics) · Hard

Q.28 [MSQ · 2 marks]

Correct answer: B, D

Concept: $P(M \leq m) = P(\text{all} \leq m) = m^3$; $P(L > l) = P(\text{all} > l) = (1 - l)^3$.

CDF and pdf of M :

$$F_M(m) = m^3, \quad f_M(m) = 3m^2, \quad 0 < m < 1$$

(A) FALSE: $3(1 - m)^2$ is the pdf of the minimum L , not of M .

(B) TRUE:

$$E[M] = \int_0^1 m \cdot 3m^2 dm = \frac{3}{4}$$

(C) FALSE: $P(L > 0.5) = (0.5)^3 = 1/8$.

(D) TRUE:

$$m^3 = \frac{1}{2} \Rightarrow m = 2^{-1/3} \approx 0.7937$$

Shortcut: General n : $E[\max] = n/(n + 1)$, $E[\min] = 1/(n + 1)$ for n i.i.d. $U(0, 1)$.

Topic: Conditional expectation of a truncated exponential · Hard

Q.29 [NAT · 2 marks]

Correct answer: 0.687 (range 0.685 to 0.689)

Concept: truncated density $f(x)/P(X < 2)$ on $(0, 2)$; memorylessness does NOT apply to conditioning on $X < a$.

$$\begin{aligned}
 P(X < 2) &= 1 - e^{-2} \\
 \int_0^2 x e^{-x} dx &= [-(1+x)e^{-x}]_0^2 = 1 - 3e^{-2} \\
 E[X | X < 2] &= \frac{1 - 3e^{-2}}{1 - e^{-2}} = 1 - \frac{2e^{-2}}{1 - e^{-2}} = 1 - \frac{2}{e^2 - 1} \\
 &= 1 - \frac{2}{6.3891} \approx 0.6870
 \end{aligned}$$

Contrast: $E[X | X > 2] = 2 + 1 = 3$ by memorylessness. Check with total expectation: $(1 - e^{-2})(0.6870) + e^{-2}(3) = 0.5940 + 0.4060 = 1 = E[X]$ (verified).

Trap: Applying 'memorylessness' backwards (e.g. answering 1 or $2 - 1$) is wrong; only the upper tail $X > a$ is a shifted copy of the original.

Topic: Sum of independent uniforms (convolution) · Medium

Q.30 [MCQ · 2 marks · -2/3]

Correct answer: C

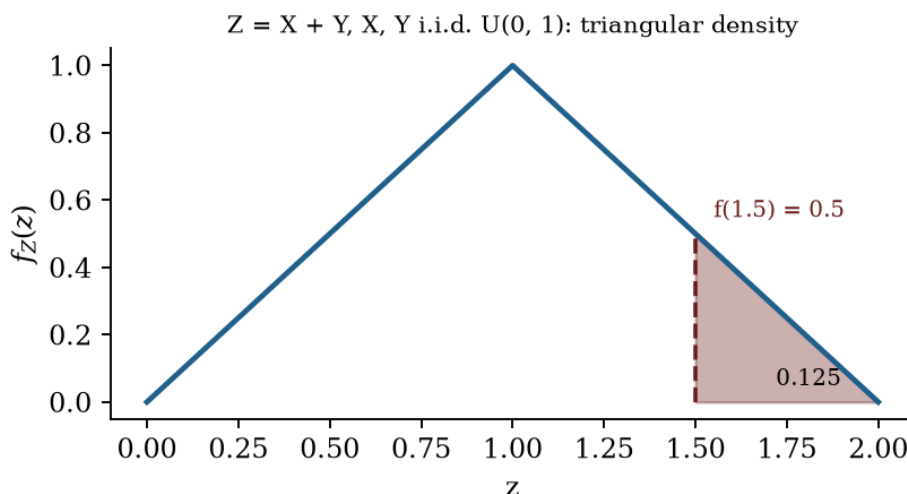
Concept: convolution $f_Z(z) = \int f_X(x) f_Y(z-x) dx$ gives the triangular density on $(0, 2)$.

For $1 \leq z \leq 2$, the integrand is 1 when $\max(0, z-1) < x < \min(1, z)$, i.e. $z-1 < x < 1$:

$$f_Z(z) = \int_{z-1}^1 1 dx = 2 - z, \quad f_Z(1.5) = 0.5$$

Tail probability = area of the small triangle:

$$P(Z > 1.5) = \int_{1.5}^2 (2 - z) dz = \frac{1}{2}(0.5)(0.5) = 0.125$$



Geometric check: $Z > 1.5$ is the corner triangle $x + y > 1.5$ in the unit square, area $(0.5)^2/2 = 1/8$.

Trap: A density value (0.5) is not a probability; the area beyond 1.5 is 0.125.

Mock Test 07 — Topic Test: Sampling, CLT, Confidence Intervals & Hypothesis Testing

An inference-intensive set: sampling distributions, the CLT, $z/t/\chi^2$ intervals and tests, errors and power.

Item	Details
Questions	30 (12 × 1-mark, 18 × 2-mark)
Maximum marks	48
Question types	MCQ: 14 · MSQ: 7 · NAT: 9
Suggested time	90 minutes
Negative marking	MCQ only: $-1/3$ (1-mark), $-2/3$ (2-mark). None for MSQ & NAT.
Calculator	Use only the GATE virtual scientific calculator

Topic focus of this set

Sampling distribution of the mean and S^2 ; CLT for sums and means; normal approximation to the binomial with continuity correction; z and t confidence intervals for means and proportions; sample-size planning; z -tests, one-sample/paired/pooled t -tests; χ^2 goodness-of-fit and independence; χ^2 interval and test for a variance; Type I/II errors and power; properties of the t and χ^2 distributions.

Instructions

- Questions 1 onwards are arranged as in GATE: all 1-mark questions first, then 2-mark questions.
- **MCQ** – exactly one correct option. **MSQ** – ONE OR MORE options are correct; credit only if you select all correct options and no wrong option (no partial marks, no negative marks).
- **NAT** – enter a numerical value using the virtual keypad. The accepted range is given in the key; round as instructed in the question.
- Statistical tables (Z , t , χ^2) are provided in the Appendix; values in solutions use those tables.
- Attempt in one sitting, then check the answer key and read every solution — including those you got right — for the alternative methods and traps.

Mock Test 07 · Question Paper

1-mark questions

Q.1 MCQ · 1 mark · -1/3

A quality engineer samples $n = 36$ cartons from a filling line whose fill volume has standard deviation $\sigma = 12$ mL. To make the standard error of the sample mean exactly half of its current value, the sample size must be

- (A) 72
- (B) 144
- (C) 108
- (D) 288

Q.2 NAT · 1 mark

The time a hospital pharmacy takes to process a prescription is normally distributed with known standard deviation $\sigma = 6$ minutes. A random sample of 25 prescriptions has mean 52.4 minutes. The **upper limit** of the 95% confidence interval for the population mean is _____ (round off to 2 decimal places). [Use $z_{0.025} = 1.96$.]

Answer: _____

Q.3 MSQ · 1 mark

From a sample of tomato plants an agronomist reports a 95% confidence interval for the mean yield per plant, μ , as (48.2, 53.8) kg, computed as $\bar{x} \pm z_{0.025}\sigma/\sqrt{n}$. Which of the following statements is/are correct?

- (A) If the sampling and interval construction were repeated many times, about 95% of the intervals so obtained would contain μ .
- (B) $P(48.2 \leq \mu \leq 53.8) = 0.95$, where μ is the fixed population mean.
- (C) The sample mean yield was 51.0 kg.
- (D) About 95% of individual plants yield between 48.2 kg and 53.8 kg.

Q.4 MCQ · 1 mark · -1/3

A random variable T follows Student's t -distribution with 6 degrees of freedom. $\text{Var}(T)$ equals

- (A) 1
- (B) 1.2
- (C) 1.5
- (D) 3

Q.5 MCQ · 1 mark · -1/3

A survey cross-classifies 600 students by **study programme** (4 categories) and **preferred mode of learning** (online, hybrid, offline). For the χ^2 test of independence on this contingency table, the degrees of freedom are

- (A) 12
- (B) 11
- (C) 6
- (D) 599

Q.6 NAT · 1 mark

If $X \sim \chi^2_{10}$ (chi-squared with 10 degrees of freedom), then $E(X^2) = \underline{\hspace{2cm}}$ (answer as an integer).

Answer:

Q.7 MCQ · 1 mark · -1/3

A two-sided test of $H_0: \mu = \mu_0$ uses the standardised statistic Z , which is $N(0, 1)$ under H_0 , and rejects H_0 when $|Z| > 2.33$. [$\Phi(2.33) = 0.9901$.] The probability of a Type I error is closest to

- (A) 0.0099
- (B) 0.0198
- (C) 0.9802
- (D) 0.0500

Q.8 MSQ · 1 mark

Let $t_{\alpha, \nu}$ denote the upper- α point of Student's t -distribution with ν degrees of freedom. Which of the following statements is/are TRUE?

- (A) The t -distribution with ν degrees of freedom is symmetric about 0.
- (B) The variance of the t -distribution with 5 degrees of freedom is $5/3$.
- (C) $t_{0.025, 10} < 1.96$
- (D) For fixed α , $t_{\alpha, \nu}$ decreases as ν increases and approaches z_{α} .

Q.9 NAT · 1 mark

A fair coin is tossed 100 times. Using the normal approximation **with continuity correction**, $P(\text{number of heads} \geq 60)$ is approximately (round off to 3 decimal places). [$\Phi(1.90) = 0.9713$.]

Answer:

Q.10 MCQ · 1 mark · -1/3

The breaking strength of a cable has known $\sigma = 15$ N. To test $H_0: \mu = 200$ N, a sample of 36 cables gives $\bar{x} = 204.5$ N. The value of the z -test statistic is

- (A) 0.30
- (B) 1.80
- (C) 0.75
- (D) 0.125

Q.11 MCQ · 1 mark · -1/3

The resting heart rate of each of 12 athletes is measured before and after a 6-week training programme. To test whether the programme lowers the mean heart rate (population SD unknown), the appropriate test and its degrees of freedom are

- (A) Two-sample pooled t -test with 22 df
- (B) Paired t -test with 11 df
- (C) Paired t -test with 12 df
- (D) Two-sample pooled t -test with 23 df

Q.12 NAT · 1 mark

A nutritionist wants to estimate the mean daily sugar intake (grams) of teenagers to within ± 2 g with 95% confidence. A pilot study suggests $\sigma = 8$ g. The minimum sample size required is _____ (answer as an integer). [Use $z_{0.025} = 1.96$.]

Answer: _____

2-mark questions**Q.13 NAT · 2 marks**

A mill fills rice bags whose weights are independent with mean 25 kg and standard deviation 0.5 kg (distribution not necessarily normal). A truck is loaded with 50 such bags. Using the central limit theorem, the probability that the total load exceeds 1252 kg is _____ (round off to 3 decimal places).

Answer: _____

Q.14 MCQ · 2 marks · -2/3

In a pre-launch survey, 248 out of 400 randomly chosen customers say they would buy a new fitness tracker. The approximate 95% confidence interval for the population proportion p is [$z_{0.025} = 1.96$]

- (A) (0.572, 0.668)
- (B) (0.580, 0.660)
- (C) (0.557, 0.683)
- (D) (0.596, 0.644)

Q.15 MCQ · 2 marks · -2/3

A new fertiliser is claimed to raise the mean wheat yield above 10 quintals per acre. On 16 randomly chosen plots the yields give $\bar{x} = 10.6$ and $s = 1.2$ (yields are normal). For $H_0: \mu = 10$ vs $H_1: \mu > 10$ [$t_{0.05,15} = 1.753$, $t_{0.01,15} = 2.602$], the correct conclusion is

- (A) Reject H_0 at both the 5% and 1% levels
- (B) Reject H_0 at the 5% level but not at the 1% level
- (C) Do not reject H_0 at the 5% level
- (D) No conclusion is possible because σ is unknown

Q.16 NAT · 2 marks

Systolic blood pressure (mmHg) of 8 patients was recorded before and after a month on a new drug:

Patient	1	2	3	4	5	6	7	8
Before	148	152	139	160	145	155	150	142
After	141	149	140	151	138	150	146	139

For testing whether the drug reduces mean blood pressure, the value of the paired t statistic (based on $d = \text{Before} - \text{After}$) is _____ (round off to 2 decimal places).

Answer: _____

Q.17 NAT · 2 marks

Two teaching methods are compared. Method I: $n_1 = 10$ students, mean score 82, sample SD 6. Method II: $n_2 = 12$ students, mean score 76, sample SD 8. Assuming normal populations with equal variances, the pooled two-sample t statistic for $H_0: \mu_1 = \mu_2$ is _____ (round off to 2 decimal places).

Answer: _____

Q.18 MCQ · 2 marks · -2/3

Mendelian theory predicts that a dihybrid pea cross produces Round-Yellow, Round-Green, Wrinkled-Yellow and Wrinkled-Green seeds in the ratio 9 : 3 : 3 : 1. Among 160 seeds the observed counts are 80, 38, 28 and 14 respectively. [$\chi^2_{0.05,3} = 7.815$, $\chi^2_{0.05,4} = 9.488$]. Which option correctly gives the test statistic, its df and the decision at the 5% level?

- (A) $\chi^2 \approx 4.98$, df = 3, do not reject the 9:3:3:1 hypothesis
 (B) $\chi^2 \approx 4.98$, df = 4, do not reject the 9:3:3:1 hypothesis
 (C) $\chi^2 \approx 4.22$, df = 3, do not reject the 9:3:3:1 hypothesis
 (D) $\chi^2 \approx 4.98$, df = 3, reject the 9:3:3:1 hypothesis

Q.19 NAT · 2 marks

In a clinical trial, 200 patients were randomised equally to two drugs and the outcome recorded:

	Improved	No change	Worse	Total
Drug A	45	30	25	100
Drug B	60	25	15	100
Total	105	55	40	200

The value of the χ^2 statistic for testing independence of drug and outcome is _____ (round off to 2 decimal places).

Answer: _____

Q.20 MSQ · 2 marks

In a vaccine field study, 400 children were followed for one season:

	Infected	Not infected	Total
Vaccinated	18	182	200
Unvaccinated	42	158	200
Total	60	340	400

For the χ^2 test of independence (without Yates' correction), which statements are correct?
 [$\chi^2_{0.01,1} = 6.635$]

- (A) The expected count for the cell (Vaccinated, Infected) is 30.
 (B) The test has 2 degrees of freedom.
 (C) The χ^2 statistic is approximately 11.29.
 (D) Independence of vaccination status and infection is NOT rejected at the 1% level.

Q.21 MCQ · 2 marks · -2/3

A pharmaceutical plant checks the variability of tablet weight. A random sample of 10 tablets has sample variance $s^2 = 4.5 \text{ mg}^2$. Assuming normality, the 95% confidence interval for the population variance σ^2 is $[\chi^2_{0.025,9} = 19.023, \chi^2_{0.975,9} = 2.700]$, where $\chi^2_{\alpha,\nu}$ is the upper- α point]

- (A) (2.13, 15.00)
- (B) (2.37, 16.67)
- (C) (1.46, 3.87)
- (D) (0.24, 1.67)

Q.22 MCQ · 2 marks · -2/3

A bottling machine is acceptable only if the SD of fill volume does not exceed 0.5 mL. A sample of 20 bottles gives $s = 0.65 \text{ mL}$. For $H_0: \sigma = 0.5$ vs $H_1: \sigma > 0.5$ (normal fills), $[\chi^2_{0.05,19} = 30.144, \chi^2_{0.01,19} = 36.191]$, which option is correct?

- (A) $\chi^2 = 32.11$; reject H_0 at 5% but not at 1%
- (B) $\chi^2 = 33.80$; reject H_0 at 5% but not at 1%
- (C) $\chi^2 = 24.70$; do not reject H_0 at 5%
- (D) $\chi^2 = 32.11$; reject H_0 at both 5% and 1%

Q.23 NAT · 2 marks

Battery life is normal with $\sigma = 10 \text{ h}$. To test $H_0: \mu = 50 \text{ h}$ against $H_1: \mu > 50 \text{ h}$, a sample of 25 batteries is used with $\alpha = 0.05$ ($z_{0.05} = 1.645$). If the true mean is 54 h, the power of the test is _____ (round off to 3 decimal places).

Answer: _____

Q.24 MCQ · 2 marks · -2/3

For the setting $H_0: \mu = \mu_0$ vs $H_1: \mu > \mu_0$ with $\sigma = 10$ and a one-sided z-test at $\alpha = 0.05$, a researcher wants power 0.90 when the true mean is $\mu_0 + 4$. [$z_{0.05} = 1.645$, $z_{0.10} = 1.2816$, $z_{0.20} = 0.8416$, $z_{0.025} = 1.96$]. The minimum sample size is

- (A) 39
- (B) 54
- (C) 66
- (D) 17

Q.25 MSQ · 2 marks

Consider a z-test about a population mean with known σ . Which of the following statements is/are TRUE?

- (A) The p-value is the probability that H_0 is true given the observed data.
- (B) For a fixed α and a fixed true alternative μ_1 , increasing the sample size increases the power.
- (C) For a fixed sample size, reducing α (e.g. from 0.05 to 0.01) increases the Type II error probability β against a fixed alternative.
- (D) If the two-sided 95% confidence interval for μ does not contain μ_0 , then the two-sided test of $H_0: \mu = \mu_0$ at $\alpha = 0.05$ rejects H_0 .

Q.26 MSQ · 2 marks

Which of the following statements about confidence intervals is/are TRUE (all other quantities held fixed)?

- (A) Quadrupling the sample size halves the width of a z-interval for μ .
- (B) From the same data, the 99% z-interval for μ is wider than the 95% z-interval.
- (C) With $n = 10$ and the same s , the t-interval ($t_{0.025,9} = 2.262$) is narrower than the interval $\bar{x} \pm 1.96 s/\sqrt{n}$.
- (D) When planning a survey for a proportion, using $p = 0.5$ in $n = z^2 p(1 - p)/E^2$ gives the smallest required sample size.

Q.27 MCQ · 2 marks · -2/3

LED bulbs have exponentially distributed lifetimes with mean 100 hours. For a random sample of 64 bulbs, the approximate probability that the sample mean lifetime is less than 90 hours is [$\Phi(0.8) = 0.7881$, $\Phi(0.1) = 0.5398$]

- (A) 0.2119
- (B) 0.7881
- (C) 0.4602
- (D) 0.5934

Q.28 MSQ · 2 marks

A machine is set to cut rods of mean length μ_0 (σ known). For a two-sided test of $H_0: \mu = \mu_0$, the observed statistic is $z = -2.17$. [$\Phi(2.17) = 0.9850$.] Which of the following is/are correct?

- (A) H_0 is rejected at the 1% level.
- (B) The two-sided p-value is approximately 0.030.
- (C) For the one-sided alternative $H_1: \mu > \mu_0$, the p-value would be approximately 0.015.
- (D) H_0 is rejected at the 5% level.

Q.29 MCQ · 2 marks · -2/3

$X_1, \dots, X_{11}$ is a random sample from $N(\mu, 9)$ and S^2 is the sample variance (divisor $n - 1$). $\text{Var}(S^2)$ equals

- (A) 16.2
- (B) 14.73
- (C) 1.62
- (D) 18.0

Q.30 MSQ · 2 marks

Let $X_1, \dots, X_{16}$ be a random sample from $N(\mu, \sigma^2)$ with sample mean $\bar{X}$ and sample variance S^2 (divisor 15). Which of the following statements is/are TRUE?

- (A) $\bar{X}$ and S^2 are independent random variables.
- (B) $(\bar{X} - \mu)/(S/4)$ has exactly the standard normal distribution.
- (C) $15S^2/\sigma^2$ has a χ^2 distribution with 16 degrees of freedom.
- (D) $\sum_{i=1}^{16} (X_i - \mu)^2/\sigma^2$ has a χ^2 distribution with 16 degrees of freedom.

Mock Test 07 · Answer Key

Q	Type	Marks	Answer	Topic
1	MCQ	1	B	Sampling distribution of the mean
2	NAT	1	54.75 (range 54.74 to 54.76)	z confidence interval for μ
3	MSQ	1	A, C	Interpreting a confidence interval
4	MCQ	1	C	Properties of the t-distribution
5	MCQ	1	C	χ^2 test of independence: degrees of freedom
6	NAT	1	120	Moments of the χ^2 distribution
7	MCQ	1	B	Type I error / size of a test
8	MSQ	1	A, B, D	Properties of the t-distribution
9	NAT	1	0.029 (range 0.028 to 0.03)	Normal approximation to binomial (continuity correction)
10	MCQ	1	B	z-test statistic
11	MCQ	1	B	Choosing the t-test (paired)
12	NAT	1	62	Sample size determination for estimating μ
13	NAT	2	0.286 (range 0.282 to 0.29)	CLT for a sum
14	MCQ	2	A	Confidence interval for a proportion
15	MCQ	2	B	One-sample t-test (one-sided)
16	NAT	2	4.20 (range 4.18 to 4.22)	Paired t-test
17	NAT	2	1.95 (range 1.94 to 1.97)	Two-sample pooled t-test
18	MCQ	2	A	χ^2 goodness-of-fit test
19	NAT	2	5.10 (range 5.08 to 5.12)	χ^2 test of independence (2×3 table)
20	MSQ	2	A, C	χ^2 test for a 2×2 table
21	MCQ	2	A	Confidence interval for a variance
22	MCQ	2	A	χ^2 test for a variance
23	NAT	2	0.639 (range 0.634 to 0.643)	Power of a one-sided z-test
24	MCQ	2	B	Sample size for a desired power
25	MSQ	2	B, C, D	Hypothesis-testing concepts (α , β , power, p-value)
26	MSQ	2	A, B	Width and margin of error of confidence intervals
27	MCQ	2	A	CLT for a sample mean (exponential lifetimes)
28	MSQ	2	B, D	p-values for a z-test
29	MCQ	2	A	Sampling distribution of S^2 (χ^2)
30	MSQ	2	A, D	Sampling from a normal population (t and χ^2 facts)

Score tracker

	Attempted	Correct	Wrong (MCQ)	Marks
1-mark				
2-mark				
Total				/ 48

Mock Test 07 · Detailed Solutions

Q.1 [MCQ · 1 mark · -1/3]

Topic: Sampling distribution of the mean · Easy

Correct answer: B

Concept: the standard error of $\bar{X}$ is $SE = \sigma/\sqrt{n}$, so SE scales as $1/\sqrt{n}$.

$$SE_{\text{now}} = \frac{12}{\sqrt{36}} = 2 \text{ mL}, \quad SE_{\text{target}} = 1 \text{ mL}$$

$$\frac{12}{\sqrt{n}} = 1 \Rightarrow \sqrt{n} = 12 \Rightarrow n = 144$$

Equivalently, halving SE requires multiplying n by $2^2 = 4$: $36 \times 4 = 144$.

Option (A) doubles n , which only reduces SE by a factor $\sqrt{2}$ (to 1.414 mL).

Key idea: Precision improves with the square root of the sample size: k -fold better precision costs k^2 -fold more data.

Q.2 [NAT · 1 mark]

Topic: z confidence interval for μ · Easy

Correct answer: 54.75 (range 54.74 to 54.76)

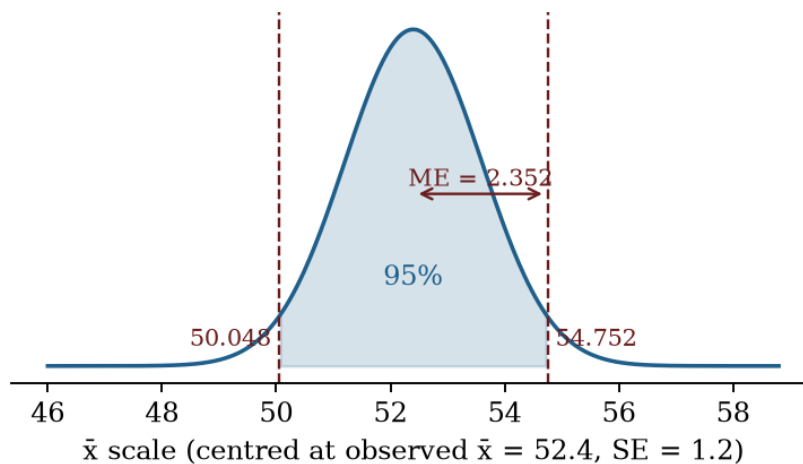
Concept: with σ known, the $100(1-\alpha)\%$ CI for μ is $\bar{x} \pm z_{\alpha/2} \sigma/\sqrt{n}$.

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{6}{\sqrt{25}} = 1.2$$

$$ME = z_{0.025} \times SE = 1.96 \times 1.2 = 2.352$$

$$\text{Upper limit} = \bar{x} + ME = 52.4 + 2.352 = 54.752 \approx 54.75$$

The full interval is (50.048, 54.752).



Key idea: Use z (not t) because σ is known and the population is normal — the sample size being only 25 does not matter here.

Q.3 [MSQ · 1 mark]**Topic: Interpreting a confidence interval · Medium****Correct answer: A, C**

Concept: the confidence level is a property of the procedure; once computed, a specific interval either contains μ or it does not.

(A) Correct. This is the frequentist meaning of '95% confidence': the random interval $\bar{X} \pm 1.96\sigma/\sqrt{n}$ covers μ with probability 0.95 before the data are seen.

(B) Incorrect. μ is a constant and (48.2, 53.8) is a fixed interval, so the probability is either 0 or 1. The 0.95 refers to the random interval, not to this realised one.

(C) Correct. A z-interval is symmetric about $\bar{x}$:

$$\bar{x} = \frac{48.2 + 53.8}{2} = 51.0 \text{ kg}, \quad ME = 2.8 \text{ kg}$$

(D) Incorrect. The interval describes where the mean lies, using $SE = \sigma/\sqrt{n}$. Individual plants vary with standard deviation σ , which is $\sqrt{n}$ times larger, so far fewer than 95% of plants fall inside this narrow interval.

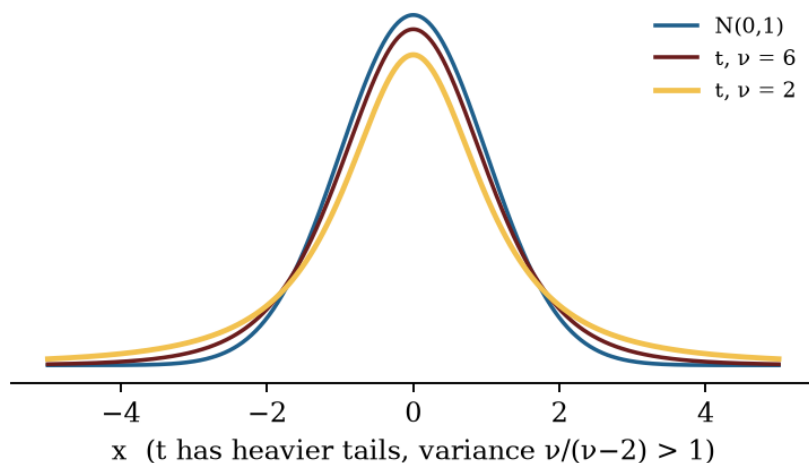
Trap: Confusing the CI for μ with a range for individual values (option D) is one of the most frequently tested traps.

Q.4 [MCQ · 1 mark · -1/3]**Topic: Properties of the t-distribution · Easy****Correct answer: C**

Concept: for $T \sim t_\nu$ with $\nu > 2$, $E(T) = 0$ and $\text{Var}(T) = \nu/(\nu - 2)$.

$$\text{Var}(T) = \frac{6}{6-2} = \frac{6}{4} = 1.5$$

Option (A) is the variance of $N(0,1)$; the t-distribution always has a variance larger than 1 because its tails are heavier. Option (B) would be $\nu/(\nu-1)$, a common misremembering.



Key idea: $\text{Var}(t_\nu) \rightarrow 1$ as $\nu \rightarrow \infty$; it is infinite for $\nu = 2$ and undefined for $\nu = 1$ (Cauchy).

Topic: χ^2 test of independence: degrees of freedom · Easy

Q.5 [MCQ · 1 mark · -1/3]

Correct answer: C

Concept: for an $r \times c$ contingency table, $df = (r - 1)(c - 1)$.

$$df = (4 - 1)(3 - 1) = 3 \times 2 = 6$$

Reason: the expected counts are fixed by the row and column totals, which impose $r + c - 1$ independent constraints on the rc cells:

$$rc - (r + c - 1) = 12 - 6 = 6$$

(B) is $rc - 1$, the df of a goodness-of-fit test over 12 categories with no estimated parameters.

Trap: The sample size (600) never enters the degrees of freedom of a χ^2 contingency test.

Q.6 [NAT · 1 mark]

Topic: Moments of the χ^2 distribution · Medium

Correct answer: 120

Concept: $X \sim \chi^2_k$ has $E(X) = k$ and $\text{Var}(X) = 2k$.

$$E(X) = 10, \quad \text{Var}(X) = 2(10) = 20$$

$$E(X^2) = \text{Var}(X) + [E(X)]^2 = 20 + 100 = 120$$

Check via sum representation: $X = Z_1^2 + \dots + Z_{10}^2$, and each Z_i^2 has mean 1 and variance $E(Z^4) - 1 = 3 - 1 = 2$.

Shortcut: Mean k , variance $2k$ — memorise both; they generate most χ^2 moment questions.

Q.7 [MCQ · 1 mark · -1/3]

Topic: Type I error / size of a test · Easy

Correct answer: B

Concept: $P(\text{Type I error}) = \alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$.

$$\alpha = P(|Z| > 2.33) = 2[1 - \Phi(2.33)] = 2(0.0099) = 0.0198$$

(A) counts only one tail — it would be the size of the one-sided test $Z > 2.33$. (C) is the probability of correctly retaining H_0 .

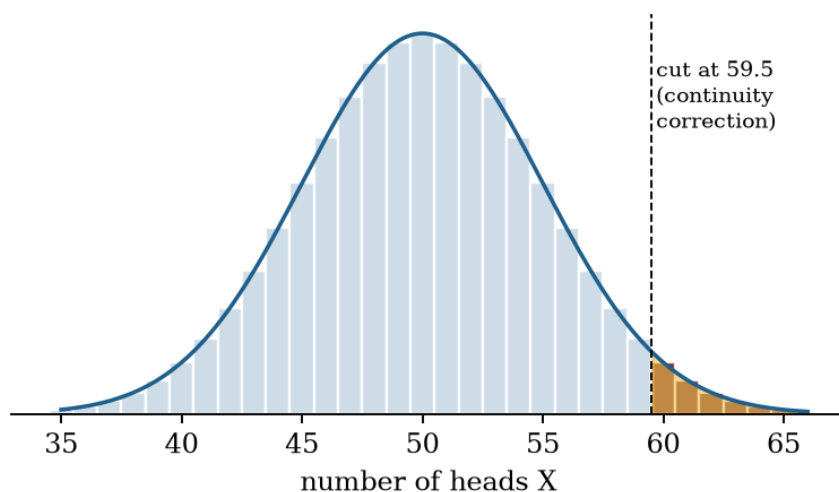
Trap: 2.33 is the one-sided 1% point; using it in a two-sided rule gives $\alpha \approx 2\%$.

Q.8 [MSQ · 1 mark]**Topic: Properties of the t-distribution · Medium****Correct answer: A, B, D****Concept:** $T = Z/\sqrt{V/\nu}$ with $Z \sim N(0,1)$, $V \sim \chi^2_\nu$ independent; the extra randomness in the denominator fattens the tails.**(A) True.** The density is proportional to $(1 + t^2/\nu)^{-(\nu+1)/2}$, an even function of t .**(B) True.**

$$\text{Var}(t_5) = \frac{5}{5-2} = \frac{5}{3}$$

(C) False. Heavier tails push critical values outward: $t_{0.025,10} = 2.228 > 1.96$.**(D) True.** As $\nu \rightarrow \infty$, $V/\nu \rightarrow 1$, so $t_\nu \rightarrow N(0,1)$ and the critical values decrease to z_α (e.g. 2.228, 2.086, 2.042, ... $\rightarrow 1.960$).**Key idea:** Every t critical value exceeds the matching z value; that is why t -intervals are wider.**Q.9** [NAT · 1 mark]**Topic: Normal approximation to binomial (continuity correction) · Medium****Correct answer: 0.029 (range 0.028 to 0.03)****Concept:** $X \sim \text{Bin}(n, p) \approx N(np, np(1-p))$; for $P(X \geq k)$ use the cut $k - 0.5$.

$$\begin{aligned} \mu &= np = 50, & \sigma &= \sqrt{np(1-p)} = \sqrt{25} = 5 \\ P(X \geq 60) &= P(X > 59.5) \approx P\left(Z > \frac{59.5 - 50}{5}\right) = P(Z > 1.90) \\ &= 1 - 0.9713 = 0.0287 \end{aligned}$$

The exact binomial value is 0.0284, so the corrected approximation is excellent. Without the correction one gets $P(Z > 2) = 0.0228$, a noticeably worse answer.**Trap:** ' ≥ 60 ' includes the bar at 60, which spans 59.5 to 60.5 — so the cut is 59.5, not 60.5.

Q.10 [MCQ · 1 mark · -1/3]**Topic: z-test statistic · Easy****Correct answer: B****Concept:** $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$.

$$z = \frac{204.5 - 200}{15/\sqrt{36}} = \frac{4.5}{2.5} = 1.80$$

(A) divides by σ instead of $\sigma/\sqrt{n}$ ($4.5/15$). (C) divides by $\sqrt{n}$ only ($4.5/6$). (D) divides by n ($4.5/36$).

Trap: Always standardise with the standard error $\sigma/\sqrt{n}$, never with σ itself.

Q.11 [MCQ · 1 mark · -1/3]**Topic: Choosing the t-test (paired) · Easy****Correct answer: B**

Concept: two measurements on the same subject are dependent; analyse the differences $d_i = \text{before}_i - \text{after}_i$ as one sample.

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \quad df = n - 1 = 12 - 1 = 11$$

(A) and (D) treat the 24 readings as two independent samples, ignoring the pairing (which removes athlete-to-athlete variability). (C) forgets that one df is used to estimate the mean difference.

Key idea: Same units measured twice → paired. Different units in two groups → two-sample.

Q.12 [NAT · 1 mark]**Topic: Sample size determination for estimating μ · Medium****Correct answer: 62**

Concept: require $z_{\alpha/2} \sigma/\sqrt{n} \leq E$, i.e. $n \geq (z_{\alpha/2} \sigma / E)^2$.

$$n \geq \left(\frac{1.96 \times 8}{2} \right)^2 = (7.84)^2 = 61.47$$

Since n must be an integer and 61 would give a margin slightly above 2 g, round **up**: $n = 62$.

$$\text{Check: } 1.96 \times 8/\sqrt{62} = 1.991 \leq 2, \quad 1.96 \times 8/\sqrt{61} = 2.008 > 2$$

Trap: Sample-size answers are always rounded UP, even $61.01 \rightarrow 62$.

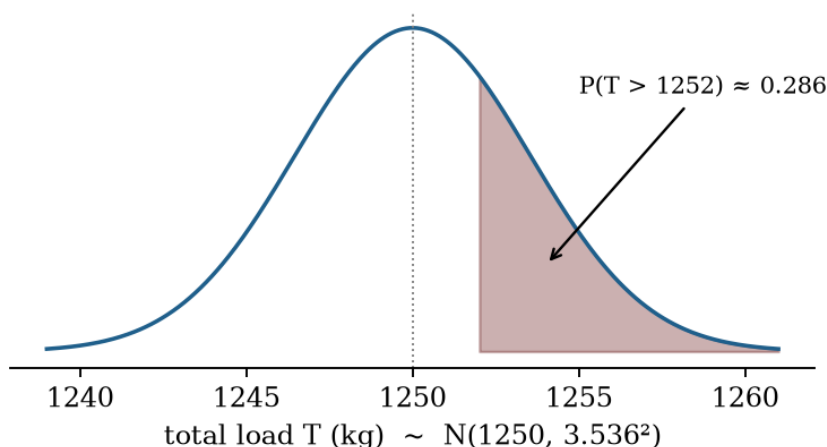
Q.13 [NAT · 2 marks]**Topic: CLT for a sum · Medium****Correct answer: 0.286 (range 0.282 to 0.29)****Concept (CLT):** for iid X_i with mean μ and variance σ^2 , the sum $T = \sum X_i$ is approximately $N(n\mu, n\sigma^2)$ for large n .

$$E(T) = 50 \times 25 = 1250 \text{ kg}$$

$$\text{Var}(T) = 50 \times 0.5^2 = 12.5, \quad SD(T) = \sqrt{12.5} = 3.536 \text{ kg}$$

$$P(T > 1252) \approx P\left(Z > \frac{1252 - 1250}{3.536}\right) = P(Z > 0.566)$$

$$= 1 - \Phi(0.566) \approx 1 - 0.7142 = 0.2858$$

With the table value $z \approx 0.57$: $1 - 0.7157 = 0.2843$ (inside the accepted range).Common errors: using $SD(T) = 50 \times 0.5 = 25$ (adding standard deviations instead of variances) gives $P \approx 0.468$; using $\sigma/\sqrt{n}$ treats the total as if it were the mean.**Key idea:** Variances add for independent variables: SD of a sum grows like $\sqrt{n}$, SD of a mean shrinks like $1/\sqrt{n}$.**Q.14** [MCQ · 2 marks · -2/3]**Topic: Confidence interval for a proportion · Medium****Correct answer: A****Concept:** for large n , $\hat{p} \pm z_{\alpha/2} \sqrt{(\hat{p}(1 - \hat{p})/n)}$.

$$\hat{p} = \frac{248}{400} = 0.62$$

$$SE = \sqrt{\frac{0.62 \times 0.38}{400}} = \sqrt{0.000589} = 0.02427$$

$$ME = 1.96 \times 0.02427 = 0.0476$$

$$CI = 0.62 \pm 0.0476 = (0.572, 0.668)$$

Option	What it computes
(A)	correct: $0.62 \pm 1.96 \text{ SE}$
(B)	uses 1.645 (a 90% interval)
(C)	uses 2.576 (a 99% interval)

Option	What it computes
(D)	$0.62 \pm 1 \text{ SE}$ (forgets the z multiplier)

Validity check: $n\hat{p} = 248$ and $n(1 - \hat{p}) = 152$ are both large, so the normal approximation is fine.

Trap: For a CI use $\hat{p}$ in the SE; for a test of $H_0: p = p_0$ use p_0 .

Q.15 [MCQ · 2 marks · -2/3]

Topic: One-sample t-test (one-sided) · Medium

Correct answer: B

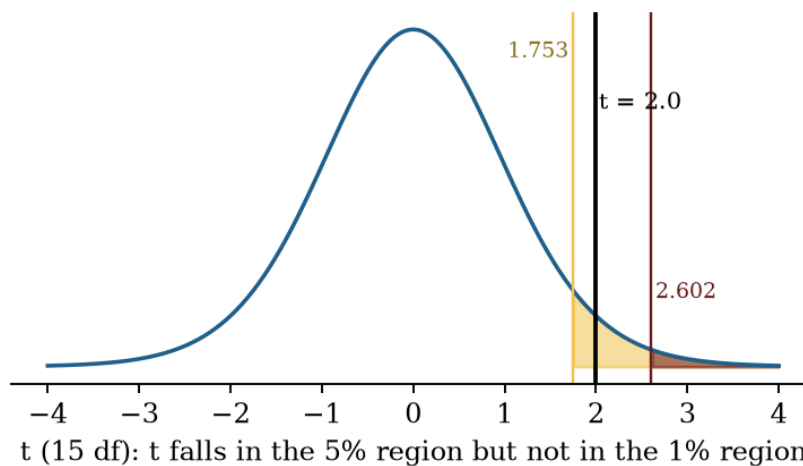
Concept: σ unknown, normal data $\rightarrow t = (\bar{x} - \mu_0)/(s/\sqrt{n})$ with $n - 1$ df.

$$t = \frac{10.6 - 10}{1.2/\sqrt{16}} = \frac{0.6}{0.3} = 2.0, \quad df = 15$$

Right-tailed test: reject H_0 when $t > t_{\alpha, 15}$.

$$2.0 > 1.753 \Rightarrow \text{reject at 5\%}; \quad 2.0 < 2.602 \Rightarrow \text{retain at 1\%}$$

Equivalently the p-value lies between 0.01 and 0.05 (it is ≈ 0.032).



(D) is wrong: an unknown σ is exactly the situation the t-test is designed for.

Trap: If 1.96 or 1.645 were (wrongly) used here the decision at 5% would be the same, but at other borderline values the t vs z choice changes the verdict.

Q.16 [NAT · 2 marks]**Topic: Paired t-test · Medium****Correct answer: 4.20 (range 4.18 to 4.22)**

Concept: paired $t = \bar{d} / (s_d/\sqrt{n})$ with $n - 1$ df, where d_i are the within-patient differences. Differences d_i : 7, 3, -1, 9, 7, 5, 4, 3.

$$\bar{d} = \frac{37}{8} = 4.625$$

$$\sum (d_i - \bar{d})^2 = 67.875 \quad (\text{or } \sum d_i^2 - 8\bar{d}^2 = 239 - 171.125)$$

$$s_d^2 = \frac{67.875}{7} = 9.6964, \quad s_d = 3.1139$$

$$SE = \frac{s_d}{\sqrt{8}} = 1.1009$$

$$t = \frac{4.625}{1.1009} = 4.201$$

With 7 df, $t_{0.05,7} = 1.895$ and $t_{0.01,7} = 2.998$, so the reduction is significant even at the 1% level.

Key idea: A two-sample t-test on these columns would give $t \approx 1.50$ — the pairing removes the large patient-to-patient variation and reveals the effect.

Q.17 [NAT · 2 marks]**Topic: Two-sample pooled t-test · Medium****Correct answer: 1.95 (range 1.94 to 1.97)**

Concept: pooled variance $s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$; $t = (\bar{x}_1 - \bar{x}_2) / [s_p \sqrt{(1/n_1 + 1/n_2)}]$.

$$s_p^2 = \frac{9(36) + 11(64)}{10 + 12 - 2} = \frac{324 + 704}{20} = 51.4$$

$$s_p = \sqrt{51.4} = 7.1694$$

$$\sqrt{\frac{1}{10} + \frac{1}{12}} = \sqrt{0.18333} = 0.4282$$

$$SE = 7.1694 \times 0.4282 = 3.0697$$

$$t = \frac{82 - 76}{3.0697} = 1.9546$$

Degrees of freedom: $n_1 + n_2 - 2 = 20$. With $t_{0.025,20} = 2.086$, a two-sided 5% test would NOT reject H_0 (but a one-sided test with $t_{0.05,20} = 1.725$ would).

Trap: Pool the variances weighted by df (9 and 11), not by n , and never average the SDs.

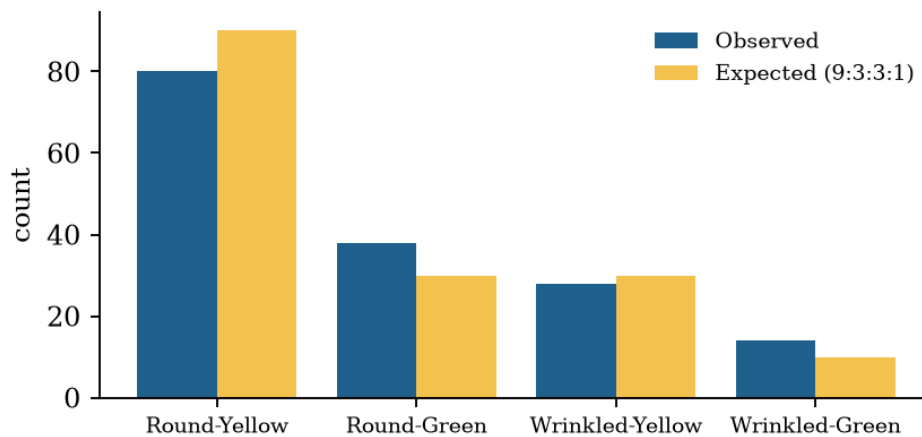
Q.18 [MCQ · 2 marks · -2/3]**Topic: χ^2 goodness-of-fit test · Medium****Correct answer: A****Concept:** $\chi^2 = \sum (O - E)^2/E$ with $df = (\text{number of categories}) - 1 - (\text{parameters estimated})$.Expected counts: $160 \times (9, 3, 3, 1)/16 = 90, 30, 30, 10$.

Class	O	E	$(O-E)^2/E$
RY	80	90	1.1111
RG	38	30	2.1333
WY	28	30	0.1333
WG	14	10	1.6000
Total	160	160	4.9778

$$\chi^2 = \frac{100}{90} + \frac{64}{30} + \frac{4}{30} + \frac{16}{10} = 4.978$$

No parameter is estimated (the ratio is fully specified), so $df = 4 - 1 = 3$.

$$4.98 < 7.815 \Rightarrow \text{do not reject } H_0$$

(B) uses the wrong df (number of categories). (C) divides by O instead of E . (D) gets the decision backwards.**Key idea:** Large $\chi^2 \Rightarrow$ poor fit $\Rightarrow$ reject; the GOF test is always upper-tailed.**Q.19** [NAT · 2 marks]**Topic: χ^2 test of independence (2×3 table) · Medium****Correct answer: 5.10 (range 5.08 to 5.12)****Concept:** $E_{ij} = (\text{row total} \times \text{column total})/n$; $\chi^2 = \sum (O - E)^2/E$ with $(r-1)(c-1)$ df .

Expected counts (identical for both rows because both row totals are 100):

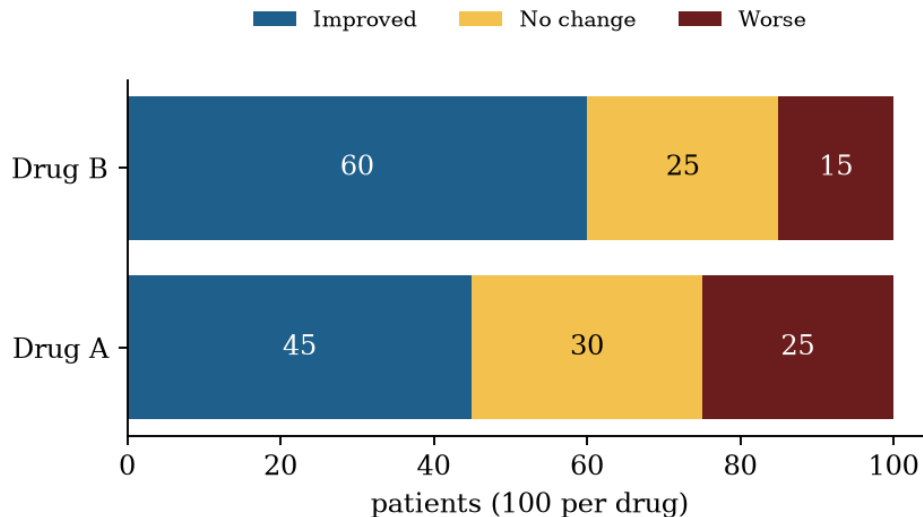
$$E_{A1} = \frac{100 \times 105}{200} = 52.5, \quad E_{A2} = \frac{100 \times 55}{200} = 27.5, \quad E_{A3} = \frac{100 \times 40}{200} = 20$$

Cell	O	E	$(O-E)^2/E$
A, Imp	45	52.5	1.0714
A, NoCh	30	27.5	0.2273
A, Worse	25	20	1.2500
B, Imp	60	52.5	1.0714

Cell	O	E	(O-E) ² /E
B, NoCh	25	27.5	0.2273
B, Worse	15	20	1.2500

$$\chi^2 = 2(1.0714 + 0.2273 + 1.25) = 5.0974$$

df = (2 - 1)(3 - 1) = 2 and $\chi^2_{0.05,2} = 5.991$, so $5.10 < 5.991$: independence is not rejected at 5% ($p \approx 0.078$).



Shortcut: When row totals are equal, the deviations in the two rows are equal and opposite — compute one row and double.

Q.20 [MSQ · 2 marks]

Topic: χ^2 test for a 2×2 table · Medium

Correct answer: A, C

Concept: for a 2×2 table $[[a, b], [c, d]]$, $\chi^2 = n(ad - bc)^2 / (r_1 r_2 c_1 c_2)$, df = 1.

(A) Correct.

$$E_{11} = \frac{200 \times 60}{400} = 30$$

(B) Incorrect. df = (2 - 1)(2 - 1) = 1.

(C) Correct.

$$ad - bc = 18(158) - 182(42) = 2844 - 7644 = -4800$$

$$\chi^2 = \frac{400 \times 4800^2}{200 \times 200 \times 60 \times 340} = \frac{9.216 \times 10^9}{8.16 \times 10^8} = 11.294$$

Check by cells: every $|O - E| = 12$, so $\chi^2 = 144(1/30 + 1/170 + 1/30 + 1/170) = 11.29$.

(D) Incorrect. $11.29 > 6.635$, so independence IS rejected at 1%: infection rate 9% vs 21% is a real difference.

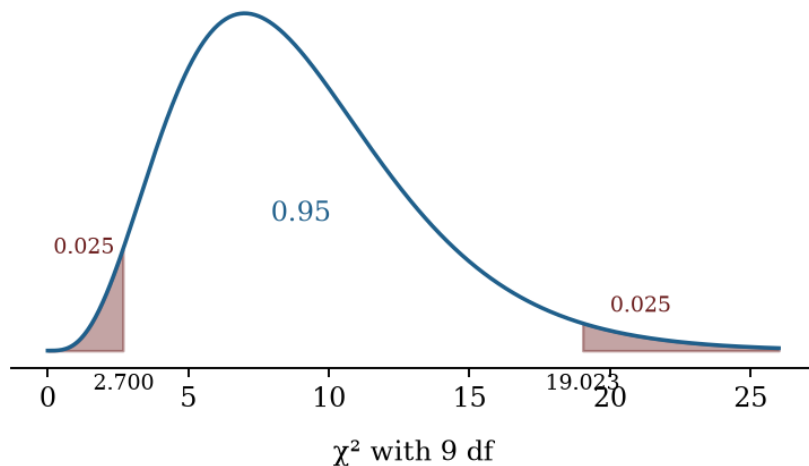
Shortcut: In a 2×2 table all four $|O - E|$ are equal — a quick consistency check.

Q.21 [MCQ · 2 marks · -2/3]**Topic: Confidence interval for a variance · Medium****Correct answer: A****Concept:** $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, so the CI for σ^2 is $[(n-1)s^2/\chi^2_{\alpha/2}, (n-1)s^2/\chi^2_{1-\alpha/2}]$.

$$(n-1)s^2 = 9 \times 4.5 = 40.5$$

$$\text{Lower} = \frac{40.5}{19.023} = 2.129, \quad \text{Upper} = \frac{40.5}{2.700} = 15.00$$

The large upper critical value gives the LOWER limit and vice versa. The interval is not symmetric about $s^2 = 4.5$ because the χ^2 density is skewed.



(B) uses $n s^2 = 45$ instead of $(n-1)s^2$. (C) is the CI for σ (square roots of A) — not what was asked. (D) mistakenly divides s^2 (not $9s^2$) by the critical values.

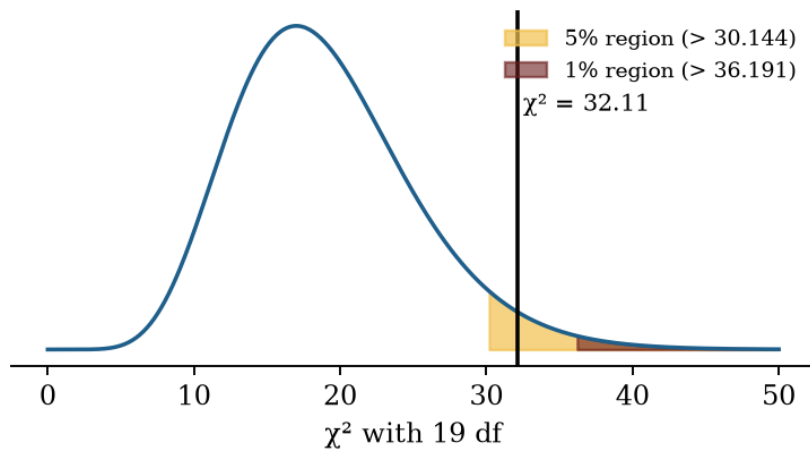
Key idea: Variance CIs come from the χ^2 pivot; they are asymmetric and the larger critical value goes in the denominator of the lower limit.

Q.22 [MCQ · 2 marks · -2/3]**Topic: χ^2 test for a variance · Hard****Correct answer: A****Concept:** under $H_0: \sigma^2 = \sigma_0^2$, the statistic $(n-1)S^2/\sigma_0^2 \sim \chi^2_{n-1}$; for $H_1: \sigma > \sigma_0$ reject in the upper tail.

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{19 \times 0.65^2}{0.5^2} = \frac{19 \times 0.4225}{0.25} = 32.110$$

$$30.144 < 32.11 < 36.191$$

So H_0 is rejected at the 5% level but retained at the 1% level.



(B) uses $n = 20$ in place of $n - 1 = 19$. (C) uses s and σ instead of s^2 and σ^2 . (D) misreads the 1% critical value.

Trap: The χ^2 variance test always works with squared quantities: s^2 and σ_0^2 .

Q.23 [NAT · 2 marks]

Topic: Power of a one-sided z-test · Hard

Correct answer: 0.639 (range 0.634 to 0.643)

Concept: power = $P(\text{reject } H_0 \mid \mu = \mu_1)$. Find the rejection cut-off on the $\bar{x}$ scale under H_0 , then compute its probability under μ_1 .

$$SE = \frac{10}{\sqrt{25}} = 2$$

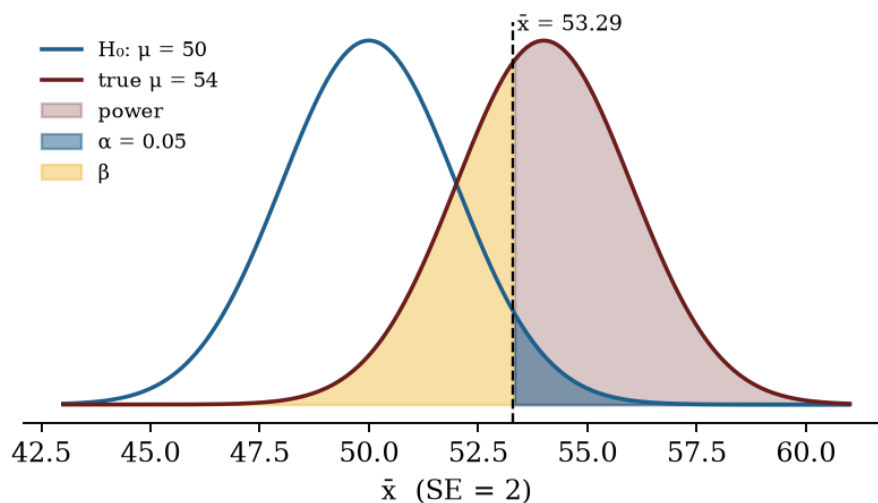
Rejection region: $\bar{x} > c$, where

$$c = 50 + 1.645 \times 2 = 53.29$$

Under $\mu = 54$:

$$\begin{aligned} \text{Power} &= P(\bar{X} > 53.29 \mid \mu = 54) = P\left(Z > \frac{53.29 - 54}{2}\right) = P(Z > -0.355) \\ &= \Phi(0.355) \approx 0.6387 \end{aligned}$$

Hence $\beta = P(\text{Type II error}) = 1 - \text{power} \approx 0.361$.



Shortcut: Power formula for an upper-tailed z-test: $\Phi(\sqrt{n}(\mu_1 - \mu_0)/\sigma - z_\alpha) = \Phi(2 - 1.645)$.

Q.24 [MCQ · 2 marks · -2/3]

Topic: Sample size for a desired power · Hard

Correct answer: B

Concept: power $1 - \beta$ requires $\sqrt{n} \delta/\sigma \geq z_\alpha + z_\beta$, so $n \geq [(z_\alpha + z_\beta)\sigma/\delta]^2$.

Derivation: reject when $\bar{x} > \mu_0 + z_\alpha \sigma/\sqrt{n}$. Power at $\mu_0 + \delta$ is

$$\Phi\left(\frac{\sqrt{n} \delta}{\sigma} - z_\alpha\right) \geq 0.90 \Leftrightarrow \frac{\sqrt{n} \delta}{\sigma} - z_\alpha \geq z_{0.10}$$

$$n \geq \left(\frac{(1.645 + 1.2816) \times 10}{4}\right)^2 = (7.316)^2 = 53.53$$

Rounding up: $n = 54$.

Option	Origin
39	power 0.80 ($z_{0.20} = 0.8416$)
54	correct
66	two-sided α (1.96 instead of 1.645)
17	ignores the power requirement ($z_\beta = 0$, i.e. power only 0.5)

Key idea: Halving δ quadruples n ; raising power from 0.80 to 0.90 costs about 40% more data.

Q.25 [MSQ · 2 marks]

Topic: Hypothesis-testing concepts (α , β , power, p-value) · Medium

Correct answer: B, C, D

Concept: α , β and power are long-run error rates of the decision rule; a p-value is a tail probability computed assuming H_0 .

(A) False. p-value = $P(\text{statistic at least as extreme as observed} \mid H_0 \text{ true})$. It is conditioned ON H_0 , so it cannot be the probability OF H_0 .

(B) True. Power = $\Phi(\sqrt{n}|\mu_1 - \mu_0|/\sigma - z_\alpha)$ (one-sided) increases with n .

(C) True. A smaller α pushes the critical value outward, enlarging the acceptance region, so β rises (power falls).

(D) True. Duality: the 95% CI is exactly the set of μ_0 values NOT rejected by the two-sided 5% test.

$$|\bar{x} - \mu_0| > 1.96 \sigma/\sqrt{n} \Leftrightarrow \mu_0 \notin \bar{x} \pm 1.96 \sigma/\sqrt{n}$$

Key idea: α and β trade off at fixed n ; only more data (or a larger effect) reduces both.

Topic: Width and margin of error of confidence intervals
· Medium

Q.26 [MSQ · 2 marks]

Correct answer: A, B

Concept: width = 2 × (critical value) × SE, with $SE \propto 1/\sqrt{n}$.

(A) True.

$$\frac{\sigma}{\sqrt{4n}} = \frac{1}{2} \cdot \frac{\sigma}{\sqrt{n}}$$

(B) True. $2.576 > 1.96$, so the 99% interval is $2.576/1.96 \approx 1.31$ times as wide.

(C) False. $2.262 > 1.96$, so the t-interval is about 15% WIDER — the price of estimating σ .

(D) False. $p(1 - p)$ is maximised at $p = 0.5$ (value 0.25), so $p = 0.5$ gives the LARGEST (most conservative) n.

$$\frac{d}{dp} p(1 - p) = 1 - 2p = 0 \Rightarrow p = 0.5, \quad \max = 0.25$$

Key idea: Conservative planning uses $p = 0.5$ precisely because it guarantees the margin for any true p .

Topic: CLT for a sample mean (exponential lifetimes) · Medium

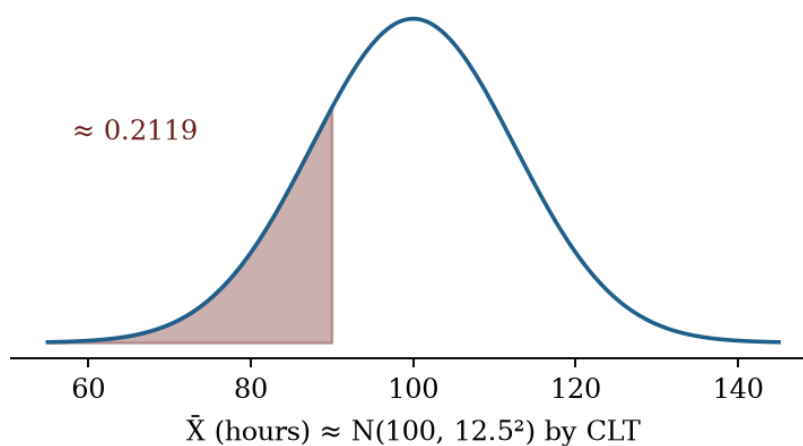
Q.27 [MCQ · 2 marks · -2/3]

Correct answer: A

Concept: for an exponential with mean θ , the SD is also θ ; by CLT $\bar{X} \approx N(\theta, \theta^2/n)$.

$$E(\bar{X}) = 100, \quad SD(\bar{X}) = \frac{100}{\sqrt{64}} = 12.5$$

$$P(\bar{X} < 90) \approx \Phi\left(\frac{90 - 100}{12.5}\right) = \Phi(-0.8) = 1 - 0.7881 = 0.2119$$

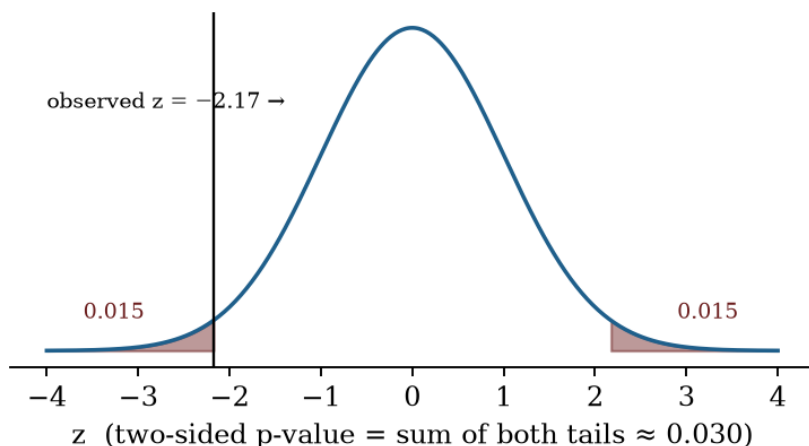


(B) is $P(X > 90)$. (C) forgets $\sqrt{n}$ ($z = -0.1$). (D) is $P(\text{single bulb} < 90) = 1 - e^{-0.9}$, which ignores averaging.

Key idea: Although each lifetime is highly skewed, $n = 64$ is ample for the CLT on the mean.

Q.28 [MSQ · 2 marks]**Topic: p-values for a z-test · Medium****Correct answer: B, D****Concept:** two-sided p-value = $2P(Z \geq |z_{\text{obs}}|)$; reject iff p-value $\leq \alpha$.

$$p = 2[1 - \Phi(2.17)] = 2(0.0150) = 0.0300$$

(B) True. $p \approx 0.030$.**(D) True.** $0.030 \leq 0.05$.**(A) False.** $0.030 > 0.01$ (equivalently $|z| = 2.17 < 2.576$).**(C) False.** For $H_1: \mu > \mu_0$, the p-value is $P(Z \geq -2.17) = 0.985$. The value 0.015 belongs to $H_1: \mu < \mu_0$, the direction the data actually point.

Trap: A one-sided p-value is half the two-sided one ONLY when the data fall on the side of the alternative.

Q.29 [MCQ · 2 marks · -2/3]**Topic: Sampling distribution of S^2 (χ^2) · Hard****Correct answer: A****Concept:** $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, and $\text{Var}(\chi^2_k) = 2k$.

$$\frac{10S^2}{9} \sim \chi^2_{10} \Rightarrow \text{Var}\left(\frac{10S^2}{9}\right) = 20$$

$$\frac{100}{81} \text{Var}(S^2) = 20 \Rightarrow \text{Var}(S^2) = \frac{20 \times 81}{100} = 16.2$$

General formula: $\text{Var}(S^2) = 2\sigma^4/(n-1) = 2(81)/10 = 16.2$.(B) uses n instead of $n-1$ ($162/11$). (C) divides by $(n-1)^2 = 100$ ($2 \times 81/100$). (D) divides by $n-2 = 9$ ($2 \times 81/9$).

Key idea: Also $E(S^2) = \sigma^2$, so S^2 is unbiased; its precision improves like $1/(n-1)$.

Topic: Sampling from a normal population (t and χ^2 facts) · Hard

Q.30 [MSQ · 2 marks]

Correct answer: A, D

Concept: for normal samples, $\bar{X} \sim N(\mu, \sigma^2/n)$, $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, and the two are independent.

(A) True. Independence of $\bar{X}$ and S^2 is a special property of the normal distribution (it characterises normality).

(B) False. Replacing σ by S gives Student's t with 15 df, not $N(0,1)$:

$$\frac{\bar{X} - \mu}{S/\sqrt{16}} = \frac{(\bar{X} - \mu)/(\sigma/4)}{\sqrt{\frac{15S^2/\sigma^2}{15}}} \sim t_{15}$$

(C) False. One df is lost in estimating μ by $\bar{X}$: $15S^2/\sigma^2 \sim \chi^2_{15}$.

(D) True. $(X_i - \mu)/\sigma$ are 16 independent $N(0,1)$ variables; the sum of their squares is χ^2_{16} .

$$\sum_{i=1}^{16} \left(\frac{X_i - \mu}{\sigma} \right)^2 = \sum_{i=1}^{16} Z_i^2 \sim \chi^2_{16}$$

Key idea: Known $\mu \rightarrow n$ df; estimated μ (centre at $\bar{X}$) $\rightarrow n - 1$ df.

Mock Test 08 — Topic Test: Descriptive Statistics, Covariance, Correlation & Mixed Review

Summary statistics, spread, shape, covariance algebra and correlation — with a mixed review of the remaining syllabus.

Item	Details
Questions	30 (12 × 1-mark, 18 × 2-mark)
Maximum marks	48
Question types	MCQ: 14 · MSQ: 7 · NAT: 9
Suggested time	90 minutes
Negative marking	MCQ only: $-1/3$ (1-mark), $-2/3$ (2-mark). None for MSQ & NAT.
Calculator	Use only the GATE virtual scientific calculator

Topic focus of this set

Mean, median, mode, variance and SD for grouped and ungrouped data; linear transformations; combining groups; sample vs population variance; quartiles, box plots and outliers; skewness; covariance and correlation (properties, $\text{Var}(aX + bY)$, linear transforms, zero correlation vs independence, covariance matrices); regression slopes via covariance. Mixed review: binomial, exponential, Poisson, normal percentiles, Bayes theorem, conditional variance, joint PDFs, mixed CDFs and counting.

Instructions

- Questions 1 onwards are arranged as in GATE: all 1-mark questions first, then 2-mark questions.
- **MCQ** – exactly one correct option. **MSQ** – ONE OR MORE options are correct; credit only if you select all correct options and no wrong option (no partial marks, no negative marks).
- **NAT** – enter a numerical value using the virtual keypad. The accepted range is given in the key; round as instructed in the question.
- Statistical tables (Z , t , χ^2) are provided in the Appendix; values in solutions use those tables.
- Attempt in one sitting, then check the answer key and read every solution — including those you got right — for the alternative methods and traps.

Mock Test 08 · Question Paper

1-mark questions

Q.1 MCQ · 1 mark · -1/3

The daily maximum temperatures in a city during May have mean 25°C and standard deviation 4°C . Using $F = 1.8C + 32$, the mean and standard deviation in $^{\circ}\text{F}$ are respectively

- (A) 77°F and 7.2°F
- (B) 77°F and 39.2°F
- (C) 45°F and 7.2°F
- (D) 77°F and 51.84°F

Q.2 NAT · 1 mark

The runs scored by a batter in 40 innings are grouped below.

Runs	0-20	20-40	40-60	60-80	80-100
Innings	5	9	14	8	4

Using class mid-points, the mean runs per innings is _____ (round off to 1 decimal place).

Answer: _____

Q.3 MSQ · 1 mark

Eleven employees of a start-up have distinct monthly salaries. A data-entry error increases the **largest** salary by 100 (thousand rupees); all other values are unchanged. Which of the following is/are TRUE about the corrupted data compared with the original?

- (A) The mean increases by $100/11$ thousand rupees.
- (B) The median is unchanged.
- (C) The standard deviation is unchanged.
- (D) The interquartile range is unchanged.

Q.4 MSQ · 1 mark

For random variables X and Y with finite, positive variances, and constants a, b , which of the following statements is/are ALWAYS TRUE?

- (A) $\text{Cov}(X, X) = \text{Var}(X)$.
- (B) $\text{Corr}(2X + 3, -Y + 1) = -\text{Corr}(X, Y)$.
- (C) If $\text{Cov}(X, Y) = 0$, then X and Y are independent.
- (D) $\text{Cov}(aX, bY) = |ab| \text{Cov}(X, Y)$.

Q.5 NAT · 1 mark

$\text{Var}(X) = 4$, $\text{Var}(Y) = 1$ and $\text{Cov}(X, Y) = -0.5$. Then $\text{Var}(2X - 3Y + 5) = \underline{\hspace{2cm}}$ (answer as an integer).

Answer: _____

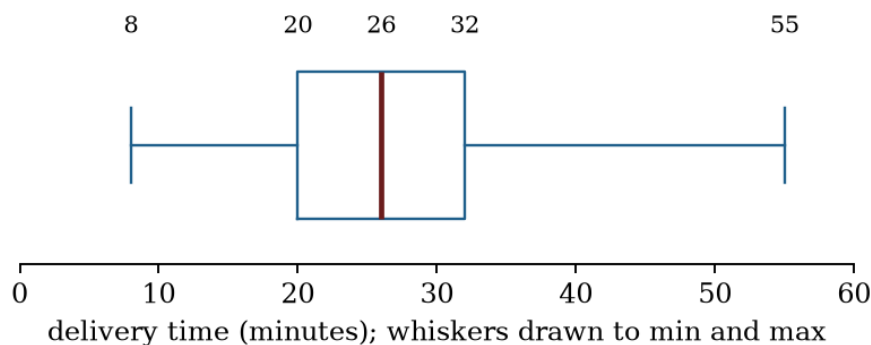
Q.6 MCQ · 1 mark · -1/3

Five repeated measurements of a reaction time (ms) are 2, 4, 6, 8, 10 (after subtracting a baseline). The unbiased sample variance s^2 (divisor $n - 1$) is

- (A) 8
- (B) 10
- (C) 2.83
- (D) 3.16

Q.7 MCQ · 1 mark · -1/3

The five-number summary of food-delivery times (minutes) is: minimum 8, $Q1 = 20$, median 26, $Q3 = 32$, maximum 55.



Using the $1.5 \times \text{IQR}$ rule, which statement is correct?

- (A) The IQR is 47 and there are no outliers.
- (B) The maximum 55 is an outlier, but the minimum 8 is not.
- (C) Both the minimum 8 and the maximum 55 are outliers.
- (D) The minimum 8 is an outlier, but the maximum 55 is not.

Q.8 MCQ · 1 mark · -1/3

For the annual household incomes in a district (in ₹ thousand), mean = 52, median = 47, mode = 40 and $SD = 10$. Which option correctly describes the shape and Pearson's coefficient $Sk = 3(\text{mean} - \text{median})/SD$?

- (A) Positively skewed; $Sk = 1.5$
- (B) Negatively skewed; $Sk = -1.5$
- (C) Positively skewed; $Sk = 1.2$
- (D) Positively skewed; $Sk = 0.5$

Q.9 NAT · 1 mark

Section A has 30 students with mean marks 62.5 and Section B has 45 students with mean marks 70. The mean marks of all 75 students together is _____ (round off to 1 decimal place).

Answer: _____

Q.10 MCQ · 1 mark · -1/3

X takes the values $-1, 0, 1$ with probability $1/3$ each, and $Y = X^2$. Which statement is correct?

- (A) $\text{Cov}(X, Y) = 0$ and X, Y are independent
- (B) $\text{Cov}(X, Y) = 0$ but X, Y are not independent
- (C) $\text{Cov}(X, Y) > 0$ since Y increases with $|X|$
- (D) $\text{Corr}(X, Y) = 1$ since Y is a function of X

Q.11 MCQ · 1 mark · -1/3

Each seed in a packet germinates independently with probability 0.3 . If 5 seeds are sown, the probability that exactly 2 germinate is

- (A) 0.3087
- (B) 0.0309
- (C) 0.8369
- (D) 0.1681

Q.12 NAT · 1 mark

The time (years) until a pump fails is exponential with rate $\lambda = 0.4$ per year. Given that a pump has worked for 2 years, the probability that it works for more than 5 years in total is _____ (round off to 3 decimal places).

Answer: _____

2-mark questions**Q.13 NAT · 2 marks**

In a factory, 40 workers on Shift 1 have mean output 50 units with standard deviation 5 , and 60 workers on Shift 2 have mean output 55 with standard deviation 4 (both SDs computed with divisor equal to the group size). The standard deviation (divisor 100) of the outputs of all 100 workers combined is _____ (round off to 2 decimal places).

Answer: _____

Q.14 MCQ · 2 marks · -2/3

Yields (kg) from 60 experimental plots are grouped as follows.

Yield	10-20	20-30	30-40	40-50	50-60
Plots	6	14	22	12	6

The median yield, by linear interpolation within the median class, is approximately

- (A) 34.55 kg
- (B) 34.77 kg
- (C) 34.44 kg
- (D) 34.67 kg

Q.15 NAT · 2 marks

Hours studied (x) and test score (y) for six students:

x	2	4	5	7	8	10
y	52	58	63	70	71	80

The Pearson correlation coefficient r between x and y is _____ (round off to 3 decimal places).

Answer: _____

Q.16 MCQ · 2 marks · -2/3

For a sample of farms, X = fertiliser used (kg/acre) and Y = yield (quintal/acre) have $\bar{x} = 20$, $\bar{y} = 50$, $\text{Var}(X) = 8$, $\text{Var}(Y) = 25$ and $\text{Cov}(X, Y) = 12$. The least-squares prediction of Y at $X = 24$ from the regression of Y on X , and the correlation coefficient, are

- (A) $\hat{y} = 56.00$, $r \approx 0.85$
- (B) $\hat{y} = 51.92$, $r \approx 0.85$
- (C) $\hat{y} = 56.00$, $r \approx 0.30$
- (D) $\hat{y} = 98.00$, $r \approx 0.85$

Q.17 MSQ · 2 marks

Which of the following 2×2 matrices (written row by row) can be the covariance matrix of some random vector (X, Y) ?

- (A) Rows (4, 2) and (2, 1)
- (B) Rows (1, 2) and (2, 1)
- (C) Rows (2, -1) and (-1, 2)
- (D) Rows (3, 1) and (2, 3)

Q.18 MSQ · 2 marks

The covariance matrix of the returns (X, Y, Z) on three stocks is

	X	Y	Z
X	4	1	-1
Y	1	9	2
Z	-1	2	1

Which of the following statements is/are correct?

- (A) $\text{Corr}(X, Y) = 1/6$
- (B) $\text{Var}(X + Y) = 15$
- (C) $\text{Var}(X - 2Y + Z) = 27$
- (D) $\text{Corr}(Y, Z) = 2/9$

Q.19 MCQ · 2 marks · -2/3

$\text{Corr}(X, Y) = 0.6$, $\text{SD}(X) = 2$ and $\text{SD}(Y) = 5$. Let $U = 3 - 2X$ and $V = 4Y + 1$. Then $\text{Corr}(U, V)$ and $\text{Var}(U + V)$ are respectively

- (A) -0.6 and 320
- (B) -0.6 and 512
- (C) 0.6 and 320
- (D) -0.6 and 416

Q.20 MSQ · 2 marks

The scores X and Y of a student on two independent sections of an aptitude test satisfy $\text{Var}(X) = 4$ and $\text{Var}(Y) = 12$. Which of the following statements is/are correct?

- (A) $\text{Corr}(X, X + Y) = 0.5$
- (B) $\text{Var}(X - Y) = \text{Var}(X + Y)$
- (C) $\text{Corr}(X + Y, X - Y) = -0.5$
- (D) $\text{Cov}(X + Y, X - Y) = 0$

Q.21 MCQ · 2 marks · -2/3

X = number of defective welds (0, 1, 2) and Y = indicator that a joint fails inspection have the joint PMF below.

	$Y = 0$	$Y = 1$
$X = 0$	0.20	0.10
$X = 1$	0.15	0.25
$X = 2$	0.05	0.25

The correlation coefficient $\rho(X, Y)$ is closest to

- (A) 0.395
- (B) 0.150
- (C) 0.625
- (D) 0.250

Q.22 NAT · 2 marks

The mean and standard deviation (divisor n) of 20 sprint times were computed as 30 and 5 seconds. It was later found that one time recorded as 24 should have been 42. The corrected standard deviation is _____ (round off to 2 decimal places).

Answer: _____

Q.23 MCQ · 2 marks · -2/3

Machines M1, M2, M3 produce 50%, 30% and 20% of a factory's bolts, with defect rates 2%, 3% and 6% respectively. A randomly chosen bolt is found defective. The probability that it came from M3 is closest to

- (A) 0.387
- (B) 0.323
- (C) 0.200
- (D) 0.060

Q.24 NAT · 2 marks

The germination probability X of a seed batch is uniformly distributed on $(0, 1)$. Given $X = x$, the number Y of seeds that germinate out of 10 is $\text{Bin}(10, x)$. Then $\text{Var}(Y) = \underline{\hspace{2cm}}$ (round off to 2 decimal places).

Answer:

Q.25 MCQ · 2 marks · -2/3

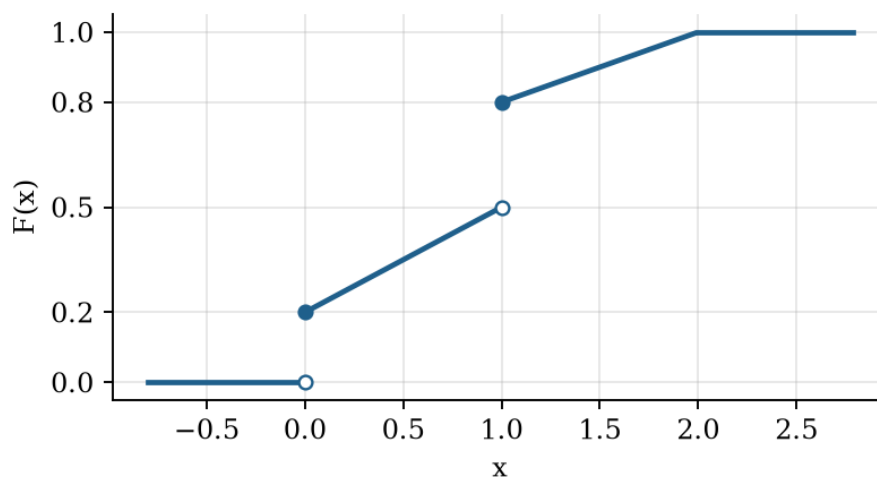
(X, Y) has joint density $f(x, y) = c(x + y)$ for $0 < x < 1$, $0 < y < 1$ and 0 elsewhere. Then $P(X + Y \leq 1)$ equals

- (A) $1/3$
- (B) $1/6$
- (C) $1/2$
- (D) $2/3$

Q.26 MSQ · 2 marks

A random variable X (waiting time, minutes) has CDF

$$F(x) = 0 \quad (x < 0); \quad 0.2 + 0.3x \quad (0 \leq x < 1); \quad 0.8 + 0.2(x - 1) \quad (1 \leq x < 2); \quad 1 \quad (x \geq 2)$$



Which of the following is/are correct?

- (A) $P(X = 0) = 0.2$
- (B) $P(X = 1) = 0.3$
- (C) $P(0.5 < X \leq 1.5) = 0.55$
- (D) $P(X < 1) = 0.8$

Q.27 MCQ · 2 marks · -2/3

A 5-member sports committee is to be chosen from 6 men and 4 women. The number of committees containing at least 2 women is

- (A) 186
- (B) 66
- (C) 120
- (D) 336

Q.28 NAT · 2 marks

Adult male heights in a region are $N(170, 8^2)$ cm. An army unit admits only men in the tallest 10%. The minimum admissible height is _____ cm (round off to 2 decimal places). [Use $z_{0.10} = 1.2816$, i.e. $\Phi(1.2816) = 0.90$.]

Answer: _____

Q.29 MCQ · 2 marks · -2/3

A call centre receives calls as a Poisson process at an average rate of 3 calls per 10 minutes. The probability of receiving at least 2 calls in a 5-minute interval is

- (A) 0.4422
- (B) 0.5578
- (C) 0.8009
- (D) 0.2231

Q.30 MSQ · 2 marks

For bivariate data on (X, Y) , the two least-squares regression lines are $3x + 2y = 26$ and $6x + y = 31$. Which of the following is/are correct?

- (A) $\bar{x} = 4$ and $\bar{y} = 7$
- (B) The correlation coefficient is $r = -0.5$
- (C) The correlation coefficient is $r = +0.5$
- (D) If $\text{Var}(X) = 9$, then $\text{Var}(Y) = 81$

Mock Test 08 · Answer Key

Q	Type	Marks	Answer	Topic
1	MCQ	1	A	Linear transformation: mean and SD
2	NAT	1	48.5 (range 48.4 to 48.6)	Mean of grouped data
3	MSQ	1	A, B, D	Robustness of summary statistics
4	MSQ	1	A, B	Properties of covariance and correlation
5	NAT	1	31	Variance of a linear combination
6	MCQ	1	B	Sample vs population variance
7	MCQ	1	B	Box plot and the $1.5 \times \text{IQR}$ rule
8	MCQ	1	A	Skewness from mean, median and mode
9	NAT	1	67.0 (range 66.9 to 67.1)	Combined mean of two groups
10	MCQ	1	B	Zero correlation vs independence
11	MCQ	1	A	Binomial PMF (review)
12	NAT	1	0.301 (range 0.3 to 0.302)	Exponential distribution: memoryless property (review)
13	NAT	2	5.06 (range 5.05 to 5.07)	Combined variance of two groups
14	MCQ	2	A	Median of grouped data
15	NAT	2	0.995 (range 0.993 to 0.997)	Pearson correlation coefficient
16	MCQ	2	A	Regression line via covariance
17	MSQ	2	A, C	Valid covariance matrices
18	MSQ	2	A, B, C	Covariance matrix computations
19	MCQ	2	A	Correlation and variance under linear transformations
20	MSQ	2	A, B, C	Covariance of sums and differences
21	MCQ	2	A	Correlation from a joint PMF
22	NAT	2	5.44 (range 5.43 to 5.45)	Correcting mean and SD after a recording error
23	MCQ	2	A	Bayes theorem (review)
24	NAT	2	10.00 (range 9.99 to 10.01)	Conditional expectation and variance (review)
25	MCQ	2	A	Joint PDF: region probability (review)
26	MSQ	2	A, B, C	Mixed-type CDF (review)
27	MCQ	2	A	Counting: combinations with constraints (review)
28	NAT	2	180.25 (range 180.2 to 180.3)	Normal percentile (review)
29	MCQ	2	A	Poisson distribution with rescaled rate (review)
30	MSQ	2	A, B, D	Two regression lines: means, r and variances

Score tracker

	Attempted	Correct	Wrong (MCQ)	Marks
1-mark				
2-mark				
Total				/ 48

Mock Test 08 · Detailed Solutions

Q.1 [MCQ · 1 mark · -1/3]

Topic: Linear transformation: mean and SD · Easy

Correct answer: A

Concept: for $Y = aX + b$, $\text{mean}(Y) = a \cdot \text{mean}(X) + b$ and $\text{SD}(Y) = |a| \cdot \text{SD}(X)$; the shift b does not affect spread.

$$\bar{F} = 1.8(25) + 32 = 77^\circ\text{F}$$

$$\text{SD}_F = |1.8| \times 4 = 7.2^\circ\text{F}$$

(B) adds 32 to the SD. (C) forgets to add 32 to the mean. (D) gives the variance $1.8^2 \times 16 = 51.84$ (units $^\circ\text{F}^2$), not the SD.

Key idea: Location shifts move the mean, median and quartiles but leave SD, variance and IQR unchanged.

Q.2 [NAT · 1 mark]

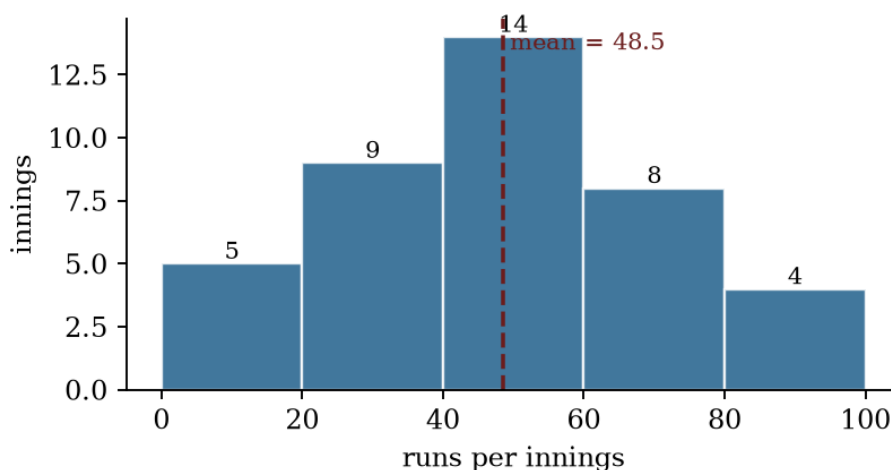
Topic: Mean of grouped data · Easy

Correct answer: 48.5 (range 48.4 to 48.6)

Concept: for grouped data, $\bar{x} = \sum f_i m_i / \sum f_i$ with m_i the class mid-point.

Class	m	f	f·m
0-20	10	5	50
20-40	30	9	270
40-60	50	14	700
60-80	70	8	560
80-100	90	4	360
Total		40	1940

$$\bar{x} = \frac{1940}{40} = 48.5$$



Shortcut: Coding $u = (m - 50)/20$ gives $\sum fu = -10 - 9 + 0 + 8 + 8 = -3$, so $\bar{x} = 50 + 20(-3/40) = 48.5$.

Q.3 [MSQ · 1 mark]**Topic: Robustness of summary statistics · Medium****Correct answer: A, B, D**

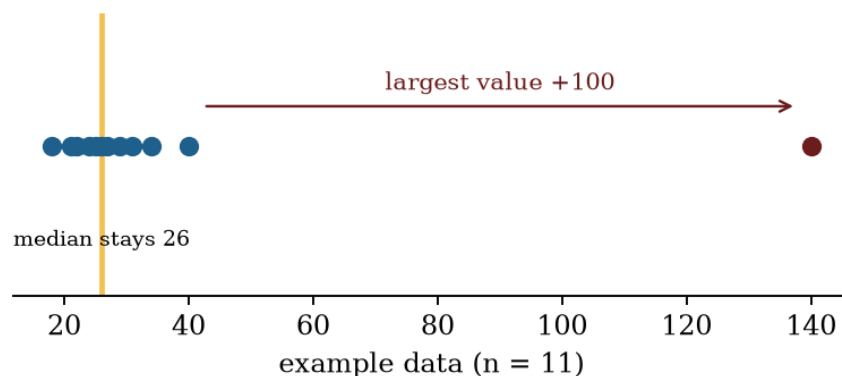
Concept: the mean and SD use every value; order statistics such as the median and quartiles depend only on ranks near the middle.

(A) True. The total rises by 100, so the mean rises by $100/11 \approx 9.09$.

(B) True. With $n = 11$ the median is the 6th ordered value; the largest (11th) value moving further right does not change any rank.

(C) False. The largest value moves farther from the mean, so $\sum(x_i - \bar{x})^2$ increases; the SD increases.

(D) True. Q1 and Q3 are determined by values around the 3rd and 9th positions, which are unchanged; the max is not involved.



Key idea: Median and IQR are resistant (robust) to outliers; mean and SD are not.

Q.4 [MSQ · 1 mark]**Topic: Properties of covariance and correlation · Medium****Correct answer: A, B**

Concept: $\text{Cov}(aX + c, bY + d) = ab \text{Cov}(X, Y)$; $\text{Corr}(aX + c, bY + d) = \text{sign}(ab) \text{Corr}(X, Y)$.

(A) True. $\text{Cov}(X, X) = E[(X - \mu)^2] = \text{Var}(X)$.

(B) True. Here $a = 2$, $b = -1$, so $ab < 0$ and the correlation flips sign; shifts 3 and 1 are irrelevant.

$$\text{Corr}(2X + 3, -Y + 1) = \frac{(2)(-1)\text{Cov}(X, Y)}{2\sigma_X \cdot 1\sigma_Y} = -\rho_{XY}$$

(C) False. Zero covariance means no LINEAR relation only (counter-example: X uniform on $\{-1, 0, 1\}$, $Y = X^2$).

(D) False. The correct factor is ab , not $|ab|$: $\text{Cov}(-X, Y) = -\text{Cov}(X, Y)$.

Trap: Absolute values appear for the SD ($\sigma_{aX} = |a|\sigma_X$), not for covariance.

Q.5 [NAT · 1 mark]**Topic: Variance of a linear combination · Easy****Correct answer: 31****Concept:** $\text{Var}(aX + bY + c) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab \text{Cov}(X, Y)$.

$$\begin{aligned}\text{Var}(2X - 3Y + 5) &= 2^2(4) + (-3)^2(1) + 2(2)(-3)(-0.5) \\ &= 16 + 9 + 6 = 31\end{aligned}$$

The constant 5 contributes nothing. Note that the negative covariance increases the variance here because the coefficients of X and Y have opposite signs.

Trap: Sign of the cross term = $\text{sign}(a) \times \text{sign}(b) \times \text{sign}(\text{Cov})$. Track all three.

Q.6 [MCQ · 1 mark · -1/3]**Topic: Sample vs population variance · Easy****Correct answer: B**

Concept: $s^2 = \sum(x_i - \bar{x})^2/(n - 1)$ is unbiased for σ^2 ; dividing by n gives the population (or MLE) variance.

$$\begin{aligned}\bar{x} &= 6, \quad \sum(x_i - \bar{x})^2 = 16 + 4 + 0 + 4 + 16 = 40 \\ s^2 &= \frac{40}{5 - 1} = 10 \quad (\text{divisor } n : 40/5 = 8)\end{aligned}$$

(A) divides by n; (C) = $\sqrt{8}$ and (D) = $\sqrt{10}$ are standard deviations, not variances.

Key idea: $E[\sum(X_i - \bar{X})^2] = (n - 1)\sigma^2$: one degree of freedom is spent on $\bar{X}$.

Q.7 [MCQ · 1 mark · -1/3]**Topic: Box plot and the 1.5 × IQR rule · Easy****Correct answer: B**

Concept: fences are $Q_1 - 1.5 \cdot \text{IQR}$ and $Q_3 + 1.5 \cdot \text{IQR}$; points beyond them are flagged as outliers.

$$\begin{aligned}\text{IQR} &= Q_3 - Q_1 = 32 - 20 = 12, \quad 1.5 \text{IQR} = 18 \\ \text{Lower fence} &= 20 - 18 = 2, \quad \text{Upper fence} = 32 + 18 = 50\end{aligned}$$

$8 > 2$, so 8 is not an outlier; $55 > 50$, so 55 is an outlier.

(A) uses the range ($55 - 8 = 47$) in place of the IQR.

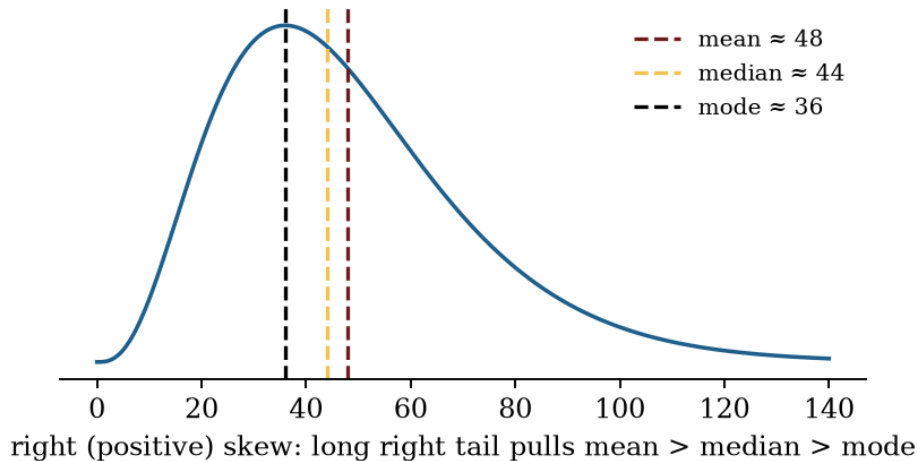
The long upper whisker relative to the lower one also indicates right skew.

Trap: $\text{IQR} = Q_3 - Q_1$, never $\text{max} - \text{min}$. The fences are measured from the quartiles, not from the median.

Q.8 [MCQ · 1 mark · -1/3]**Topic: Skewness from mean, median and mode · Easy****Correct answer: A****Concept:** mean > median > mode indicates a long right tail (positive skew).

$$Sk = \frac{3(52 - 47)}{10} = \frac{15}{10} = 1.5$$

(C) is (mean – mode)/SD = 1.2 (Pearson's first coefficient — not the formula asked). (D) forgets the factor 3.



Key idea: Empirical rule for moderate skew: mean – mode ≈ 3(mean – median). Here 12 vs 15 — close.

Q.9 [NAT · 1 mark]**Topic: Combined mean of two groups · Easy****Correct answer: 67.0 (range 66.9 to 67.1)****Concept:** the combined mean is the size-weighted average of group means.

$$\bar{x} = \frac{30(62.5) + 45(70)}{75} = \frac{1875 + 3150}{75} = \frac{5025}{75} = 67.0$$

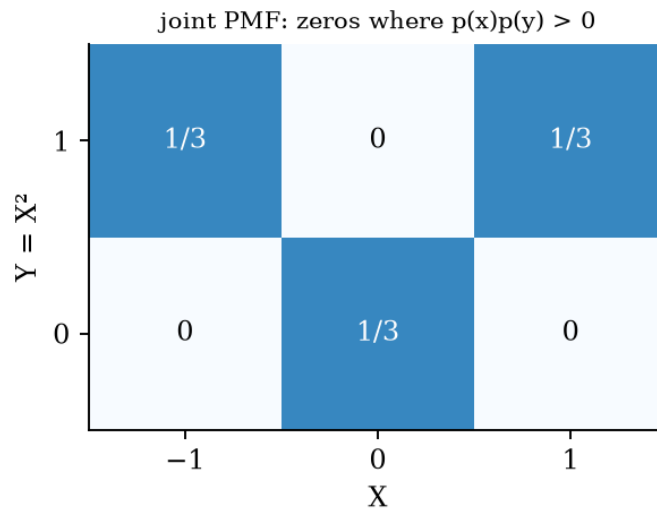
The simple average $(62.5 + 70)/2 = 66.25$ is wrong because the groups have unequal sizes.

Key idea: The combined mean always lies between the group means, closer to the larger group.

Q.10 [MCQ · 1 mark · -1/3]**Topic: Zero correlation vs independence · Medium****Correct answer: B****Concept:** independence $\Rightarrow$ zero covariance, but not conversely.

$$E(X) = 0, \quad E(XY) = E(X^3) = \frac{(-1) + 0 + 1}{3} = 0$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0 - 0 = 0$$

Dependence: $P(X = 0, Y = 1) = 0$ but $P(X = 0)P(Y = 1) = (1/3)(2/3) = 2/9 \neq 0$.

Trap: Correlation measures LINEAR association only; a perfect quadratic relation can have $\rho = 0$.

Q.11 [MCQ · 1 mark · -1/3]**Topic: Binomial PMF (review) · Easy****Correct answer: A****Concept:** $X \sim \text{Bin}(n, p)$: $P(X = k) = C(n, k) p^k (1 - p)^{n-k}$.

$$P(X = 2) = \binom{5}{2} (0.3)^2 (0.7)^3 = 10 \times 0.09 \times 0.343 = 0.3087$$

(B) omits $C(5, 2)$. (C) is $P(X \leq 2)$. (D) is $P(X = 0)$.

Trap: The binomial coefficient counts the $C(5, 2) = 10$ orders in which 2 successes can occur.

Topic: Exponential distribution: memoryless property
(review) · Easy

Q.12 [NAT · 1 mark]

Correct answer: 0.301 (range 0.3 to 0.302)

Concept: $P(X > s + t \mid X > s) = P(X > t) = e^{-\lambda t}$ for an exponential.

$$P(X > 5 \mid X > 2) = \frac{e^{-0.4(5)}}{e^{-0.4(2)}} = e^{-0.4(3)} = e^{-1.2} = 0.3012$$

Using $e^{-0.4 \times 5} = 0.1353$ ignores the conditioning information.

Key idea: Only the REMAINING time (3 years) matters for an exponential lifetime.

Q.13 [NAT · 2 marks]

Topic: Combined variance of two groups · Medium

Correct answer: 5.06 (range 5.05 to 5.07)

Concept: total variance = mean of within-group variances + variance of group means (about the combined mean).

$$\begin{aligned}\bar{x} &= \frac{40(50) + 60(55)}{100} = 53 \\ d_1 &= 50 - 53 = -3, \quad d_2 = 55 - 53 = 2 \\ \sigma^2 &= \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2} \\ &= \frac{40(25 + 9) + 60(16 + 4)}{100} = \frac{1360 + 1200}{100} = 25.6 \\ \sigma &= \sqrt{25.6} = 5.0596\end{aligned}$$

Checks: within part = $(40 \cdot 25 + 60 \cdot 16)/100 = 19.6$; between part = $(40 \cdot 9 + 60 \cdot 4)/100 = 6.0$; sum 25.6.

A common error is to average variances only (19.6, SD 4.43), ignoring the spread between the means.

Key idea: This is the data version of $\text{Var}(Y) = E[\text{Var}(Y|G)] + \text{Var}(E[Y|G])$.

Q.14 [MCQ · 2 marks · -2/3]

Topic: Median of grouped data · Medium

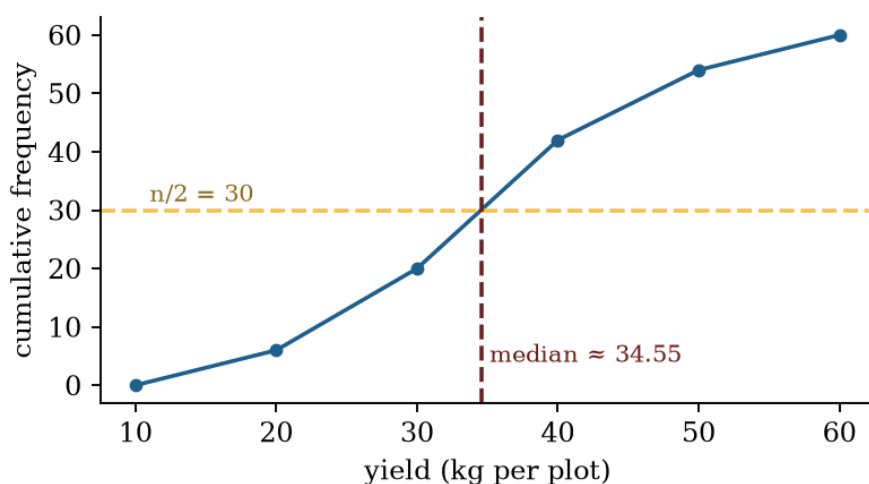
Correct answer: A

Concept: median = $L + [(n/2 - CF)/f] \times h$, where L, f, h are the lower limit, frequency and width of the median class and CF the cumulative frequency before it.

Class	f	Cumulative
10-20	6	6
20-30	14	20
30-40	22	42
40-50	12	54
50-60	6	60

$n/2 = 30$ first falls in 30-40 (cumulative 20 → 42). So $L = 30$, $CF = 20$, $f = 22$, $h = 10$.

$$\text{Median} = 30 + \frac{30 - 20}{22} \times 10 = 30 + 4.545 = 34.55$$



(B) uses $(n + 1)/2 = 30.5$ — the ungrouped-data rule. (C) is the grouped MODE $30 + 8/(44 - 14 - 12) \times 10 = 34.44$. (D) is the grouped MEAN.

Key idea: Mean (34.67) > median (34.55) > mode (34.44): a mild positive skew.

Q.15 [NAT · 2 marks]

Topic: Pearson correlation coefficient · Medium

Correct answer: 0.995 (range 0.993 to 0.997)

Concept: $r = S_{xy} / \sqrt{(S_{xx} S_{yy})}$ with $S_{xy} = \sum (x - \bar{x})(y - \bar{y})$.

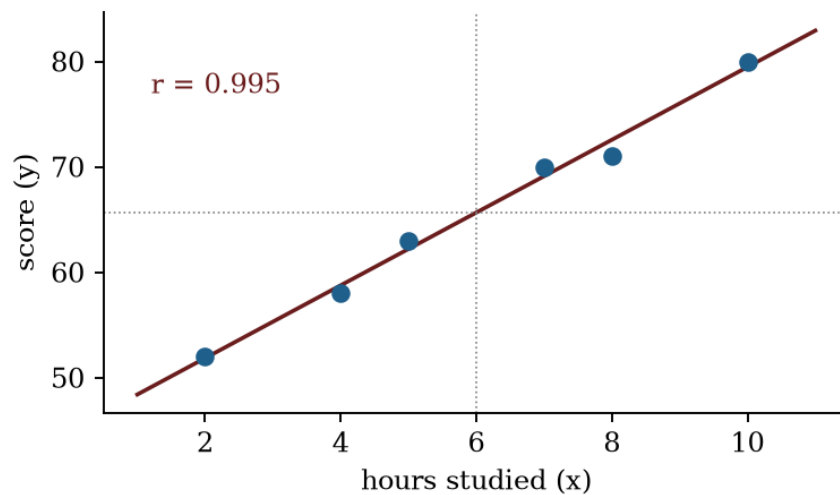
$$\bar{x} = \frac{36}{6} = 6, \quad \bar{y} = \frac{394}{6} = 65.6667$$

$x - \bar{x}$	-4	-2	-1	1	2	4
$y - \bar{y}$	-13.667	-7.667	-2.667	4.333	5.333	14.333

$$S_{xx} = 42, \quad S_{yy} = 505.333, \quad S_{xy} = 145$$

(Shortcut: $S_{xy} = \sum xy - n \bar{x} \bar{y} = 2509 - 6 \cdot 6 \cdot 65.6667 = 145$; $S_{xx} = \sum x^2 - n \bar{x}^2 = 258 - 216 = 42$.)

$$r = \frac{145}{\sqrt{42 \times 505.333}} = \frac{145}{145.6846} = 0.9953$$



The fitted slope is $S_{xy}/S_{xx} = 3.452$ marks per hour.

Key idea: r is unit-free: rescaling hours to minutes or marks to percentages leaves r unchanged.

Q.16 [MCQ · 2 marks · -2/3]

Topic: Regression line via covariance · Medium

Correct answer: A

Concept: the Y-on-X line passes through $(\bar{x}, \bar{y})$ with slope $b_{YX} = \text{Cov}(X, Y)/\text{Var}(X)$; $r = \text{Cov}/(\sigma_X \sigma_Y)$.

$$b_{YX} = \frac{12}{8} = 1.5$$

$$\hat{y} = \bar{y} + b_{YX}(x - \bar{x}) = 50 + 1.5(24 - 20) = 56$$

$$r = \frac{12}{\sqrt{8} \times \sqrt{25}} = \frac{12}{14.142} = 0.849$$

(B) uses $\text{Cov}/\text{Var}(Y) = 0.48$ — the slope of the regression of X on Y. (C) divides Cov by $\text{Var}(X) \cdot \text{SD}(Y)$ instead of $\text{SD}(X) \cdot \text{SD}(Y)$. (D) uses the covariance itself as slope.

Consistency check: $b_{YX} = r \sigma_Y / \sigma_X = 0.849 \times 5 / 2.828 = 1.5$. (verified)

Trap: Slope of Y on X divides by $\text{Var}(X)$ — the variance of the PREDICTOR.

Q.17 [MSQ · 2 marks]

Topic: Valid covariance matrices · Medium

Correct answer: A, C

Concept: a covariance matrix must be symmetric and positive semi-definite; for 2×2 this means diagonal entries ≥ 0 and determinant ≥ 0 , i.e. $|\text{Cov}| \leq \sigma_X \sigma_Y$ ($|\rho| \leq 1$).

(A) Valid. $\det = 4 - 4 = 0 \geq 0$, $\rho = 2/(2 \cdot 1) = 1$. Degenerate ($Y = X/2 + c$) but allowed.

(B) Invalid. $\det = 1 - 4 = -3 < 0$; it would need $\rho = 2/(1 \cdot 1) = 2 > 1$.

(C) Valid. $\det = 4 - 1 = 3 > 0$, $\rho = -1/2$.

$$\text{eigenvalues of (C)} : 2 \pm 1 = \{1, 3\} > 0$$

(D) Invalid. Not symmetric ($\text{Cov}(X, Y)$ must equal $\text{Cov}(Y, X)$).

Key idea: PSD $\Leftrightarrow \text{Var}(aX + bY) = (a, b)\Sigma(a, b)^T \geq 0$ for all a, b — a negative determinant would allow a negative variance.

Q.18 [MSQ · 2 marks]

Topic: Covariance matrix computations · Medium

Correct answer: A, B, C

Concept: $\text{Var}(a^T V) = a^T \Sigma a = \sum a_i^2 \sigma_{ii} + 2 \sum_{i < j} a_i a_j \sigma_{ij}$.

(A) Correct.

$$\rho_{XY} = \frac{1}{\sqrt{4}\sqrt{9}} = \frac{1}{6}$$

(B) Correct. $\text{Var}(X + Y) = 4 + 9 + 2(1) = 15$.

(C) Correct. With $a = (1, -2, 1)$:

$$\begin{aligned} &1(4) + 4(9) + 1(1) + 2[(1)(-2)(1) + (1)(1)(-1) + (-2)(1)(2)] \\ &= 41 + 2(-2 - 1 - 4) = 41 - 14 = 27 \end{aligned}$$

(D) Incorrect. $\rho_{YZ} = 2/(\sqrt{9} \cdot \sqrt{1}) = 2/3$. (2/9 wrongly divides by $\text{Var}(Y)$.)

Trap: Divide covariances by STANDARD DEVIATIONS, not variances, to get correlations.

Q.19 [MCQ · 2 marks · -2/3]

Topic: Correlation and variance under linear transformations · Medium

Correct answer: A

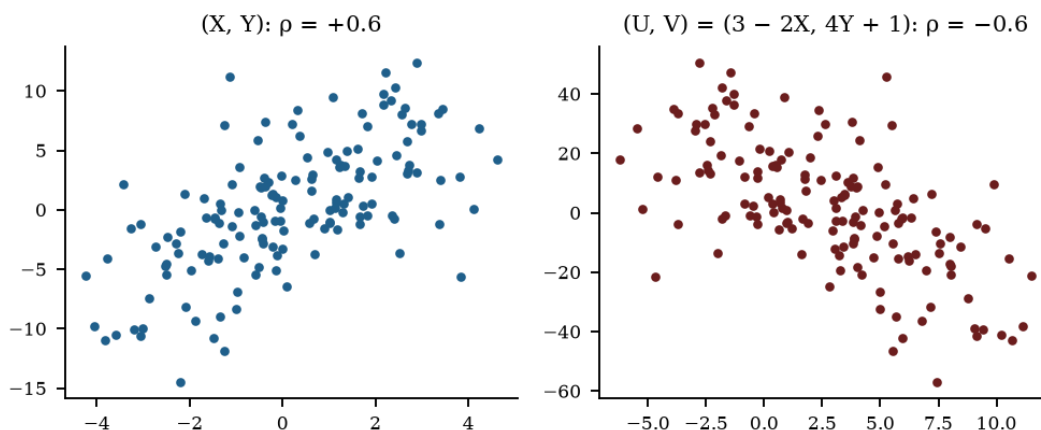
Concept: $\text{Cov}(aX + c, bY + d) = ab \text{Cov}(X, Y)$; correlation keeps its magnitude and takes $\text{sign}(ab)$.

$$\text{Cov}(X, Y) = 0.6 \times 2 \times 5 = 6$$

$$\text{Cov}(U, V) = (-2)(4)(6) = -48, \quad \text{Corr}(U, V) = -0.6$$

$$\text{Var}(U) = 4 \times 4 = 16, \quad \text{Var}(V) = 16 \times 25 = 400$$

$$\text{Var}(U + V) = 16 + 400 + 2(-48) = 320$$



(B) adds +96 (sign error). (C) ignores the sign flip. (D) ignores the covariance term.

Key idea: A negative scale on one variable flips the sign of ρ ; scaling both negatively leaves it.

Q.20 [MSQ · 2 marks]**Topic: Covariance of sums and differences · Medium****Correct answer: A, B, C**

Concept: covariance is bilinear: $\text{Cov}(aX + bY, cX + dY) = ac \text{Var } X + bd \text{Var } Y + (ad + bc) \text{Cov}(X, Y)$, and $\text{Cov}(X, Y) = 0$ under independence.

(A) Correct.

$$\text{Cov}(X, X + Y) = \text{Var}(X) = 4, \quad \text{Var}(X + Y) = 4 + 12 = 16$$

$$\text{Corr}(X, X + Y) = \frac{4}{\sqrt{4}\sqrt{16}} = \frac{4}{8} = 0.5$$

(B) Correct. With $\text{Cov}(X, Y) = 0$ both equal $4 + 12 = 16$ (the cross term $\pm 2\text{Cov}$ vanishes).**(C) Correct.**

$$\text{Cov}(X + Y, X - Y) = \text{Var}(X) - \text{Var}(Y) = 4 - 12 = -8$$

$$\text{Corr}(X + Y, X - Y) = \frac{-8}{\sqrt{16}\sqrt{16}} = -0.5$$

(D) Incorrect. It is -8 ; the sum and difference are uncorrelated only when $\text{Var}(X) = \text{Var}(Y)$.

Shortcut: $\text{Cov}(X + Y, X - Y) = \text{Var } X - \text{Var } Y$ — a frequent one-line GATE fact.

Q.21 [MCQ · 2 marks · -2/3]**Topic: Correlation from a joint PMF · Medium****Correct answer: A**

Concept: $\rho = [E(XY) - E(X)E(Y)] / \sqrt{(\text{Var } X \cdot \text{Var } Y)}$, using marginals from row/column sums.

Marginals: $P(X = 0, 1, 2) = 0.30, 0.40, 0.30$; $P(Y = 1) = 0.60$.

$$E(X) = 1.0, \quad E(X^2) = 0.40 + 4(0.30) = 1.6, \quad \text{Var}(X) = 0.60$$

$$E(Y) = 0.6, \quad \text{Var}(Y) = 0.6(0.4) = 0.24$$

$$E(XY) = 1(0.25) + 2(0.25) = 0.75, \quad \text{Cov} = 0.75 - 1.0(0.6) = 0.15$$

$$\rho = \frac{0.15}{\sqrt{0.6 \times 0.24}} = \frac{0.15}{0.3795} = 0.3953$$

mass shifts to $Y = 1$ as X grows $\rightarrow \rho > 0$

		Y = 0	Y = 1
X = 2		0.05	0.25
X = 1		0.15	0.25
X = 0		0.20	0.10

(B) is the covariance. (C) and (D) divide by a variance instead of the product of SDs.

Shortcut: In $E(XY)$ only cells with $x \neq 0$ and $y \neq 0$ contribute — skip the zeros.

Topic: Correcting mean and SD after a recording error · Medium

Q.22 [NAT · 2 marks]

Correct answer: 5.44 (range 5.43 to 5.45)

Concept: recover $\sum x$ and $\sum x^2$ from the wrong summary, correct them, then recompute.

$$\begin{aligned}\sum x &= 20 \times 30 = 600, & \sum x^2 &= n(\sigma^2 + \bar{x}^2) = 20(25 + 900) = 18500 \\ \sum x_{\text{new}} &= 600 - 24 + 42 = 618, & \bar{x}_{\text{new}} &= 30.9 \\ \sum x_{\text{new}}^2 &= 18500 - 576 + 1764 = 19688 \\ \sigma_{\text{new}}^2 &= \frac{19688}{20} - 30.9^2 = 984.4 - 954.81 = 29.59 \\ \sigma_{\text{new}} &= \sqrt{29.59} = 5.4397\end{aligned}$$

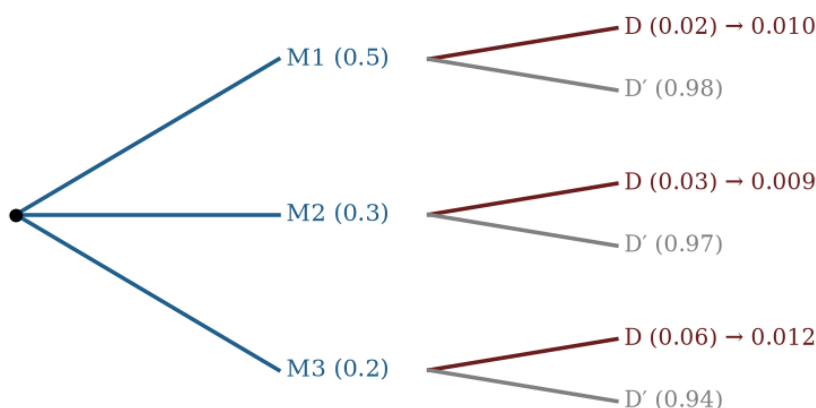
Trap: Correct BOTH $\sum x$ and $\sum x^2$; then remember the new mean also changes before subtracting $\bar{x}^2$.

Q.23 [MCQ · 2 marks · -2/3]

Topic: Bayes theorem (review) · Medium

Correct answer: A

Concept: $P(M_i | D) = P(M_i)P(D | M_i) / \sum_j P(M_j)P(D | M_j)$.



$$P(D) = 0.5(0.02) + 0.3(0.03) + 0.2(0.06) = 0.010 + 0.009 + 0.012 = 0.031$$

$$P(M_3 | D) = \frac{0.012}{0.031} = 0.3871$$

For comparison $P(M_1 | D) = 0.010/0.031 = 0.3226$ and $P(M_2 | D) = 0.009/0.031 = 0.2903$; the three posteriors sum to 1.

(B) is $P(M_1 | D)$. (C) is the prior $P(M_3)$ — it ignores the evidence. (D) is the likelihood $P(D | M_3)$, the reverse conditional.

Key idea: M3 makes only 20% of bolts but, being the least reliable, accounts for nearly 39% of the defectives.

Topic: Conditional expectation and variance (review) · Hard

Q.24 [NAT · 2 marks]

Correct answer: 10.00 (range 9.99 to 10.01)

Concept (law of total variance): $\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E[Y | X])$.

$$E[Y|X] = 10X, \quad \text{Var}(Y|X) = 10X(1 - X)$$

$$E[10X(1 - X)] = 10(E[X] - E[X^2]) = 10\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{10}{6}$$

$$\text{Var}(10X) = 100 \times \frac{1}{12} = \frac{100}{12}$$

$$\text{Var}(Y) = \frac{10}{6} + \frac{100}{12} = \frac{20 + 100}{12} = 10$$

Also $E(Y) = E[10X] = 5$. (In fact Y is uniform on $\{0, 1, \dots, 10\}$, whose variance is $(11^2 - 1)/12 = 10$. (verified))

A Monte Carlo simulation with 400 000 draws gives $\text{Var}(Y) \approx 10.0$.

Trap: Plugging in $E(X) = 0.5$ to get $\text{Bin}(10, 0.5)$ variance 2.5 ignores the uncertainty in X — the between-batch term $\text{Var}(E[Y|X])$ dominates.

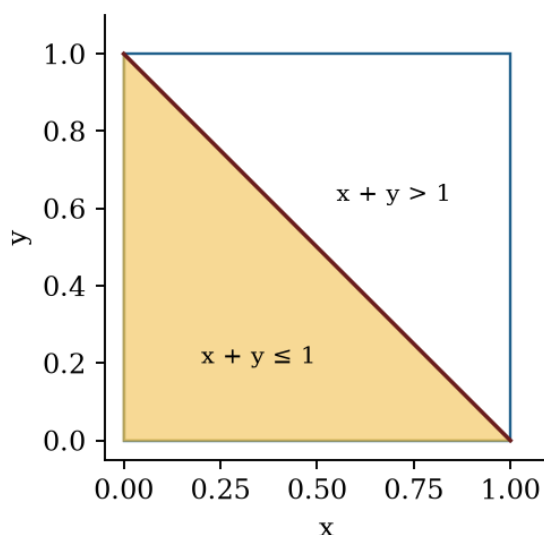
Q.25 [MCQ · 2 marks · -2/3]

Topic: Joint PDF: region probability (review) · Medium

Correct answer: A

Concept: normalise first, then integrate the density over the region.

$$\int_0^1 \int_0^1 c(x + y) dy dx = c \left(\frac{1}{2} + \frac{1}{2} \right) = c = 1$$



$$P(X + Y \leq 1) = \int_0^1 \int_0^{1-x} (x + y) dy dx$$

$$= \int_0^1 \left[x(1 - x) + \frac{(1 - x)^2}{2} \right] dx = \int_0^1 \frac{1 - x^2}{2} dx$$

$$= \frac{1}{2} \left(1 - \frac{1}{3} \right) = \frac{1}{3}$$

Alternative: $S = X + Y$ on the triangle; the density grows with $x + y$, so the lower triangle (where $x + y$ is small) gets LESS than half the mass — ruling out (C) and (D).

Trap: (C) = $1/2$ is the triangle's AREA — correct only for a uniform density.

Q.26 [MSQ · 2 marks]

Topic: Mixed-type CDF (review) · Medium

Correct answer: A, B, C

Concept: $P(X = a) = F(a) - F(a^-)$ (jump size); $P(a < X \leq b) = F(b) - F(a)$; $P(X < a) = F(a^-)$.

(A) Correct. $F(0) - F(0^-) = 0.2 - 0 = 0.2$ (customers served immediately).

(B) Correct.

$$F(1) - F(1^-) = 0.8 - (0.2 + 0.3) = 0.3$$

(C) Correct.

$$F(1.5) - F(0.5) = [0.8 + 0.1] - [0.2 + 0.15] = 0.90 - 0.35 = 0.55$$

(This includes the jump of 0.3 at $x = 1$.)

(D) Incorrect. $P(X < 1) = F(1^-) = 0.5$; the value 0.8 is $P(X \leq 1)$.

Trap: For mixed distributions, $<$ versus $\leq$ matters exactly at the jump points.

Q.27 [MCQ · 2 marks · -2/3]

Topic: Counting: combinations with constraints (review)
· Medium

Correct answer: A

Concept: split 'at least' into disjoint cases, or use the complement.

$$\binom{4}{2}\binom{6}{3} + \binom{4}{3}\binom{6}{2} + \binom{4}{4}\binom{6}{1} = 120 + 60 + 6 = 186$$

Complement check: total minus (0 or 1 woman):

$$\binom{10}{5} - \binom{6}{5} - \binom{4}{1}\binom{6}{4} = 252 - 6 - 60 = 186$$

(B) is the complement count (0 or 1 woman). (C) counts EXACTLY 2 women only. (D) is the over-count $C(4,2) \cdot C(8,3) = 6 \times 56 = 336$ from 'choose 2 women first, then any 3 of the remaining 8'.

Trap: Never 'fix the required women then choose the rest from everyone' — it double-counts committees with 3 or 4 women.

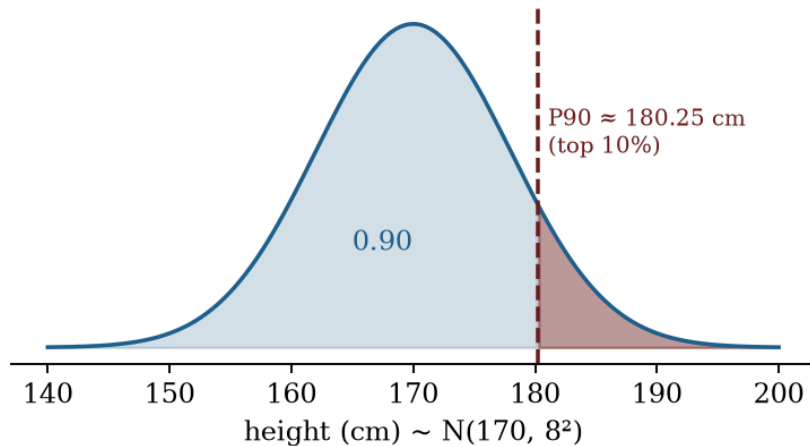
Q.28 [NAT · 2 marks]

Topic: Normal percentile (review) · Medium

Correct answer: 180.25 (range 180.2 to 180.3)**Concept:** the p -th quantile of $N(\mu, \sigma^2)$ is $\mu + z \sigma$ with $\Phi(z) = p$.

$$P(X > h) = 0.10 \Rightarrow \frac{h - 170}{8} = 1.2816$$

$$h = 170 + 1.2816 \times 8 = 170 + 10.2528 = 180.2528$$



Using 1.645 (the 5% point) gives 183.16 cm — the cut-off for the top 5%, not 10%.

Key idea: Top 10% $\Rightarrow$ upper-tail area 0.10 $\Rightarrow z = +1.2816$; bottom 10% would use -1.2816 .**Q.29** [MCQ · 2 marks · -2/3]

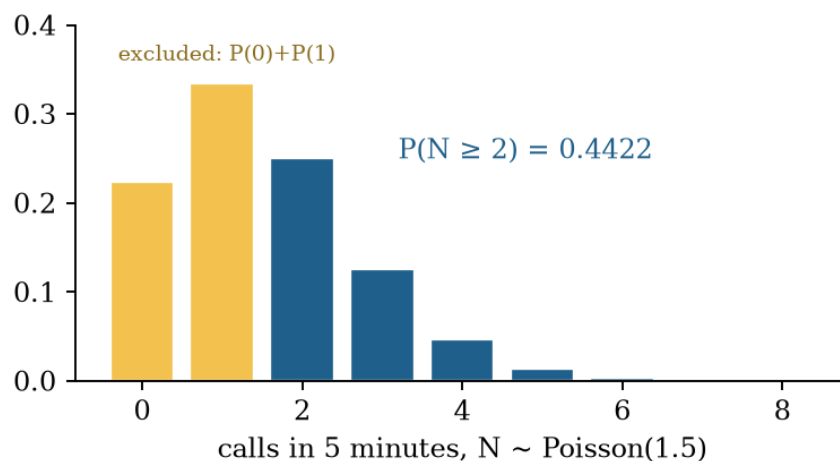
Topic: Poisson distribution with rescaled rate (review) · Medium

Correct answer: A**Concept:** counts in an interval of length t are $\text{Poisson}(\lambda t)$; rescale the rate first.

$$\lambda t = 3 \times \frac{5}{10} = 1.5$$

$$P(N \geq 2) = 1 - P(0) - P(1) = 1 - e^{-1.5}(1 + 1.5)$$

$$= 1 - 2.5 \times 0.22313 = 1 - 0.55783 = 0.4422$$

(B) is $P(N \leq 1)$. (C) forgets to rescale (uses mean 3). (D) is $P(N = 0)$.

Trap: Always convert the rate to the interval in the question before using the PMF.

Topic: Two regression lines: means, r and variances · Hard

Q.30 [MSQ · 2 marks]

Correct answer: A, B, D

Concept: both regression lines pass through $(\bar{x}, \bar{y})$; $r^2 = b_{YX} b_{XY}$ with r having the common sign of the slopes, and $b_{YX} b_{XY}$ must be ≤ 1 .

(A) Correct. Solve simultaneously: from $6x + y = 31$, $y = 31 - 6x$; then $3x + 62 - 12x = 26 \Rightarrow x = 4$, $y = 7$.

Identify the lines. Try $3x + 2y = 26$ as Y on X and $6x + y = 31$ as X on Y :

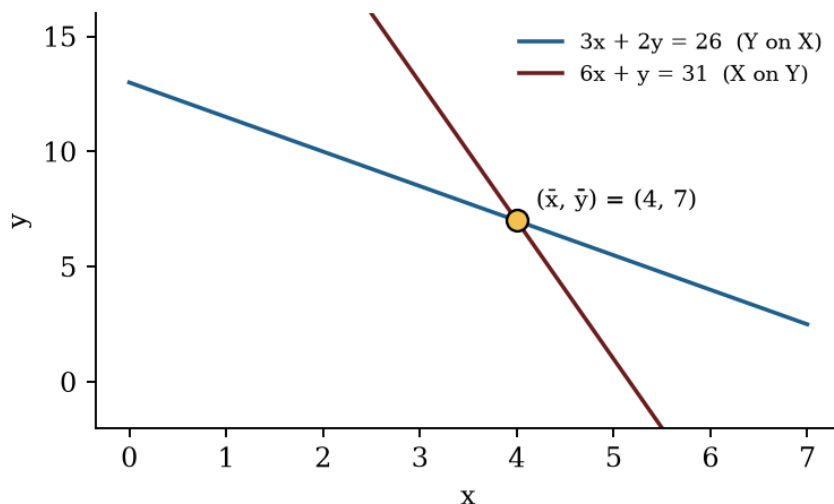
$$b_{YX} = -\frac{3}{2}, \quad b_{XY} = -\frac{1}{6}, \quad b_{YX}b_{XY} = \frac{1}{4} \leq 1 \quad \checkmark$$

(The other assignment gives $b_{YX} = -6$, $b_{XY} = -2/3$, product $4 > 1$ — impossible.)

(B) Correct, (C) Incorrect. $r^2 = 1/4$ and both slopes are negative, so $r = -0.5$.

(D) Correct.

$$b_{YX} = r \frac{\sigma_Y}{\sigma_X} \Rightarrow -1.5 = -0.5 \frac{\sigma_Y}{3} \Rightarrow \sigma_Y = 9, \text{ Var}(Y) = 81$$



Shortcut: Test both assignments of the lines: the valid one has slope product ≤ 1 .

Mock Test 09 — Advanced Full-Syllabus Test I

Hardest GATE DA level — multi-concept questions across the entire Probability & Statistics syllabus.

Item	Details
Questions	30 (12 × 1-mark, 18 × 2-mark)
Maximum marks	48
Question types	MCQ: 14 · MSQ: 7 · NAT: 9
Suggested time	90 minutes
Negative marking	MCQ only: $-1/3$ (1-mark), $-2/3$ (2-mark). None for MSQ & NAT.
Calculator	Use only the GATE virtual scientific calculator

Topic focus of this set

A final-stage readiness check covering the WHOLE syllabus at the top end of GATE DA difficulty: counting with restrictions, independence subtleties, mixed CDFs, order statistics of uniforms, minima of exponentials, Poisson thinning, random sums and conditional variance, joint pdfs on triangular supports, geometric probability, normal mixtures, covariance of indicator sums, Bayes with binomial and naive-Bayes likelihoods, CLT-based sample-size design, z-test power, paired t-test, t-based CI from raw sums, and a chi-squared goodness-of-fit test whose expected counts must first be derived (Hardy-Weinberg). Almost every 2-mark question chains two or more ideas — attempt it under strict 90-minute conditions and treat any slip as a revision flag.

Instructions

- Questions 1 onwards are arranged as in GATE: all 1-mark questions first, then 2-mark questions.
- **MCQ** – exactly one correct option. **MSQ** – ONE OR MORE options are correct; credit only if you select all correct options and no wrong option (no partial marks, no negative marks).
- **NAT** – enter a numerical value using the virtual keypad. The accepted range is given in the key; round as instructed in the question.
- Statistical tables (Z , t , χ^2) are provided in the Appendix; values in solutions use those tables.
- Attempt in one sitting, then check the answer key and read every solution — including those you got right — for the alternative methods and traps.

Mock Test 09 · Question Paper

1-mark questions

Q.1 MCQ · 1 mark · -1/3

The number of distinct arrangements of the letters of the word **STATISTICS** in which the three S's do **not** all appear together (as a block SSS) is

- (A) 50400
- (B) 47040
- (C) 43680
- (D) 3360

Q.2 MSQ · 1 mark

Two fair coins are tossed. Let $A = \{\text{first coin shows H}\}$, $B = \{\text{second coin shows H}\}$ and $C = \{\text{exactly one head}\}$. Which of the following statements is/are TRUE?

- (A) A, B, C are pairwise independent
- (B) A, B, C are mutually independent
- (C) $P(A \cap B \cap C) = 0$
- (D) A and C^c are independent

Q.3 NAT · 1 mark

A sensor node works only if all three of its components work (series system). The component lifetimes are independent exponential random variables with failure rates 0.1, 0.2 and 0.2 per year. The probability that the node is still working after 3 years is _____ (round off to 3 decimal places).

Answer: _____

Q.4 MCQ · 1 mark · -1/3

X_1, X_2, X_3 are i.i.d. $\text{Uniform}(0, 1)$ and M is their median (the middle value). $P(M \leq 1/4)$ equals

- (A) 5/32
- (B) 1/64
- (C) 9/64
- (D) 1/4

Q.5 MSQ · 1 mark

Which of the following statements is/are TRUE?

- (A) If $Z_1, \dots, Z_k$ are i.i.d. $N(0, 1)$, then $\sum Z_i^2$ has mean k and variance $2k$
- (B) For every finite ν , the t_ν density has heavier tails than $N(0, 1)$
- (C) If $T \sim t_\nu$, then $T^2 \sim \chi^2_1$
- (D) The χ^2_k density is symmetric about its mean k

Q.6 NAT · 1 mark

E-mails arrive at an inbox as a Poisson process with rate 12 per hour. Independently, each e-mail is spam with probability 0.25. The probability that no spam arrives in a 20-minute window is _____ (round off to 3 decimal places).

Answer: _____

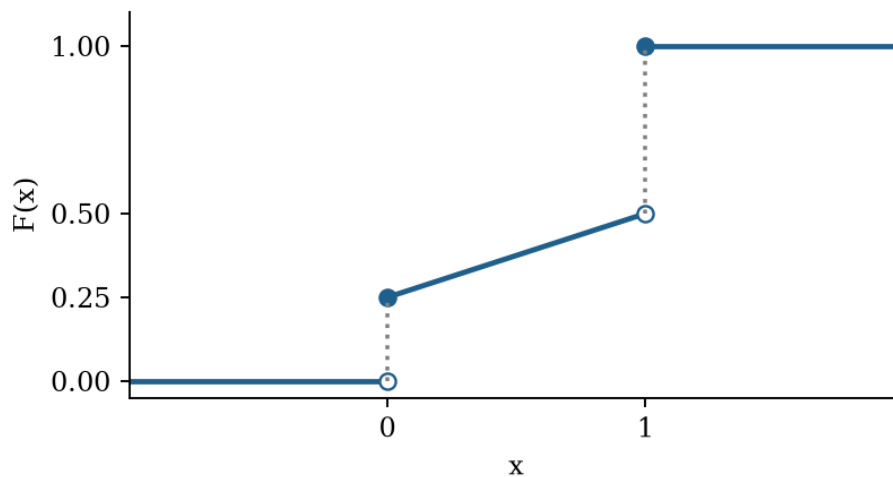
Q.7 MCQ · 1 mark · -1/3

Ten observations have mean 20 and standard deviation 4 (computed with divisor n). One observation whose value is exactly 20 is deleted. The standard deviation (divisor n) of the remaining 9 observations is

- (A) 4.000
- (B) 4.216
- (C) 3.795
- (D) 4.444

Q.8 MCQ · 1 mark · -1/3

A random variable X has CDF $F(x) = 0$ for $x < 0$, $F(x) = (x + 1)/4$ for $0 \leq x < 1$ and $F(x) = 1$ for $x \geq 1$ (plotted below). $E[X]$ equals



- (A) $1/8$
- (B) $3/8$
- (C) $1/2$
- (D) $5/8$

Q.9 MCQ · 1 mark · -1/3

$\text{Var}(X) = 4$, $\text{Var}(Y) = 9$ and the correlation coefficient of X and Y is -0.5 . $\text{Var}(2X - 3Y + 5)$ equals

- (A) 61
- (B) 97
- (C) 133
- (D) 138

Q.10 NAT · 1 mark

A fair die is rolled; if it shows N , then N fair coins are tossed. Let H be the number of heads obtained. $\text{Var}(H)$ is _____ (round off to 2 decimal places).

Answer: _____

Q.11 MSQ · 1 mark

Which of the following statements is/are TRUE?

- (A) The p-value is the probability that H_0 is true given the data
- (B) If the 95% two-sided z-interval for μ excludes μ_0 , the two-sided z-test of $H_0: \mu = \mu_0$ rejects at $\alpha = 0.05$
- (C) For a fixed α and a fixed true $\mu \neq \mu_0$, the power of the z-test increases with n
- (D) $P(\text{Type I error}) = 1 - \text{power}$

Q.12 MCQ · 1 mark · -1/3

$X \sim N(\mu, \sigma^2)$ with $P(X < 10) = 0.1587$ and $P(X > 16) = 0.0228$. Using $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$ and $\Phi(0.5) = 0.6915$, $P(X > 13)$ equals

- (A) 0.6915
- (B) 0.3085
- (C) 0.1915
- (D) 0.4013

2-mark questions**Q.13 NAT · 2 marks**

A bag has three coins: two are fair and one is biased with $P(H) = 0.8$. A coin is drawn at random and tossed 5 times, giving exactly 4 heads. The same coin is tossed once more. The probability that this sixth toss is a head is _____ (round off to 3 decimal places).

Answer: _____

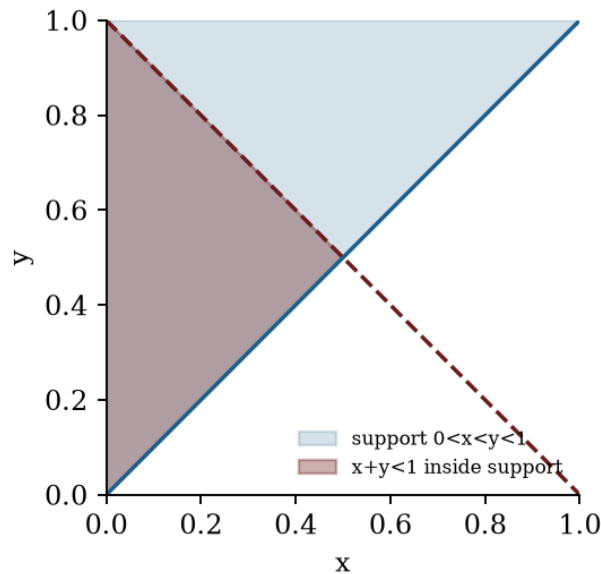
Q.14 MCQ · 2 marks · -2/3

The number of insurance claims in a month, N , is Poisson with mean 4. Claim sizes $X_1, X_2, \dots$ are i.i.d. exponential with mean 2 (lakh rupees), independent of N . Let $S = X_1 + \dots + X_N$ ($S = 0$ if $N = 0$). $\text{Var}(S)$ equals

- (A) 8
- (B) 16
- (C) 32
- (D) 64

Q.15 MSQ · 2 marks

(X, Y) has joint pdf $f(x, y) = 2$ for $0 < x < y < 1$ and 0 otherwise (support shaded below). Which of the following is/are TRUE?



- (A) X and Y are independent
- (B) $E[Y | X = x] = (1 + x)/2$ for $0 < x < 1$
- (C) $P(X + Y < 1) = 1/2$
- (D) $\text{Cov}(X, Y) = 1/36$

Q.16 NAT · 2 marks

Two analysts A and B arrive at a meeting room independently, each at a time uniformly distributed between 10:00 and 11:00. A waits 10 minutes for B and then leaves; B waits 20 minutes for A and then leaves. The probability that they meet is _____ (round off to 3 decimal places).

Answer: _____

Q.17 MCQ · 2 marks · -2/3

An ML team wants to estimate the misclassification rate p of a model on live traffic with a 95% confidence interval of half-width (margin of error) at most 0.03. A pilot study shows that p is certainly no larger than 0.20. Using the normal approximation ($z_{0.025} = 1.96$), the smallest sample size that guarantees the requirement is

- (A) 683
- (B) 1068
- (C) 682
- (D) 601

Q.18 NAT · 2 marks

Genotypes at a locus with alleles A and a are recorded for 100 individuals:

Genotype	AA	Aa	aa	Total
Count	50	30	20	100

To test Hardy-Weinberg equilibrium (expected proportions p^2 , $2p(1 - p)$, $(1 - p)^2$), the allele frequency p is estimated from the data by allele counting. The value of the Pearson χ^2 statistic is _____ (round off to 2 decimal places).

Answer: _____

Q.19 MCQ · 2 marks · -2/3

Ten distinguishable balls are thrown independently and uniformly into 5 boxes. Let X be the number of empty boxes. $\text{Var}(X)$ is closest to

- (A) 0.250
- (B) 0.537
- (C) 0.479
- (D) 0.370

Q.20 MSQ · 2 marks

In an entrance test, 60% of candidates come from population 1 with scores $N(50, 10^2)$ and 40% from population 2 with scores $N(70, 10^2)$. Let X be the score of a randomly chosen candidate. Using $\Phi(2) = 0.9772$, which of the following is/are TRUE?

- (A) $E[X] = 58$
- (B) The standard deviation of X is 14
- (C) $X \sim N(58, 14^2)$
- (D) $P(X > 70) \approx 0.214$

Q.21 MCQ · 2 marks · -2/3

A new caching layer is tested on 6 servers. The reduction in median latency (ms) on each server, d = before – after, is:

Server	1	2	3	4	5	6
d	3	5	-1	4	6	2

Test $H_0: \mu_d = 0$ against $H_1: \mu_d \neq 0$ at $\alpha = 0.05$ ($t_{0.025,5} = 2.571$). Which is correct?

- (A) $t \approx 3.42$; reject H_0
- (B) $t \approx 1.28$; do not reject H_0
- (C) $t \approx 3.12$; reject H_0
- (D) $t \approx 2.85$; reject H_0

Q.22 NAT · 2 marks

Daily page-load counts are normal with $\sigma = 15$. To test $H_0: \mu = 100$ against $H_1: \mu > 100$, a z -test at $\alpha = 0.05$ ($z_{0.05} = 1.645$) is based on $n = 36$ days. If in fact $\mu = 106$, the power of the test is _____ (round off to 2 decimal places).

Answer: _____

Q.23 MCQ · 2 marks · -2/3

The numbers of bug reports from Android and iOS users in a day are independent Poisson random variables with means 3 and 5 respectively. Given that a total of 8 reports arrived, the probability that exactly 3 came from Android users is

- (A) 0.2816
- (B) 0.2240
- (C) 0.3750
- (D) 0.2188

Q.24 MSQ · 2 marks

$T_1 \sim \text{Exp}(\text{rate } 1)$ and $T_2 \sim \text{Exp}(\text{rate } 2)$ are independent lifetimes (in years) of two redundant disks. Which of the following is/are TRUE?

- (A) $P(T_1 < T_2) = 1/3$
- (B) $\min(T_1, T_2)$ is exponential with mean $1/3$
- (C) $E[\max(T_1, T_2)] = 7/6$
- (D) $P(T_1 > 2 \mid T_1 > 1) = e^{-2}$

Q.25 NAT · 2 marks

A spam filter assumes word occurrences are conditionally independent given the class. Prior: $P(\text{spam}) = 0.3$. The word 'free' appears in 40% of spam and 5% of ham e-mails; the word 'meeting' appears in 2% of spam and 20% of ham e-mails. An e-mail contains 'free' but does NOT contain 'meeting'. The posterior probability that it is spam is _____ (round off to 3 decimal places).

Answer: _____

Q.26 MCQ · 2 marks · -2/3

$X \sim \text{Uniform}(0, 6)$ and, given $X = x$, $Y \sim N(x, 1)$. The correlation coefficient between X and Y is

- (A) $1/2$
- (B) $\sqrt{3}/2$
- (C) $3/4$
- (D) $1/\sqrt{3}$

Q.27 MCQ · 2 marks · -2/3

X has pdf $f(x) = x e^{-x}$ for $x > 0$. Which of the following correctly describes its mode, median and mean?

- (A) mode = 1 < mean = 2 < median ≈ 2.20
- (B) mode = 1 < median ≈ 1.68 < mean = 2
- (C) median ≈ 1.39 < mode = 1.5 < mean = 2
- (D) mode = median = mean = 2

Q.28 NAT · 2 marks

API response times (ms) of $n = 16$ randomly sampled requests give $\sum x_i = 1920$ and $\sum x_i^2 = 231540$. Assuming normality, the WIDTH (upper minus lower limit) of the 95% confidence interval for the mean response time is _____ ms (round off to 2 decimal places). Use $t_{0.025,15} = 2.131$.

Answer: _____

Q.29 MSQ · 2 marks

Which of the following statements is/are TRUE?

- (A) For a χ^2 test of independence on a 3×4 contingency table, the degrees of freedom are 6
- (B) A Pearson χ^2 goodness-of-fit statistic can be negative when observed counts are below expected counts
- (C) For a random sample of size n from $N(\mu, \sigma^2)$, $(n - 1)S^2/\sigma^2 \sim \chi^2_{n-1}$
- (D) For $n = 10$ the two-sided 5% critical value of the one-sample t-test exceeds 1.96

Q.30 MCQ · 2 marks · -2/3

Five cards are dealt from a well-shuffled standard deck of 52. The probability of getting exactly 'two pairs' (two cards of one rank, two cards of another rank, and a fifth card of a third rank) is closest to

- (A) 0.0951
- (B) 0.0238
- (C) 0.0211
- (D) 0.0475

Mock Test 09 · Answer Key

Q	Type	Marks	Answer	Topic
1	MCQ	1	B	Permutations with repetition
2	MSQ	1	A, C, D	Pairwise vs mutual independence
3	NAT	1	0.223 (range 0.222 to 0.224)	Minimum of exponentials (series reliability)
4	MCQ	1	A	Order statistics of uniforms
5	MSQ	1	A, B	Chi-squared, t and normal relationships
6	NAT	1	0.368 (range 0.367 to 0.369)	Poisson thinning
7	MCQ	1	B	Standard deviation (update)
8	MCQ	1	D	Mixed CDF and expectation
9	MCQ	1	C	Variance of a linear combination
10	NAT	1	1.60 (range 1.59 to 1.61)	Conditional variance (random number of trials)
11	MSQ	1	B, C	Hypothesis-testing concepts
12	MCQ	1	B	Normal distribution (parameter recovery)
13	NAT	2	0.670 (range 0.668 to 0.672)	Bayes theorem with binomial likelihood
14	MCQ	2	C	Random sums (compound Poisson)
15	MSQ	2	B, C, D	Joint pdf on a triangle; conditional expectation; covariance
16	NAT	2	0.431 (range 0.429 to 0.432)	Geometric probability (joint uniform)
17	MCQ	2	A	CLT-based sample size for a proportion
18	NAT	2	11.60 (range 11.55 to 11.66)	Chi-squared goodness of fit with estimated parameter (genetics)
19	MCQ	2	D	Covariance of indicator sums
20	MSQ	2	A, B, D	Normal mixtures
21	MCQ	2	C	Paired t-test
22	NAT	2	0.77 (range 0.77 to 0.78)	Power of a one-sided z-test
23	MCQ	2	A	Conditioning a Poisson sum
24	MSQ	2	A, B, C	Competing exponentials; memorylessness
25	NAT	2	0.808 (range 0.806 to 0.809)	Naive Bayes classifier (Bayes theorem)
26	MCQ	2	B	Conditional distributions; correlation
27	MCQ	2	B	Mean, median and mode of a continuous distribution
28	NAT	2	9.29 (range 9.25 to 9.33)	t confidence interval from raw sums
29	MSQ	2	A, C, D	Sampling distributions & Chi-squared / t tests
30	MCQ	2	D	Combinatorial probability (card game)

Score tracker

	Attempted	Correct	Wrong (MCQ)	Marks
1-mark				
2-mark				
Total				/ 48

Mock Test 09 · Detailed Solutions

Q.1 [MCQ · 1 mark · -1/3]

Topic: Permutations with repetition · Medium

Correct answer: B

Concept: arrangements of a multiset = $n! / (\text{product of factorials of repeat counts})$; use the complement 'all S together'.

Letter counts: S = 3, T = 3, I = 2, A = 1, C = 1 (total 10).

$$\text{Total} = \frac{10!}{3! 3! 2!} = \frac{3628800}{72} = 50400$$

Glue the three S's into one super-letter [SSS]. Now there are 8 objects: [SSS], T, T, T, I, I, A, C.

$$\text{SSS together} = \frac{8!}{3! 2!} = \frac{40320}{12} = 3360$$

$$\text{Required} = 50400 - 3360 = 47040$$

Option (C) divides 8! only by 3! (forgets that the two I's are identical); (D) is the complementary count itself.

Trap: When you glue identical letters into a block, the block is a single object — do not multiply by 3! for internal order (the S's are identical), but keep dividing by the factorials of the OTHER repeated letters.

Q.2 [MSQ · 1 mark]

Topic: Pairwise vs mutual independence · Medium

Correct answer: A, C, D

Concept: mutual independence needs every sub-collection to factor, including the triple; pairwise independence does not imply it.

Sample space {HH, HT, TH, TT}, each 1/4. A = {HH, HT}, B = {HH, TH}, C = {HT, TH}; each has probability 1/2.

Pair	Intersection	Probability	Product of marginals
A, B	{HH}	1/4	1/4
A, C	{HT}	1/4	1/4
B, C	{TH}	1/4	1/4

(A) **TRUE** — every pair factorises.

(C) **TRUE** — $A \cap B = \{HH\}$ has two heads, so it cannot have exactly one head: $A \cap B \cap C = \emptyset$.

$$P(A \cap B \cap C) = 0 \neq P(A)P(B)P(C) = \frac{1}{8}$$

(B) **FALSE** — the triple product fails.

(D) **TRUE** — if A and C are independent then so are A and C^c : $P(A \cap C^c) = P(A) - P(A \cap C) = 1/2 - 1/4 = 1/4 = P(A)P(C^c)$.

Key idea: Knowing any two of A, B, C determines the third completely — that is why the triple fails while every pair is fine.

Topic: Minimum of exponentials (series reliability) · Medium

Q.3 [NAT · 1 mark]

Correct answer: 0.223 (range 0.222 to 0.224)

Concept: the minimum of independent exponentials is exponential with the sum of the rates. System lifetime $T = \min(T_1, T_2, T_3)$.

$$P(T > t) = \prod_{i=1}^3 P(T_i > t) = e^{-0.1t} e^{-0.2t} e^{-0.2t} = e^{-0.5t}$$

$$P(T > 3) = e^{-1.5} \approx 0.2231$$

Shortcut: The expected life of the series node is $1/0.5 = 2$ years, and the probability that component 1 is the first to fail is $0.1/0.5 = 0.2$ — both follow from the same 'competing exponentials' fact.

Q.4 [MCQ · 1 mark · -1/3]

Topic: Order statistics of uniforms · Medium

Correct answer: A

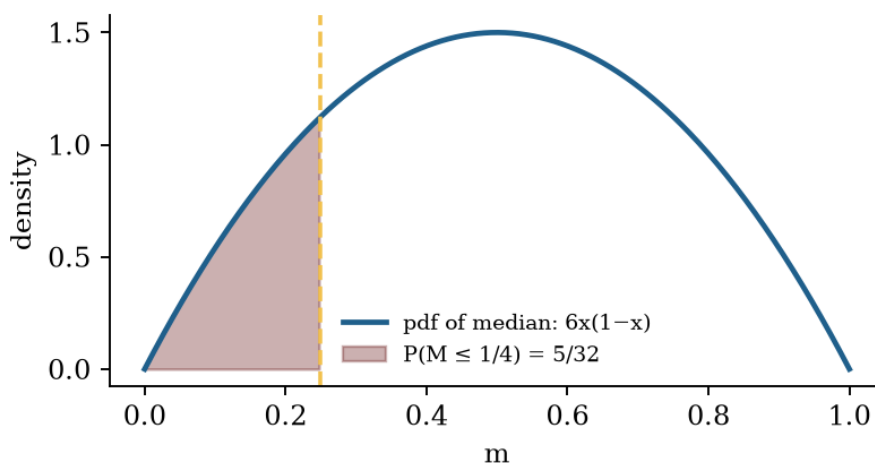
Concept: the k -th order statistic is $\leq m$ iff at least k of the n observations are $\leq m$ — a binomial count.

Let N = number of $X_i \leq 1/4$. Then $N \sim \text{Bin}(3, 1/4)$. The median is $\leq 1/4$ iff $N \geq 2$.

$$P(N \geq 2) = \binom{3}{2} \left(\frac{1}{4}\right)^2 \frac{3}{4} + \left(\frac{1}{4}\right)^3 = \frac{9}{64} + \frac{1}{64} = \frac{5}{32}$$

Cross-check with the pdf of the median, $f_M(m) = 6m(1 - m)$:

$$\int_0^{1/4} 6m(1 - m) dm = [3m^2 - 2m^3]_0^{1/4} = \frac{3}{16} - \frac{1}{32} = \frac{5}{32}$$



(B) $1/64$ is $P(\text{all three} \leq 1/4)$, i.e. the maximum; (C) $9/64$ counts exactly two only.

Trap: 'At least k successes' — never 'exactly k ' — is what translates an order statistic into a binomial event.

Q.5 [MSQ · 1 mark]**Topic: Chi-squared, t and normal relationships · Medium****Correct answer: A, B****Concept:** χ^2_k = sum of k squared standard normals; $t_\nu = Z/\sqrt{V/\nu}$ with $V \sim \chi^2_\nu$ independent of Z.**(A) TRUE** — $E[Z^2] = 1$ and $\text{Var}(Z^2) = E[Z^4] - 1 = 3 - 1 = 2$, so the sum has mean k, variance 2k.**(B) TRUE** — the random denominator $\sqrt{V/\nu}$ inflates the spread; $P(|T| > c) > P(|Z| > c)$ for large c, e.g. $t_{0.025,10} = 2.228 > 1.96$.**(C) FALSE** — $T^2 = Z^2/(V/\nu)$ is a ratio of two independent χ^2 's scaled by their df, i.e. an $F(1, \nu)$ variable, not χ^2_1 . Only as $\nu \rightarrow \infty$ does it approach χ^2_1 .**(D) FALSE** — χ^2_k is supported on $(0, \infty)$ and right-skewed; mean k exceeds its mode $k - 2$.**Trap:** Option (C) is the classic 'limit statement passed off as exact' trap.**Q.6 [NAT · 1 mark]****Topic: Poisson thinning · Medium****Correct answer: 0.368 (range 0.367 to 0.369)****Concept:** independently marking each event of a $\text{Poisson}(\lambda)$ stream with probability p gives a $\text{Poisson}(\lambda p)$ stream (thinning).Spam stream rate = $12 \times 0.25 = 3$ per hour. In 20 min = $1/3$ hour the spam count is Poisson with mean

$$\mu = 3 \times \frac{1}{3} = 1, \quad P(\text{no spam}) = e^{-1} \approx 0.368$$

Long route (same answer): condition on the total $N \sim \text{Poisson}(4)$:

$$\sum_{n=0}^{\infty} e^{-4} \frac{4^n}{n!} (0.75)^n = e^{-4} e^3 = e^{-1}$$

Shortcut: Thinning lets you ignore the non-spam e-mails entirely.**Q.7 [MCQ · 1 mark · -1/3]****Topic: Standard deviation (update) · Medium****Correct answer: B****Concept:** $SD = \sqrt{(\text{sum of squared deviations} / n)}$; removing a value at the mean leaves the mean and the sum of squared deviations unchanged but reduces n.

$$\sum_{i=1}^{10} (x_i - 20)^2 = n\sigma^2 = 10 \times 16 = 160$$

Removing $x = 20$ removes a zero deviation, and the mean of the remaining 9 values is still $(200 - 20)/9 = 20$. Hence

$$\sigma_{\text{new}} = \sqrt{\frac{160}{9}} = \frac{4\sqrt{10}}{3} \approx 4.216$$

(A) assumes SD is unchanged; (C) multiplies instead of divides by 9/10; (D) is $40/9 = 160/36$, a mis-scaling.

Key idea: The SD INCREASES when you delete a point sitting exactly at the mean — the remaining points are, on average, further from the centre.

Q.8 [MCQ · 1 mark · -1/3]

Topic: Mixed CDF and expectation · Hard

Correct answer: D

Concept: jumps in a CDF are point masses; the sloped part is a density. For $X \geq 0$, $E[X] = \int_0^\infty (1 - F(x)) dx$.

Jump at 0: $P(X = 0) = F(0) - F(0^-) = 1/4$. Jump at 1: $P(X = 1) = 1 - F(1^-) = 1 - 2/4 = 1/2$.

Continuous part on $(0, 1)$: density $F'(x) = 1/4$ (total mass $1/4$). Check: $1/4 + 1/2 + 1/4 = 1$.

$$E[X] = 0 \cdot \frac{1}{4} + \int_0^1 x \cdot \frac{1}{4} dx + 1 \cdot \frac{1}{2} = \frac{1}{8} + \frac{1}{2} = \frac{5}{8}$$

Tail-integral check:

$$\int_0^1 \left(1 - \frac{x+1}{4}\right) dx = \int_0^1 \frac{3-x}{4} dx = \frac{3 - 1/2}{4} = \frac{5}{8}$$

(A) $1/8$ uses only the continuous part and forgets the atom at 1.

Trap: Always read off the jumps first — a mixed distribution's mean needs both the atoms and the density.

Q.9 [MCQ · 1 mark · -1/3]

Topic: Variance of a linear combination · Medium

Correct answer: C

Concept: $\text{Var}(aX + bY + c) = a^2 \text{Var } X + b^2 \text{Var } Y + 2ab \text{Cov}(X, Y)$; constants do not matter.

$$\text{Cov}(X, Y) = \rho \sigma_X \sigma_Y = (-0.5)(2)(3) = -3$$

$$\text{Var} = 4(4) + 9(9) + 2(2)(-3)(-3) = 16 + 81 + 36 = 133$$

(A) $61 = 97 - 36$ uses the wrong sign for the cross term; (B) 97 ignores correlation; (D) adds the constant 5.

Trap: Negative correlation combined with a minus sign in front of Y gives a POSITIVE cross term: $2ab \cdot \text{Cov} = 2(2)(-3)(-3) = +36$.

Q.10 [NAT · 1 mark]

Topic: Conditional variance (random number of trials) · Hard

Correct answer: 1.60 (range 1.59 to 1.61)

Concept: law of total variance $\text{Var}(H) = E[\text{Var}(H | N)] + \text{Var}(E[H | N])$.

Given $N, H \sim \text{Bin}(N, 1/2)$: $E[H | N] = N/2$ and $\text{Var}(H | N) = N/4$.

$$\begin{aligned} E[N] &= 3.5, & \text{Var}(N) &= \frac{6^2 - 1}{12} = \frac{35}{12} \\ \text{Var}(H) &= E\left[\frac{N}{4}\right] + \text{Var}\left(\frac{N}{2}\right) = \frac{3.5}{4} + \frac{35/12}{4} \\ &= 0.875 + 0.7292 = 1.6042 \approx 1.60 \end{aligned}$$

Trap: Forgetting the second term gives 0.875 — the variability of N itself adds 0.729.

Q.11 [MSQ · 1 mark]

Topic: Hypothesis-testing concepts · Medium

Correct answer: B, C

Concept: duality of CIs and tests; power = $P(\text{reject } H_0 \mid H_1 \text{ true})$.

(A) FALSE — the p-value is $P(\text{data at least this extreme} \mid H_0)$, computed assuming H_0 ; it is not a posterior probability.

(B) TRUE — μ_0 is outside $\bar{x} \pm 1.96\sigma/\sqrt{n} \iff |\bar{x} - \mu_0|/(\sigma/\sqrt{n}) > 1.96$.

(C) TRUE — the standardised shift $(\mu - \mu_0)\sqrt{n}/\sigma$ grows with n , so the alternative distribution moves deeper into the rejection region.

(D) FALSE — $1 - \text{power} = \beta = P(\text{Type II error})$. α and power are computed under different hypotheses.

Key idea: α is fixed under H_0 ; β and power live under H_1 .

Q.12 [MCQ · 1 mark · -1/3]

Topic: Normal distribution (parameter recovery) · Medium

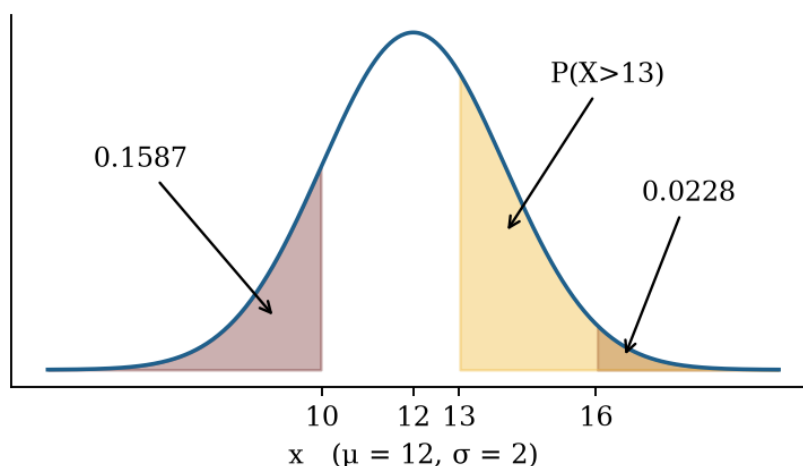
Correct answer: B

Concept: each tail probability fixes one z-score; two equations give μ and σ .

$P(X < 10) = 0.1587 = \Phi(-1) \Rightarrow (10 - \mu)/\sigma = -1$. $P(X > 16) = 0.0228 \Rightarrow (16 - \mu)/\sigma = 2$.

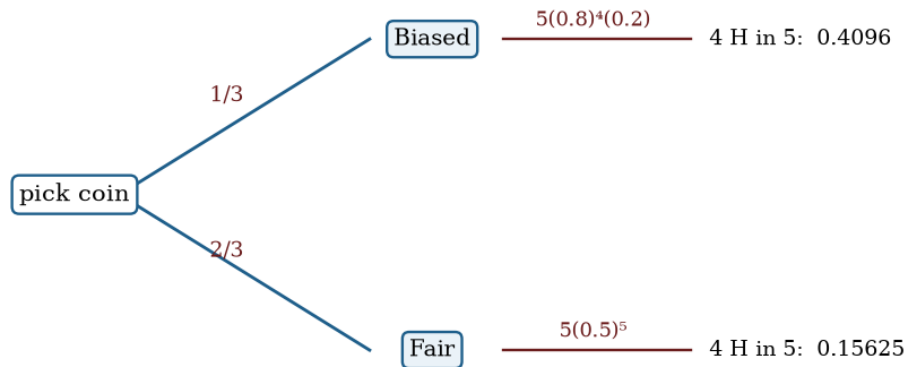
$$\mu - \sigma = 10, \quad \mu + 2\sigma = 16 \Rightarrow \sigma = 2, \quad \mu = 12$$

$$P(X > 13) = P\left(Z > \frac{13 - 12}{2}\right) = 1 - \Phi(0.5) = 0.3085$$



(A) is $P(X < 13)$; (C) is $P(12 < X < 13)$.

Shortcut: Write both tail conditions as linear equations in μ and σ — never guess σ from one tail only.

Q.13 [NAT · 2 marks]**Topic: Bayes theorem with binomial likelihood · Hard****Correct answer: 0.670 (range 0.668 to 0.672)****Concept:** update the prior on the coin with the binomial likelihood (Bayes), then average the next-toss probability over the posterior (law of total probability).

Likelihoods of '4 heads in 5 tosses':

$$L_B = \binom{5}{4}(0.8)^4(0.2) = 0.4096, \quad L_F = \binom{5}{4}(0.5)^5 = 0.15625$$

$$P(B | 4H) = \frac{\frac{1}{3}(0.4096)}{\frac{1}{3}(0.4096) + \frac{2}{3}(0.15625)} = \frac{0.4096}{0.4096 + 0.3125}$$

$$P(B | 4H) = \frac{0.4096}{0.7221} \approx 0.5672$$

Given the coin, the tosses are independent, so

$$\begin{aligned} P(H_6 | 4H) &= 0.8P(B | 4H) + 0.5P(F | 4H) \\ &= 0.8(0.5672) + 0.5(0.4328) \approx 0.6702 \end{aligned}$$

Monte-Carlo (10^6 runs) gives ≈ 0.669 – 0.670 , confirming the value.

Key idea: The tosses are NOT unconditionally independent — learning '4 heads' raises the chance the coin is biased, so the 6th-toss probability (0.670) is neither 0.5 nor the prior average 0.6.

Q.14 [MCQ · 2 marks · –2/3]**Topic: Random sums (compound Poisson) · Hard****Correct answer: C****Concept:** for a random sum, $E[S] = E[N]E[X]$ and $\text{Var}(S) = E[N]\text{Var}(X) + \text{Var}(N)(E[X])^2$.Conditional moments: $E[S | N] = 2N$ and $\text{Var}(S | N) = N \cdot \text{Var}(X) = 4N$ (Var of Exp with mean 2 is $2^2 = 4$).

$$\begin{aligned} \text{Var}(S) &= E[\text{Var}(S | N)] + \text{Var}(E[S | N]) = E[4N] + \text{Var}(2N) \\ &= 4(4) + 4(4) = 32 \end{aligned}$$

Compound-Poisson shortcut: $\text{Var}(S) = \lambda E[X^2] = 4 \times (\text{Var } X + (E X)^2) = 4 \times 8 = 32$.Distractors: $16 = E[N]\text{Var}(X)$ only (forgets randomness of N); $64 = (E N)^2 \text{Var}(X)$, treating S as 4 copies of the same claim; $8 = E[S]$.

Shortcut: For compound Poisson, $\text{Var}(S) = \lambda E[X^2]$ — the SECOND RAW moment, not the variance.

Topic: Joint pdf on a triangle; conditional expectation; covariance · Hard

Q.15 [MSQ · 2 marks]

Correct answer: B, C, D

Concept: marginals by integrating out; conditional pdf = joint/marginal; a non-rectangular support rules out independence.

Marginals:

$$f_X(x) = \int_x^1 2 dy = 2(1 - x), \quad f_Y(y) = \int_0^y 2 dx = 2y$$

(A) FALSE — $f_X(x)f_Y(y) = 4y(1 - x) \neq 2$, and the support is a triangle (e.g. $f = 0$ at $(0.8, 0.2)$ though both marginals are positive there).

(B) TRUE — for fixed x :

$$f_{Y|X}(y|x) = \frac{2}{2(1 - x)} = \frac{1}{1 - x}, \quad x < y < 1 \Rightarrow Y|X = x \sim U(x, 1)$$

so $E[Y | X = x] = (x + 1)/2$.

(C) TRUE — inside the support, $x + y < 1$ means $x < y < 1 - x$ with $x < 1/2$: a triangle with vertices $(0,0)$, $(0,1)$, $(1/2, 1/2)$ of area $1/4$.

$$P(X + Y < 1) = 2 \times \frac{1}{4} = \frac{1}{2}$$

(D) TRUE:

$$E[X] = \int_0^1 2x(1 - x) dx = \frac{1}{3}, \quad E[Y] = \int_0^1 2y^2 dy = \frac{2}{3}$$

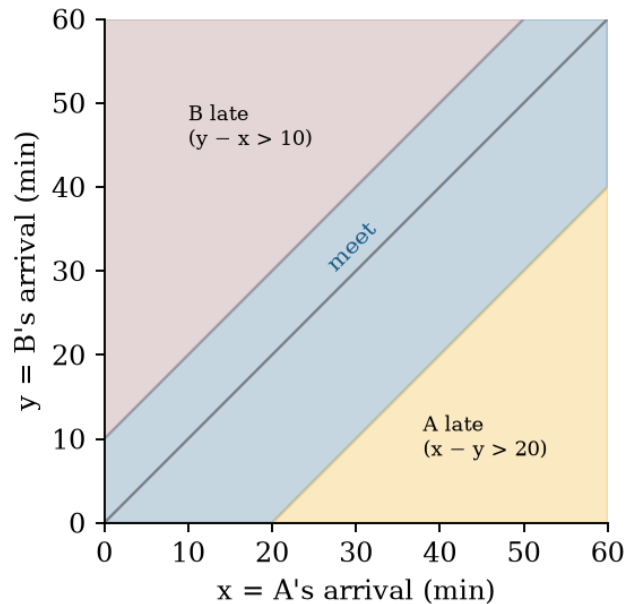
$$E[XY] = \int_0^1 \int_0^y 2xy dx dy = \int_0^1 y^3 dy = \frac{1}{4}$$

$$\text{Cov}(X, Y) = \frac{1}{4} - \frac{1}{3} \cdot \frac{2}{3} = \frac{9 - 8}{36} = \frac{1}{36}$$

Key idea: This is exactly the (min, max) of two i.i.d. $U(0,1)$ variables — that is why $E[X] = 1/3$ and $E[Y] = 2/3$.

Q.16 [NAT · 2 marks]**Topic: Geometric probability (joint uniform) · Hard****Correct answer: 0.431 (range 0.429 to 0.432)****Concept:** for independent uniforms the joint pdf is constant, so probability = favourable area / total area.Let $X, Y \in (0, 60)$ be the arrival minutes of A and B. They meet iff (A first and B arrives within 10 min) or (B first and A arrives within 20 min):

$$0 \leq Y - X \leq 10 \quad \text{or} \quad 0 \leq X - Y \leq 20$$



Non-meeting regions are two right triangles:

$$Y - X > 10 : \frac{50^2}{2} = 1250, \quad X - Y > 20 : \frac{40^2}{2} = 800$$

$$P(\text{meet}) = 1 - \frac{1250 + 800}{3600} = \frac{1550}{3600} \approx 0.4306$$

Trap: Attach each waiting time to the person who arrives FIRST: A's 10-minute patience matters only when A is early ($Y - X$), B's 20 minutes only when B is early ($X - Y$). Swapping them gives the same number here by symmetry of the square — but the triangles must be labelled correctly in an exam figure.

Q.17 [MCQ · 2 marks · -2/3]**Topic: CLT-based sample size for a proportion · Hard****Correct answer: A**

Concept: margin $E = z_{\alpha/2} \sqrt{p(1-p)/n} \Rightarrow n \geq z^2 p(1-p)/E^2$, using the worst-case p over the allowed range.

$p(1-p)$ increases on $[0, 0.5]$, so over $p \leq 0.2$ its maximum is at $p = 0.2$: $0.2 \times 0.8 = 0.16$.

$$n \geq \frac{1.96^2(0.16)}{0.03^2} = \frac{3.8416 \times 0.16}{0.0009} = 682.95$$

n must be an integer AND satisfy the inequality, so round UP: $n = 683$.

(B) 1068 uses the worst case $p = 0.5$ — valid but not the smallest given the pilot information;

(C) 682 rounds down and violates the margin.

Trap: Sample-size answers are always rounded UP, and the 'worst case' p is the value in the allowed range CLOSEST to 0.5.

Q.18 [NAT · 2 marks]**Topic: Chi-squared goodness of fit with estimated parameter (genetics) · Hard****Correct answer: 11.60 (range 11.55 to 11.66)**

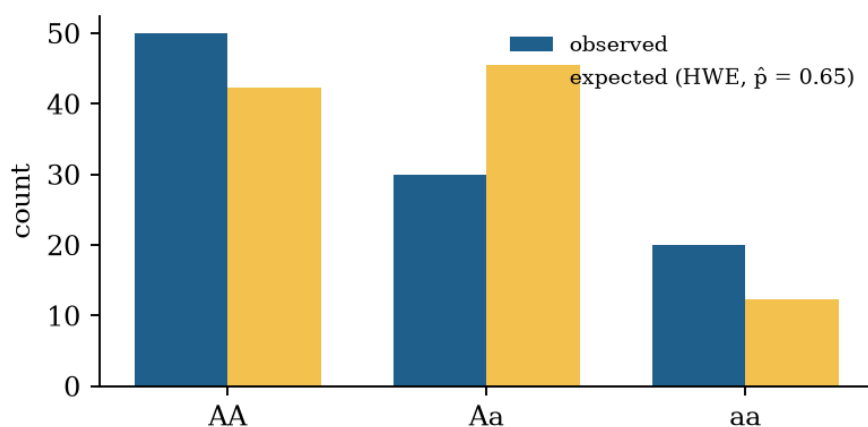
Concept: $\chi^2 = \sum (O - E)^2/E$ with E derived from the model; each estimated parameter costs one degree of freedom.

Allele counting: each AA contributes 2 A alleles, each Aa one; total alleles = 200.

$$\hat{p} = \frac{2(50) + 30}{200} = \frac{130}{200} = 0.65$$

$$E_{AA} = 100(0.65)^2 = 42.25, E_{Aa} = 100(2)(0.65)(0.35) = 45.5, E_{aa} = 100(0.35)^2 = 12.25$$

Genotype	O	E	$(O - E)^2/E$
AA	50	42.25	1.4216
Aa	30	45.50	5.2802
aa	20	12.25	4.9031
Total	100	100	11.6049



$$\chi^2 = \frac{7.75^2}{42.25} + \frac{15.5^2}{45.5} + \frac{7.75^2}{12.25} \approx 11.60$$

Degrees of freedom = 3 categories – 1 – 1 (estimated p) = 1. Since $11.60 > \chi^2_{0.05,1} = 3.841$ (even $> \chi^2_{0.001,1} = 10.83$), HWE is rejected — there is a heterozygote deficit.

Key idea: The deviations are ± 7.75 for both homozygotes and -15.5 for heterozygotes: with one estimated parameter the deviations are forced to cancel, which is the $df = 1$ constraint you can see.

Q.19 [MCQ · 2 marks · -2/3]

Topic: Covariance of indicator sums · Hard

Correct answer: D

Concept: write $X = \sum I_j$ with $I_j = 1$ if box j is empty; then $\text{Var}(X) = \sum \text{Var}(I_j) + \sum_{j \neq k} \text{Cov}(I_j, I_k)$.

$$P(I_j = 1) = \left(\frac{4}{5}\right)^{10} = 0.10737, \quad P(I_j I_k = 1) = \left(\frac{3}{5}\right)^{10} = 0.00605$$

$$E[X] = 5(0.8)^{10} = 0.5369$$

$$E[X^2] = \sum_j E[I_j] + \sum_{j \neq k} E[I_j I_k] = 5(0.8)^{10} + 20(0.6)^{10}$$

$$\begin{aligned} \text{Var}(X) &= 5(0.8)^{10} + 20(0.6)^{10} - 25(0.8)^{20} \\ &= 0.5369 + 0.1209 - 0.2882 = 0.3696 \end{aligned}$$

There are $5 \times 4 = 20$ ordered pairs (j, k) , $j \neq k$. A simulation of 2×10^5 throws gives $\text{Var} \approx 0.37$.

(C) $0.479 = 5p(1 - p)$ ignores the (negative) covariances between boxes; (B) 0.537 is $E[X]$.

Key idea: Indicators of 'box j empty' are negatively correlated: if one box is empty the balls crowd into the others, making another empty box less likely.

Q.20 [MSQ · 2 marks]

Topic: Normal mixtures · Hard

Correct answer: A, B, D

Concept: condition on the population label G ; use total expectation, total variance and total probability.

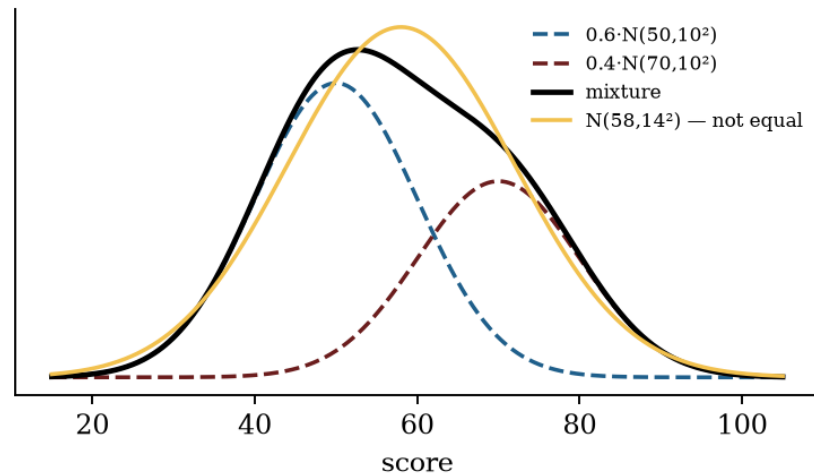
(A) **TRUE:** $E[X] = 0.6(50) + 0.4(70) = 58$.

(B) **TRUE:** $\text{Var}(X) = E[\text{Var}(X | G)] + \text{Var}(E[X | G])$.

$$E[\text{Var}(X | G)] = 100, \quad \text{Var}(E[X | G]) = 0.6(0.4)(70 - 50)^2 = 96$$

$$\text{Var}(X) = 196 \Rightarrow \text{SD} = 14$$

(C) **FALSE:** a mixture of normals is not normal (it is a weighted sum of densities, not of random variables). Its density below is visibly skewed / flattened compared with $N(58, 14^2)$.



(D) TRUE:

$$\begin{aligned}
 P(X > 70) &= 0.6 P\left(Z > \frac{70 - 50}{10}\right) + 0.4 P(Z > 0) = 0.6(0.0228) + 0.4(0.5) \\
 &= 0.01368 + 0.2 = 0.2137
 \end{aligned}$$

(Using $N(58, 14^2)$ would give $P(Z > 0.857) \approx 0.196$ — wrong.)

Trap: Sum of independent normals = normal; MIXTURE of normals = generally not normal. The between-group term $p(1 - p)(\Delta\mu)^2$ is the part students forget.

Q.21 [MCQ · 2 marks · -2/3]

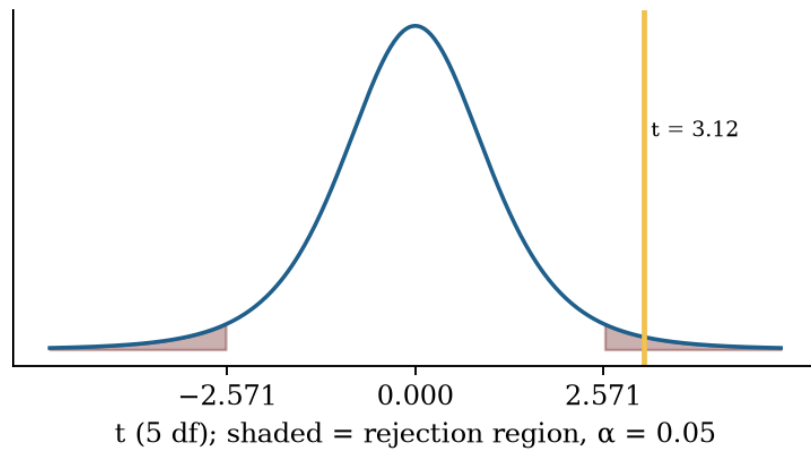
Topic: Paired t-test · Hard

Correct answer: C

Concept: paired data reduce to a one-sample t-test on the differences: $t = \bar{d}/(s_d/\sqrt{n})$ with $n - 1$ df.

$$\begin{aligned}
 \bar{d} &= \frac{3 + 5 - 1 + 4 + 6 + 2}{6} = \frac{19}{6} = 3.1667 \\
 \sum d_i^2 &= 91, \quad \sum (d_i - \bar{d})^2 = 91 - 6(3.1667)^2 = 30.833 \\
 s_d &= \sqrt{\frac{30.833}{5}} = 2.4833, \quad \frac{s_d}{\sqrt{6}} = 1.0138 \\
 t &= \frac{3.1667}{1.0138} = 3.124
 \end{aligned}$$

$|t| = 3.12 > 2.571 \Rightarrow$ reject H_0 : the caching layer changes latency.



(A) uses divisor n in s (3.42); (B) forgets the $\sqrt{n}$ (d/s); (D) divides by $\sqrt{(n - 1)}$ instead of $\sqrt{n}$.

Trap: Treating the before/after columns as two independent samples throws away the pairing and usually inflates the standard error.

Q.22 [NAT · 2 marks]

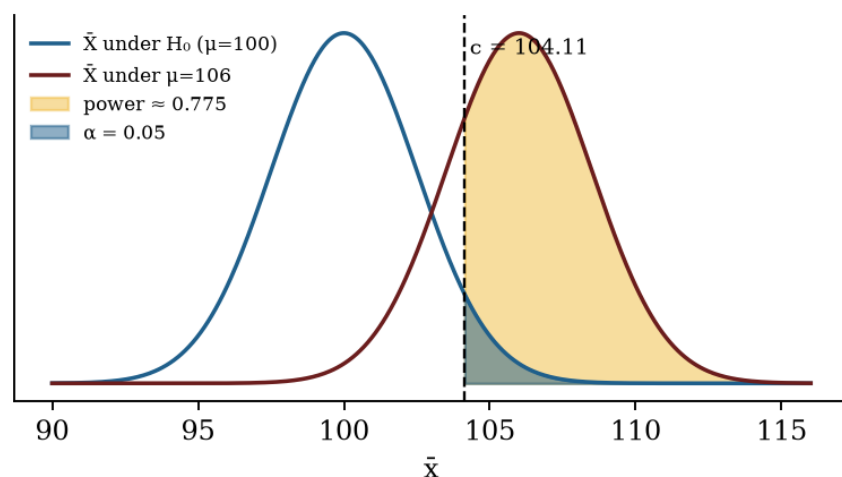
Topic: Power of a one-sided z-test · Hard

Correct answer: 0.77 (range 0.77 to 0.78)

Concept: find the rejection cut-off on the $\bar{x}$ scale under H_0 , then compute the probability of exceeding it under the alternative.

$$\sigma_{\bar{X}} = \frac{15}{\sqrt{36}} = 2.5, \quad c = 100 + 1.645(2.5) = 104.1125$$

$$\begin{aligned} \text{Power} &= P(\bar{X} > 104.1125 \mid \mu = 106) = P\left(Z > \frac{104.1125 - 106}{2.5}\right) \\ &= P(Z > -0.755) = \Phi(0.755) \approx 0.7749 \end{aligned}$$



$\beta = P(\text{Type II error}) \approx 0.225$.

Shortcut: Power formula in one line: $\Phi((\mu_1 - \mu_0)\sqrt{n}/\sigma - z_\alpha) = \Phi(2.4 - 1.645) = \Phi(0.755)$.

Q.23 [MCQ · 2 marks · -2/3]**Topic: Conditioning a Poisson sum · Hard****Correct answer: A****Concept:** if $X \sim \text{Poi}(\lambda_1)$, $Y \sim \text{Poi}(\lambda_2)$ independent, then $X | X + Y = n \sim \text{Bin}(n, \lambda_1/(\lambda_1 + \lambda_2))$.

Derivation:

$$\begin{aligned}
 P(X = 3 | X + Y = 8) &= \frac{P(X = 3)P(Y = 5)}{P(X + Y = 8)} = \frac{e^{-3} \frac{3^3}{3!} e^{-5} \frac{5^5}{5!}}{e^{-8} \frac{8^8}{8!}} \\
 &= \frac{8!}{3! 5!} \cdot \frac{3^3 5^5}{8^8} = \binom{8}{3} \left(\frac{3}{8}\right)^3 \left(\frac{5}{8}\right)^5 \\
 &= 56 \times 0.052734 \times 0.095367 \approx 0.2816
 \end{aligned}$$

(B) is the unconditional $P(X = 3)$; (C) is $3/8$ alone; (D) uses $p = 1/2$.

Key idea: This is the reverse of Poisson thinning: conditioning a superposition on its total gives a binomial split.

Q.24 [MSQ · 2 marks]**Topic: Competing exponentials; memorylessness · Hard****Correct answer: A, B, C****Concept:** competing exponentials — the minimum is $\text{Exp}(\lambda_1 + \lambda_2)$ and the winner is i with probability $\lambda_i/(\lambda_1 + \lambda_2)$.**(A) TRUE:**

$$P(T_1 < T_2) = \int_0^\infty e^{-t} e^{-2t} dt = \frac{1}{1+2} = \frac{1}{3}$$

(B) TRUE: $P(\min > t) = e^{-t} e^{-2t} = e^{-3t}$, mean $1/3$.**(C) TRUE:** $\max + \min = T_1 + T_2$, so

$$E[\max] = E[T_1] + E[T_2] - E[\min] = 1 + \frac{1}{2} - \frac{1}{3} = \frac{7}{6}$$

(D) FALSE: by memorylessness $P(T_1 > 2 | T_1 > 1) = P(T_1 > 1) = e^{-1}$.

Shortcut: $\max + \min = \text{sum of the two}$ — the fastest way to $E[\max]$ for any pair.

Q.25 [NAT · 2 marks]**Topic: Naive Bayes classifier (Bayes theorem) · Hard****Correct answer: 0.808 (range 0.806 to 0.809)****Concept:** Bayes theorem with a likelihood that factorises over features (conditional independence); absent words contribute $(1 - p)$.

$$\begin{aligned}
 P(x | S) &= 0.40 \times (1 - 0.02) = 0.392, & P(x | H) &= 0.05 \times (1 - 0.20) = 0.040 \\
 P(S | x) &= \frac{0.3(0.392)}{0.3(0.392) + 0.7(0.040)} = \frac{0.1176}{0.1176 + 0.028} \\
 &= \frac{0.1176}{0.1456} \approx 0.8077
 \end{aligned}$$

If you ignored the absent word 'meeting' you would get $0.12/(0.12 + 0.035) = 0.774$ — wrong.

Trap: Absence of a word is evidence too: it multiplies the likelihood by $(1 - p_{\text{word}})$.

Q.26 [MCQ · 2 marks · -2/3]

Topic: Conditional distributions; correlation · Hard

Correct answer: B**Concept:** use total variance for $\text{Var}(Y)$ and $\text{Cov}(X, Y) = \text{Cov}(X, E[Y | X])$.

$$\text{Var}(X) = \frac{6^2}{12} = 3, \quad E[Y | X] = X, \quad \text{Var}(Y | X) = 1$$

$$\text{Var}(Y) = E[1] + \text{Var}(X) = 1 + 3 = 4$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = E[XE[Y | X]] - E[X]^2 = \text{Var}(X) = 3$$

$$\rho = \frac{3}{\sqrt{3} \cdot 2} = \frac{\sqrt{3}}{2} \approx 0.866$$

(C) $3/4$ is ρ^2 (the fraction of $\text{Var } Y$ explained by X) — the square root was forgotten; (A) would need $\text{Var}(Y) = 12$, and (D) would need $\text{Var}(Y) = 9$ — both come from mis-adding the conditional and between-group variances.

Shortcut: When $Y = X + \text{noise}$, $\rho^2 = \text{signal variance} / \text{total variance} = 3/4$, so $\rho = \sqrt{3}/2$.

Q.27 [MCQ · 2 marks · -2/3]

Topic: Mean, median and mode of a continuous distribution · Hard

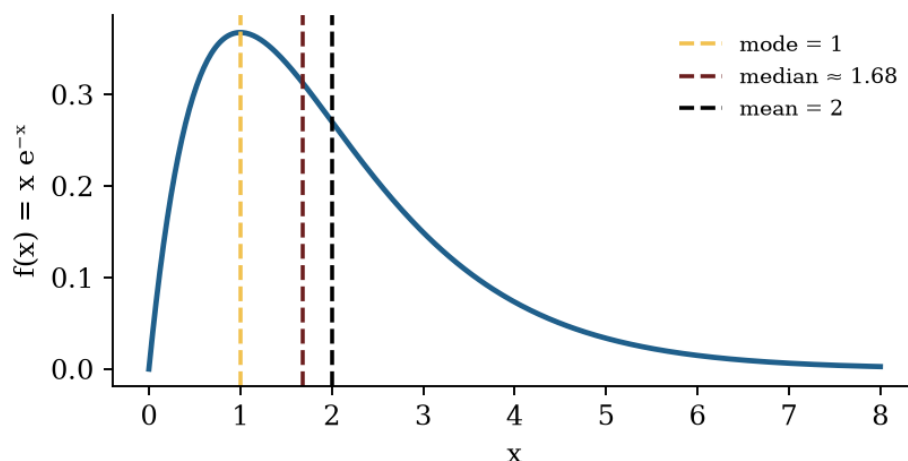
Correct answer: B**Concept:** mode maximises f ; median solves $F(m) = 1/2$; mean $= \int x f(x) dx$.Mode: $f'(x) = e^{-x}(1 - x) = 0 \Rightarrow x = 1$.

$$E[X] = \int_0^{\infty} x^2 e^{-x} dx = 2! = 2$$

CDF by integration by parts:

$$F(x) = \int_0^x t e^{-t} dt = 1 - (1 + x)e^{-x}$$

$$F(m) = \frac{1}{2} \Rightarrow (1 + m)e^{-m} = \frac{1}{2} \Rightarrow m \approx 1.678$$

Check: $(1 + 1.678)e^{-1.678} = 2.678 \times 0.1867 \approx 0.500$.Right-skewed $\Rightarrow \text{mode} < \text{median} < \text{mean}$, as in (B).

Key idea: Empirical rule for moderately skewed distributions: mean – mode $\approx 3(\text{mean} - \text{median})$: here $1 \text{ vs } 3(0.32) = 0.97$.

Q.28 [NAT · 2 marks]

Topic: t confidence interval from raw sums · Hard

Correct answer: 9.29 (range 9.25 to 9.33)

Concept: σ unknown, normal data $\Rightarrow \bar{x} \pm t_{\alpha/2, n-1} s/\sqrt{n}$ with s computed with divisor $n - 1$.

$$\begin{aligned}\bar{x} &= \frac{1920}{16} = 120 \\ s^2 &= \frac{\sum x_i^2 - n\bar{x}^2}{n-1} = \frac{231540 - 16(14400)}{15} = \frac{1140}{15} = 76 \\ s &= \sqrt{76} = 8.7178, \quad \frac{s}{\sqrt{n}} = \frac{8.7178}{4} = 2.1794 \\ \text{Width} &= 2(2.131)(2.1794) \approx 9.29\end{aligned}$$

The interval itself is 120 ± 4.64 , i.e. (115.36, 124.64).

Using $z = 1.96$ instead of t would give 8.54 — too narrow; using divisor n would give $s = \sqrt{71.25}$.

Trap: The question asks for the WIDTH ($2 \times$ margin), not the margin — a frequent point-loser.

Q.29 [MSQ · 2 marks]

Topic: Sampling distributions & Chi-squared / t tests · Hard

Correct answer: A, C, D

Concept: df for $r \times c$ independence = $(r - 1)(c - 1)$; sample variance of normal data is scaled χ^2 ; t critical values exceed z critical values.

(A) TRUE: $(3 - 1)(4 - 1) = 6$.

(B) FALSE: each term $(O - E)^2/E \geq 0$, so $\chi^2 \geq 0$ always.

(C) TRUE: standard result (Cochran); one df is lost because $\bar{X}$ estimates μ .

(D) TRUE: $t_{0.025, 9} = 2.262 > 1.96$.

Key idea: χ^2 tests are always right-tailed: large values signal disagreement in either direction.

Q.30 [MCQ · 2 marks · -2/3]**Topic: Combinatorial probability (card game) · Hard****Correct answer: D****Concept:** count by choosing ranks first (unordered when the roles are identical), then suits.Choose the 2 pair-ranks: $C(13, 2)$ — unordered, because both pairs play the same role.Suits for each pair: $C(4, 2)$ each. Fifth card: any of 11 remaining ranks $\times$ 4 suits = 44.

$$N = \binom{13}{2} \binom{4}{2}^2 (44) = 78 \times 36 \times 44 = 123552$$

$$P = \frac{123552}{\binom{52}{5}} = \frac{123552}{2598960} \approx 0.0475$$

(A) chooses the pair ranks as 13×12 (ordered), double counting; (C) 0.0211 is three-of-a-kind.**Trap:** Ask: are the chosen ranks interchangeable? Two pairs — yes (combination). Full house (triple + pair) — no (13×12).

Mock Test 10 — Advanced Full-Syllabus Test II (Grand Mock)

The final dress rehearsal — hardest GATE DA level, every syllabus topic, multi-step reasoning throughout.

Item	Details
Questions	30 (12 × 1-mark, 18 × 2-mark)
Maximum marks	48
Question types	MCQ: 14 · MSQ: 7 · NAT: 9
Suggested time	90 minutes
Negative marking	MCQ only: $-1/3$ (1-mark), $-2/3$ (2-mark). None for MSQ & NAT.
Calculator	Use only the GATE virtual scientific calculator

Topic focus of this set

The Grand Mock: a final-stage readiness check that sweeps the ENTIRE syllabus at peak GATE DA difficulty. Counting with upper bounds (inclusion-exclusion), probability bounds, indicator expectations, range of uniform order statistics, half-normal conditional mean, memoryless lifetimes, CLT for sums, reliability block diagrams with a Bayes twist, random sums with a geometric count, joint PMF tables, non-uniform joint pdfs over sub-regions, queue waiting times via minima of exponentials, hypergeometric variance through indicator covariances, Poisson thinning, a two-proportion z-test for A/B testing, t-test/CI duality, Poisson-mixture Bayes, pooled standard deviations, CLT sample-size design, carrier-risk genetics, and χ^2 tests of independence and goodness of fit with model-derived expected counts. Score above 60% here and you are exam-ready; anything less tells you exactly which chapters to revisit.

Instructions

- Questions 1 onwards are arranged as in GATE: all 1-mark questions first, then 2-mark questions.
- **MCQ** – exactly one correct option. **MSQ** – ONE OR MORE options are correct; credit only if you select all correct options and no wrong option (no partial marks, no negative marks).
- **NAT** – enter a numerical value using the virtual keypad. The accepted range is given in the key; round as instructed in the question.
- Statistical tables (Z, t, χ^2) are provided in the Appendix; values in solutions use those tables.
- Attempt in one sitting, then check the answer key and read every solution — including those you got right — for the alternative methods and traps.

Mock Test 10 · Question Paper

1-mark questions

Q.1 MCQ · 1 mark · -1/3

The number of non-negative integer solutions of $x_1 + x_2 + x_3 + x_4 = 12$ with every $x_i \leq 5$ is

- (A) 455
- (B) 131
- (C) 125
- (D) 119

Q.2 MSQ · 1 mark

Events A and B have $P(A) = 0.7$ and $P(B) = 0.6$. Which of the following is/are necessarily TRUE or possible as stated?

- (A) $P(A \cap B) \geq 0.3$ necessarily
- (B) $P(A \cap B) \leq 0.6$ necessarily
- (C) A and B can be mutually exclusive
- (D) $P(A \cup B) = 0.9$ is possible

Q.3 NAT · 1 mark

A fair die is rolled 6 times. The expected number of distinct faces that appear is _____ (round off to 2 decimal places).

Answer: _____

Q.4 MCQ · 1 mark · -1/3

$X_1, \dots, X_4$ are i.i.d. Uniform(0, 1). The expected value of the sample range $R = \max X_i - \min X_i$ is

- (A) $1/2$
- (B) $3/5$
- (C) $4/5$
- (D) $2/5$

Q.5 MCQ · 1 mark · -1/3

The number of defects in a 1 m length of optical fibre is Poisson. If $P(X = 1) = P(X = 2)$, then $P(X \geq 2)$ is

- (A) 0.865
- (B) 0.594
- (C) 0.323
- (D) 0.406

Q.6 NAT · 1 mark

$Z \sim N(0, 1)$. The value of $E[Z \mid Z > 0]$ is _____ (round off to 3 decimal places).

Answer: _____

Q.7 MSQ · 1 mark

Which of the following statements is/are TRUE?

- (A) $\text{Cov}(X, Y) = 0$ implies that X and Y are independent
- (B) $\text{Corr}(aX + b, cY + d) = \text{Corr}(X, Y)$ whenever $a > 0$ and $c > 0$
- (C) $|\text{Corr}(X, Y)| \leq 1$ for any X, Y with finite positive variances
- (D) If $X \sim U(-1, 1)$ and $Y = X^2$, then $\text{Corr}(X, Y) = 0$

Q.8 MCQ · 1 mark · -1/3

$X \sim \text{Binomial}(n, p)$ has mean 6 and variance 2.4. The most probable value (mode) of X is

- (A) 6
- (B) 7
- (C) 5
- (D) both 6 and 7

Q.9 MCQ · 1 mark · -1/3

$Z_1, \dots, Z_4$ are i.i.d. $N(0, 4)$ (variance 4). Let $Y = Z_1^2 + \dots + Z_4^2$. $\text{Var}(Y)$ equals

- (A) 8
- (B) 32
- (C) 64
- (D) 128

Q.10 NAT · 1 mark

LED lifetimes are exponential with mean 1000 hours. A particular LED has already worked for 500 hours. The median of its TOTAL lifetime, given this information, is _____ hours (round off to the nearest integer).

Answer: _____

Q.11 MCQ · 1 mark · -1/3

S is the sum of 100 i.i.d. $\text{Uniform}(0, 1)$ random variables. Using the CLT ($\Phi(1.73) \approx 0.9582$), $P(S > 55)$ is approximately

- (A) 0.042
- (B) 0.084
- (C) 0.023
- (D) 0.309

Q.12 MSQ · 1 mark

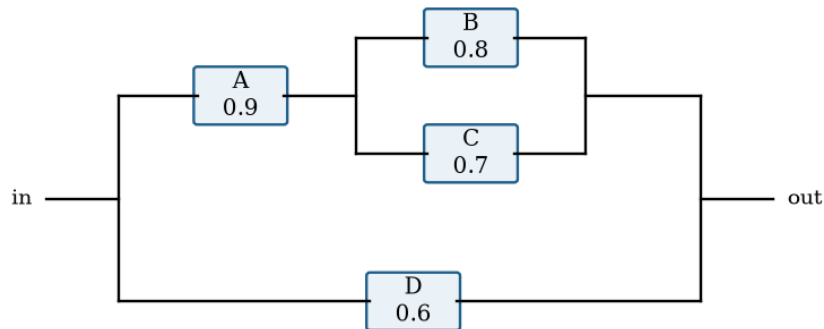
For confidence intervals for a population mean, which of the following is/are TRUE?

- (A) For fixed n and σ , a 99% interval is wider than a 95% interval
- (B) To halve the width of a z -interval, the sample size must be doubled
- (C) Once computed, a 95% interval (e.g. (48.1, 53.9)) contains μ with probability 0.95
- (D) With the same n and $s = \sigma$, the t -interval is wider than the z -interval

2-mark questions

Q.13 NAT · 2 marks

In the network below, signal passes from 'in' to 'out' if there is a path of working components. A is in series with the parallel pair (B, C), and this branch is in parallel with D. Components fail independently; their reliabilities are shown.



Given that the system is working, the probability that component D has FAILED is _____ (round off to 3 decimal places).

Answer: _____

Q.14 MCQ · 2 marks · -2/3

A player keeps playing rounds of a game until the first loss; each round is lost with probability 0.25, independently, and N is the total number of rounds played (including the losing round). In each round the player earns an amount with mean 2 and variance 1, independently of everything else. Let S be the total earnings over the N rounds. $\text{Var}(S)$ is

- (A) 4
- (B) 48
- (C) 52
- (D) 64

Q.15 MSQ · 2 marks

The joint PMF of X (number of retries) and Y (indicator of a timeout) is:

	$X = 0$	$X = 1$	$X = 2$
$Y = 0$	0.10	0.20	0.10
$Y = 1$	0.20	0.10	0.30

Which of the following is/are TRUE?

- (A) X and Y are NOT independent
- (B) $\text{Cov}(X, Y) = 0.04$
- (C) $E[Y | X = 2] = 0.75$
- (D) $P(X > Y) = 0.5$

Q.16 NAT · 2 marks

(X, Y) has joint pdf $f(x, y) = x + y$ for $0 < x < 1$, $0 < y < 1$, and 0 otherwise. $P(Y > 2X)$ is _____ (round off to 3 decimal places).

Answer: _____

Q.17 MCQ · 2 marks · -2/3

A help desk has 3 agents; service times are i.i.d. exponential with mean 10 minutes. You arrive to find all 3 agents busy and nobody else waiting. Your expected total time in the system (waiting + your own service) is

- (A) 10 min
- (B) $40/3$ min
- (C) $50/3$ min
- (D) 20 min

Q.18 NAT · 2 marks

Two classifiers are evaluated on 100 hold-out samples each; errors are graded as follows:

Model	Correct	Minor error	Major error	Total
A	60	25	15	100
B	40	35	25	100

The Pearson χ^2 statistic for testing whether outcome grade is independent of model is _____ (round off to 2 decimal places).

Answer: _____

Q.19 MCQ · 2 marks · -2/3

A class has 20 students of whom 8 are women. A committee of 5 is chosen at random without replacement. If X is the number of women on the committee, $\text{Var}(X)$ is closest to

- (A) 1.200
- (B) 0.947
- (C) 0.900
- (D) 1.520

Q.20 MSQ · 2 marks

Requests reach a load balancer as a Poisson process with rate 6 per minute. Each request is routed to server A with probability $1/3$ and to server B with probability $2/3$, independently. Which of the following is/are TRUE?

- (A) Arrivals at A form a Poisson process with rate 2 per minute
- (B) The number of requests reaching B in a 30-second interval has variance 4
- (C) Given that exactly 9 requests arrived in a minute, the number sent to A has mean 3 and variance 2
- (D) The mean time between consecutive arrivals at A is $1/6$ minute

Q.21 MCQ · 2 marks · -2/3

In an A/B test, the old page converts 120 of 1000 visitors and the new page 155 of 1000 visitors. Using the pooled two-proportion z-test of $H_0: p_A = p_B$ vs $H_1: p_A \neq p_B$ ($z_{0.025} = 1.96$, $z_{0.005} = 2.576$), which is correct?

- (A) $z \approx 2.27$; reject H_0 at 5% but not at 1%
- (B) $z \approx 2.27$; reject H_0 at both 5% and 1%
- (C) $z \approx 3.21$; reject H_0 at both 5% and 1%
- (D) $z \approx 1.61$; do not reject H_0 at 5%

Q.22 NAT · 2 marks

The inference latencies (ms) of two independent models are $X \sim N(10, 4)$ and $Y \sim N(12, 9)$ (second parameter = variance). The probability that model X is SLOWER than model Y on a request, $P(X > Y)$, is _____ (round off to 3 decimal places).

Answer: _____

Q.23 MCQ · 2 marks · -2/3

(X, Y) has joint pdf $f(x, y) = e^{-y}$ for $0 < x < y < \infty$. The correlation coefficient between X and Y is

- (A) $1/2$
- (B) $1/\sqrt{3}$
- (C) $1/\sqrt{2}$
- (D) $\sqrt{(2/3)}$

Q.24 MSQ · 2 marks

A sample of $n = 25$ battery lives gives $\bar{x} = 52$ h and $s = 5$ h (normal population). Consider $H_0: \mu = 50$. Given $t_{0.025, 24} = 2.064$ and $t_{0.05, 24} = 1.711$, which of the following is/are TRUE?

- (A) The two-sided test rejects H_0 at $\alpha = 0.05$
- (B) The one-sided test against $H_1: \mu > 50$ rejects H_0 at $\alpha = 0.05$
- (C) The 95% two-sided t-interval for μ contains 50
- (D) The two-sided p-value lies between 0.05 and 0.10

Q.25 NAT · 2 marks

An e-commerce site has two customer segments: 30% are 'heavy' buyers whose monthly number of purchases is Poisson with mean 4, and 70% are 'light' buyers with Poisson mean 1. A randomly chosen customer made exactly 3 purchases last month. The probability that the customer is a heavy buyer is _____ (round off to 3 decimal places).

Answer: _____

Q.26 MCQ · 2 marks · -2/3

X and Y are independent Uniform(0, 1). $P(XY \leq 1/2)$ equals

- (A) $\ln 2$
- (B) $(1 + \ln 2)/2$
- (C) $3/4$
- (D) $1/2$

Q.27 MCQ · 2 marks · -2/3

Section 1 has 20 students with mean score 50 and SD 4; section 2 has 30 students with mean 60 and SD 6 (SDs computed with divisor n). The SD (divisor n) of all 50 scores combined is closest to

- (A) 5.20
- (B) 5.29
- (C) 7.21
- (D) 10.00

Q.28 MSQ · 2 marks

Job durations on a cluster are i.i.d. exponential with unknown mean θ . An engineer wants the sample mean $\bar{X}$ of n jobs to satisfy $P(|\bar{X} - \theta| \leq 0.1\theta) \geq 0.95$, using the CLT ($z_{0.025} = 1.96$, $z_{0.005} = 2.576$). Which of the following is/are TRUE?

- (A) The smallest such n is 385
- (B) If the tolerance is tightened to 0.05 θ (still 95%), the smallest n is 1537
- (C) For tolerance 0.1 θ at 99% confidence, the smallest n is 664
- (D) The required n cannot be found without a pilot estimate of θ

Q.29 MCQ · 2 marks · -2/3

A recessive disorder requires two copies of allele a. A woman's brother is affected, both her parents are unaffected, and she herself is unaffected. She has three children with a man known to be a carrier (Aa), and all three children are unaffected. The probability that she is a carrier is

- (A) $2/3$
- (B) $1/2$
- (C) $27/59$
- (D) $9/32$

Q.30 NAT · 2 marks

The number of goals per match in 100 league matches is summarised below. A pundit claims goals per match follow a Poisson distribution with mean 1.5.

Goals	0	1	2	3 or more	Total
Matches	20	35	25	20	100

The Pearson χ^2 goodness-of-fit statistic (using the four categories shown, with expected counts from the Poisson(1.5) model) is _____ (round off to 2 decimal places).

Answer: _____

Mock Test 10 · Answer Key

Q	Type	Marks	Answer	Topic
1	MCQ	1	C	Counting with upper bounds (inclusion–exclusion)
2	MSQ	1	A, B, D	Probability axioms and bounds
3	NAT	1	3.99 (range 3.98 to 4.0)	Expectation via indicators
4	MCQ	1	B	Order statistics (range of uniforms)
5	MCQ	1	B	Poisson distribution
6	NAT	1	0.798 (range 0.797 to 0.799)	Conditional expectation (half-normal)
7	MSQ	1	B, C, D	Covariance and correlation properties
8	MCQ	1	A	Binomial: parameters and mode
9	MCQ	1	D	Chi-squared distribution (scaling)
10	NAT	1	1193 (range 1192 to 1194)	Exponential: memorylessness and median
11	MCQ	1	A	Central limit theorem
12	MSQ	1	A, D	Confidence-interval properties
13	NAT	2	0.361 (range 0.359 to 0.362)	System reliability with Bayes
14	MCQ	2	C	Random sum with geometric count
15	MSQ	2	A, B, C	Joint PMF, covariance, conditional expectation
16	NAT	2	0.208 (range 0.207 to 0.209)	Joint pdf over a sub-region
17	MCQ	2	B	Queues: minima of exponentials
18	NAT	2	8.17 (range 8.12 to 8.21)	Chi-squared test of independence
19	MCQ	2	B	Hypergeometric variance via indicator covariances
20	MSQ	2	A, C	Poisson thinning and conditioning
21	MCQ	2	A	Two-proportion z-test (A/B testing)
22	NAT	2	0.290 (range 0.287 to 0.292)	Linear combinations of normals
23	MCQ	2	C	Conditional pdf; covariance from conditional expectation
24	MSQ	2	B, C, D	One-sample t-test and CI duality
25	NAT	2	0.577 (range 0.576 to 0.579)	Bayes theorem with Poisson likelihoods
26	MCQ	2	B	Geometric probability (product of uniforms)
27	MCQ	2	C	Combined mean and standard deviation
28	MSQ	2	A, B, C	CLT sample-size design
29	MCQ	2	C	Bayes theorem in genetics (carrier risk)
30	NAT	2	0.35 (range 0.34 to 0.36)	Chi-squared goodness of fit to a Poisson model

Score tracker

	Attempted	Correct	Wrong (MCQ)	Marks
1-mark				
2-mark				
Total				/ 48

Mock Test 10 · Detailed Solutions

Q.1 [MCQ · 1 mark · -1/3]

Topic: Counting with upper bounds (inclusion-exclusion)
· Hard

Correct answer: C

Concept: stars and bars gives $C(n + k - 1, k - 1)$; subtract solutions violating the caps by inclusion-exclusion.

$$\text{Unrestricted} = \binom{12+3}{3} = \binom{15}{3} = 455$$

If $x_1 \geq 6$, put $x_1 = 6 + y_1$: $y_1 + x_2 + x_3 + x_4 = 6 \Rightarrow C(9, 3) = 84$ solutions; 4 choices of the variable.

Two variables ≥ 6 uses up all 12: remaining sum 0 $\Rightarrow C(3, 3) = 1$ solution; $C(4, 2) = 6$ pairs.

Three variables ≥ 6 is impossible ($18 > 12$).

$$N = 455 - 4(84) + 6(1) = 455 - 336 + 6 = 125$$

(D) 119 = 455 - 336 forgets to add back the double violations; (B) adds them back twice.

Trap: With caps, always ask 'can two (or three) caps be broken simultaneously?' — here exactly two can, contributing +6.

Q.2 [MSQ · 1 mark]

Topic: Probability axioms and bounds · Medium

Correct answer: A, B, D

Concept: Bonferroni lower bound $P(A \cap B) \geq P(A) + P(B) - 1$ and $P(A \cap B) \leq \min(P(A), P(B))$.

(A) TRUE: $P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq 0.7 + 0.6 - 1 = 0.3$.

(B) TRUE: $A \cap B \subset B$, so $P(A \cap B) \leq 0.6$.

(C) FALSE: mutual exclusivity needs $P(A \cap B) = 0$, which violates (A); equivalently $P(A \cup B)$ would be $1.3 > 1$.

(D) TRUE: $P(A \cup B) = 0.9$ corresponds to $P(A \cap B) = 0.4 \in [0.3, 0.6]$ — admissible (e.g. a uniform sample space with suitable intervals).

Key idea: Feasible range: $0.3 \leq P(A \cap B) \leq 0.6 \Leftrightarrow 0.7 \leq P(A \cup B) \leq 1$.

Q.3 [NAT · 1 mark]

Topic: Expectation via indicators · Medium

Correct answer: 3.99 (range 3.98 to 4.0)

Concept: linearity of expectation over indicators — no independence needed.

Let $I_j = 1$ if face j appears at least once ($j = 1, \dots, 6$).

$$E[I_j] = 1 - \left(\frac{5}{6}\right)^6 = 1 - 0.33490 = 0.66510$$

$$E[\text{distinct}] = 6 \times 0.66510 \approx 3.9906$$

Key idea: The indicators are dependent, but expectation is linear regardless — dependence would matter only for the variance.

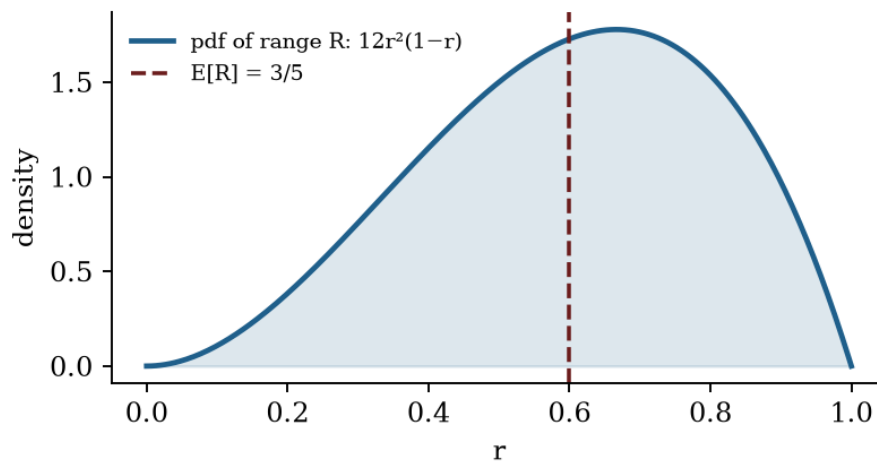
Q.4 [MCQ · 1 mark · -1/3]**Topic: Order statistics (range of uniforms) · Medium****Correct answer: B****Concept:** for n i.i.d. $U(0, 1)$, $E[\max] = n/(n + 1)$ and $E[\min] = 1/(n + 1)$.

$$E[\max] = \frac{4}{5}, \quad E[\min] = \frac{1}{5}$$

$$E[R] = \frac{4}{5} - \frac{1}{5} = \frac{3}{5}$$

Check via the pdf of the range, $f_R(r) = n(n - 1)r^{n-2}(1 - r) = 12r^2(1 - r)$:

$$E[R] = \int_0^1 12r^3(1 - r) dr = 12 \left(\frac{1}{4} - \frac{1}{5} \right) = \frac{3}{5}$$

(C) is $E[\max]$ alone; (A) is the mean of a single uniform.**Shortcut:** General result: $E[R] = (n - 1)/(n + 1)$.**Q.5** [MCQ · 1 mark · -1/3]**Topic: Poisson distribution · Medium****Correct answer: B****Concept:** equate the PMF values to find λ , then use the complement.

$$e^{-\lambda}\lambda = e^{-\lambda}\frac{\lambda^2}{2} \Rightarrow \lambda = 2$$

$$P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-2}(1 + 2) = 1 - 3e^{-2} \approx 0.5940$$

(A) is $P(X \geq 1)$; (C) is $P(X \geq 3)$; (D) is $P(X \leq 1)$.**Shortcut:** For Poisson, $P(k)/P(k - 1) = \lambda/k$ — the ratio method finds λ instantly.

Q.6 [NAT · 1 mark]**Topic: Conditional expectation (half-normal) · Hard****Correct answer: 0.798 (range 0.797 to 0.799)****Concept:** $E[Z | A] = E[Z \cdot 1_A] / P(A)$; for the normal density, $\phi'(z) = -z\phi(z)$.

$$E[Z 1_{\{Z > 0\}}] = \int_0^{\infty} z \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz = \frac{1}{\sqrt{2\pi}} [-e^{-z^2/2}]_0^{\infty} = \frac{1}{\sqrt{2\pi}}$$

$$E[Z | Z > 0] = \frac{1/\sqrt{2\pi}}{1/2} = \sqrt{\frac{2}{\pi}} \approx 0.7979$$

Key idea: Equivalently $E|Z| = \sqrt{2/\pi} \approx 0.798$ — the mean absolute deviation of a normal is about 0.8σ .

Q.7 [MSQ · 1 mark]**Topic: Covariance and correlation properties · Medium****Correct answer: B, C, D****Concept:** correlation measures LINEAR dependence only, is scale/shift invariant (up to sign) and bounded by Cauchy-Schwarz.**(A) FALSE:** zero covariance does not imply independence — (D) is a counter-example.**(B) TRUE:** $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$ and the SDs scale by $|a|$, $|c|$; ratio is unchanged when $ac > 0$ (sign flips if $ac < 0$).**(C) TRUE:** Cauchy-Schwarz $|\text{Cov}(X, Y)| \leq \sigma_X \sigma_Y$.**(D) TRUE:** $\text{Cov}(X, X^2) = E[X^3] - E[X]E[X^2] = 0 - 0 = 0$ by symmetry, though Y is a function of X .

Key idea: $Y = X^2$ with symmetric X is the standard 'uncorrelated but totally dependent' example.

Q.8 [MCQ · 1 mark · -1/3]**Topic: Binomial: parameters and mode · Medium****Correct answer: A****Concept:** np = mean, $np(1 - p)$ = variance; the binomial mode is $\lfloor (n + 1)p \rfloor$ (two modes only when $(n + 1)p$ is an integer).

$$1 - p = \frac{2.4}{6} = 0.4 \Rightarrow p = 0.6, \quad n = \frac{6}{0.6} = 10$$

$$(n + 1)p = 6.6 \Rightarrow \text{mode} = \lfloor 6.6 \rfloor = 6$$

$$\frac{P(7)}{P(6)} = \frac{(10 - 6)}{7} \cdot \frac{0.6}{0.4} = \frac{6}{7} < 1, \quad \frac{P(6)}{P(5)} = \frac{5}{6} \cdot 1.5 = 1.25 > 1$$

Trap: 'Both 6 and 7' would need $(n + 1)p = 7$ exactly.

Q.9 [MCQ · 1 mark · -1/3]

Topic: Chi-squared distribution (scaling) · Medium

Correct answer: D**Concept:** standardise first: $Z_i = 2W_i$ with $W_i \sim N(0, 1)$, so $Y = 4\sum W_i^2 = 4V$ with $V \sim \chi^2_4$.

$$\text{Var}(V) = 2k = 8, \quad \text{Var}(Y) = 4^2 \text{Var}(V) = 16 \times 8 = 128$$

(A) treats Y as χ^2_4 ; (B) and (C) apply the wrong scale factor (4 or 8) instead of $4^2 = 16$.**Trap:** A constant multiplier c on a random variable multiplies its variance by c^2 — here $c = \sigma^2 = 4$.**Q.10** [NAT · 1 mark]

Topic: Exponential: memorylessness and median · Medium

Correct answer: 1193 (range 1192 to 1194)**Concept:** memorylessness — the remaining life after survival to 500 h is again $\text{Exp}(\text{mean } 1000)$.Median m of $\text{Exp}(\text{mean } \theta)$: $e^{-m/\theta} = 1/2 \Rightarrow m = \theta \ln 2$.

$$m_{\text{remaining}} = 1000 \ln 2 \approx 693.1$$

$$m_{\text{total}} = 500 + 693.1 \approx 1193.1$$

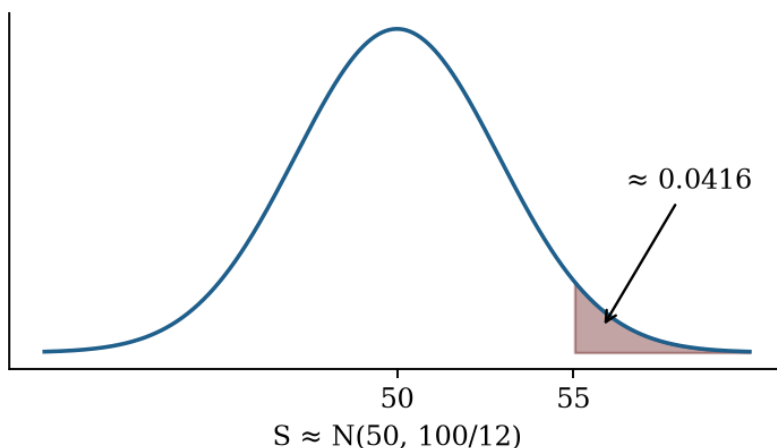
Key idea: The median of an exponential (0.6930) is smaller than its mean θ — right skew.**Q.11** [MCQ · 1 mark · -1/3]

Topic: Central limit theorem · Medium

Correct answer: A**Concept:** $S \approx N(n\mu, n\sigma^2)$ with $\mu = 1/2$ and $\sigma^2 = 1/12$ for $U(0, 1)$.

$$E[S] = 50, \quad \text{Var}(S) = \frac{100}{12} = 8.333, \quad \text{SD} = 2.887$$

$$P(S > 55) \approx P\left(Z > \frac{5}{2.887}\right) = P(Z > 1.732) \approx 0.042$$

(B) is the two-sided probability; (C) corresponds to $z = 2$, i.e. taking $\text{SD} = 2.5$; (D) $\approx P(Z > 0.5)$.**Trap:** Variance of $U(a, b)$ is $(b - a)^2/12$ — the 12 is the most-forgotten constant in CLT questions.

Q.12 [MSQ · 1 mark]**Topic: Confidence-interval properties · Medium****Correct answer: A, D****Concept:** width = $2 \times \text{critical value} \times \sigma/\sqrt{n}$; the confidence level refers to the procedure, not a computed interval.**(A) TRUE:** $2.576 > 1.96$.**(B) FALSE:** width $\propto 1/\sqrt{n}$, so halving the width needs 4 times the sample size.**(C) FALSE:** μ is fixed; a realised interval either contains it or not. '95%' is the long-run coverage of the method.**(D) TRUE:** $t_{\alpha/2, n-1} > z_{\alpha/2}$ for every finite n .**Shortcut:** Width scales like $1/\sqrt{n}$ — quadrupling n halves the width.**Q.13** [NAT · 2 marks]**Topic: System reliability with Bayes · Hard****Correct answer: 0.361 (range 0.359 to 0.362)****Concept:** series $\Rightarrow$ multiply reliabilities; parallel $\Rightarrow 1 - \text{product of failure probabilities}$; then Bayes on the event 'system works'.

$$R_{BC} = 1 - (0.2)(0.3) = 0.94, \quad R_1 = R_A R_{BC} = 0.9(0.94) = 0.846$$

$$R_{\text{sys}} = 1 - (1 - 0.846)(1 - 0.6) = 1 - 0.154(0.4) = 0.9384$$

If D has failed, the system works only through the upper branch:

$$P(D^c \cap \text{works}) = P(D^c) R_1 = 0.4(0.846) = 0.3384$$

$$P(D^c | \text{works}) = \frac{0.3384}{0.9384} \approx 0.3606$$

Monte-Carlo with 10^6 runs gives 0.3610.**Key idea:** Conditioning on success lowers the failure probability of D from 0.4 to 0.361 — but not by much, because the upper branch is itself very reliable.**Q.14** [MCQ · 2 marks · -2/3]**Topic: Random sum with geometric count · Hard****Correct answer: C****Concept:** $\text{Var}(S) = E[N]\text{Var}(X) + \text{Var}(N)(E[X])^2$ for a random sum with N independent of the X 's. N is geometric on $\{1, 2, \dots\}$ with success (loss) probability $p = 0.25$:

$$E[N] = \frac{1}{p} = 4, \quad \text{Var}(N) = \frac{1-p}{p^2} = \frac{0.75}{0.0625} = 12$$

$$\text{Var}(S) = E[N] \cdot 1 + \text{Var}(N) \cdot 2^2 = 4 + 48 = 52$$

 $E[S] = 4 \times 2 = 8$ for reference.(A) keeps only the first term; (B) only the second; (D) $= (E[N])^2(E[X])^2 = E[S]^2$.**Key idea:** The second term usually dominates: uncertainty in HOW MANY rounds matters more than uncertainty in each round's payoff.

Topic: Joint PMF, covariance, conditional expectation · Hard

Q.15 [MSQ · 2 marks]

Correct answer: A, B, C

Concept: marginals from row/column sums; independence needs EVERY cell to factor.

joint PMF $p(x, y)$

$y=1$	0.20	0.10	0.30
$y=0$	0.10	0.20	0.10
	$x=0$	$x=1$	$x=2$

Marginals: $P(X = 0, 1, 2) = 0.3, 0.3, 0.4$ and $P(Y = 0, 1) = 0.4, 0.6$.

(A) TRUE: $P(X = 0, Y = 0) = 0.10$ but $P(X = 0)P(Y = 0) = 0.12$.

(B) TRUE:

$$E[X] = 0.3 + 2(0.4) = 1.1, \quad E[Y] = 0.6, \quad E[XY] = 1(0.10) + 2(0.30) = 0.7$$

$$\text{Cov}(X, Y) = 0.7 - 1.1(0.6) = 0.7 - 0.66 = 0.04$$

(C) TRUE:

$$E[Y|X = 2] = P(Y = 1 | X = 2) = \frac{0.30}{0.40} = 0.75$$

(D) FALSE: $X > Y$ in cells $(1,0), (2,0), (2,1)$:

$$P(X > Y) = 0.20 + 0.10 + 0.30 = 0.60$$

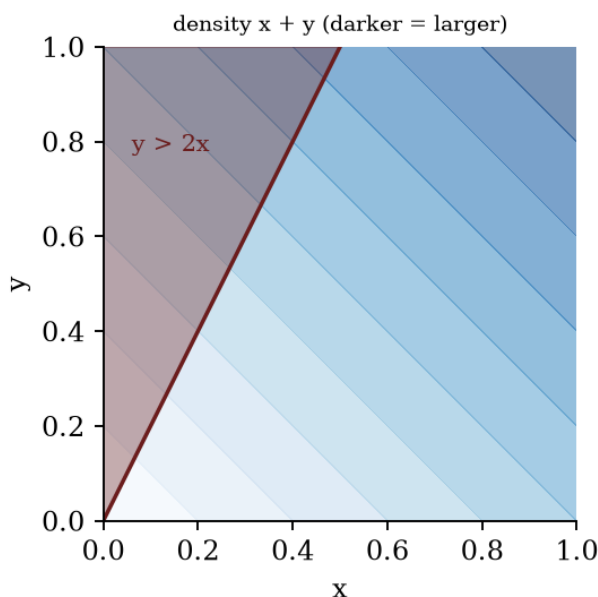
Shortcut: A single non-factoring cell is enough to rule out independence; you never need to check all six.

Q.16 [NAT · 2 marks]

Topic: Joint pdf over a sub-region · Hard

Correct answer: 0.208 (range 0.207 to 0.209)

Concept: integrate the joint pdf over the region; draw it first to set the limits.



$Y > 2X$ inside the unit square requires $x < 1/2$ and $2x < y < 1$.

$$P = \int_0^{1/2} \int_{2x}^1 (x+y) dy dx$$

$$\int_{2x}^1 (x+y) dy = x(1-2x) + \frac{1-4x^2}{2} = \frac{1}{2} + x - 4x^2$$

$$P = \left[\frac{x}{2} + \frac{x^2}{2} - \frac{4x^3}{3} \right]_0^{1/2} = \frac{1}{4} + \frac{1}{8} - \frac{1}{6} = \frac{5}{24}$$

$$P = \frac{5}{24} \approx 0.2083$$

Check: f is linear, so $P = (\text{area}) \times f(\text{centroid})$. The triangle has area $1/4$ and centroid $(1/6, 2/3)$, where $f = 5/6$; hence $P = (1/4)(5/6) = 5/24$. The region lies where x is small, so the density there is below its average value 1.

Trap: Using area alone (0.25) is valid only for UNIFORM joint pdfs.

Q.17 [MCQ · 2 marks · -2/3]

Topic: Queues: minima of exponentials · Hard

Correct answer: B

Concept: by memorylessness the three ongoing services restart afresh; the first completion is the minimum of 3 i.i.d. $\text{Exp}(\text{rate } 1/10)$, i.e. $\text{Exp}(\text{rate } 3/10)$.

$$E[W] = \frac{1}{3/10} = \frac{10}{3} \text{ min}$$

Your own service then takes $\text{Exp}(\text{mean } 10)$, independent of the wait.

$$E[\text{total}] = \frac{10}{3} + 10 = \frac{40}{3} \approx 13.33 \text{ min}$$

(C) $50/3 = 10/3 + 10 + 10/3$ double-counts the wait; (D) 20 treats the wait as a full service time, ignoring that 3 agents are competing to free up first; (A) ignores the wait altogether.

Key idea: Bonus: when you start service, the other two are still busy and memoryless, so $P(\text{you are the last of these three to finish}) = (2/3)(1/2) = 1/3$.

Q.18 [NAT · 2 marks]

Topic: Chi-squared test of independence · Hard

Correct answer: 8.17 (range 8.12 to 8.21)

Concept: expected count $E_{ij} = (\text{row total})(\text{column total})/n$; $\chi^2 = \sum (O - E)^2/E$ with $(r - 1)(c - 1)$ df.

Column totals: 100, 60, 40; row totals 100 each; $n = 200 \Rightarrow$ expected counts 50, 30, 20 in both rows.

Cell	O	E	$(O - E)^2/E$
A, correct	60	50	2.0000
A, minor	25	30	0.8333
A, major	15	20	1.2500
B, correct	40	50	2.0000
B, minor	35	30	0.8333
B, major	25	20	1.2500

$$\chi^2 = 2 \left(\frac{10^2}{50} + \frac{5^2}{30} + \frac{5^2}{20} \right) = 2(2 + 0.8333 + 1.25) = 8.1667$$

df = $(2 - 1)(3 - 1) = 2$. Since $8.17 > \chi^2_{0.05,2} = 5.991$ (but $< \chi^2_{0.01,2} = 9.210$), independence is rejected at 5% but not at 1%.

Shortcut: With equal row totals the two rows have identical expected counts and mirror-image deviations — compute one row and double it.

Q.19 [MCQ · 2 marks · -2/3]

Topic: Hypergeometric variance via indicator covariances · Hard

Correct answer: B

Concept: $X = \sum I_k$ over the 5 seats; the I_k are exchangeable but negatively correlated because sampling is without replacement.

$$P(I_k = 1) = \frac{8}{20} = 0.4, \quad \text{Var}(I_k) = 0.4(0.6) = 0.24$$

$$P(I_j I_k = 1) = \frac{8}{20} \cdot \frac{7}{19} = \frac{56}{380}, \quad \text{Cov}(I_j, I_k) = \frac{56}{380} - 0.16 = -\frac{0.24}{19}$$

$$\text{Var}(X) = 5(0.24) + 20 \left(-\frac{0.24}{19} \right) = 1.2 - 0.2526 = 0.9474$$

This matches the hypergeometric formula with finite-population correction:

$$\text{Var}(X) = np(1-p) \frac{N-n}{N-1} = 5(0.4)(0.6) \frac{15}{19} = 0.9474$$

(A) is the binomial (with-replacement) variance; (C) uses $(N - n)/N$; (D) inverts the correction.

Key idea: Without replacement $\Rightarrow$ negative covariances $\Rightarrow$ smaller variance than binomial.

Q.20 [MSQ · 2 marks]**Topic: Poisson thinning and conditioning · Hard****Correct answer: A, C**

Concept: thinning a Poisson(λ) process with probability p gives independent Poisson(λp) processes; conditioning on the total gives a binomial split.

(A) TRUE: rate $6 \times 1/3 = 2$ per minute.

(B) FALSE: B has rate 4 per minute; in 0.5 min the count is Poisson with mean 2, so its variance is 2, not 4.

(C) TRUE: given $N = 9$, the A-count $\sim \text{Bin}(9, 1/3)$:

$$E = 9 \cdot \frac{1}{3} = 3, \quad \text{Var} = 9 \cdot \frac{1}{3} \cdot \frac{2}{3} = 2$$

(D) FALSE: inter-arrival times at A are Exp(rate 2), mean 1/2 minute (1/6 is for the combined stream).

Trap: Poisson $\Rightarrow$ variance = mean; once you condition on the total, the variance shrinks to binomial $np(1 - p)$.

Q.21 [MCQ · 2 marks · -2/3]**Topic: Two-proportion z-test (A/B testing) · Hard****Correct answer: A**

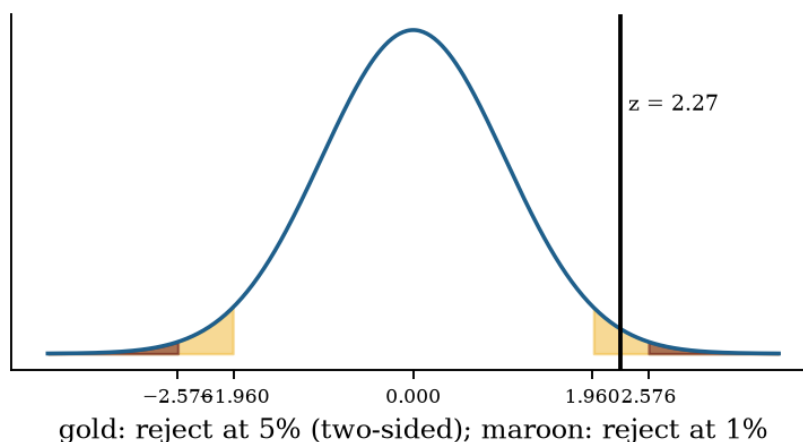
Concept: under H_0 the common p is estimated by pooling; $z = (\hat{p}_B - \hat{p}_A) / \sqrt{(\hat{p}(1 - \hat{p}))(1/n_A + 1/n_B)}$.

$$\hat{p} = \frac{120 + 155}{2000} = 0.1375$$

$$SE = \sqrt{0.1375(0.8625) \left(\frac{1}{1000} + \frac{1}{1000} \right)} = 0.01540$$

$$z = \frac{0.155 - 0.120}{0.01540} \approx 2.273$$

$1.96 < 2.27 < 2.576 \Rightarrow$ reject at 5%, not at 1% (two-sided p-value ≈ 0.023).



(C) forgets the factor $(1/n_A + 1/n_B)$, using only $1/1000$; (D) divides by an extra $\sqrt{2}$.

Trap: The SE of a DIFFERENCE adds the two variances — it is $\sqrt{2}$ times the one-sample SE when $n_A = n_B$.

Q.22 [NAT · 2 marks]

Topic: Linear combinations of normals · Hard

Correct answer: 0.290 (range 0.287 to 0.292)**Concept:** a difference of independent normals is normal with variances ADDED.

$$\begin{aligned}
 D &= X - Y \sim N(10 - 12, 4 + 9) = N(-2, 13) \\
 P(X > Y) &= P(D > 0) = P\left(Z > \frac{0 - (-2)}{\sqrt{13}}\right) = P(Z > 0.5547) \\
 &= 1 - \Phi(0.5547) \approx 0.2895
 \end{aligned}$$

(Table at $z = 0.55$ gives 0.2912, inside the accepted range.)**Trap:** $\text{Var}(X - Y) = \text{Var } X + \text{Var } Y$, never $\text{Var } X - \text{Var } Y$ (which would be negative here!).**Q.23** [MCQ · 2 marks · -2/3]

Topic: Conditional pdf; covariance from conditional expectation · Hard

Correct answer: C**Concept:** find marginals, use $X | Y = y$ to get $E[XY]$ by iterated expectation.

$$f_X(x) = \int_x^\infty e^{-y} dy = e^{-x} \Rightarrow X \sim \text{Exp}(1): E[X] = 1, \text{Var}(X) = 1$$

$$f_Y(y) = \int_0^y e^{-y} dx = ye^{-y}: E[Y] = 2, E[Y^2] = 6, \text{Var}(Y) = 2$$

Given $Y = y$, $f(x | y) = e^{-y}/(ye^{-y}) = 1/y$ on $(0, y)$: $X | Y = y \sim U(0, y)$.

$$E[XY] = E[Y E[X | Y]] = E\left[\frac{Y^2}{2}\right] = 3$$

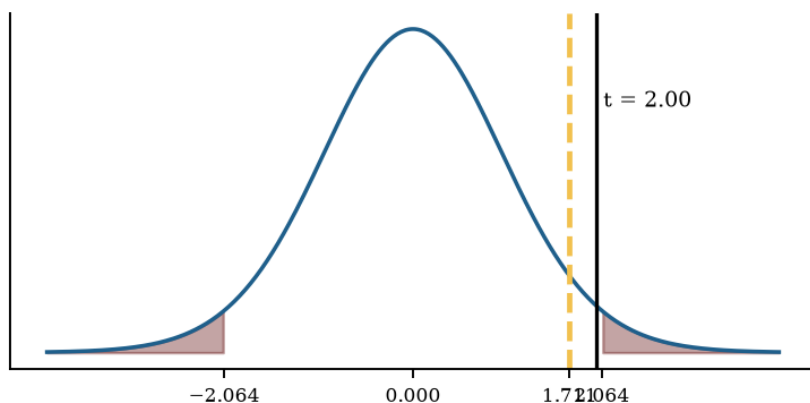
$$\text{Cov} = 3 - 1(2) = 1, \quad \rho = \frac{1}{\sqrt{1}\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.707$$

Interpretation: $Y = X + W$ where $W \sim \text{Exp}(1)$ is independent of X (memorylessness), so $\rho = \sqrt{(\text{Var } X / \text{Var } Y)} = 1/\sqrt{2}$.**Shortcut:** Spotting $Y = X +$ independent noise gives ρ without any integration.**Q.24** [MSQ · 2 marks]

Topic: One-sample t-test and CI duality · Hard

Correct answer: B, C, D**Concept:** $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$ with $n - 1$ df; a two-sided level- α test rejects iff μ_0 lies outside the $(1 - \alpha)$ interval.

$$t = \frac{52 - 50}{5/\sqrt{25}} = \frac{2}{1} = 2.00$$



t (24 df): maroon = two-sided 5% region; gold line = one-sided 5% cut-off

(A) **FALSE**: $|t| = 2.00 < 2.064$.

(B) **TRUE**: $t = 2.00 > 1.711$.

(C) **TRUE**: $52 \pm 2.064(1) = (49.936, 54.064) \ni 50$ — consistent with (A) failing.

(D) **TRUE**: one-sided $p \in (0.025, 0.05)$ because $1.711 < 2 < 2.064$, so the two-sided $p \in (0.05, 0.10)$. (Exact: 0.0569.)

Trap: The same data can be significant one-sided and not two-sided — the direction must be fixed BEFORE seeing the data.

Q.25 [NAT · 2 marks]

Topic: Bayes theorem with Poisson likelihoods · Hard

Correct answer: 0.577 (range 0.576 to 0.579)

Concept: posterior $\propto$ prior $\times$ likelihood, with Poisson likelihoods $e^{-\lambda} \lambda^k / k!$.

$$\begin{aligned}
 P(3 | H) &= e^{-4} \frac{4^3}{3!} = 0.19537, & P(3 | L) &= e^{-1} \frac{1}{3!} = 0.06131 \\
 P(H | 3) &= \frac{0.3(0.19537)}{0.3(0.19537) + 0.7(0.06131)} \\
 &= \frac{0.05861}{0.05861 + 0.04292} \approx 0.5773
 \end{aligned}$$

Ratio form (3! cancels): posterior odds = $(0.3/0.7) \times 64e^{-3} = 0.4286 \times 3.1864 = 1.3656 = P = 1.3656/2.3656 \approx 0.577$.

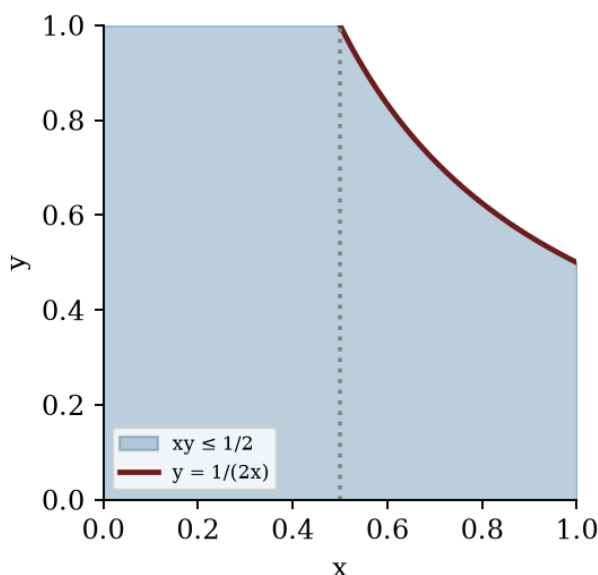
Shortcut: Cancel common factors (k!) before computing — fewer chances for arithmetic slips.

Topic: Geometric probability (product of uniforms) · Hard

Q.26 [MCQ · 2 marks · -2/3]

Correct answer: B

Concept: $P = \text{area of } \{(x, y) \in \text{unit square: } y \leq 1/(2x)\}$; split where the curve meets $y = 1$.



For $x \leq 1/2$, $1/(2x) \geq 1$ so the whole vertical strip counts (area $1/2$). For $x > 1/2$ the allowed height is $1/(2x)$:

$$P = \frac{1}{2} + \int_{1/2}^1 \frac{1}{2x} dx = \frac{1}{2} + \frac{1}{2} \ln 2 = \frac{1 + \ln 2}{2} \approx 0.8466$$

Monte-Carlo (10^6 points) gives 0.8465.

(A) $\ln 2 \approx 0.693 = \int_{1/2}^1 dx/x$ drops both the factor $1/2$ and the full strip $x \leq 1/2$; (D) wrongly assumes XY is symmetric about $1/2$.

Trap: Always find where the boundary curve leaves the unit square — that is where the integral must be split.

Q.27 [MCQ · 2 marks · -2/3]

Topic: Combined mean and standard deviation · Hard

Correct answer: C

Concept: total variance = average within-group variance + variance of group means (law of total variance for data).

$$\begin{aligned} \bar{x} &= \frac{20(50) + 30(60)}{50} = 56 \\ \text{within} &= \frac{20(16) + 30(36)}{50} = \frac{320 + 1080}{50} = 28 \\ \text{between} &= \frac{20(50 - 56)^2 + 30(60 - 56)^2}{50} = \frac{720 + 480}{50} = 24 \\ \sigma &= \sqrt{28 + 24} = \sqrt{52} \approx 7.211 \end{aligned}$$

(A) is the weighted average of SDs; (B) $\sqrt{28}$ ignores the between-group spread.

Trap: Never average standard deviations — combine VARIANCES and add the between-means term.

Q.28 [MSQ · 2 marks]

Topic: CLT sample-size design · Hard

Correct answer: A, B, C

Concept: for $\text{Exp}(\text{mean } \theta)$, $\text{SD} = \theta$, so $\text{SD}(X) = \theta/\sqrt{n}$; by the CLT the condition becomes $z_{\alpha/2} \theta/\sqrt{n} \leq c\theta$.

$$P\left(|Z| \leq \frac{0.1\theta}{\theta/\sqrt{n}}\right) \geq 0.95 \Rightarrow 0.1\sqrt{n} \geq 1.96$$

$$(A) n \geq \left(\frac{1.96}{0.1}\right)^2 = 384.16 \Rightarrow n = 385 \quad \text{TRUE}$$

$$(B) n \geq \left(\frac{1.96}{0.05}\right)^2 = 1536.64 \Rightarrow n = 1537 \quad \text{TRUE}$$

$$(C) n \geq \left(\frac{2.576}{0.1}\right)^2 = 663.58 \Rightarrow n = 664 \quad \text{TRUE}$$

(D) FALSE: θ cancels because both the tolerance and the SD are proportional to θ — the coefficient of variation of an exponential is 1.

Key idea: A RELATIVE tolerance with a distribution whose $\text{SD} \propto \text{mean}$ (exponential) makes the sample size parameter-free.

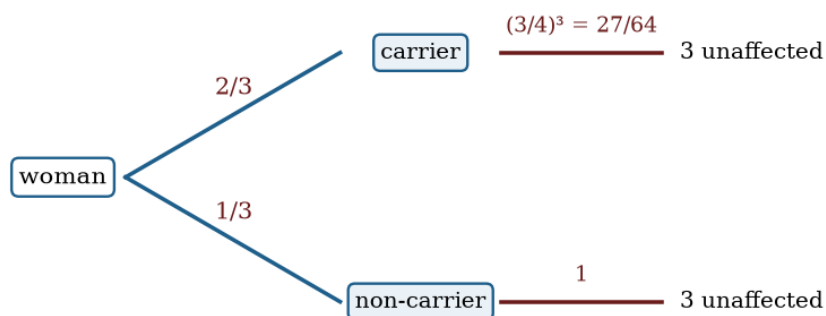
Q.29 [MCQ · 2 marks · -2/3]

Topic: Bayes theorem in genetics (carrier risk) · Hard

Correct answer: C

Concept: prior from Mendelian segregation conditioned on her being unaffected, then update with the likelihood of three unaffected children.

Affected brother $\Rightarrow$ both parents are Aa. Her genotype: AA : Aa : aa = 1 : 2 : 1, but she is unaffected so aa is excluded $\Rightarrow P(\text{carrier}) = 2/3$ (prior).



If she is Aa (and he is Aa), each child is affected with probability 1/4, so $P(3 \text{ unaffected}) = (3/4)^3 = 27/64$. If she is AA, no child can be aa, so $P(3 \text{ unaffected}) = 1$.

$$P(\text{carrier} | \text{data}) = \frac{\frac{2}{3} \cdot \frac{27}{64}}{\frac{2}{3} \cdot \frac{27}{64} + \frac{1}{3} \cdot 1} = \frac{54}{54 + 64} = \frac{27}{59}$$

$$\approx 0.4576$$

(A) is the prior (ignores the children); (B) is the naive 'Aa × AA' guess; (D) 9/32 is the joint numerator (2/3)(27/64) without normalising.

Key idea: Unaffected children are evidence AGAINST her being a carrier, lowering 2/3 to ≈ 0.458 .

Topic: Chi-squared goodness of fit to a Poisson model ·
Hard

Q.30 [NAT · 2 marks]

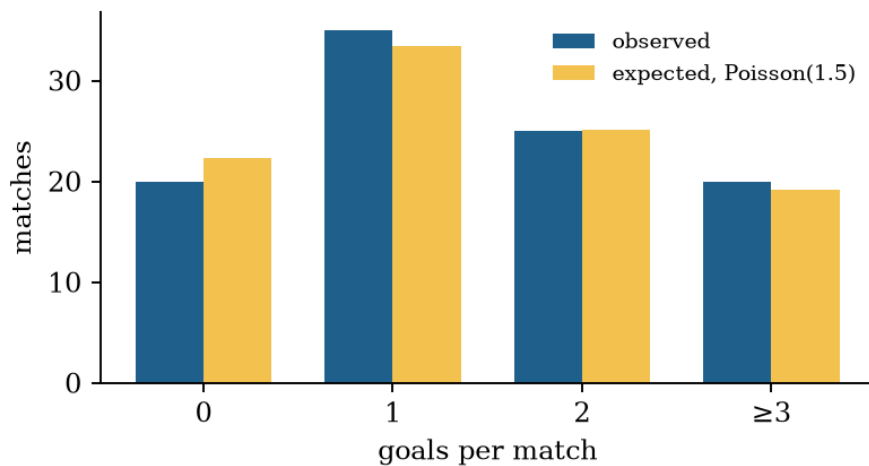
Correct answer: 0.35 (range 0.34 to 0.36)

Concept: derive the expected counts from the hypothesised PMF (lumping the tail into '3 or more'), then $\chi^2 = \sum (O - E)^2/E$ with $(k - 1)$ df (no parameter estimated).

$$p_0 = e^{-1.5} = 0.22313, \quad p_1 = 1.5e^{-1.5} = 0.33470, \quad p_2 = \frac{1.5^2}{2}e^{-1.5} = 0.25102$$

$$p_{3+} = 1 - (p_0 + p_1 + p_2) = 0.19115$$

Goals	O	E = 100p	(O - E) ² /E
0	20	22.313	0.2398
1	35	33.470	0.0700
2	25	25.102	0.0004
≥3	20	19.115	0.0409
Total	100	100	0.3511



$$\chi^2 \approx 0.351$$

df = 4 - 1 = 3 (λ was given, not estimated); $\chi^2_{0.05,3} = 7.815$, so the Poisson(1.5) claim is not rejected — the fit is good.

Trap: Had λ been estimated from the data, df would drop to 2. Always lump the tail so the probabilities sum to 1.

Part C · Appendix: Statistical Tables

C1. Standard normal CDF $\Phi(z) = P(Z \leq z)$

Row gives z to one decimal; column adds the second decimal. For negative z use $\Phi(-z) = 1 - \Phi(z)$. Example: $\Phi(1.96) = 0.9750$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

C2. Student t critical values $t_{\alpha, \nu}$ ($P(T > t_{\alpha, \nu}) = \alpha$)

For a two-sided test / CI at level α use the column $\alpha/2$. Example: 95% CI with $n = 10$ uses $t_{0.025, 9} = 2.262$.

$\nu \setminus \alpha$ (one tail)	0.1	0.05	0.025	0.01	0.005	0.001
1	3.078	6.314	12.706	31.821	63.657	318.309
2	1.886	2.920	4.303	6.965	9.925	22.327
3	1.638	2.353	3.182	4.541	5.841	10.215
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.893
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144
11	1.363	1.796	2.201	2.718	3.106	4.025
12	1.356	1.782	2.179	2.681	3.055	3.930
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
16	1.337	1.746	2.120	2.583	2.921	3.686
17	1.333	1.740	2.110	2.567	2.898	3.646
18	1.330	1.734	2.101	2.552	2.878	3.610
19	1.328	1.729	2.093	2.539	2.861	3.579
20	1.325	1.725	2.086	2.528	2.845	3.552
21	1.323	1.721	2.080	2.518	2.831	3.527
22	1.321	1.717	2.074	2.508	2.819	3.505
23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450
26	1.315	1.706	2.056	2.479	2.779	3.435
27	1.314	1.703	2.052	2.473	2.771	3.421
28	1.313	1.701	2.048	2.467	2.763	3.408
29	1.311	1.699	2.045	2.462	2.756	3.396
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
50	1.299	1.676	2.009	2.403	2.678	3.261
60	1.296	1.671	2.000	2.390	2.660	3.232
80	1.292	1.664	1.990	2.374	2.639	3.195
100	1.290	1.660	1.984	2.364	2.626	3.174
120	1.289	1.658	1.980	2.358	2.617	3.160
∞ (z)	1.282	1.645	1.960	2.326	2.576	3.090

C3. Chi-squared critical values $\chi^2_{\alpha, \nu}$ ($P(\chi^2 > \chi^2_{\alpha, \nu}) = \alpha$)

Columns give the area to the RIGHT. Example: $\chi^2_{0.05, 4} = 9.488$; for a variance CI with $\nu = 9$ at 95% use 19.023 and 2.700.

$\nu \setminus$ upper tail	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

Overall Performance Tracker

Record each attempt. A consistent score above 70% across Tests 07–10 indicates exam readiness for this section.

Test	Date	Time used	Attempted	Correct	Wrong	Score	%	Weak topics
01						/48		
02						/48		
03						/48		
04						/48		
05						/48		
06						/48		
07						/48		
08						/48		
09						/48		
10						/48		

Error log

Test · Q	Topic	Why I got it wrong (concept / calculation / misread / time)	Fix